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(54) **YEAST EXPRESSING A HETEROLOGOUS
TREHALASE FOR ETHANOL PRODUCTION**

(71) Applicant: **Novozymes A/S**, Bagsvaerd (DK)

(72) Inventors: **Michael Glenn Catlett**, West
Sacramento, CA (US); **Monica Tassone**,
West Sacramento, CA (US); **Paul**
Vincent Harris, Carnation, WA (US);
Robert Lyle Osborne, Raleigh, NC
(US); **Shiro Fukuyama**, Chiba (JP);
Tomoko Matsui, Chiba (JP); **Ryoko**
Kataoka, Longmont, CO (US)

(73) Assignee: **Novozymes A/S**, Bagsvaerd (DK)

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C12P 7/06 (2006.01)

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CPC **C12P 7/06** (2013.01); **C12P 2203/00**
(2013.01); **C12Y 302/01001** (2013.01); **C12Y**
302/01093 (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

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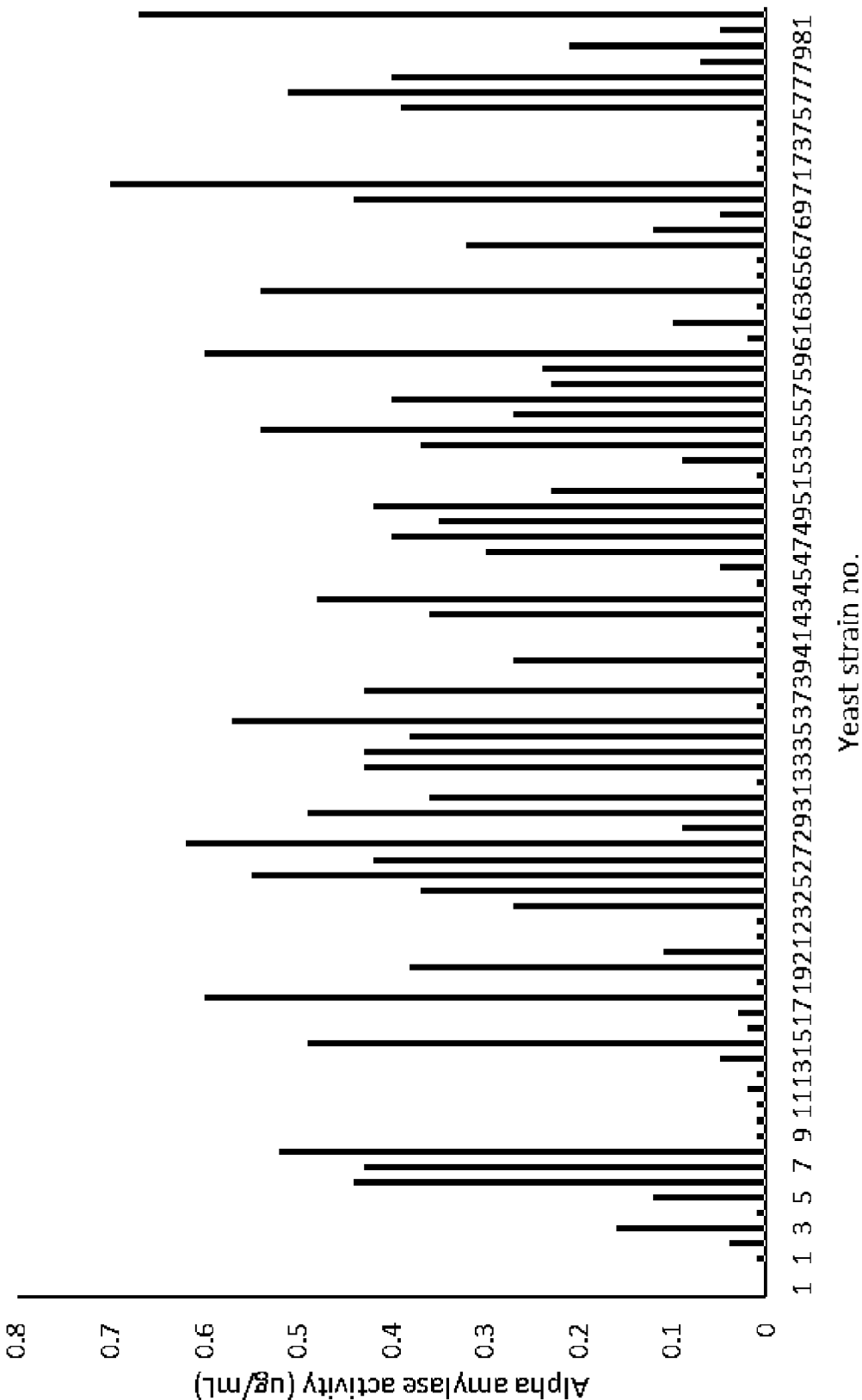
(74) *Attorney, Agent, or Firm* — Eric J. Fechter

(57) **ABSTRACT**

Described herein are recombinant fermenting organisms
having a heterologous polynucleotide encoding an alpha-
amylase and/or a heterologous polynucleotide encoding a
trehalase. Also described are processes for producing a
fermentation product, such as ethanol, from starch or cellu-
losic-containing material with the recombinant fermenting
organisms.

13 Claims, 3 Drawing Sheets

Specification includes a Sequence Listing.



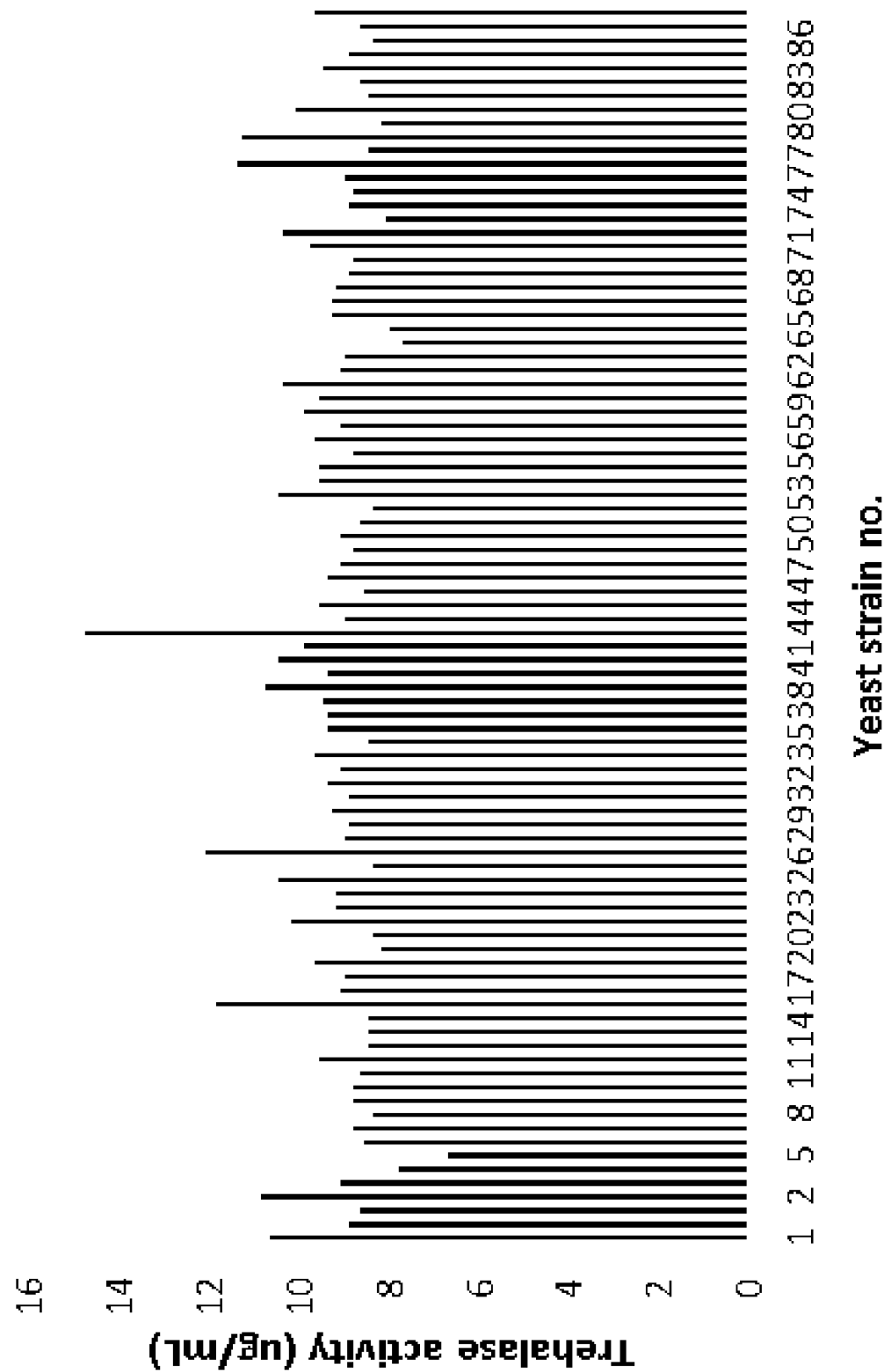


FIG. 2

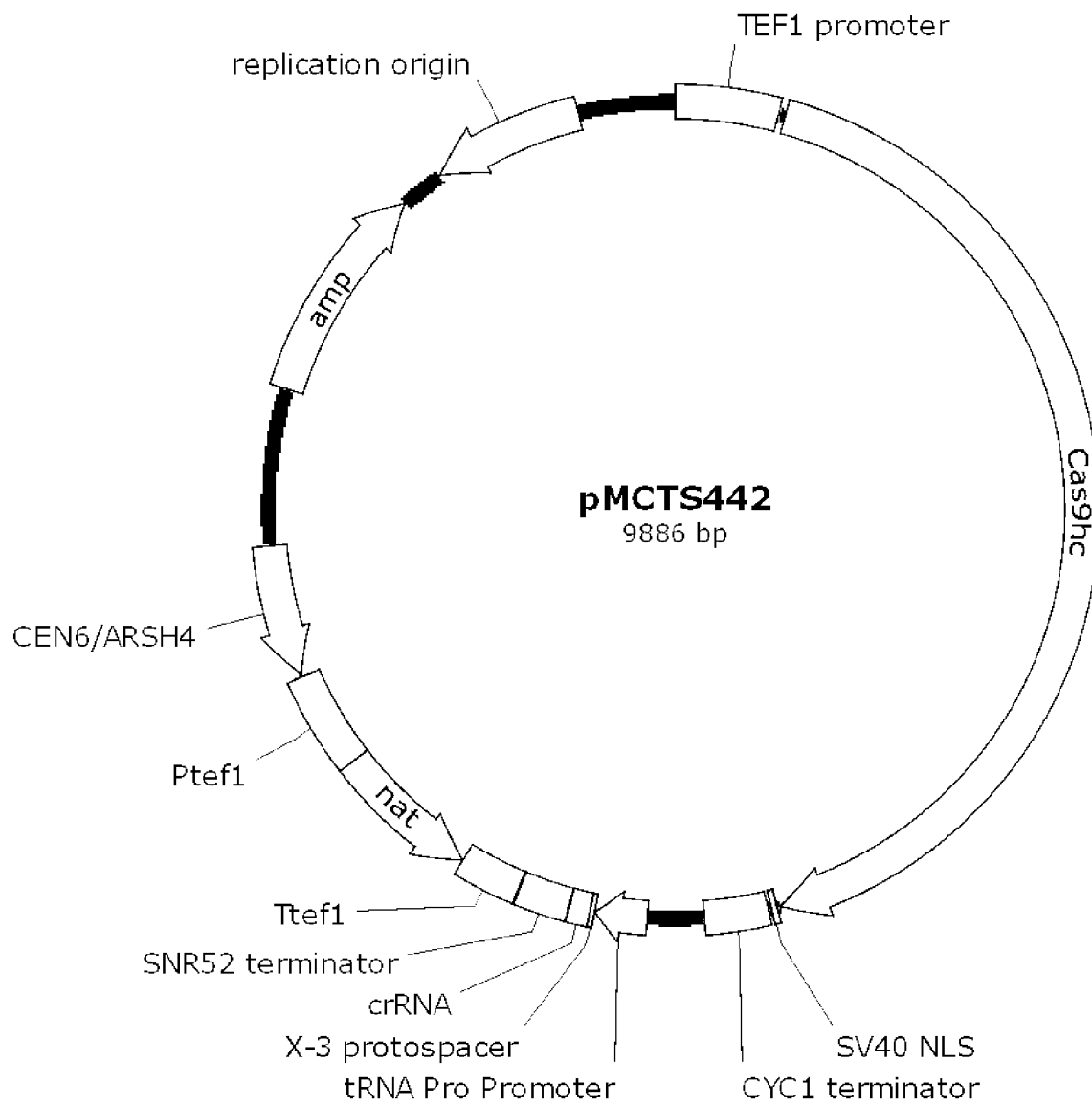


FIG. 3

YEAST EXPRESSING A HETEROLOGOUS TREHALASE FOR ETHANOL PRODUCTION

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a divisional of U.S. patent application Ser. No. 17/260,516, which is a 35 U.S.C. 371 national application of PCT/US2019/042870, filed Jul. 22, 2019, which claims priority or the benefit from U.S. Provisional Application Ser. No. 62/703,103, filed Jul. 25, 2018. The contents of these applications are fully incorporated herein by reference.

REFERENCE TO A SEQUENCE LISTING

This application contains a Sequence Listing in computer readable form created on Nov. 21, 2023, named SQ_ST26.xml and 335 KB in size, which is incorporated herein by reference.

BACKGROUND

Production of ethanol from starch and cellulosic containing materials is well-known in the art.

The most commonly industrially used commercial process for starch-containing material, often referred to as a “conventional process”, includes liquefying gelatinized starch at high temperature (about 85° C.) using typically a bacterial alpha-amylase, followed by simultaneous saccharification and fermentation (SSF) carried out anaerobically in the presence of typically a glucoamylase and a *Saccharomyces cerevisiae* yeast.

Yeasts which are used for production of ethanol for use as fuel, such as in the corn ethanol industry, require several characteristics to ensure cost effective production of the ethanol. These characteristics include ethanol tolerance, low by-product yield, rapid fermentation, and the ability to limit the amount of residual sugars remaining in the ferment. Such characteristics have a marked effect on the viability of the industrial process.

Yeast of the genus *Saccharomyces* exhibits many of the characteristics required for production of ethanol. In particular, strains of *Saccharomyces cerevisiae* are widely used for the production of ethanol in the fuel ethanol industry. Strains of *Saccharomyces cerevisiae* that are widely used in the fuel ethanol industry have the ability to produce high yields of ethanol under fermentation conditions found in, for example, the fermentation of corn mash. An example of such a strain is the yeast used in commercially available ethanol yeast product called ETHANOL RED®.

Saccharomyces cerevisiae yeast have been genetically engineered to express alpha-amylase and/or glucoamylase to improve yield and decrease the amount of exogenously added enzymes necessary during SSF (e.g., WO2018/098381, WO2017/087330, WO2017/037614, WO2011/128712, WO2011/153516, US2018/0155744). Yeast have also been engineered to express trehalase in an attempt to increase fermentation yield by breaking down residual trehalose (e.g., WO2017/077504).

Despite significant improvement of ethanol production processes over the past decade there is still a desire and need for providing improved processes of ethanol fermentation from starch and cellulosic containing material in an economically and commercially relevant scale.

SUMMARY

Described herein are, inter alia, methods for producing a fermentation product, such as ethanol, from starch or cellu-

losic-containing material, and yeast suitable for use in such processes. The Applicant has surprisingly found that yeast expressing certain alpha-amylases and/or trehalases provide beneficial properties that may be useful for ethanol fermentation.

A first aspect relates to methods of producing a fermentation product from a starch-containing or cellulosic-containing material comprising: (a) saccharifying the starch-containing or cellulosic-containing material; and (b) fermenting the saccharified material of step (a) with a fermenting organism; wherein the fermenting organism comprises a heterologous polynucleotide encoding an alpha-amylase or a heterologous polynucleotide encoding a trehalase.

In some embodiments of the methods, fermentation and saccharification are performed simultaneously in a simultaneous saccharification and fermentation (SSF). In other embodiments, fermentation and saccharification are performed sequentially (SHF).

In some embodiments of the methods, the method comprises recovering the fermentation product from the fermentation (e.g., by distillation).

In some embodiments of the methods, the fermentation product is ethanol.

In some embodiments of the methods, fermentation is performed under reduced nitrogen conditions (e.g., less than 1000 ppm urea or ammonium hydroxide, such as less than 750 ppm, less than 500 ppm, less than 400 ppm, less than 300 ppm, less than 250 ppm, less than 200 ppm, less than 150 ppm, less than 100 ppm, less than 75 ppm, less than 50 ppm, less than 25 ppm, or less than 10 ppm).

In some embodiments of the methods, the alpha-amylase has a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of any one of SEQ ID NOS: 76-101, 121-174 and 231. In some embodiments of the methods, the heterologous polynucleotide encodes an alpha-amylase having a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of SEQ ID NOS: 76-101, 121-174 and 231. In some embodiments of the methods, the heterologous polynucleotide encodes an alpha-amylase having a mature polypeptide sequence comprising or consisting of the amino acid sequence of any one of SEQ ID NOS: SEQ ID NOS: 76-101, 121-174 and 231.

In some embodiments of the methods, the trehalase has mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of any one of SEQ ID NOS: 175-226. In some embodiments of the methods, the heterologous polynucleotide encodes a trehalase having a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of SEQ ID NOS: 175-226. In some embodiments of the methods, the heterologous polynucleotide encodes a trehalase having a mature polypeptide sequence comprising or consisting of the amino acid sequence of any one of SEQ ID NOS: SEQ ID NOS: 175-226.

In some embodiments of the methods, saccharification of step (a) occurs on a starch-containing material, and wherein the starch-containing material is either gelatinized or ungelatinized starch.

In some embodiments of the methods, the method comprises liquefying the starch-containing material by contacting the material with an alpha-amylase prior to saccharification.

In some embodiments of the methods, liquefying the starch-containing material and/or saccharifying the starch-containing material is conducted in presence of exogenously added protease.

In some embodiments of the methods, the fermenting organism comprises a heterologous polynucleotide encoding a glucoamylase, such as a glucoamylase having a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of a *Pycnoporus* glycoamylase (e.g., a *Pycnoporus sanguineus* glucoamylase of SEQ ID NO: 229), a *Gloeophyllum* glucoamylase (e.g. a *Gloeophyllum sepiarium* of SEQ ID NO: 8), or a glucoamylase of any one of SEQ ID NOs: 102-113 (e.g., a *Saccharomycopsis fibuligera* glucoamylase of SEQ ID NO: 103 or 104, or a *Trichoderma reesei* glucoamylase of SEQ ID NO: 230).

In some embodiments of the methods, the fermenting organism comprises a heterologous polynucleotide encoding a protease, such as a protease having a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence of any one of SEQ ID NOs: 9-73 (e.g., any one of SEQ ID NOs: 9, 14, 16, 21, 22, 33, 41, 45, 61, 62, 66, 67, and 69; such as any one of SEQ NOs: 9, 14, 16, and 69).

In some embodiments of the methods, saccharification of step (a) occurs on a cellulosic-containing material, and wherein the cellulosic-containing material is pretreated (e.g. a dilute acid pretreatment).

In some embodiments of the methods, saccharification occurs on a cellulosic-containing material, and wherein the enzyme composition comprises one or more enzymes selected from a cellulase (e.g., endoglucanase, a cellobiohydrolase, or a beta-glucosidase), an AA9 polypeptide, a hemicellulase (e.g., a xylanase, an acetylxyylan esterase, a feruloyl esterase, an arabinofuranosidase, a xylosidase, or a glucuronidase), a CIP, an esterase, an expansin, a ligninolytic enzyme, an oxidoreductase, a pectinase, a protease, and a swollenin.

In some embodiments of the methods, the fermenting organism is a *Saccharomyces*, *Rhodotorula*, *Schizosaccharomyces*, *Kluyveromyces*, *Pichia*, *Hansenula*, *Rhodospiridium*, *Candida*, *Yarrowia*, *Lipomyces*, *Cryptococcus*, or *Dekkera* sp. cell. In some embodiments, the fermenting organism is a *Saccharomyces cerevisiae* cell.

Another aspect relates to a recombinant yeast cell comprising a heterologous polynucleotide encoding an alpha-amylase or a heterologous polynucleotide encoding a trehalase.

In some embodiments, the recombinant yeast cell is a *Saccharomyces*, *Rhodotorula*, *Schizosaccharomyces*, *Kluyveromyces*, *Pichia*, *Hansenula*, *Rhodospiridium*, *Candida*, *Yarrowia*, *Lipomyces*, *Cryptococcus*, or *Dekkera* sp. cell. In some embodiments, the recombinant yeast cell is a *Saccharomyces cerevisiae* cell.

In some embodiments of the yeast cell, the alpha-amylase has a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or

100% sequence identity, to the amino acid sequence of any one of SEQ ID NOs: 76-101, 121-174 and 231. In some embodiments of the methods, the heterologous polynucleotide encodes an alpha-amylase having a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of SEQ ID NOs: 76-101, 121-174 and 231. In some embodiments of the methods, the heterologous polynucleotide encodes an alpha-amylase having a mature polypeptide sequence comprising or consisting of the amino acid sequence of any one of SEQ ID NOs: SEQ ID NOs: 76-101, 121-174 and 231.

In some embodiments of the yeast cell, the trehalase has mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of any one of SEQ ID NOs: 175-226. In some embodiments of the methods, the heterologous polynucleotide encodes a trehalase having a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of SEQ ID NOs: 175-226. In some embodiments of the methods, the heterologous polynucleotide encodes a trehalase having a mature polypeptide sequence comprising or consisting of the amino acid sequence of any one of SEQ ID NOs: SEQ ID NOs: 175-226.

In some embodiments of the yeast cell, the fermenting organism comprises a heterologous polynucleotide encoding a glucoamylase, such as a glucoamylase having a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of a *Pycnoporus* glycoamylase (e.g., a *Pycnoporus sanguineus* glucoamylase of SEQ ID NO: 229), a *Gloeophyllum* glucoamylase (e.g. a *Gloeophyllum sepiarium* of SEQ ID NO: 8), or a glucoamylase of any one of SEQ ID NOs: 102-113 (e.g., a *Saccharomycopsis fibuligera* glucoamylase of SEQ ID NO: 103 or 104, or a *Trichoderma reesei* glucoamylase of SEQ ID NO: 230).

In some embodiments of the yeast cell, the fermenting organism comprises a heterologous polynucleotide encoding a protease, such as a protease having a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence of any one of SEQ ID NOs: 9-73 (e.g., any one of SEQ ID NOs: 9, 14, 16, 21, 22, 33, 41, 45, 61, 62, 66, 67, and 69; such as any one of SEQ NOs: 9, 14, 16, and 69).

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 shows alpha-amylase activity for strains constructed in Example 1.

FIG. 2 shows trehalase activity for strains constructed in Example 4.

FIG. 3 shows a plasmid map for pMcTs442.

DEFINITIONS

Unless defined otherwise or clearly indicated by context, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art.

Allelic variant: The term “allelic variant” means any of two or more alternative forms of a gene occupying the same chromosomal locus. Allelic variation arises naturally through mutation, and may result in polymorphism within populations. Gene mutations can be silent (no change in the encoded polypeptide) or may encode polypeptides having altered amino acid sequences. An allelic variant of a polypeptide is a polypeptide encoded by an allelic variant of a gene.

Alpha-amylase: The term “alpha amylase” means an 1,4-alpha-D-glucan glucanohydrolase, EC. 3.2.1.1, which catalyze hydrolysis of starch and other linear and branched 1,4-glucosidic oligo- and polysaccharides. For purposes of the present invention, alpha amylase activity can be determined using an alpha amylase assay described in the examples section below.

Auxiliary Activity 9: The term “Auxiliary Activity 9” or “AA9” means a polypeptide classified as a lytic polysaccharide monooxygenase (Quinlan et al., 2011, *Proc. Natl. Acad. Sci. USA* 208: 15079-15084; Phillips et al., 2011, *ACS Chem. Biol.* 6: 1399-1406; Lin et al., 2012, *Structure* 20: 1051-1061). AA9 polypeptides were formerly classified into the glycoside hydrolase Family 61 (GH61) according to Henrissat, 1991, *Biochem. J.* 280: 309-316, and Henrissat and Bairoch, 1996, *Biochem. J.* 316: 695-696.

AA9 polypeptides enhance the hydrolysis of a cellulosic-containing material by an enzyme having cellulolytic activity. Cellulolytic enhancing activity can be determined by measuring the increase in reducing sugars or the increase of the total of cellobiose and glucose from the hydrolysis of a cellulosic-containing material by cellulolytic enzyme under the following conditions: 1-50 mg of total protein/g of cellulose in pretreated corn stover (PCS), wherein total protein is comprised of 50-99.5% w/w cellulolytic enzyme protein and 0.5-50% w/w protein of an AA9 polypeptide for 1-7 days at a suitable temperature, such as 40 C-80° C., e.g., 50° C., 55° C., 60° C., 65° C., or 70° C., and a suitable pH, such as 4-9, e.g., 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 7.5, 8.0, or 8.5, compared to a control hydrolysis with equal total protein loading without cellulolytic enhancing activity (1-50 mg of cellulolytic protein/g of cellulose in PCS).

AA9 polypeptide enhancing activity can be determined using a mixture of CELLULAST® 1.5 L (Novozymes A/S, Bagsvaerd, Denmark) and beta-glucosidase as the source of the cellulolytic activity, wherein the beta-glucosidase is present at a weight of at least 2-5% protein of the cellulase protein loading. In one embodiment, the beta-glucosidase is an *Aspergillus oryzae* beta-glucosidase (e.g., recombinantly produced in *Aspergillus oryzae* according to WO02/095014). In another embodiment, the beta-glucosidase is an *Aspergillus fumigatus* beta-glucosidase (e.g., recombinantly produced in *Aspergillus oryzae* as described in WO02/095014).

AA9 polypeptide enhancing activity can also be determined by incubating an AA9 polypeptide with 0.5% phosphoric acid swollen cellulose (PASC), 100 mM sodium acetate pH 5, 1 mM MnSO₄, 0.1% gallic acid, 0.025 mg/ml of *Aspergillus fumigatus* beta-glucosidase, and 0.01% TRITON® X-100 (4-(1,1,3,3-tetramethylbutyl)phenyl-polyethylene glycol) for 24-96 hours at 40° C. followed by determination of the glucose released from the PASO.

AA9 polypeptide enhancing activity can also be determined according to WO2013/028928 for high temperature compositions.

AA9 polypeptides enhance the hydrolysis of a cellulosic-containing material catalyzed by enzyme having cellulolytic activity by reducing the amount of cellulolytic enzyme

required to reach the same degree of hydrolysis preferably at least 1.01-fold, e.g., at least 1.05-fold, at least 1.10-fold, at least 1.25-fold, at least 1.5-fold, at least 2-fold, at least 3-fold, at least 4-fold, at least 5-fold, at least 10-fold, or at least 20-fold.

Beta-glucosidase: The term “beta-glucosidase” means a beta-D-glucoside glucohydrolase (E.C. 3.2.1.21) that catalyzes the hydrolysis of terminal non-reducing beta-D-glucose residues with the release of beta-D-glucose. Beta-glucosidase activity can be determined using p-nitrophenyl-beta-D-glucopyranoside as substrate according to the procedure of Venturi et al., 2002, *J. Basic Microbiol.* 42: 55-66. One unit of beta-glucosidase is defined as 1.0 μmole of p-nitrophenolate anion produced per minute at 25° C., pH 4.8 from 1 mM p-nitrophenyl-beta-D-glucopyranoside as substrate in 50 mM sodium citrate containing 0.01% TWEEN® 20.

Beta-xylosidase: The term “beta-xylosidase” means a beta-D-xyloside xylohydrolase (E.C. 3.2.1.37) that catalyzes the exo-hydrolysis of short beta (1→4)-xylooligosaccharides to remove successive D-xylose residues from non-reducing termini. Beta-xylosidase activity can be determined using 1 mM p-nitrophenyl-beta-D-xyloside as substrate in 100 mM sodium citrate containing 0.01% TWEEN® 20 at pH 5, 40° C. One unit of beta-xylosidase is defined as 1.0 μmole of p-nitrophenolate anion produced per minute at 40° C., pH 5 from 1 mM p-nitrophenyl-beta-D-xyloside in 100 mM sodium citrate containing 0.01% TWEEN® 20.

Catalase: The term “catalase” means a hydrogen-peroxide:hydrogen-peroxide oxidoreductase (EC 1.11.1.6) that catalyzes the conversion of 2 H₂O₂ to O₂+2 H₂O. For purposes of the present invention, catalase activity is determined according to U.S. Pat. No. 5,646,025. One unit of catalase activity equals the amount of enzyme that catalyzes the oxidation of 1 μmole of hydrogen peroxide under the assay conditions.

Catalytic domain: The term “catalytic domain” means the region of an enzyme containing the catalytic machinery of the enzyme.

Cellobiohydrolase: The term “cellobiohydrolase” means a 1,4-beta-D-glucan cellobiohydrolase (E.C. 3.2.1.91 and E.C. 3.2.1.176) that catalyzes the hydrolysis of 1,4-beta-D-glucosidic linkages in cellulose, cellooligosaccharides, or any beta-1,4-linked glucose containing polymer, releasing cellobiose from the reducing end (cellobiohydrolase I) or non-reducing end (cellobiohydrolase II) of the chain (Teeri, 1997, *Trends in Biotechnology* 15: 160-167; Teeri et al., 1998, *Biochem. Soc. Trans.* 26: 173-178). Cellobiohydrolase activity can be determined according to the procedures described by Lever et al., 1972, *Anal. Biochem.* 47: 273-279; van Tilbeurgh et al., 1982, *FEBS Letters* 149: 152-156; van Tilbeurgh and Claeyssens, 1985, *FEBS Letters* 187: 283-288; and Tomme et al., 1988, *Eur. J. Biochem.* 170: 575-581.

Cellulolytic enzyme or cellulase: The term “cellulolytic enzyme” or “cellulase” means one or more (e.g., several) enzymes that hydrolyze a cellulosic-containing material. Such enzymes include endoglucanase(s), cellobiohydrolase(s), beta-glucosidase(s), or combinations thereof. The two basic approaches for measuring cellulolytic enzyme activity include: (1) measuring the total cellulolytic enzyme activity, and (2) measuring the individual cellulolytic enzyme activities (endoglucanases, cellobiohydrolases, and beta-glucosidases) as reviewed in Zhang et al., 2006, *Biotechnology Advances* 24: 452-481. Total cellulolytic enzyme activity can be measured using insoluble substrates, including Whatman N21 filter paper, microcrystalline cellulose,

bacterial cellulose, algal cellulose, cotton, pretreated ligno-cellulose, etc. The most common total cellulolytic activity assay is the filter paper assay using Whatman No 1 filter paper as the substrate. The assay was established by the International Union of Pure and Applied Chemistry (IUPAC) (Ghose, 1987, *Pure Appl. Chem.* 59: 257-68).

Cellulolytic enzyme activity can be determined by measuring the increase in production/release of sugars during hydrolysis of a cellulosic-containing material by cellulolytic enzyme(s) under the following conditions: 1-50 mg of cellulolytic enzyme protein/g of cellulose in pretreated corn stover (PCS) (or other pretreated cellulosic-containing material) for 3-7 days at a suitable temperature such as 40° C.-80° C., e.g., 50° C., 55° C., 60° C., 65° C., or 70° C., and a suitable pH such as 4-9, e.g., 5.0, 5.5, 6.0, 6.5, or 7.0, compared to a control hydrolysis without addition of cellulolytic enzyme protein. Typical conditions are 1 ml reactions, washed or unwashed PCS, 5% insoluble solids (dry weight), 50 mM sodium acetate pH 5, 1 mM MnSO₄, 50° C., 55° C., or 60° C., 72 hours, sugar analysis by AMINEX® HPX-87H column chromatography (Bio-Rad Laboratories, Inc., Hercules, CA, USA).

Coding sequence: The term "coding sequence" or "coding region" means a polynucleotide sequence, which specifies the amino acid sequence of a polypeptide. The boundaries of the coding sequence are generally determined by an open reading frame, which usually begins with the ATG start codon or alternative start codons such as GTG and TTG and ends with a stop codon such as TAA, TAG, and TGA. The coding sequence may be a sequence of genomic DNA, cDNA, a synthetic polynucleotide, and/or a recombinant polynucleotide.

Control sequence: The term "control sequence" means a nucleic acid sequence necessary for polypeptide expression. Control sequences may be native or foreign to the polynucleotide encoding the polypeptide, and native or foreign to each other. Such control sequences include, but are not limited to, a leader sequence, polyadenylation sequence, propeptide sequence, promoter sequence, signal peptide sequence, and transcription terminator sequence. The control sequences may be provided with linkers for the purpose of introducing specific restriction sites facilitating ligation of the control sequences with the coding region of the polynucleotide encoding a polypeptide.

Disruption: The term "disruption" means that a coding region and/or control sequence of a referenced gene is partially or entirely modified (such as by deletion, insertion, and/or substitution of one or more nucleotides) resulting in the absence (inactivation) or decrease in expression, and/or the absence or decrease of enzyme activity of the encoded polypeptide. The effects of disruption can be measured using techniques known in the art such as detecting the absence or decrease of enzyme activity using from cell-free extract measurements referenced herein; or by the absence or decrease of corresponding mRNA (e.g., at least 25% decrease, at least 50% decrease, at least 60% decrease, at least 70% decrease, at least 80% decrease, or at least 90% decrease); the absence or decrease in the amount of corresponding polypeptide having enzyme activity (e.g., at least 25% decrease, at least 50% decrease, at least 60% decrease, at least 70% decrease, at least 80% decrease, or at least 90% decrease); or the absence or decrease of the specific activity of the corresponding polypeptide having enzyme activity (e.g., at least 25% decrease, at least 50% decrease, at least 60% decrease, at least 70% decrease, at least 80% decrease, or at least 90% decrease). Disruptions of a particular gene of interest can be generated by methods known in the art, e.g.,

by directed homologous recombination (see *Methods in Yeast Genetics* (1997 edition), Adams, Gottschling, Kaiser, and Stems, Cold Spring Harbor Press (1998)).

Endogenous gene: The term "endogenous gene" means a gene that is native to the referenced host cell. "Endogenous gene expression" means expression of an endogenous gene.

Endoglucanase: The term "endoglucanase" means a 4-(1, 3;1,4)-beta-D-glucan 4-glucanohydrolase (E.C. 3.2.1.4) that catalyzes endohydrolysis of 1,4-beta-D-glycosidic linkages in cellulose, cellulose derivatives (such as carboxymethyl cellulose and hydroxyethyl cellulose), lichenin, beta-1,4 bonds in mixed beta-1,3-1,4 glucans such as cereal beta-D-glucans or xyloglucans, and other plant material containing cellulosic components. Endoglucanase activity can be determined by measuring reduction in substrate viscosity or increase in reducing ends determined by a reducing sugar assay (Zhang et al., 2006, *Biotechnology Advances* 24: 452-481). Endoglucanase activity can also be determined using carboxymethyl cellulose (CMC) as substrate according to the procedure of Ghose, 1987, *Pure and Appl. Chem.* 59: 257-268, at pH 5, 40° C.

Expression: The term "expression" includes any step involved in the production of the polypeptide including, but not limited to, transcription, post-transcriptional modification, translation, post-translational modification, and secretion. Expression can be measured—for example, to detect increased expression—by techniques known in the art, such as measuring levels of mRNA and/or translated polypeptide.

Expression vector: The term "expression vector" means a linear or circular DNA molecule that comprises a polynucleotide encoding a polypeptide and is operably linked to control sequences that provide for its expression.

Fermentable medium: The term "fermentable medium" or "fermentation medium" refers to a medium comprising one or more (e.g., two, several) sugars, such as glucose, fructose, sucrose, cellobiose, xylose, xylulose, arabinose, mannose, galactose, and/or soluble oligosaccharides, wherein the medium is capable, in part, of being converted (fermented) by a host cell into a desired product, such as ethanol. In some instances, the fermentation medium is derived from a natural source, such as sugar cane, starch, or cellulose, and may be the result of pretreating the source by enzymatic hydrolysis (saccharification). The term fermentation medium is understood herein to refer to a medium before the fermenting organism is added, such as, a medium resulting from a saccharification process, as well as a medium used in a simultaneous saccharification and fermentation process (SSF).

Glucoamylase: The term "glucoamylase" (1,4-alpha-D-glucan glucohydrolase, EC 3.2.1.3) is defined as an enzyme that catalyzes the release of D-glucose from the non-reducing ends of starch or related oligo- and polysaccharide molecules. For purposes of the present invention, glucoamylase activity may be determined according to the procedure described in the Examples herein.

Hemicellulolytic enzyme or hemicellulase: The term "hemicellulolytic enzyme" or "hemicellulase" means one or more (e.g., several) enzymes that hydrolyze a hemicellulosic material. See, for example, Shallom and Shoham, 2003, *Current Opinion In Microbiology* 6(3): 219-228). Hemicellulases are key components in the degradation of plant biomass. Examples of hemicellulases include, but are not limited to, an acetylmannan esterase, an acetylxyylan esterase, an arabinanase, an arabinofuranosidase, a coumaric acid esterase, a feruloyl esterase, a galactosidase, a glucuronidase, a glucuronoyl esterase, a mannanase, a mannosidase, a xylanase, and a xylosidase. The substrates for

these enzymes, hemicelluloses, are a heterogeneous group of branched and linear polysaccharides that are bound via hydrogen bonds to the cellulose microfibrils in the plant cell wall, crosslinking them into a robust network. Hemicelluloses are also covalently attached to lignin, forming together with cellulose a highly complex structure. The variable structure and organization of hemicelluloses require the concerted action of many enzymes for its complete degradation. The catalytic modules of hemicellulases are either glycoside hydrolases (GHs) that hydrolyze glycosidic bonds, or carbohydrate esterases (CEs), which hydrolyze ester linkages of acetate or ferulic acid side groups. These catalytic modules, based on homology of their primary sequence, can be assigned into GH and CE families. Some families, with an overall similar fold, can be further grouped into clans, marked alphabetically (e.g., GH-A). A most informative and updated classification of these and other carbohydrate active enzymes is available in the Carbohydrate-Active Enzymes (CAZy) database. Hemicellulolytic enzyme activities can be measured according to Ghose and Bisaria, 1987, *Pure & Appl. Chem.* 59: 1739-1752, at a suitable temperature such as 40° C.-80° C., e.g., 50° C., 55° C., 60° C., 65° C., or 70° C., and a suitable pH such as 4-9, e.g., 5.0, 5.5, 6.0, 6.5, or 7.0. Heterologous polynucleotide: The term "heterologous polynucleotide" is defined herein as a polynucleotide that is not native to the host cell; a native polynucleotide in which structural modifications have been made to the coding region; a native polynucleotide whose expression is quantitatively altered as a result of a manipulation of the DNA by recombinant DNA techniques, e.g., a different (foreign) promoter; or a native polynucleotide in a host cell having one or more extra copies of the polynucleotide to quantitatively alter expression. A "heterologous gene" is a gene comprising a heterologous polynucleotide.

High stringency conditions: The term "high stringency conditions" means for probes of at least 100 nucleotides in length, prehybridization and hybridization at 42° C. in 5×SSPE, 0.3% SDS, 200 micrograms/ml sheared and denatured salmon sperm DNA, and 50% formamide, following standard Southern blotting procedures for 12 to 24 hours. The carrier material is finally washed three times each for 15 minutes using 0.2×SSC, 0.2% SDS at 65° C.

Host cell: The term "host cell" means any cell type that is susceptible to transformation, transfection, transduction, and the like with a nucleic acid construct or expression vector comprising a polynucleotide described herein (e.g., a polynucleotide encoding an alpha-amylase and/or trehalase). The term "host cell" encompasses any progeny of a parent cell that is not identical to the parent cell due to mutations that occur during replication. The term "recombinant cell" is defined herein as a non-naturally occurring host cell comprising one or more (e.g., two, several) heterologous polynucleotides.

Low stringency conditions: The term "low stringency conditions" means for probes of at least 100 nucleotides in length, prehybridization and hybridization at 42° C. in 5×SSPE, 0.3% SDS, 200 micrograms/ml sheared and denatured salmon sperm DNA, and 25% formamide, following standard Southern blotting procedures for 12 to 24 hours. The carrier material is finally washed three times each for 15 minutes using 0.2×SSC, 0.2% SDS at 50° C.

Mature polypeptide: The term "mature polypeptide" is defined herein as a polypeptide having biological activity that is in its final form following translation and any post-translational modifications, such as N-terminal processing, C-terminal truncation, glycosylation, phosphorylation, etc. The mature polypeptide sequence lacks a signal sequence,

which may be determined using techniques known in the art (See, e.g., Zhang and Henzel, 2004, *Protein Science* 13: 2819-2824).

Medium stringency conditions: The term "medium stringency conditions" means for probes of at least 100 nucleotides in length, prehybridization and hybridization at 42° C. in 5×SSPE, 0.3% SDS, 200 micrograms/ml sheared and denatured salmon sperm DNA, and 35% formamide, following standard Southern blotting procedures for 12 to 24 hours. The carrier material is finally washed three times each for 15 minutes using 0.2×SSC, 0.2% SDS at 55° C.

Medium-high stringency conditions: The term "medium-high stringency conditions" means for probes of at least 100 nucleotides in length, prehybridization and hybridization at 42° C. in 5×SSPE, 0.3% SDS, 200 micrograms/ml sheared and denatured salmon sperm DNA, and 35% formamide, following standard Southern blotting procedures for 12 to 24 hours. The carrier material is finally washed three times each for 15 minutes using 0.2×SSC, 0.2% SDS at 60° C.

Nucleic acid construct: The term "nucleic acid construct" means a polynucleotide comprises one or more (e.g., two, several) control sequences. The polynucleotide may be single-stranded or double-stranded, and may be isolated from a naturally occurring gene, modified to contain segments of nucleic acids in a manner that would not otherwise exist in nature, or synthetic.

Operably linked: The term "operably linked" means a configuration in which a control sequence is placed at an appropriate position relative to the coding sequence of a polynucleotide such that the control sequence directs expression of the coding sequence.

Pretreated corn stover: The term "Pretreated Corn Stover" or "PCS" means a cellulosic-containing material derived from corn stover by treatment with heat and dilute sulfuric acid, alkaline pretreatment, neutral pretreatment, or any pretreatment known in the art.

Protease: The term "protease" is defined herein as an enzyme that hydrolyses peptide bonds. It includes any enzyme belonging to the EC 3.4 enzyme group (including each of the thirteen subclasses thereof). The EC number refers to Enzyme Nomenclature 1992 from NC-IUBMB, Academic Press, San Diego, California, including supplements 1-5 published in *Eur. J. Biochem.* 223: 1-5 (1994); *Eur. J. Biochem.* 232: 1-6 (1995); *Eur. J. Biochem.* 237: 1-5 (1996); *Eur. J. Biochem.* 250: 1-6 (1997); and *Eur. J. Biochem.* 264: 610-650 (1999); respectively. The term "subtilases" refer to a sub-group of serine protease according to Siezen et al., 1991, *Protein Engng.* 4: 719-737 and Siezen et al., 1997, *Protein Science* 6: 501-523. Serine proteases or serine peptidases is a subgroup of proteases characterised by having a serine in the active site, which forms a covalent adduct with the substrate. Further the subtilases (and the serine proteases) are characterised by having two active site amino acid residues apart from the serine, namely a histidine and an aspartic acid residue. The subtilases may be divided into 6 sub-divisions, i.e. the Subtilisin family, the Thermolysin family, the Proteinase K family, the Lantibiotic peptidase family, the Kexin family and the Pyrolysins family. The term "protease activity" means a proteolytic activity (EC 3.4). Protease activity may be determined using methods described in the art (e.g., US 2015/0125925) or using commercially available assay kits (e.g., Sigma-Aldrich).

Pullulanase: The term "pullulanase" means a starch debranching enzyme having pullulan 6-glucano-hydrolase activity (EC 3.2.1.41) that catalyzes the hydrolysis the α -1,6-glycosidic bonds in pullulan, releasing maltotriose with reducing carbohydrate ends. For purposes of the pres-

ent invention, pullulanase activity can be determined according to a PHADEBAS assay or the sweet potato starch assay described in WO2016/087237.

Sequence Identity: The relatedness between two amino acid sequences or between two nucleotide sequences is described by the parameter “sequence identity”.

For purposes described herein, the degree of sequence identity between two amino acid sequences is determined using the Needleman-Wunsch algorithm (Needleman and Wunsch, *J. Mol. Biol.* 1970, 48, 443-453) as implemented in the Needle program of the EMBOSS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., *Trends Genet* 2000, 16, 276-277), preferably version 3.0.0 or later. The optional parameters used are gap open penalty of 10, gap extension penalty of 0.5, and the EBLOSUM62 (EMBOSS version of BLOSUM62) substitution matrix. The output of Needle labeled “longest identity” (obtained using the -nobrief option) is used as the percent identity and is calculated as follows:

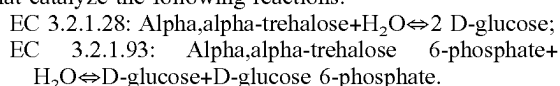
$$\frac{(\text{Identical Residues} \times 100)}{(\text{Length of the Referenced Sequence} - \text{Total Number of Gaps in Alignment})}$$

For purposes described herein, the degree of sequence identity between two deoxyribonucleotide sequences is determined using the Needleman-Wunsch algorithm (Needleman and Wunsch, 1970, supra) as implemented in the Needle program of the EMBOSS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., 2000, supra), preferably version 3.0.0 or later. The optional parameters used are gap open penalty of 10, gap extension penalty of 0.5, and the EDNAFULL (EMBOSS version of NCBI NUC4.4) substitution matrix. The output of Needle labeled “longest identity” (obtained using the -nobrief option) is used as the percent identity and is calculated as follows:

$$\frac{(\text{Identical Deoxyribonucleotides} \times 100)}{(\text{Length of Referenced Sequence} - \text{Total Number of Gaps in Alignment})}$$

Signal peptide: The term “signal peptide” is defined herein as a peptide linked (fused) in frame to the amino terminus of a polypeptide having biological activity and directs the polypeptide into the cell’s secretory pathway. Signal sequences may be determined using techniques known in the art (See, e.g., Zhang and Henzel, 2004, *Protein Science* 13: 2819-2824). The polypeptides described herein may comprise any suitable signal peptide known in the art, or any signal peptide described herein (e.g., the *S. cerevisiae* MF α 1 signal peptide of SEQ ID NO: 7, the *S. cerevisiae* EXG1 signal peptide of SEQ ID NO: 227, or the *S. cerevisiae* AG2 signal peptide of SEQ ID NO: 234, or a signal peptide having at least 80%, 85%, 90%, 95%, 96%, 97%, 98% or 99% sequence identity thereof).

Trehalase: The term “trehalase” means an enzyme which degrades trehalose into its unit monosaccharides (i.e., glucose). Trehalases are classified in EC 3.2.1.28 (alpha,alpha-trehalase) and EC. 3.2.1.93 (alpha,alpha-phosphotrehalase). The EC classes are based on recommendations of the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology (IUBMB). Description of EC classes can be found on the internet, e.g., on “http://www.expasy.org/enzyme/”. Trehalases are enzymes that catalyze the following reactions:



For purposes of the present invention, trehalase activity may be determined according to the trehalase activity assay described herein in the experimental section.

Very high stringency conditions: The term “very high stringency conditions” means for probes of at least 100 nucleotides in length, prehybridization and hybridization at 42° C. in 5 $\times$ SSPE, 0.3% SDS, 200 micrograms/ml sheared and denatured salmon sperm DNA, and 50% formamide, following standard Southern blotting procedures for 12 to 24 hours. The carrier material is finally washed three times each for 15 minutes using 0.2 $\times$ SSC, 0.2% SDS at 70° C.

Very low stringency conditions: The term “very low stringency conditions” means for probes of at least 100 nucleotides in length, prehybridization and hybridization at 42° C. in 5 $\times$ SSPE, 0.3% SDS, 200 micrograms/ml sheared and denatured salmon sperm DNA, and 25% formamide, following standard Southern blotting procedures for 12 to 24 hours. The carrier material is finally washed three times each for 15 minutes using 0.2 $\times$ SSC, 0.2% SDS at 45° C.

Xylanase: The term “xylanase” means a 1,4-beta-D-xylan-xylohydrolase (E.C. 3.2.1.8) that catalyzes the endohydrolysis of 1,4-beta-D-xylosidic linkages in xylans. Xylanase activity can be determined with 0.2% AZCL-arabinoxylan as substrate in 0.01% TRITON® X-100 and 200 mM sodium phosphate pH 6 at 37° C. One unit of xylanase activity is defined as 1.0 μ mole of azurine produced per minute at 37° C., pH 6 from 0.2% AZCL-arabinoxylan as substrate in 200 mM sodium phosphate pH 6.

Xylose Isomerase: The term “Xylose Isomerase” or “XI” means an enzyme which can catalyze D-xylose into D-xylose in vivo, and convert D-glucose into D-fructose in vitro. Xylose isomerase is also known as “glucose isomerase” and is classified as E.C. 5.3.1.5. As the structure of the enzyme is very stable, the xylose isomerase is a good model for studying the relationships between protein structure and functions (Karimaki et al., *Protein Eng Des Sel*, 12004, 17 (12):861-869). Xylose Isomerase activity may be determined using techniques known in the art (e.g., a coupled enzyme assay using D-sorbitol dehydrogenase, as described by Verhoeven et. al., 2017, *Sci Rep* 7, 46155).

Reference to “about” a value or parameter herein includes embodiments that are directed to that value or parameter per se. For example, description referring to “about X” includes the embodiment “X”. When used in combination with measured values, “about” includes a range that encompasses at least the uncertainty associated with the method of measuring the particular value, and can include a range of plus or minus two standard deviations around the stated value.

Likewise, reference to a gene or polypeptide that is “derived from” another gene or polypeptide X, includes the gene or polypeptide X.

As used herein and in the appended claims, the singular forms “a,” “or,” and “the” include plural referents unless the context clearly dictates otherwise.

It is understood that the embodiments described herein include “consisting” and/or “consisting essentially of” embodiments. As used herein, except where the context requires otherwise due to express language or necessary implication, the word “comprise” or variations such as “comprises” or “comprising” is used in an inclusive sense, i.e. to specify the presence of the stated features but not to preclude the presence or addition of further features in various embodiments.

DETAILED DESCRIPTION

Described herein, inter alia, are methods for producing a fermentation product, such as ethanol, from starch or cellulosic containing material.

During industrial scale fermentation, yeast encounter various physiological challenges including variable concentrations of sugars, high concentrations of yeast metabolites such as ethanol, glycerol, organic acids, osmotic stress, as well as potential competition from contaminating microbes such as wild yeasts and bacteria. As a consequence, many yeasts are not suitable for use in industrial fermentation. The most widely used commercially available industrial strain of *Saccharomyces* (i.e. for industrial scale fermentation) is the *Saccharomyces cerevisiae* strain used, for example, in the product ETHANOL RED®. This strain is well suited to industrial ethanol production; however, it remains unclear how modifications to the yeast will impact performance. In particular, the functional expression of heterologous enzymes by an industrially-relevant *Saccharomyces cerevisiae* yeast is uncertain (See, for example U.S. Pat. No. 9,206,444 where the applicant was unable to functionally express numerous enzymes/enzyme classes).

The Applicant has surprisingly found that yeast expressing certain alpha-amylases and/or trehalases provide beneficial properties that may be useful for ethanol fermentation.

In one aspect is a method of producing a fermentation product from a starch-containing or cellulosic-containing material comprising:

- (a) saccharifying the starch-containing or cellulosic-containing material; and
- (b) fermenting the saccharified material of step (a) with a fermenting organism;

wherein the fermenting organism comprises a heterologous polynucleotide encoding an alpha-amylase or a heterologous polynucleotide encoding a trehalase.

Steps of saccharifying and fermenting are carried out either sequentially or simultaneously (SSF). In one embodiment, steps of saccharifying and fermenting are carried out simultaneously (SSF). In another embodiment, steps of saccharifying and fermenting are carried out sequentially. Fermenting Organism

The fermenting organism described herein may be derived from any host cell known to the skilled artisan capable of producing a fermentation product, such as ethanol. As used herein, a "derivative" of strain is derived from a referenced strain, such as through mutagenesis, recombinant DNA technology, mating, cell fusion, or cytoduction between yeast strains. Those skilled in the art will understand that the genetic alterations, including metabolic modifications exemplified herein, may be described with reference to a suitable host organism and their corresponding metabolic reactions or a suitable source organism for desired genetic material such as genes for a desired metabolic pathway. However, given the complete genome sequencing of a wide variety of organisms and the high level of skill in the area of genomics, those skilled in the art can apply the teachings and guidance provided herein to other organisms. For example, the metabolic alterations exemplified herein can readily be applied to other species by incorporating the same or analogous encoding nucleic acid from species other than the referenced species.

The host cells for preparing the recombinant cells described herein can be from any suitable host, such as a yeast strain, including, but not limited to, a *Saccharomyces*, *Rhodotorula*, *Schizosaccharomyces*, *Kluyveromyces*, *Pichia*, *Hansenula*, *Rhodospiridium*, *Candida*, *Yarrowia*, *Lipomyces*, *Cryptococcus*, or *Dekkera* sp. cell. In particular, *Saccharomyces* host cells are contemplated, such as *Saccharomyces cerevisiae*, *bayanus* or *carlsbergensis* cells. Preferably, the yeast cell is a *Saccharomyces cerevisiae* cell.

Suitable cells can, for example, be derived from commercially available strains and polyploid or aneuploid industrial strains, including but not limited to those from Superstart™, THERMOSACC®, C5 FUEL™, XyloFerm®, etc. (Lallemand); RED STAR and ETHANOL RED® (Fermentis/Lesaffre); FALI (AB Mauri); Baker's Best Yeast, Baker's Compressed Yeast, etc. (Fleishmann's Yeast); BIOFERM AFT, XP, CF, and XR (North American Bioproducts Corp.); Turbo Yeast (Gert Strand AB); and FERMOL® (DSM Specialties). Other useful yeast strains are available from biological depositories such as the American Type Culture Collection (ATCC) or the Deutsche Sammlung von Mikroorganismen und Zellkulturen GmbH (DSMZ), such as, e.g., BY4741 (e.g., ATCC 201388); Y108-1 (ATCC PTA. 10567) and NRRL YB-1952 (ARS Culture Collection). Still other *S. cerevisiae* strains suitable as host cells DBY746, [Alpha] [Eta]22, 5150-2B, GPY55-15Ba, CEN.PK, USM21, TMB3500, TMB3400, VTT-A-63015, VTT-A-85068, VTT-c-79093 and their derivatives as well as *Saccharomyces* sp. 1400, 424A (LNH-ST), 259A (LNH-ST) and derivatives thereof. In one embodiment, the recombinant cell is a derivative of a strain *Saccharomyces cerevisiae* CIBTS1260 (deposited under Accession No. NRRL Y-50973 at the Agricultural Research Service Culture Collection (NRRL), Illinois 61604 U.S.A.).

The fermenting organism may be *Saccharomyces* strain, e.g., *Saccharomyces cerevisiae* strain produced using the method described and concerned in U.S. Pat. No. 8,257,959-BB.

The strain may also be a derivative of *Saccharomyces cerevisiae* strain NMI V14/004037 (See, WO2015/143324 and WO2015/143317 each incorporated herein by reference), strain nos. V15/004035, V15/004036, and V15/004037 (See, WO2016/153924 incorporated herein by reference), strain nos. V15/001459, V15/001460, V15/001461 (See, WO2016/138437 incorporated herein by reference), strain no. NRRL Y67342 (See, WO2017/063159 incorporated herein by reference), or any strain described in WO2017/087330 (incorporated herein by reference).

The fermenting organisms according to the invention have been generated in order to improve fermentation yield and to improve process economy by cutting enzyme costs since part or all of the necessary enzymes needed to improve method performance are produced by the fermenting organism.

The fermenting organisms described herein may utilize expression vectors comprising the coding sequence of one or more (e.g., two, several) heterologous genes linked to one or more control sequences that direct expression in a suitable cell under conditions compatible with the control sequence(s). Such expression vectors may be used in any of the cells and methods described herein. The polynucleotides described herein may be manipulated in a variety of ways to provide for expression of a desired polypeptide. Manipulation of the polynucleotide prior to its insertion into a vector may be desirable or necessary depending on the expression vector. The techniques for modifying polynucleotides utilizing recombinant DNA methods are well known in the art.

A construct or vector (or multiple constructs or vectors) comprising the one or more (e.g., two, several) heterologous genes may be introduced into a cell so that the construct or vector is maintained as a chromosomal integrant or as a self-replicating extra-chromosomal vector as described earlier.

The various nucleotide and control sequences may be joined together to produce a recombinant expression vector that may include one or more (e.g., two, several) convenient

restriction sites to allow for insertion or substitution of the polynucleotide at such sites. Alternatively, the polynucleotide(s) may be expressed by inserting the polynucleotide(s) or a nucleic acid construct comprising the sequence into an appropriate vector for expression. In creating the expression vector, the coding sequence is located in the vector so that the coding sequence is operably linked with the appropriate control sequences for expression.

The recombinant expression vector may be any vector (e.g., a plasmid or virus) that can be conveniently subjected to recombinant DNA procedures and can bring about expression of the polynucleotide. The choice of the vector will typically depend on the compatibility of the vector with the host cell into which the vector is to be introduced. The vector may be a linear or closed circular plasmid.

The vector may be an autonomously replicating vector, i.e., a vector that exists as an extrachromosomal entity, the replication of which is independent of chromosomal replication, e.g., a plasmid, an extrachromosomal element, a minichromosome, or an artificial chromosome. The vector may contain any means for assuring self-replication. Alternatively, the vector may be one that, when introduced into the host cell, is integrated into the genome and replicated together with the chromosome(s) into which it has been integrated. Furthermore, a single vector or plasmid or two or more vectors or plasmids that together contain the total DNA to be introduced into the genome of the cell, or a transposon, may be used.

The expression vector may contain any suitable promoter sequence that is recognized by a cell for expression of a gene described herein. The promoter sequence contains transcriptional control sequences that mediate the expression of the polypeptide. The promoter may be any polynucleotide that shows transcriptional activity in the cell of choice including mutant, truncated, and hybrid promoters, and may be obtained from genes encoding extracellular or intracellular polypeptides either homologous or heterologous to the cell.

Each heterologous polynucleotide described herein may be operably linked to a promoter that is foreign to the polynucleotide. For example, in one embodiment, the heterologous polynucleotide encoding the hexose transporter is operably linked to a promoter foreign to the polynucleotide. The promoters may be identical to or share a high degree of sequence identity (e.g., at least about 80%, at least about 85%, at least about 90%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, or at least about 99%) with a selected native promoter.

Examples of suitable promoters for directing the transcription of the nucleic acid constructs in a yeast cells, include, but are not limited to, the promoters obtained from the genes for enolase (e.g., *S. cerevisiae* enolase or *I. orientalis* enolase (ENO1)), galactokinase (e.g., *S. cerevisiae* galactokinase or *I. orientalis* galactokinase (GAL1)), alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase (e.g., *S. cerevisiae* alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase or *I. orientalis* alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase (ADH1, ADH2/GAP)), triose phosphate isomerase (e.g., *S. cerevisiae* triose phosphate isomerase or *I. orientalis* triose phosphate isomerase (TPI)), metallothionein (e.g., *S. cerevisiae* metallothionein or *I. orientalis* metallothionein (CUP1)), 3-phosphoglycerate kinase (e.g., *S. cerevisiae* 3-phosphoglycerate kinase or *I. orientalis* 3-phosphoglycerate kinase (PGK)), PDC1, xylose reductase (XR), xylitol dehydrogenase (XDH), L-(+)-lactate-cytochrome c oxidoreductase (CYB2), translation elongation factor-1 (TEF1), translation elongation factor-2 (TEF2),

glyceraldehyde-3-phosphate dehydrogenase (GAPDH), and orotidine 5'-phosphate decarboxylase (URA3) genes. Other suitable promoters may be obtained from *S. cerevisiae* TDH3, HXT7, PGK1, RPL18B and CCW12 genes. Additional useful promoters for yeast host cells are described by Romanos et al., 1992, *Yeast* 8: 423-488.

The control sequence may also be a suitable transcription terminator sequence, which is recognized by a host cell to terminate transcription. The terminator sequence is operably linked to the 3'-terminus of the polynucleotide encoding the polypeptide. Any terminator that is functional in the yeast cell of choice may be used. The terminator may be identical to or share a high degree of sequence identity (e.g., at least about 80%, at least about 85%, at least about 90%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, or at least about 99%) with the selected native terminator.

Suitable terminators for yeast host cells may be obtained from the genes for enolase (e.g., *S. cerevisiae* or *I. orientalis* enolase cytochrome C (e.g., *S. cerevisiae* or *I. orientalis* cytochrome (CYC1)), glyceraldehyde-3-phosphate dehydrogenase (e.g., *S. cerevisiae* or *I. orientalis* glyceraldehyde-3-phosphate dehydrogenase (gpd)), PDC1, XR, XDH, transaldolase (TAL), transketolase (TKL), ribose 5-phosphate ketol-isomerase (RKI), CYB2, and the galactose family of genes (especially the GAL10 terminator). Other suitable terminators may be obtained from *S. cerevisiae* ENO2 or TEF1 genes. Additional useful terminators for yeast host cells are described by Romanos et al., 1992, *supra*.

The control sequence may also be an mRNA stabilizer region downstream of a promoter and upstream of the coding sequence of a gene which increases expression of the gene.

Examples of suitable mRNA stabilizer regions are obtained from a *Bacillus thuringiensis* cryIIIA gene (WO94/25612) and a *Bacillus subtilis* SP82 gene (Hue et al., 1995, *Journal of Bacteriology* 177: 3465-3471).

The control sequence may also be a suitable leader sequence, when transcribed is a nontranslated region of an mRNA that is important for translation by the host cell. The leader sequence is operably linked to the 5'-terminus of the polynucleotide encoding the polypeptide. Any leader sequence that is functional in the yeast cell of choice may be used.

Suitable leaders for yeast host cells are obtained from the genes for enolase (e.g., *S. cerevisiae* or *I. orientalis* enolase (ENO-1)), 3-phosphoglycerate kinase (e.g., *S. cerevisiae* or *I. orientalis* 3-phosphoglycerate kinase), alpha-factor (e.g., *S. cerevisiae* or *I. orientalis* alpha-factor), and alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase (e.g., *S. cerevisiae* or *I. orientalis* alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase (ADH2/GAP)).

The control sequence may also be a polyadenylation sequence; a sequence operably linked to the 3'-terminus of the polynucleotide and, when transcribed, is recognized by the host cell as a signal to add polyadenosine residues to transcribed mRNA. Any polyadenylation sequence that is functional in the host cell of choice may be used. Useful polyadenylation sequences for yeast cells are described by Guo and Sherman, 1995, *Mol. Cellular Biol.* 15: 5983-5990.

It may also be desirable to add regulatory sequences that allow the regulation of the expression of the polypeptide relative to the growth of the host cell. Examples of regulatory systems are those that cause the expression of the gene to be turned on or off in response to a chemical or physical stimulus, including the presence of a regulatory compound.

Regulatory systems in prokaryotic systems include the lac, tac, and trp operator systems. In yeast, the ADH2 system or GAL1 system may be used.

The vectors may contain one or more (e.g., two, several) selectable markers that permit easy selection of transformed, transfected, transduced, or the like cells. A selectable marker is a gene the product of which provides for biocide or viral resistance, resistance to heavy metals, prototrophy to auxotrophs, and the like. Suitable markers for yeast host cells include, but are not limited to, ADE2, HIS3, LEU2, LYS2, MET3, TRP1, and URA3.

The vectors may contain one or more (e.g., two, several) elements that permit integration of the vector into the host cell's genome or autonomous replication of the vector in the cell independent of the genome.

For integration into the host cell genome, the vector may rely on the polynucleotide's sequence encoding the polypeptide or any other element of the vector for integration into the genome by homologous or non-homologous recombination. Alternatively, the vector may contain additional polynucleotides for directing integration by homologous recombination into the genome of the host cell at a precise location(s) in the chromosome(s). To increase the likelihood of integration at a precise location, the integrational elements should contain a sufficient number of nucleic acids, such as 100 to 10,000 base pairs, 400 to 10,000 base pairs, and 800 to 10,000 base pairs, which have a high degree of sequence identity to the corresponding target sequence to enhance the probability of homologous recombination. The integrational elements may be any sequence that is homologous with the target sequence in the genome of the host cell. Furthermore, the integrational elements may be non-encoding or encoding polynucleotides. On the other hand, the vector may be integrated into the genome of the host cell by non-homologous recombination. Potential integration loci include those described in the art (e.g., See US2012/0135481).

For autonomous replication, the vector may further comprise an origin of replication enabling the vector to replicate autonomously in the yeast cell. The origin of replication may be any plasmid replicator mediating autonomous replication that functions in a cell. The term "origin of replication" or "plasmid replicator" means a polynucleotide that enables a plasmid or vector to replicate in vivo. Examples of origins of replication for use in a yeast host cell are the 2 micron origin of replication, ARS1, ARS4, the combination of ARS1 and CEN3, and the combination of ARS4 and CEN6.

More than one copy of a polynucleotide described herein may be inserted into a host cell to increase production of a polypeptide. An increase in the copy number of the polynucleotide can be obtained by integrating at least one additional copy of the sequence into the yeast cell genome or by including an amplifiable selectable marker gene with the polynucleotide where cells containing amplified copies of the selectable marker gene, and thereby additional copies of the polynucleotide, can be selected for by cultivating the cells in the presence of the appropriate selectable agent.

The procedures used to ligate the elements described above to construct the recombinant expression vectors described herein are well known to one skilled in the art (see, e.g., Sambrook et al., 1989, *Molecular Cloning, A Laboratory Manual*, 2d edition, Cold Spring Harbor, New York).

Additional procedures and techniques known in the art for the preparation of recombinant cells for ethanol fermentation, are described in, e.g., WO2016/045569, the content of which is hereby incorporated by reference.

The fermenting organism may be in the form of a composition comprising a fermenting organism (e.g., a yeast strain described herein) and a naturally occurring and/or a nonnaturally occurring component.

The fermenting organism described herein may be in any viable form, including crumbled, dry, including active dry and instant, compressed, cream (liquid) form etc. In one embodiment, the fermenting organism (e.g., a *Saccharomyces cerevisiae* yeast strain) is dry yeast, such as active dry yeast or instant yeast. In one embodiment, the fermenting organism (e.g., a *Saccharomyces cerevisiae* yeast strain) is crumbled yeast. In one embodiment, the fermenting organism (e.g., a *Saccharomyces cerevisiae* yeast strain) is compressed yeast. In one embodiment, the fermenting organism (e.g., a *Saccharomyces cerevisiae* yeast strain) is cream yeast.

In one embodiment is a composition comprising a fermenting organism described herein (e.g., a *Saccharomyces cerevisiae* yeast strain), and one or more of the component selected from the group consisting of: surfactants, emulsifiers, gums, swelling agent, and antioxidants and other processing aids.

The compositions described herein may comprise a fermenting organism described herein (e.g., a *Saccharomyces cerevisiae* yeast strain) and any suitable surfactants. In one embodiment, the surfactant(s) is/are an anionic surfactant, cationic surfactant, and/or nonionic surfactant.

The compositions described herein may comprise a fermenting organism described herein (e.g., a *Saccharomyces cerevisiae* yeast strain) and any suitable emulsifier. In one embodiment, the emulsifier is a fatty-acid ester of sorbitan. In one embodiment, the emulsifier is selected from the group of sorbitan monostearate (SMS), citric acid esters of monodiglycerides, polyglycerolester, fatty acid esters of propylene glycol.

In one embodiment, the composition comprises a fermenting organism described herein (e.g., a *Saccharomyces cerevisiae* yeast strain), and Olindronal SMS, Olindronal SK, or Olindronal SPL including composition concerned in European Patent No. 1,724,336 (hereby incorporated by reference). These products are commercially available from Bussetti, Austria, for active dry yeast.

The compositions described herein may comprise a fermenting organism described herein (e.g., a *Saccharomyces cerevisiae* yeast strain) and any suitable gum. In one embodiment, the gum is selected from the group of carob, guar, tragacanth, arabic, xanthan and acacia gum, in particular for cream, compressed and dry yeast.

The compositions described herein may comprise a fermenting organism described herein (e.g., a *Saccharomyces cerevisiae* yeast strain) and any suitable swelling agent. In one embodiment, the swelling agent is methyl cellulose or carboxymethyl cellulose.

The compositions described herein may comprise a fermenting organism described herein (e.g., a *Saccharomyces cerevisiae* yeast strain) and any suitable anti-oxidant. In one embodiment, the antioxidant is butylated hydroxyanisole (BHA) and/or butylated hydroxytoluene (BHT), or ascorbic acid (vitamin C), particular for active dry yeast.

Gene Disruptions

The fermenting organisms described herein may also comprise one or more (e.g., two, several) gene disruptions, e.g., to divert sugar metabolism from undesired products to ethanol. In some aspects, the recombinant host cells produce a greater amount of ethanol compared to the cell without the one or more disruptions when cultivated under identical

conditions. In some aspects, one or more of the disrupted endogenous genes is inactivated.

In certain embodiments, the fermenting organism provided herein comprises a disruption of one or more endogenous genes encoding enzymes involved in producing alternate fermentative products such as glycerol or other byproducts such as acetate or diols. For example, the cells provided herein may comprise a disruption of one or more of glycerol 3-phosphate dehydrogenase (GPD, catalyzes reaction of dihydroxyacetone phosphate to glycerol 3-phosphate), glycerol 3-phosphatase (GPP, catalyzes conversion of glycerol-3 phosphate to glycerol), glycerol kinase (catalyzes conversion of glycerol 3-phosphate to glycerol), dihydroxyacetone kinase (catalyzes conversion of dihydroxyacetone phosphate to dihydroxyacetone), glycerol dehydrogenase (catalyzes conversion of dihydroxyacetone to glycerol), and aldehyde dehydrogenase (ALD, e.g., converts acetaldehyde to acetate).

Modeling analysis can be used to design gene disruptions that additionally optimize utilization of the pathway. One exemplary computational method for identifying and designing metabolic alterations favoring biosynthesis of a desired product is the OptKnock computational framework, Burgard et al., 2003, *Biotechnol. Bioeng.* 84: 647-657.

The fermenting organisms comprising a gene disruption may be constructed using methods well known in the art, including those methods described herein. A portion of the gene can be disrupted such as the coding region or a control sequence required for expression of the coding region. Such a control sequence of the gene may be a promoter sequence or a functional part thereof, i.e., a part that is sufficient for affecting expression of the gene. For example, a promoter sequence may be inactivated resulting in no expression or a weaker promoter may be substituted for the native promoter sequence to reduce expression of the coding sequence. Other control sequences for possible modification include, but are not limited to, a leader, propeptide sequence, signal sequence, transcription terminator, and transcriptional activator.

The fermenting organisms comprising a gene disruption may be constructed by gene deletion techniques to eliminate or reduce expression of the gene. Gene deletion techniques enable the partial or complete removal of the gene thereby eliminating their expression. In such methods, deletion of the gene is accomplished by homologous recombination using a plasmid that has been constructed to contiguously contain the 5' and 3' regions flanking the gene.

The fermenting organisms comprising a gene disruption may also be constructed by introducing, substituting, and/or removing one or more (e.g., two, several) nucleotides in the gene or a control sequence thereof required for the transcription or translation thereof. For example, nucleotides may be inserted or removed for the introduction of a stop codon, the removal of the start codon, or a frame-shift of the open reading frame. Such a modification may be accomplished by site-directed mutagenesis or PCR generated mutagenesis in accordance with methods known in the art. See, for example, Botstein and Shortle, 1985, *Science* 229: 4719; Lo et al., 1985, *Proc. Natl. Acad. Sci. U.S.A.* 81: 2285; Higuchi et al., 1988, *Nucleic Acids Res* 16: 7351; Shimada, 1996, *Meth. Mol. Biol.* 57: 157; Ho et al., 1989, *Gene* 77: 61; Horton et al., 1989, *Gene* 77: 61; and Sarkar and Sommer, 1990, *BioTechniques* 8: 404.

The fermenting organisms comprising a gene disruption may also be constructed by inserting into the gene a disruptive nucleic acid construct comprising a nucleic acid fragment homologous to the gene that will create a dupli-

cation of the region of homology and incorporate construct DNA between the duplicated regions. Such a gene disruption can eliminate gene expression if the inserted construct separates the promoter of the gene from the coding region or interrupts the coding sequence such that a non-functional gene product results. A disrupting construct may be simply a selectable marker gene accompanied by 5' and 3' regions homologous to the gene. The selectable marker enables identification of transformants containing the disrupted gene.

The fermenting organisms comprising a gene disruption may also be constructed by the process of gene conversion (see, for example, Iglesias and Trautner, 1983, *Molecular General Genetics* 189: 73-76). For example, in the gene conversion method, a nucleotide sequence corresponding to the gene is mutagenized in vitro to produce a defective nucleotide sequence, which is then transformed into the recombinant strain to produce a defective gene. By homologous recombination, the defective nucleotide sequence replaces the endogenous gene. It may be desirable that the defective nucleotide sequence also comprises a marker for selection of transformants containing the defective gene.

The fermenting organisms comprising a gene disruption may be further constructed by random or specific mutagenesis using methods well known in the art, including, but not limited to, chemical mutagenesis (see, for example, Hopwood, *The Isolation of Mutants in Methods in Microbiology* (J. R. Norris and D. W. Ribbons, eds.) pp. 363-433, Academic Press, New York, 1970). Modification of the gene may be performed by subjecting the parent strain to mutagenesis and screening for mutant strains in which expression of the gene has been reduced or inactivated. The mutagenesis, which may be specific or random, may be performed, for example, by use of a suitable physical or chemical mutagenizing agent, use of a suitable oligonucleotide, or subjecting the DNA sequence to PCR generated mutagenesis. Furthermore, the mutagenesis may be performed by use of any combination of these mutagenizing methods.

Examples of a physical or chemical mutagenizing agent suitable for the present purpose include ultraviolet (UV) irradiation, hydroxylamine, N-methyl-N'-nitro-N-nitrosoguanidine (MNNG), N-methyl-N'-nitrosoguanidine (NTG) O-methyl hydroxylamine, nitrous acid, ethyl methane sulphonate (EMS), sodium bisulphite, formic acid, and nucleotide analogues. When such agents are used, the mutagenesis is typically performed by incubating the parent strain to be mutagenized in the presence of the mutagenizing agent of choice under suitable conditions, and selecting for mutants exhibiting reduced or no expression of the gene.

A nucleotide sequence homologous or complementary to a gene described herein may be used from other microbial sources to disrupt the corresponding gene in a recombinant strain of choice.

In one aspect, the modification of a gene in the recombinant cell is unmarked with a selectable marker. Removal of the selectable marker gene may be accomplished by culturing the mutants on a counter-selection medium. Where the selectable marker gene contains repeats flanking its 5' and 3' ends, the repeats will facilitate the looping out of the selectable marker gene by homologous recombination when the mutant strain is submitted to counter-selection. The selectable marker gene may also be removed by homologous recombination by introducing into the mutant strain a nucleic acid fragment comprising 5' and 3' regions of the defective gene, but lacking the selectable marker gene, followed by selecting on the counter-selection medium. By homologous recombination, the defective gene containing

the selectable marker gene is replaced with the nucleic acid fragment lacking the selectable marker gene. Other methods known in the art may also be used.

Methods Using a Starch-Containing Material

In some aspects, the methods described herein produce a fermentation product from a starch-containing material. Starch-containing material is well-known in the art, containing two types of homopolysaccharides (amylose and amylopectin) and is linked by alpha-(1-4)-D-glycosidic bonds. Any suitable starch-containing starting material may be used. The starting material is generally selected based on the desired fermentation product, such as ethanol. Examples of starch-containing starting materials include cereal, tubers or grains. Specifically, the starch-containing material may be corn, wheat, barley, rye, milo, sago, cassava, tapioca, sorghum, oat, rice, peas, beans, or sweet potatoes, or mixtures thereof. Contemplated are also waxy and non-waxy types of corn and barley.

In one embodiment, the starch-containing starting material is corn. In one embodiment, the starch-containing starting material is wheat. In one embodiment, the starch-containing starting material is barley. In one embodiment, the starch-containing starting material is rye. In one embodiment, the starch-containing starting material is milo. In one embodiment, the starch-containing starting material is sago. In one embodiment, the starch-containing starting material is cassava. In one embodiment, the starch-containing starting material is tapioca. In one embodiment, the starch-containing starting material is sorghum. In one embodiment, the starch-containing starting material is rice. In one embodiment, the starch-containing starting material is peas. In one embodiment, the starch-containing starting material is beans. In one embodiment, the starch-containing starting material is sweet potatoes. In one embodiment, the starch-containing starting material is oats.

The methods using a starch-containing material may include a conventional process (e.g., including a liquefaction step described in more detail below) or a raw starch hydrolysis process. In some embodiments using a starch-containing material, saccharification of the starch-containing material is at a temperature above the initial gelatinization temperature. In some embodiments using a starch-containing material, saccharification of the starch-containing material is at a temperature below the initial gelatinization temperature.

Liquefaction

In aspects using a starch-containing material, the methods may further comprise a liquefaction step carried out by subjecting the starch-containing material at a temperature above the initial gelatinization temperature to an alpha-amylase and optionally a protease and/or a glucoamylase. Other enzymes such as a pullulanase and phytase may also be present and/or added in liquefaction. In some embodiments, the liquefaction step is carried out prior to steps a) and b) of the described methods.

Liquefaction step may be carried out for 0.5-5 hours, such as 1-3 hours, such as typically about 2 hours.

The term "initial gelatinization temperature" means the lowest temperature at which gelatinization of the starch-containing material commences. In general, starch heated in water begins to gelatinize between about 50° C. and 75° C.; the exact temperature of gelatinization depends on the specific starch and can readily be determined by the skilled artisan. Thus, the initial gelatinization temperature may vary according to the plant species, to the particular variety of the plant species as well as with the growth conditions. The initial gelatinization temperature of a given starch-contain-

ing material may be determined as the temperature at which birefringence is lost in 5% of the starch granules using the method described by Gorinstein and Lii, 1992, *Starch/Stärke* 44(12): 461-466.

Liquefaction is typically carried out at a temperature in the range from 70-100° C. In one embodiment, the temperature in liquefaction is between 75-95° C., such as between 75-90° C., between 80-90° C., or between 82-88° C., such as about 85° C.

A jet-cooking step may be carried out prior to liquefaction in step, for example, at a temperature between 110-145° C., 120-140° C., 125-135° C., or about 130° C. for about 1-15 minutes, for about 3-10 minutes, or about 5 minutes.

The pH during liquefaction may be between 4 and 7, such as pH 4.5-6.5, pH 5.0-6.5, pH 5.0-6.0, pH 5.2-6.2, or about 5.2, about 5.4, about 5.6, or about 5.8.

In one embodiment, the process further comprises, prior to liquefaction, the steps of:

- i) reducing the particle size of the starch-containing material, preferably by dry milling;
- ii) forming a slurry comprising the starch-containing material and water.

The starch-containing starting material, such as whole grains, may be reduced in particle size, e.g., by milling, in order to open up the structure, to increase surface area, and allowing for further processing. Generally, there are two types of processes: wet and dry milling. In dry milling whole kernels are milled and used. Wet milling gives a good separation of germ and meal (starch granules and protein). Wet milling is often applied at locations where the starch hydrolysate is used in production of, e.g., syrups. Both dry milling and wet milling are well known in the art of starch processing. In one embodiment the starch-containing material is subjected to dry milling. In one embodiment, the particle size is reduced to between 0.05 to 3.0 mm, e.g., 0.1-0.5 mm, or so that at least 30%, at least 50%, at least 70%, or at least 90% of the starch-containing material fit through a sieve with a 0.05 to 3.0 mm screen, e.g., 0.1-0.5 mm screen. In another embodiment, at least 50%, e.g., at least 70%, at least 80%, or at least 90% of the starch-containing material fit through a sieve with #6 screen.

The aqueous slurry may contain from 10-55 w/w-% dry solids (DS), e.g., 25-45 w/w-% dry solids (DS), or 30-40 w/w-% dry solids (DS) of starch-containing material.

The alpha-amylase, optionally a protease, and optionally a glucoamylase may initially be added to the aqueous slurry to initiate liquefaction (thinning). In one embodiment, only a portion of the enzymes (e.g., about 1/3) is added to the aqueous slurry, while the rest of the enzymes (e.g., about 2/3) are added during liquefaction step.

A non-exhaustive list of alpha-amylases used in liquefaction can be found below in the "Alpha-Amylases" section. Examples of suitable proteases used in liquefaction include any protease described supra in the "Proteases" section. Examples of suitable glucoamylases used in liquefaction include any glucoamylase found in the "Glucoamylases" section.

Saccharification and Fermentation of Starch-Containing Material

In aspects using a starch-containing material, a glucoamylase may be present and/or added in saccharification step a) and/or fermentation step b) or simultaneous saccharification and fermentation (SSF). The glucoamylase of the saccharification step a) and/or fermentation step b) or simultaneous saccharification and fermentation (SSF) is typically different from the glucoamylase optionally added to any liquefaction

step described supra. In one embodiment, the glucoamylase is present and/or added together with a fungal alpha-amylase.

In some aspects, the fermenting organism comprises a heterologous polynucleotide encoding a glucoamylase, for example, as described in WO2017/087330, the content of which is hereby incorporated by reference.

Examples of glucoamylases can be found in the "Glucoamylases" section below.

When doing sequential saccharification and fermentation, saccharification step a) may be carried out under conditions well-known in the art. For instance, saccharification step a) may last up to from about 24 to about 72 hours. In one embodiment, pre-saccharification is done. Pre-saccharification is typically done for 40-90 minutes at a temperature between 30-65° C., typically about 60° C. Pre-saccharification is, in one embodiment, followed by saccharification during fermentation in simultaneous saccharification and fermentation (SSF). Saccharification is typically carried out at temperatures from 20-75° C., preferably from 40-70° C., typically about 60° C., and typically at a pH between 4 and 5, such as about pH 4.5.

Fermentation is carried out in a fermentation medium, as known in the art and, e.g., as described herein. The fermentation medium includes the fermentation substrate, that is, the carbohydrate source that is metabolized by the fermenting organism. With the processes described herein, the fermentation medium may comprise nutrients and growth stimulator(s) for the fermenting organism(s). Nutrient and growth stimulators are widely used in the art of fermentation and include nitrogen sources, such as ammonia; urea, vitamins and minerals, or combinations thereof.

Generally, fermenting organisms such as yeast, including *Saccharomyces cerevisiae* yeast, require an adequate source of nitrogen for propagation and fermentation. Many sources of supplemental nitrogen, if necessary, can be used and such sources of nitrogen are well known in the art. The nitrogen source may be organic, such as urea, DDGs, wet cake or corn mash, or inorganic, such as ammonia or ammonium hydroxide. In one embodiment, the nitrogen source is urea.

Fermentation can be carried out under low nitrogen conditions, e.g., when using a protease-expressing yeast. In some embodiments, the fermentation step is conducted with less than 1000 ppm supplemental nitrogen (e.g., urea or ammonium hydroxide), such as less than 750 ppm, less than 500 ppm, less than 400 ppm, less than 300 ppm, less than 250 ppm, less than 200 ppm, less than 150 ppm, less than 100 ppm, less than 75 ppm, less than 50 ppm, less than 25 ppm, or less than 10 ppm, supplemental nitrogen. In some embodiments, the fermentation step is conducted with no supplemental nitrogen.

Simultaneous saccharification and fermentation ("SSF") is widely used in industrial scale fermentation product production processes, especially ethanol production processes. When doing SSF the saccharification step a) and the fermentation step b) are carried out simultaneously. There is no holding stage for the saccharification, meaning that a fermenting organism, such as yeast, and enzyme(s), may be added together. However, it is also contemplated to add the fermenting organism and enzyme(s) separately. SSF is typically carried out at a temperature from 25° C. to 40° C., such as from 28° C. to 35° C., such as from 30° C. to 34° C., or about 32° C. In one embodiment, fermentation is ongoing for 6 to 120 hours, in particular 24 to 96 hours. In one embodiment, the pH is between 4-5.

In one embodiment, a cellulolytic enzyme composition is present and/or added in saccharification, fermentation or

simultaneous saccharification and fermentation (SSF). Examples of such cellulolytic enzyme compositions can be found in the "Cellulolytic Enzymes and Compositions" section below. The cellulolytic enzyme composition may be present and/or added together with a glucoamylase, such as one disclosed in the "Glucoamylases" section below.

Alpha-Amylases

The expressed and/or exogenous alpha-amylase may be any alpha-amylase that is suitable for the host cells and/or the methods described herein, such as a naturally occurring alpha-amylase (e.g., a native alpha-amylase from another species or an endogenous alpha-amylase expressed from a modified expression vector) or a variant thereof that retains alpha-amylase activity. Any alpha-amylase contemplated for expression by a fermenting organism described below is also contemplated for aspects of the invention involving exogenous addition of an alpha-amylase.

In some embodiments, the fermenting organism comprises a heterologous polynucleotide encoding an alpha-amylase, for example, as described in WO2017/087330, the content of which is hereby incorporated by reference. Any alpha-amylase described or referenced herein is contemplated for expression in the fermenting organism.

In some embodiments, the fermenting organism comprising a heterologous polynucleotide encoding an alpha-amylase has an increased level of alpha-amylase activity compared to the host cells without the heterologous polynucleotide encoding the alpha-amylase, when cultivated under the same conditions. In some embodiments, the fermenting organism has an increased level of alpha-amylase activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the fermenting organism without the heterologous polynucleotide encoding the alpha-amylase, when cultivated under the same conditions.

Exemplary alpha-amylases that can be used with the host cells and/or the methods described herein include bacterial, yeast, or filamentous fungal alpha-amylases, e.g., derived from any of the microorganisms described or referenced herein.

The term "bacterial alpha-amylase" means any bacterial alpha-amylase classified under EC 3.2.1.1. A bacterial alpha-amylase used herein may, e.g., be derived from a strain of the genus *Bacillus*, which is sometimes also referred to as the genus *Geobacillus*. In one embodiment, the *Bacillus* alpha-amylase is derived from a strain of *Bacillus amyloliquefaciens*, *Bacillus licheniformis*, *Bacillus stearothermophilus*, or *Bacillus subtilis*, but may also be derived from other *Bacillus* sp.

Specific examples of bacterial alpha-amylases include the *Bacillus stearothermophilus* alpha-amylase (BSG) of SEQ ID NO: 3 in WO99/19467, the *Bacillus amyloliquefaciens* alpha-amylase (BAN) of SEQ ID NO: 5 in WO99/19467, and the *Bacillus licheniformis* alpha-amylase (BLA) of SEQ ID NO: 4 in WO99/19467 (all sequences are hereby incorporated by reference). In one embodiment, the alpha-amylase may be an enzyme having a mature polypeptide sequence with a degree of identity of at least 60%, e.g., at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98% or at least 99% to any of the sequences shown in SEQ ID NOs: 3, 4 or 5, in WO99/19467.

In one embodiment, the alpha-amylase is derived from *Bacillus stearothermophilus*. The *Bacillus stearothermophilus* alpha-amylase may be a mature wild-type or a mature variant thereof. The mature *Bacillus stearothermophilus* alpha-amylases may naturally be truncated during recombi-

nant production. For instance, the *Bacillus stearothermophilus* alpha-amylase may be a truncated at the C-terminal, so that it is from 480-495 amino acids long, such as about 491 amino acids long, e.g., so that it lacks a functional starch binding domain (compared to SEQ ID NO: 3 in WO99/19467).

The *Bacillus* alpha-amylase may also be a variant and/or hybrid. Examples of such a variant can be found in any of WO96/23873, WO96/23874, WO97/41213, WO99/19467, WO00/60059, and WO02/10355 (each hereby incorporated by reference). Specific alpha-amylase variants are disclosed in U.S. Pat. Nos. 6,093,562, 6,187,576, 6,297,038, and 7,713,723 (hereby incorporated by reference) and include *Bacillus stearothermophilus* alpha-amylase (often referred to as BSG alpha-amylase) variants having a deletion of one or two amino acids at positions R179, G180, I181 and/or G182, preferably a double deletion disclosed in WO96/23873—see, e.g., page 20, lines 1-10 (hereby incorporated by reference), such as corresponding to deletion of positions I181 and G182 compared to the amino acid sequence of *Bacillus stearothermophilus* alpha-amylase set forth in SEQ ID NO: 3 disclosed in WO99/19467 or the deletion of amino acids R179 and G180 using SEQ ID NO: 3 in WO99/19467 for numbering (which reference is hereby incorporated by reference). In some embodiments, the *Bacillus* alpha-amylases, such as *Bacillus stearothermophilus* alpha-amylases, have a double deletion corresponding to a deletion of positions 181 and 182 and further optionally comprise a N193F substitution (also denoted I181*+G182*+N193F) compared to the wild-type BSG alpha-amylase amino acid sequence set forth in SEQ ID NO: 3 disclosed in WO99/19467. The bacterial alpha-amylase may also have a substitution in a position corresponding to S239 in the *Bacillus licheniformis* alpha-amylase shown in SEQ ID NO: 4 in WO99/19467, or a S242 and/or E188P variant of the *Bacillus stearothermophilus* alpha-amylase of SEQ ID NO: 3 in WO99/19467.

In one embodiment, the variant is a S242A, E or Q variant, e.g., a S242Q variant, of the *Bacillus stearothermophilus* alpha-amylase.

In one embodiment, the variant is a position E188 variant, e.g., E188P variant of the *Bacillus stearothermophilus* alpha-amylase.

The bacterial alpha-amylase may, in one embodiment, be a truncated *Bacillus* alpha-amylase. In one embodiment, the truncation is so that, e.g., the *Bacillus stearothermophilus* alpha-amylase shown in SEQ ID NO: 3 in WO99/19467, is about 491 amino acids long, such as from 480 to 495 amino acids long, or so it lacks a functional starch bind domain. The bacterial alpha-amylase may also be a hybrid bacterial alpha-amylase, e.g., an alpha-amylase comprising 445 C-terminal amino acid residues of the *Bacillus licheniformis* alpha-amylase (shown in SEQ ID NO: 4 of WO99/19467) and the 37 N-terminal amino acid residues of the alpha-amylase derived from *Bacillus amyloliquefaciens* (shown in SEQ ID NO: 5 of WO99/19467). In one embodiment, this hybrid has one or more, especially all, of the following substitutions: G48A+T49I+G107A+H156Y+A181T+N190F+I201F+A209V+Q264S (using the *Bacillus licheniformis* numbering in SEQ ID NO: 4 of WO99/19467). In some embodiments, the variants have one or more of the following mutations (or corresponding mutations in other *Bacillus* alpha-amylases): H154Y, A181T, N190F, A209V and Q264S and/or the deletion of two residues between positions 176 and 179, e.g., deletion of E178 and G179 (using SEQ ID NO: 5 of WO99/19467 for position numbering).

In one embodiment, the bacterial alpha-amylase is the mature part of the chimeric alpha-amylase disclosed in Richardson et al. (2002), The Journal of Biological Chemistry, Vol. 277, No. 29, Issue 19 July, pp. 267501-26507, referred to as BD5088 or a variant thereof. This alpha-amylase is the same as the one shown in SEQ ID NO: 2 in WO2007/134207. The mature enzyme sequence starts after the initial "Met" amino acid in position 1.

The alpha-amylase may be a thermostable alpha-amylase, such as a thermostable bacterial alpha-amylase, e.g., from *Bacillus stearothermophilus*. In one embodiment, the alpha-amylase used in a process described herein has a T_{1/2} (min) at pH 4.5, 85° C., 0.12 mM CaCl₂ of at least 10 determined as described in Example 1 of WO2018/098381.

In one embodiment, the thermostable alpha-amylase has a T_{1/2} (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, of at least 15. In one embodiment, the thermostable alpha-amylase has a T_{1/2} (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, of at least 20. In one embodiment, the thermostable alpha-amylase has a T_{1/2} (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, of at least 25. In one embodiment, the thermostable alpha-amylase has a T_{1/2} (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, of at least 30. In one embodiment, the thermostable alpha-amylase has a T_{1/2} (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, of at least 40.

In one embodiment, the thermostable alpha-amylase has a T_{1/2} (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, of at least 50. In one embodiment, the thermostable alpha-amylase has a T_{1/2} (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, of at least 60. In one embodiment, the thermostable alpha-amylase has a T_{1/2} (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 10-70. In one embodiment, the thermostable alpha-amylase has a T_{1/2} (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 15-70. In one embodiment, the thermostable alpha-amylase has a T_{1/2} (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 20-70. In one embodiment, the thermostable alpha-amylase has a T_{1/2} (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 25-70. In one embodiment, the thermostable alpha-amylase has a T_{1/2} (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 30-70. In one embodiment, the thermostable alpha-amylase has a T_{1/2} (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 40-70. In one embodiment, the thermostable alpha-amylase has a T_{1/2} (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 50-70. In one embodiment, the thermostable alpha-amylase has a T_{1/2} (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 60-70.

In one embodiment, the alpha-amylase is a bacterial alpha-amylase, e.g., derived from the genus *Bacillus*, such as a strain of *Bacillus stearothermophilus*, e.g., the *Bacillus stearothermophilus* as disclosed in WO99/019467 as SEQ ID NO: 3 with one or two amino acids deleted at positions R179, G180, I181 and/or G182, in particular with R179 and G180 deleted, or with I181 and G182 deleted, with mutations in below list of mutations.

In some embodiment, the *Bacillus stearothermophilus* alpha-amylases have double deletion I181+G182, and optional substitution N193F, further comprising one of the following substitutions or combinations of substitutions:

V59A+Q89R+G112D+E129V+K177L+R179E+K220P+N224L+Q254S;
V59A+Q89R+E129V+K177L+R179E+H208Y+K220P+N224L+Q254S;
V59A+Q89R+E129V+K177L+R179E+K220P+N224L+Q254S+D269E+D281N;
V59A+Q89R+E129V+K177L+R179E+K220P+N224L+Q254S+I270L;
V59A+Q89R+E129V+K177L+R179E+K220P+N224L+Q254S+H274K;

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V59A+Q89R+E129V+K177L+R179E+K220P+N224L+Q254S+Y276F;
 V59A+E129V+R157Y+K177L+R179E+K220P+N224L+S242Q+Q254S;
 V59A+E129V+K177L+R179E+H208Y+K220P+N224L+S242Q+Q254S;
 V59A+E129V+K177L+R179E+K220P+N224L+S242Q+Q254S;
 V59A+E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+H274K;
 V59A+E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+Y276F;
 V59A+E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+D281N;
 V59A+E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+M284T;
 V59A+E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+G416V;
 V59A+E129V+K177L+R179E+K220P+N224L+Q254S;
 V59A+E129V+K177L+R179E+K220P+N224L+Q254S+M284T;
 A91L+M96I+E129V+K177L+R179E+K220P+N224L+S242Q+Q254S;
 E129V+K177L+R179E;
 E129V+K177L+R179E+K220P+N224L+S242Q+Q254S;
 E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+Y276F+L427M;
 E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+M284T;
 E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+N376*+I377*;
 E129V+K177L+R179E+K220P+N224L+Q254S;
 E129V+K177L+R179E+K220P+N224L+Q254S+M284T;
 E129V+K177L+R179E+S242Q;
 E129V+K177L+R179V+K220P+N224L+S242Q+Q254S;
 K220P+N224L+S242Q+Q254S;
 M284V;
 V59A+Q89R+E129V+K177L+R179E+Q254S+M284V;
 and
 V59A+E129V+K177L+R179E+Q254S+M284V;

In one embodiment, the alpha-amylase is selected from the group of *Bacillus stearothermophilus* alpha-amylase variants with double deletion I181*+G182*, and optionally substitution N193F, and further one of the following substitutions or combinations of substitutions:

E129V+K177L+R179E;
 V59A+Q89R+E129V+K177L+R179E+H208Y+K220P+N224L+Q254S;
 V59A+Q89R+E129V+K177L+R179E+Q254S+M284V;
 V59A+E129V+K177L+R179E+Q254S+M284V; and
 E129V+K177L+R179E+K220P+N224L+S242Q+Q254S (using SEQ ID NO: 1 herein for numbering).

It should be understood that when referring to *Bacillus stearothermophilus* alpha-amylase and variants thereof they are normally produced in truncated form. In particular, the truncation may be so that the *Bacillus stearothermophilus* alpha-amylase shown in SEQ ID NO: 3 in WO99/19467, or variants thereof, are truncated in the C-terminal and are typically from 480-495 amino acids long, such as about 491 amino acids long, e.g., so that it lacks a functional starch binding domain.

In one embodiment, the alpha-amylase variant may be an enzyme having a mature polypeptide sequence with a degree of identity of at least 60%, e.g., at least 70%, at least 80%,

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at least 90%, at least 95%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98% or at least 99%, but less than 100% to the sequence shown in SEQ ID NO: 3 in WO99/19467.

5 In one embodiment, the bacterial alpha-amylase, e.g., *Bacillus* alpha-amylase, such as especially *Bacillus stearothermophilus* alpha-amylase, or variant thereof, is dosed to liquefaction in a concentration between 0.01-10 KNU-A/g DS, e.g., between 0.02 and 5 KNU-A/g DS, such as 0.03 and 10 3 KNU-A, preferably 0.04 and 2 KNU-A/g DS, such as especially 0.01 and 2 KNU-A/g DS. In one embodiment, the bacterial alpha-amylase, e.g., *Bacillus* alpha-amylase, such as especially *Bacillus stearothermophilus* alpha-amylases, or variant thereof, is dosed to liquefaction in a concentration 15 of between 0.0001-1 mg EP (Enzyme Protein)/g DS, e.g., 0.0005-0.5 mg EP/g DS, such as 0.001-0.1 mg EP/g DS.

In one embodiment, the bacterial alpha-amylase is derived from the *Bacillus subtilis* alpha-amylase of SEQ ID NO: 76, the *Bacillus subtilis* alpha-amylase of SEQ ID NO: 82, the *Bacillus subtilis* alpha-amylase of SEQ ID NO: 83, the *Bacillus subtilis* alpha-amylase of SEQ ID NO: 84, or the *Bacillus licheniformis* alpha-amylase of SEQ ID NO: 85, the *Clostridium phytofermentans* alpha-amylase of SEQ ID NO: 89, the *Clostridium phytofermentans* alpha-amylase of SEQ ID NO: 90, the *Clostridium phytofermentans* alpha-amylase of SEQ ID NO: 91, the *Clostridium phytofermentans* alpha-amylase of SEQ ID NO: 92, the *Clostridium phytofermentans* alpha-amylase of SEQ ID NO: 93, the *Clostridium phytofermentans* alpha-amylase of SEQ ID NO: 94, the *Clostridium thermocellum* alpha-amylase of SEQ ID NO: 95, the *Thermobifida fusca* alpha-amylase of SEQ ID NO: 96, the *Thermobifida fusca* alpha-amylase of SEQ ID NO: 97, the *Anaerocellum thermophilum* of SEQ ID NO: 98, the *Anaerocellum thermophilum* of SEQ ID NO: 99, the *Anaerocellum thermophilum* of SEQ ID NO: 100, the *Streptomyces avermitilis* of SEQ ID NO: 101, or the *Streptomyces avermitilis* of SEQ ID NO: 88.

In one embodiment, the alpha-amylase is derived from *Bacillus amyloliquefaciens*, such as the *Bacillus amyloliquefaciens* alpha-amylase of SEQ ID NO: 231 (e.g., as described in WO2018/002360, or variants thereof as described in WO2017/037614).

In one embodiment, the alpha-amylase is derived from a yeast alpha-amylase, such as the *Saccharomycopsis fibuligera* alpha-amylase of SEQ ID NO: 77, the *Debaryomyces occidentalis* alpha-amylase of SEQ ID NO: 78, the *Debaryomyces occidentalis* alpha-amylase of SEQ ID NO: 79, the *Lipomyces kononenkoae* alpha-amylase of SEQ ID NO: 80, the *Lipomyces kononenkoae* alpha-amylase of SEQ ID NO: 81.

In one embodiment, the alpha-amylase is derived from a filamentous fungal alpha-amylase, such as the *Aspergillus niger* alpha-amylase of SEQ ID NO: 86, or the *Aspergillus niger* alpha-amylase of SEQ ID NO: 87.

55 Additional alpha-amylases that may be expressed with the fermenting organisms and used with the methods described herein are described in the examples, and include, but are not limited to alpha-amylases shown in Table 1 (or derivatives thereof).

TABLE 1

Donor Organism (catalytic domain)	SEQ ID NO: (mature polypeptide)
<i>Rhizomucor pusillus</i>	121
<i>Bacillus licheniformis</i>	122

TABLE 1-continued

Donor Organism (catalytic domain)	SEQ ID NO: (mature polypeptide)
<i>Aspergillus niger</i>	123
<i>Aspergillus tamarii</i>	124
<i>Acidomyces richmondensis</i>	125
<i>Aspergillus bombycis</i>	126
<i>Alternaria</i> sp	127
<i>Rhizopus microsporus</i>	128
<i>Syncephalastrum racemosum</i>	129
<i>Rhizomucor pusillus</i>	130
<i>Dichotomocladium hesseltinei</i>	131
<i>Lichtheimia ramosa</i>	132
<i>Penicillium aethiopicum</i>	133
<i>Subulispora</i> sp	134
<i>Trichoderma paraviridescens</i>	135
<i>Byssosascus striatosporus</i>	136
<i>Aspergillus brasiliensis</i>	137
<i>Penicillium subspinosum</i>	138
<i>Penicillium antarcticum</i>	139
<i>Penicillium coprophilum</i>	140
<i>Penicillium olsonii</i>	141
<i>Penicillium vasconiae</i>	142
<i>Penicillium</i> sp	143
<i>Heterocephalum aurantiacum</i>	144
<i>Neosartorya massa</i>	145
<i>Penicillium janthinellum</i>	146
<i>Aspergillus brasiliensis</i>	147
<i>Aspergillus westerdijkiae</i>	148
<i>Hamigera avellanea</i>	149
<i>Hamigera avellanea</i>	150
<i>Meripilus giganteus</i>	151
<i>Cerrena unicolor</i>	152
<i>Physalacria cryptomeriae</i>	153
<i>Lenzites betulinus</i>	154
<i>Trametes ljubarskyi</i>	155
<i>Bacillus subtilis</i>	156
<i>Bacillus subtilis</i> subsp. <i>subtilis</i>	157
<i>Schwanniomyces occidentalis</i>	158
<i>Rhizomucor pusillus</i>	159
<i>Aspergillus niger</i>	160
<i>Bacillus stearothermophilus</i>	161
<i>Bacillus halmapalus</i>	162
<i>Aspergillus oryzae</i>	163
<i>Bacillus amyloliquefaciens</i>	164
<i>Rhizomucor pusillus</i>	165
<i>Kionochaeta ivoriensis</i>	166
<i>Aspergillus niger</i>	167
<i>Aspergillus oryzae</i>	168
<i>Penicillium canescens</i>	169
<i>Acidomyces acidothermus</i>	170
<i>Kionochaeta ivoriensis</i>	171
<i>Aspergillus terreus</i>	172
<i>Thamnidium elegans</i>	173
<i>Meripilus giganteus</i>	174

Additional alpha-amylases contemplated for use with the present invention can be found in WO2011/153516 (the content of which is incorporated herein).

Additional polynucleotides encoding suitable alpha-amylases may be obtained from microorganisms of any genus, including those readily available within the UniProtKB database (www.uniprot.org).

As described supra, the alpha-amylase may be a bacterial alpha-amylase. For example, the alpha-amylase may be derived from a Gram-positive bacterium such as a *Bacillus*, *Clostridium*, *Enterococcus*, *Geobacillus*, *Lactobacillus*, *Lactococcus*, *Oceanobacillus*, *Staphylococcus*, *Streptococcus*, or *Streptomyces*, or a Gram-negative bacterium such as a *Campylobacter*, *E. coli*, *Flavobacterium*, *Fusobacterium*, *Helicobacter*, *Ilyobacter*, *Neisseria*, *Pseudomonas*, *Salmonella*, or *Ureaplasma*.

In one embodiment, the alpha-amylase is derived from *Bacillus alkalophilus*, *Bacillus amyloliquefaciens*, *Bacillus brevis*, *Bacillus circulans*, *Bacillus clausii*, *Bacillus coagu-*

lans, *Bacillus firmus*, *Bacillus lautus*, *Bacillus lentus*, *Bacillus licheniformis*, *Bacillus megaterium*, *Bacillus pumilus*, *Bacillus stearothermophilus*, *Bacillus subtilis*, or *Bacillus thuringiensis*.

In another embodiment, the alpha-amylase is derived from *Streptococcus equisimilis*, *Streptococcus pyogenes*, *Streptococcus uberis*, or *Streptococcus equi* subsp. *Zooepidemicus*.

In another embodiment, the alpha-amylase is derived from *Streptomyces achromogenes*, *Streptomyces avermitilis*, *Streptomyces coelicolor*, *Streptomyces griseus*, or *Streptomyces lividans*.

The alpha-amylase may be a fungal alpha-amylase. For example, the alpha-amylase may be derived from a yeast such as a *Candida*, *Kluyveromyces*, *Pichia*, *Saccharomyces*, *Schizosaccharomyces*, *Yarrowia* or *Issatchenkia*; or derived from a filamentous fungus such as an *Acremonium*, *Agaricus*, *Alternaria*, *Aspergillus*, *Aureobasidium*, *Botryosphaeria*, *Ceriporiopsis*, *Chaetomidium*, *Chrysosporium*, *Claviceps*, *Cochliobolus*, *Coprinopsis*, *Coptotermes*, *Corynascus*, *Cryphonectria*, *Cryptococcus*, *Diplodia*, *Exidia*, *Filibasidium*, *Fusarium*, *Gibberella*, *Holomastigotoides*, *Humicola*, *Irpex*, *Lentinula*, *Leptosphaeria*, *Magnaporthe*, *Melanocarpus*, *Meripilus*, *Mucor*, *Myceliophthora*, *Neocallimastix*, *Neurospora*, *Paecilomyces*, *Penicillium*, *Phanerochaete*, *Piromyces*, *Poitrasia*, *Pseudoplectania*, *Pseudotrichonympha*, *Rhizomucor*, *Schizophyllum*, *Scytalidium*, *Talaromyces*, *Thermoascus*, *Thielavia*, *Tolypocladium*, *Trichoderma*, *Trichophaea*, *Verticillium*, *Volvariella*, or *Xylaria*.

In another embodiment, the alpha-amylase is derived from *Saccharomyces carlsbergensis*, *Saccharomyces cerevisiae*, *Saccharomyces diastaticus*, *Saccharomyces douglasii*, *Saccharomyces kluyveri*, *Saccharomyces norbensis*, or *Saccharomyces oviformis*.

In another embodiment, the alpha-amylase is derived from *Acremonium cellulolyticus*, *Aspergillus aculeatus*, *Aspergillus awamori*, *Aspergillus foetidus*, *Aspergillus fumigatus*, *Aspergillus japonicus*, *Aspergillus nidulans*, *Aspergillus niger*, *Aspergillus oryzae*, *Chrysosporium inops*, *Chrysosporium keratinophilum*, *Chrysosporium lucknowense*, *Chrysosporium merdarium*, *Chrysosporium pannicola*, *Chrysosporium queenslandicum*, *Chrysosporium tropicum*, *Chrysosporium zonatum*, *Fusarium bactridioides*, *Fusarium cerealis*, *Fusarium crookwellense*, *Fusarium culmorum*, *Fusarium graminearum*, *Fusarium graminum*, *Fusarium heterosporum*, *Fusarium negundi*, *Fusarium oxysporum*, *Fusarium reticulatum*, *Fusarium roseum*, *Fusarium sambucinum*, *Fusarium sarcochroum*, *Fusarium sporotrichioides*, *Fusarium sulphureum*, *Fusarium torulosum*, *Fusarium trichothecioides*, *Fusarium venenatum*, *Humicola grisea*, *Humicola insolens*, *Humicola lanuginosa*, *Irpex lacteus*, *Mucor miehei*, *Myceliophthora thermophila*, *Neurospora crassa*, *Penicillium funiculosum*, *Penicillium purpurogenum*, *Phanerochaete chrysosporium*, *Thielavia achromatica*, *Thielavia albomyces*, *Thielavia albopilosa*, *Thielavia australeinsis*, *Thielavia fimeti*, *Thielavia microspora*, *Thielavia ovispora*, *Thielavia peruviana*, *Thielavia setosa*, *Thielavia spededonium*, *Thielavia subthermophila*, *Thielavia terrestris*, *Trichoderma harzianum*, *Trichoderma koningii*, *Trichoderma longibrachiatum*, *Trichoderma reesei*, or *Trichoderma viride*.

It will be understood that for the aforementioned species, the invention encompasses both the perfect and imperfect states, and other taxonomic equivalents, e.g., anamorphs,

regardless of the species name by which they are known. Those skilled in the art will readily recognize the identity of appropriate equivalents.

Strains of these species are readily accessible to the public in a number of culture collections, such as the American Type Culture Collection (ATCC), Deutsche Sammlung von Mikroorganismen und Zellkulturen GmbH (DSMZ), Centraalbureau Voor Schimmelcultures (CBS), and Agricultural Research Service Patent Culture Collection, Northern Regional Research Center (NRRL).

The alpha-amylase coding sequences described or referenced herein, or a subsequence thereof, as well as the alpha-amylases described or referenced herein, or a fragment thereof, may be used to design nucleic acid probes to identify and clone DNA encoding an alpha-amylase from strains of different genera or species according to methods well known in the art. In particular, such probes can be used for hybridization with the genomic DNA or cDNA of a cell of interest, following standard Southern blotting procedures, in order to identify and isolate the corresponding gene therein. Such probes can be considerably shorter than the entire sequence, but should be at least 15, e.g., at least 25, at least 35, or at least 70 nucleotides in length. Preferably, the nucleic acid probe is at least 100 nucleotides in length, e.g., at least 200 nucleotides, at least 300 nucleotides, at least 400 nucleotides, at least 500 nucleotides, at least 600 nucleotides, at least 700 nucleotides, at least 800 nucleotides, or at least 900 nucleotides in length. Both DNA and RNA probes can be used. The probes are typically labeled for detecting the corresponding gene (for example, with ^{32}P , ^3H , ^{35}S , biotin, or avidin).

A genomic DNA or cDNA library prepared from such other strains may be screened for DNA that hybridizes with the probes described above and encodes a parent. Genomic or other DNA from such other strains may be separated by agarose or polyacrylamide gel electrophoresis, or other separation techniques. DNA from the libraries or the separated DNA may be transferred to and immobilized on nitrocellulose or other suitable carrier material. In order to identify a clone or DNA that hybridizes with a coding sequence, or a subsequence thereof, the carrier material is used in a Southern blot.

In one embodiment, the nucleic acid probe is a polynucleotide, or subsequence thereof, that encodes the alpha-amylase of any one of SEQ ID NOs: 76-101, 121-174 and 231, or a fragment thereof.

For purposes of the probes described above, hybridization indicates that the polynucleotide hybridizes to a labeled nucleic acid probe, or the full-length complementary strand thereof, or a subsequence of the foregoing; under very low to very high stringency conditions. Molecules to which the nucleic acid probe hybridizes under these conditions can be detected using, for example, X-ray film. Stringency and washing conditions are defined as described supra.

In one embodiment, the alpha-amylase is encoded by a polynucleotide that hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence for any one of the alpha-amylases described or referenced herein (e.g., the coding sequence that encodes any one of SEQ ID NOs: 76-101, 121-174 and 231). (Sambrook et al., 1989, *Molecular Cloning, A Laboratory Manual*, 2d edition, Cold Spring Harbor, New York).

The alpha-amylase may also be identified and obtained from other sources including microorganisms isolated from

nature (e.g., soil, composts, water, silage, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, silage, etc.) using the above-mentioned probes. Techniques for isolating microorganisms and DNA directly from natural habitats are well known in the art. The polynucleotide encoding an alpha-amylase may then be derived by similarly screening a genomic or cDNA library of another microorganism or mixed DNA sample.

Once a polynucleotide encoding an alpha-amylase has been detected with a suitable probe as described herein, the sequence may be isolated or cloned by utilizing techniques that are known to those of ordinary skill in the art (see, e.g., Sambrook et al., 1989, *Molecular Cloning, A Laboratory Manual*, 2d edition, Cold Spring Harbor, New York). Techniques used to isolate or clone polynucleotides encoding alpha-amylases include isolation from genomic DNA, preparation from cDNA, or a combination thereof. The cloning of the polynucleotides from such genomic DNA can be effected, e.g., by using the well-known polymerase chain reaction (PCR) or antibody screening of expression libraries to detect cloned DNA fragments with shares structural features. See, e.g., Innis et al., 1990, *PCR: A Guide to Methods and Application*, Academic Press, New York. Other nucleic acid amplification procedures such as ligase chain reaction (LCR), ligated activated transcription (LAT) and nucleotide sequence-based amplification (NASBA) may be used.

In one embodiment, the alpha-amylase has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any one of the alpha-amylases described or referenced herein (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231). In another embodiment, the alpha-amylase has a mature polypeptide sequence that is a fragment of the any one of the alpha-amylases described or referenced herein (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231). In one embodiment, the number of amino acid residues in the fragment is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of amino acid residues in referenced full length alpha-amylase (e.g. any one of SEQ ID NOs: 76-101, 121-174 and 231). In other embodiments, the alpha-amylase may comprise the catalytic domain of any alpha-amylase described or referenced herein (e.g., the catalytic domain of any one of SEQ ID NOs: 76-101, 121-174 and 231).

The alpha-amylase may be a variant of any one of the alpha-amylases described supra (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231). In one embodiment, the alpha-amylase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to any one of the alpha-amylases described supra (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231).

In one embodiment, the alpha-amylase has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of the alpha-amylases described supra (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231). In one embodiment, the alpha-amylase has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) of amino acid sequence of any one of the alpha-amylases described supra (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231). In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

The amino acid changes are generally of a minor nature, that is conservative amino acid substitutions or insertions that do not significantly affect the folding and/or activity of the protein; small deletions, typically of one to about 30 amino acids; small amino-terminal or carboxyl-terminal extensions, such as an amino-terminal methionine residue; a small linker peptide of up to about 20-25 residues; or a small extension that facilitates purification by changing net charge or another function, such as a poly-histidine tract, an antigenic epitope or a binding domain.

Examples of conservative substitutions are within the group of basic amino acids (arginine, lysine and histidine), acidic amino acids (glutamic acid and aspartic acid), polar amino acids (glutamine and asparagine), hydrophobic amino acids (leucine, isoleucine and valine), aromatic amino acids (phenylalanine, tryptophan and tyrosine), and small amino acids (glycine, alanine, serine, threonine and methionine). Amino acid substitutions that do not generally alter specific activity are known in the art and are described, for example, by H. Neurath and R. L. Hill, 1979, In, *The Proteins*, Academic Press, New York. The most commonly occurring exchanges are Ala/Ser, Val/Ile, Asp/Glu, Thr/Ser, Ala/Gly, Ala/Thr, Ser/Asn, Ala/Val, Ser/Gly, Tyr/Phe, Ala/Pro, Lys/Arg, Asp/Asn, Leu/Ile, Leu/Val, Ala/Glu, and Asp/Gly.

Alternatively, the amino acid changes are of such a nature that the physico-chemical properties of the polypeptides are altered. For example, amino acid changes may improve the thermal stability of the alpha-amylase, alter the substrate specificity, change the pH optimum, and the like.

Essential amino acids can be identified according to procedures known in the art, such as site-directed mutagenesis or alanine-scanning mutagenesis (Cunningham and Wells, 1989, *Science* 244: 1081-1085). In the latter technique, single alanine mutations are introduced at every residue in the molecule, and the resultant mutant molecules are tested for activity to identify amino acid residues that are critical to the activity of the molecule. See also, Hilton et al., 1996, *J. Biol. Chem.* 271: 4699-4708. The active site or other biological interaction can also be determined by physical analysis of structure, as determined by such techniques as nuclear magnetic resonance, crystallography, electron diffraction, or photoaffinity labeling, in conjunction with mutation of putative contact site amino acids. See, for example, de Vos et al., 1992, *Science* 255: 306-312; Smith et al., 1992, *J. Mol. Biol.* 224: 899-904; Wlodaver et al., 1992, *FEBS Lett.* 309: 59-64. The identities of essential amino acids can also be inferred from analysis of identities with other alpha-amylases that are related to the referenced alpha-amylase.

Additional guidance on the structure-activity relationship of the polypeptides herein can be determined using multiple sequence alignment (MSA) techniques well-known in the art. Based on the teachings herein, the skilled artisan could make similar alignments with any number of alpha-amylases described herein or known in the art. Such alignments aid the skilled artisan to determine potentially relevant domains (e.g., binding domains or catalytic domains), as well as which amino acid residues are conserved and not conserved among the different alpha-amylase sequences. It is appreciated in the art that changing an amino acid that is conserved at a particular position between disclosed polypeptides will more likely result in a change in biological activity (Bowie et al., 1990, *Science* 247: 1306-1310: "Residues that are directly involved in protein functions such as binding or catalysis will certainly be among the most conserved"). In contrast, substituting an amino acid that is not highly con-

served among the polypeptides will not likely or significantly alter the biological activity.

Even further guidance on the structure-activity relationship for the skilled artisan can be found in published x-ray crystallography studies known in the art.

Single or multiple amino acid substitutions, deletions, and/or insertions can be made and tested using known methods of mutagenesis, recombination, and/or shuffling, followed by a relevant screening procedure, such as those disclosed by Reidhaar-Olson and Sauer, 1988, *Science* 241: 53-57; Bowie and Sauer, 1989, *Proc. Natl. Acad. Sci. USA* 86: 2152-2156; WO95/17413; or WO95/22625. Other methods that can be used include error-prone PCR, phage display (e.g., Lowman et al., 1991, *Biochemistry* 30: 10832-10837; U.S. Pat. No. 5,223,409; WO92/06204), and region-directed mutagenesis (Derbyshire et al., 1986, *Gene* 46: 145; Ner et al., 1988, *DNA* 7: 127).

Mutagenesis/shuffling methods can be combined with high-throughput, automated screening methods to detect activity of cloned, mutagenized polypeptides expressed by host cells (Ness et al., 1999, *Nature Biotechnology* 17: 893-896). Mutagenized DNA molecules that encode active alpha-amylases can be recovered from the host cells and rapidly sequenced using standard methods in the art. These methods allow the rapid determination of the importance of individual amino acid residues in a polypeptide.

In some embodiments, the alpha-amylase has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the alpha-amylase activity of any alpha-amylase described or referenced herein (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231) under the same conditions.

In one embodiment, the alpha-amylase coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any alpha-amylase described or referenced herein (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231). In one embodiment, the alpha-amylase coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any alpha-amylase described or referenced herein (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231).

In one embodiment, the alpha-amylase comprises the coding sequence of any alpha-amylase described or referenced herein (any one of SEQ ID NOs: 76-101, 121-174 and 231). In one embodiment, the alpha-amylase comprises a coding sequence that is a subsequence of the coding sequence from any alpha-amylase described or referenced herein, wherein the subsequence encodes a polypeptide having alpha-amylase activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

The referenced coding sequence of any related aspect or embodiment described herein can be the native coding sequence or a degenerate sequence, such as a codon-optimized coding sequence designed for use in a particular host cell (e.g., optimized for expression in *Saccharomyces cerevisiae*).

The alpha-amylase may be a fused polypeptide or cleavable fusion polypeptide in which another polypeptide is

fused at the N-terminus or the C-terminus of the alpha-amylase. A fused polypeptide may be produced by fusing a polynucleotide encoding another polypeptide to a polynucleotide encoding the alpha-amylase. Techniques for producing fusion polypeptides are known in the art, and include ligating the coding sequences encoding the polypeptides so that they are in frame and that expression of the fused polypeptide is under control of the same promoter(s) and terminator. Fusion proteins may also be constructed using intein technology in which fusions are created post-translationally (Cooper et al., 1993, *EMBO J.* 12: 2575-2583; Dawson et al., 1994, *Science* 266: 776-779).

Trehalases

The expressed and/or exogenous trehalase can be any trehalase that is suitable for the fermenting organisms and/or their methods of use described herein, such as a naturally occurring trehalase or a variant thereof that retains trehalase activity. Any trehalase contemplated for expression by a fermenting organism described below is also contemplated for aspects of the invention involving exogenous addition of a trehalase (e.g., added before, during or after liquefaction and/or saccharification).

In some embodiments, the fermenting organism comprising a heterologous polynucleotide encoding a trehalase has an increased level of trehalase activity compared to the host cells without the heterologous polynucleotide encoding the trehalase, when cultivated under the same conditions. In some embodiments, the fermenting organism has an increased level of trehalase activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the fermenting organism without the heterologous polynucleotide encoding the trehalase, when cultivated under the same conditions.

Trehalases that may be expressed with the fermenting organisms and used with the methods described herein include, but are not limited to, trehalases shown in Table 2 (or derivatives thereof).

TABLE 2

Donor Organism (catalytic domain)	SEQ ID NO: (mature polypeptide)
<i>Chaetomium megalocarpum</i>	175
<i>Lecanicillium psalliotae</i>	176
<i>Doratomyces</i> sp	177
<i>Mucor moelleri</i>	178
<i>Phialophora cyclaminis</i>	179
<i>Thielavia arenaria</i>	180
<i>Thielavia antarctica</i>	181
<i>Chaetomium</i> sp	182
<i>Chaetomium nigricolor</i>	183
<i>Chaetomium jodhpurens</i>	184
<i>Chaetomium piluliferum</i>	185
<i>Myceliophthora himulea</i>	186
<i>Cladidium virescens</i>	187
<i>Gelasinospora cratophora</i>	188
<i>Acidobacteriaceae bacterium</i>	189
<i>Acidobacterium capsulatum</i>	190
<i>Acidovorax wautersii</i>	191
<i>Xanthomonas arboricola</i>	192
<i>Kosakonia sacchari</i>	193
<i>Enterobacter</i> sp	194
<i>Saitozyma flava</i>	195
<i>Phaeotremella skinneri</i>	196
<i>Trichoderma asperellum</i>	197
<i>Corynascus sepedonium</i>	198
<i>Myceliophthora thermophila</i>	199
<i>Trichoderma reesei</i>	200
<i>Chaetomium virescens</i>	201
<i>Rhodothermus marinus</i>	202

TABLE 2-continued

Donor Organism (catalytic domain)	SEQ ID NO: (mature polypeptide)
<i>Myceliophthora sepedonium</i>	203
<i>Moelleriella libera</i>	204
<i>Acremonium dichromosporum</i>	205
<i>Fusarium sambucinum</i>	206
<i>Phoma</i> sp	207
<i>Lentinus similis</i>	208
<i>Diaporthe nobilis</i>	209
<i>Solicozozyma terricola</i>	210
<i>Dioszegia cryoxerica</i>	211
<i>Talaromyces funiculosus</i>	212
<i>Hamigera avellanea</i>	213
<i>Talaromyces ruber</i>	214
<i>Trichoderma lixii</i>	215
<i>Aspergillus cervinus</i>	216
<i>Rasamsonia brevistipitata</i>	217
<i>Acremonium curvulum</i>	218
<i>Talaromyces piceae</i>	219
<i>Penicillium</i> sp	220
<i>Talaromyces aurantiacus</i>	221
<i>Talaromyces pinophilus</i>	222
<i>Talaromyces leycettanus</i>	223
<i>Talaromyces variabilis</i>	224
<i>Aspergillus niger</i>	225
<i>Trichoderma reesei</i>	226

Additional polynucleotides encoding suitable trehalases may be derived from microorganisms of any suitable genus, including those readily available within the UniProtKB database (www.uniprot.org).

The trehalase coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding trehalases from strains of different genera or species, as described supra.

The polynucleotides encoding trehalases may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc.) as described supra.

Techniques used to isolate or clone polynucleotides encoding trehalases are described supra.

In one embodiment, the trehalase has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any one of the trehalases described or referenced herein (e.g., any one of SEQ ID NOs: 175-226). In another embodiment, the trehalase has a mature polypeptide sequence that is a fragment of the any one of the trehalases described or referenced herein (e.g., any one of SEQ ID NOs: 175-226). In one embodiment, the number of amino acid residues in the fragment is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of amino acid residues in referenced full length trehalase (e.g. any one of SEQ ID NOs: 175-226). In other embodiments, the trehalase may comprise the catalytic domain of any trehalase described or referenced herein (e.g., the catalytic domain of any one of SEQ ID NOs: 175-226).

The trehalase may be a variant of any one of the trehalases described supra (e.g., any one of SEQ ID NOs: 175-226). In one embodiment, the trehalase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to any one of the trehalases described supra (e.g., any one of SEQ ID NOs: 175-226).

In one embodiment, the trehalase has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino

acid sequence of any one of the trehalases described supra (e.g., any one of SEQ ID NOs: 175-226). In one embodiment, the trehalase has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) of amino acid sequence of any one of the trehalases described supra (e.g., any one of SEQ ID NOs: 175-226). In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

In some embodiments, the trehalase has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the trehalase activity of any trehalase described or referenced herein (e.g., any one of SEQ ID NOs: 175-226) under the same conditions.

In one embodiment, the trehalase coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any trehalase described or referenced herein (e.g., any one of SEQ ID NOs: 175-226). In one embodiment, the trehalase coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any trehalase described or referenced herein (e.g., any one of SEQ ID NOs: 175-226).

In one embodiment, the trehalase comprises the coding sequence of any trehalase described or referenced herein (any one of SEQ ID NOs: 175-226). In one embodiment, the trehalase comprises a coding sequence that is a subsequence of the coding sequence from any trehalase described or referenced herein, wherein the subsequence encodes a polypeptide having trehalase activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

The referenced coding sequence of any related aspect or embodiment described herein can be the native coding sequence or a degenerate sequence, such as a codon-optimized coding sequence designed for use in a particular host cell (e.g., optimized for expression in *Saccharomyces cerevisiae*).

The trehalase can also include fused polypeptides or cleavable fusion polypeptides, as described supra.

Glucoamylases

The expressed and/or exogenous glucoamylase can be any glucoamylase that is suitable for the fermenting organisms and/or their methods of use described herein, such as a naturally occurring glucoamylase or a variant thereof that retains glucoamylase activity. Any glucoamylase contemplated for expression by a fermenting organism described below is also contemplated for aspects of the invention involving exogenous addition of a glucoamylase (e.g., added before, during or after liquefaction and/or saccharification).

In some embodiments, the fermenting organism comprises a heterologous polynucleotide encoding a glucoamylase, for example, as described in WO2017/087330, the content of which is hereby incorporated by reference. Any glucoamylase described or referenced herein is contemplated for expression in the fermenting organism.

In some embodiments, the fermenting organism comprising a heterologous polynucleotide encoding an glucoamylase has an increased level of glucoamylase activity com-

pared to the host cells without the heterologous polynucleotide encoding the glucoamylase, when cultivated under the same conditions. In some embodiments, the fermenting organism has an increased level of glucoamylase activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the fermenting organism without the heterologous polynucleotide encoding the glucoamylase, when cultivated under the same conditions.

Exemplary glucoamylases that can be used with the host cells and/or the methods described herein include bacterial, yeast, or filamentous fungal glucoamylases, e.g., obtained from any of the microorganisms described or referenced herein, as described supra under the sections related to alpha-amylases.

The glucoamylase may be derived from any suitable source, e.g., derived from a microorganism or a plant. Preferred glucoamylases are of fungal or bacterial origin, selected from the group consisting of *Aspergillus* glucoamylases, in particular *Aspergillus niger* G1 or G2 glucoamylase (Boel et al. (1984), EMBO J. 3 (5), p. 1097-1102), or variants thereof, such as those disclosed in WO92/00381, WO00/04136 and WO01/04273 (from Novozymes, Denmark); the *A. awamori* glucoamylase disclosed in WO84/02921, *Aspergillus oryzae* glucoamylase (Agric. Biol. Chem. (1991), 55 (4), p. 941-949), or variants or fragments thereof. Other *Aspergillus* glucoamylase variants include variants with enhanced thermal stability: G137A and G139A (Chen et al. (1996), Prot. Eng. 9, 499-505); D257E and D293E/Q (Chen et al. (1995), Prot. Eng. 8, 575-582); N182 (Chen et al. (1994), Biochem. J. 301, 275-281); disulphide bonds, A246C (Fierobe et al. (1996), Biochemistry, 35, 8698-8704; and introduction of Pro residues in position A435 and S436 (Li et al. (1997), Protein Eng. 10, 1199-1204).

Other glucoamylases include *Athelia rolfsii* (previously denoted *Corticium rolfsii*) glucoamylase (see U.S. Pat. No. 4,727,026 and (Nagasaka et al. (1998) "Purification and properties of the raw-starch-degrading glucoamylases from *Corticium rolfsii*, Appl Microbiol Biotechnol 50:323-330), *Talaromyces* glucoamylases, in particular derived from *Talaromyces emersonii* (WO99/28448), *Talaromyces leycettanus* (U.S. Pat. No. Re. 32,153), *Talaromyces dupontii*, *Talaromyces thermophilus* (U.S. Pat. No. 4,587,215). In one embodiment, the glucoamylase used during saccharification and/or fermentation is the *Talaromyces emersonii* glucoamylase disclosed in WO99/28448.

Bacterial glucoamylases contemplated include glucoamylases from the genus *Clostridium*, in particular *C. thermoamylolyticum* (EP 135,138), and *C. thermohydrosulfuricum* (WO86/01831).

Contemplated fungal glucoamylases include *Trametes cingulate*, *Pachykytospora papyracea*; and *Leucopaxillus giganteus* all disclosed in WO2006/069289; or *Peniophora rufomarginata* disclosed in WO2007/124285; or a mixture thereof. Also hybrid glucoamylase are contemplated. Examples include the hybrid glucoamylases disclosed in WO2005/045018.

In one embodiment, the glucoamylase is derived from a strain of the genus *Pycnoporus*, in particular a strain of *Pycnoporus* as described in WO2011/066576 (SEQ ID NO: 2, 4 or 6 therein), including the *Pycnoporus sanguineus* glucoamylase, or from a strain of the genus *Gloeophyllum*, such as a strain of *Gloeophyllum sepiarium* or *Gloeophyllum trabeum*, in particular a strain of *Gloeophyllum* as described in WO2011/068803 (SEQ ID NO: 2, 4, 6, 8, 10, 12, 14 or 16

therein). In one embodiment, the glucoamylase is SEQ ID NO: 2 in WO2011/068803 (i.e. *Gloeophyllum sepiarium* glucoamylase). In one embodiment, the glucoamylase is the *Gloeophyllum sepiarium* glucoamylase of SEQ ID NO: 8. In one embodiment, the glucoamylase is the *Pycnoporus sanguineus* glucoamylase of SEQ ID NO: 229.

In one embodiment, the glucoamylase is a *Gloeophyllum trabeum* glucoamylase (disclosed as SEQ ID NO: 3 in WO2014/177546). In another embodiment, the glucoamylase is derived from a strain of the genus *Nigrospora*, in particular a strain of *Nigrospora* sp. disclosed in WO2012/064351 (disclosed as SEQ ID NO: 2 therein).

Also contemplated are glucoamylases with a mature polypeptide sequence which exhibit a high identity to any of the above mentioned glucoamylases, i.e., at least 60%, such as at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% or even 100% identity to any one of the mature polypeptide sequences mentioned above.

Glucoamylases may be added to the saccharification and/or fermentation in an amount of 0.0001-20 AGU/g DS, such as 0.001-10 AGU/g DS, 0.01-5 AGU/g DS, or 0.1-2 AGU/g DS.

Glucoamylases may be added to the saccharification and/or fermentation in an amount of 1-1,000 µg EP/g DS, such as 10-500 µg/g DS, or 25-250 µg/g DS.

Glucoamylases may be added to liquefaction in an amount of 0.1-100 µg EP/g DS, such as 0.5-50 µg EP/g DS, 1-25 µg EP/g DS, or 2-12 µg EP/g DS.

In one embodiment, the glucoamylase is added as a blend further comprising an alpha-amylase (e.g., any alpha-amylase described herein). In one embodiment, the alpha-amylase is a fungal alpha-amylase, especially an acid fungal alpha-amylase. The alpha-amylase is typically a side activity.

In one embodiment, the glucoamylase is a blend comprising *Talaromyces emersonii* glucoamylase disclosed in WO99/28448 as SEQ ID NO: 34 and *Trametes cingulata* glucoamylase disclosed as SEQ ID NO: 2 in WO06/069289.

In one embodiment, the glucoamylase is a blend comprising *Talaromyces emersonii* glucoamylase disclosed in WO99/28448, *Trametes cingulata* glucoamylase disclosed as SEQ ID NO: 2 in WO06/69289, and an alpha-amylase.

In one embodiment, the glucoamylase is a blend comprising *Talaromyces emersonii* glucoamylase disclosed in WO99/28448, *Trametes cingulata* glucoamylase disclosed in WO06/69289, and *Rhizomucor pusillus* alpha-amylase with *Aspergillus niger* glucoamylase linker and SBD disclosed as V039 in Table 5 in WO2006/069290.

In one embodiment, the glucoamylase is a blend comprising *Gloeophyllum sepiarium* glucoamylase shown as SEQ ID NO: 2 in WO2011/068803 and an alpha-amylase, in particular *Rhizomucor pusillus* alpha-amylase with an *Aspergillus niger* glucoamylase linker and starch-binding domain (SBD), disclosed SEQ ID NO: 3 in WO2013/006756, in particular with the following substitutions: G128D+D143N.

In one embodiment, the alpha-amylase may be derived from a strain of the genus *Rhizomucor*, preferably a strain the *Rhizomucor pusillus*, such as the one shown in SEQ ID NO: 3 in WO2013/006756, or the genus *Meripilus*, preferably a strain of *Meripilus giganteus*. In one embodiment, the alpha-amylase is derived from a *Rhizomucor pusillus* with an *Aspergillus niger* glucoamylase linker and starch-binding domain (SBD), disclosed as V039 in Table 5 in WO2006/069290.

In one embodiment, the *Rhizomucor pusillus* alpha-amylase or the *Rhizomucor pusillus* alpha-amylase with an *Aspergillus niger* glucoamylase linker and starch-binding domain (SBD) has at least one of the following substitutions or combinations of substitutions: D165M; Y141W; Y141R; K136F; K192R; P224A; P224R; S123H+Y141W; G20S+Y141W; A76G+Y141W; G128D+Y141W; G128D+D143N; P219C+Y141W; N142D+D143N; Y141W+K192R; Y141W+D143N; Y141W+N383R; Y141W+P219C+
A265C; Y141W+N142D+D143N; Y141W+K192R V410A; G128D+Y141W+D143N; Y141W+D143N+P219C; Y141W+D143N+K192R; G128D+D143N+K192R; Y141W+D143N+K192R+P219C; and G128D+Y141W+D143N+K192R; or G128D+Y141W+D143N+K192R+P219C (using SEQ ID NO: 3 in WO2013/006756 for numbering).

In one embodiment, the glucoamylase blend comprises *Gloeophyllum sepiarium* glucoamylase (e.g., SEQ ID NO: 2 in WO2011/068803) and *Rhizomucor pusillus* alpha-amylase.

In one embodiment, the glucoamylase blend comprises *Gloeophyllum sepiarium* glucoamylase shown as SEQ ID NO: 2 in WO2011/068803 and *Rhizomucor pusillus* with an *Aspergillus niger* glucoamylase linker and starch-binding domain (SBD), disclosed SEQ ID NO: 3 in WO2013/006756 with the following substitutions: G128D+D143N.

Commercially available compositions comprising glucoamylase include AMG 200L; AMG 300 L; SANTM SUPER, SANTM EXTRA L, SPIRIZYME[®] PLUS, SPIRIZYME[®] FUEL, SPIRIZYME[®] B4U, SPIRIZYME[®] ULTRA, SPIRIZYME[®] EXCEL, SPIRIZYME ACHIEVE[®], and AMG[®] E (from Novozymes A/S); OPTIDEXTM 300, GC480, GC417 (from DuPont-Danisco); AMIGASETM and AMIGASETM PLUS (from DSM); G-ZYMETM G900, G-ZYMETM and G990 ZR (from DuPont-Danisco).

In one embodiment, the glucoamylase is derived from the *Debaryomyces occidentalis* glucoamylase of SEQ ID NO: 102. In one embodiment, the glucoamylase is derived from the *Saccharomycopsis fibuligera* glucoamylase of SEQ ID NO: 103. In one embodiment, the glucoamylase is derived from the *Saccharomycopsis fibuligera* glucoamylase of SEQ ID NO: 104. In one embodiment, the glucoamylase is derived from the *Saccharomyces cerevisiae* glucoamylase of SEQ ID NO: 105. In one embodiment, the glucoamylase is derived from the *Aspergillus niger* glucoamylase of SEQ ID NO: 106. In one embodiment, the glucoamylase is derived from the *Aspergillus oryzae* glucoamylase of SEQ ID NO: 107. In one embodiment, the glucoamylase is derived from the *Rhizopus oryzae* glucoamylase of SEQ ID NO: 108. In one embodiment, the glucoamylase is derived from the *Clostridium thermocellum* glucoamylase of SEQ ID NO: 109. In one embodiment, the glucoamylase is derived from the *Clostridium thermocellum* glucoamylase of SEQ ID NO: 110. In one embodiment, the glucoamylase is derived from the *Arxula adeninivorans* glucoamylase of SEQ ID NO: 111. In one embodiment, the glucoamylase is derived from the *Hormoconis resinae* glucoamylase of SEQ ID NO: 112. In one embodiment, the glucoamylase is derived from the *Aureobasidium pullulans* glucoamylase of SEQ ID NO: 113.

In one embodiment, the glucoamylase is a *Trichoderma reesei* glucoamylase, such as the *Trichoderma reesei* glucoamylase of SEQ ID NO: 230.

In one embodiment, the glucoamylase has a Relative Activity heat stability at 85° C. of at least 20%, at least 30%, or at least 35% determined as described in Example 4 of WO2018/098381 (heat stability).

In one embodiment, the glucoamylase has a relative activity pH optimum at pH 5.0 of at least 90%, e.g., at least 95%, at least 97%, or 100% determined as described in Example 4 of WO2018/098381 (pH optimum).

In one embodiment, the glucoamylase has a pH stability at pH 5.0 of at least 80%, at least 85%, at least 90% determined as described in Example 4 of WO2018/098381 (pH stability).

In one embodiment, the glucoamylase used in liquefaction, such as a *Penicillium oxalicum* glucoamylase variant, has a thermostability determined as DSC Td at pH 4.0 as described in Example 15 of WO2018/098381 of at least 70° C., preferably at least 75° C., such as at least 80° C., such as at least 81° C., such as at least 82° C., such as at least 83° C., such as at least 84° C., such as at least 85° C., such as at least 86° C., such as at least 87%, such as at least 88° C., such as at least 89° C., such as at least 90° C. In one embodiment, the glucoamylase, such as a *Penicillium oxalicum* glucoamylase variant, has a thermostability determined as DSC Td at pH 4.0 as described in Example 15 of WO2018/098381 in the range between 70° C. and 95° C., such as between 80° C. and 90° C.

In one embodiment, the glucoamylase, such as a *Penicillium oxalicum* glucoamylase variant, used in liquefaction has a thermostability determined as DSC Td at pH 4.8 as described in Example 15 of WO2018/098381 of at least 70° C., preferably at least 75° C., such as at least 80° C., such as at least 81° C., such as at least 82° C., such as at least 83° C., such as at least 84° C., such as at least 85° C., such as at least 86° C., such as at least 87%, such as at least 88° C., such as at least 89° C., such as at least 90° C., such as at least 91° C. In one embodiment, the glucoamylase, such as a *Penicillium oxalicum* glucoamylase variant, has a thermostability determined as DSC Td at pH 4.8 as described in Example 15 of WO2018/098381 in the range between 70° C. and 95° C., such as between 80° C. and 90° C.

In one embodiment, the glucoamylase, such as a *Penicillium oxalicum* glucoamylase variant, used in liquefaction has a residual activity determined as described in Example 16 of WO2018/098381, of at least 100% such as at least 105%, such as at least 110%, such as at least 115%, such as at least 120%, such as at least 125%. In one embodiment, the glucoamylase, such as a *Penicillium oxalicum* glucoamylase variant, has a thermostability determined as residual activity as described in Example 16 of WO2018/098381, in the range between 100% and 130%.

In one embodiment, the glucoamylase, e.g., of fungal origin such as a filamentous fungi, from a strain of the genus *Penicillium*, e.g., a strain of *Penicillium oxalicum*, in particular the *Penicillium oxalicum* glucoamylase disclosed as SEQ ID NO: 2 in WO2011/127802 (which is hereby incorporated by reference).

In one embodiment, the glucoamylase has a mature polypeptide sequence of at least 80%, e.g., at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% or 100% identity to the mature polypeptide shown in SEQ ID NO: 2 in WO2011/127802.

In one embodiment, the glucoamylase is a variant of the *Penicillium oxalicum* glucoamylase disclosed as SEQ ID NO: 2 in WO2011/127802, having a K79V substitution. The K79V glucoamylase variant has reduced sensitivity to protease degradation relative to the parent as disclosed in WO2013/036526 (which is hereby incorporated by reference).

In one embodiment, the glucoamylase is derived from *Penicillium oxalicum*.

In one embodiment, the glucoamylase is a variant of the *Penicillium oxalicum* glucoamylase disclosed as SEQ ID NO: 2 in WO2011/127802. In one embodiment, the *Penicillium oxalicum* glucoamylase is the one disclosed as SEQ ID NO: 2 in WO2011/127802 having Val (V) in position 79.

Contemplated *Penicillium oxalicum* glucoamylase variants are disclosed in WO2013/053801 which is hereby incorporated by reference.

In one embodiment, these variants have reduced sensitivity to protease degradation.

In one embodiment, these variant have improved thermostability compared to the parent.

In one embodiment, the glucoamylase has a K79V substitution (using SEQ ID NO: 2 of WO2011/127802 for numbering), corresponding to the PE001 variant, and further comprises one of the following alterations or combinations of alterations

T65A; Q327F; E501V; Y504T; Y504*; T65A+Q327F; T65A+E501V; T65A+Y504T; T65A+Y504*; Q327F+E501V; Q327F+Y504T; Q327F+Y504*; E501V+Y504T; E501V+Y504*; T65A+Q327F+E501V; T65A+Q327F+Y504T; T65A+E501V+Y504T; Q327F+E501V+Y504T; T65A+Q327F+Y504*; T65A+E501V+Y504*; Q327F+E501V+Y504*; T65A+Q327F+E501V+Y504T; T65A+Q327F+E501V+Y504*; E501V+Y504T; T65A+K161S; T65A+Q405T; T65A+Q327W; T65A+Q327F; T65A+Q327Y; P11F+T65A+Q327F; R1K+D3W+K5Q+G7V+N8S+T10K+P11S+T65A+Q327F; P2N+P4S+P11F+T65A+Q327F; P11F+D26C+K33C+T65A+Q327F; P2N+P4S+P11F+T65A+Q327W+E501V+Y504T; R1E+D3N+P4G+G6R+G7A+N8A+T10D+P11D+T65A+Q327F; P11F+T65A+Q327W; P2N+P4S+P11F+T65A+Q327F+E501V+Y504T; P11F+T65A+Q327W+E501V+Y504T; T65A+Q327F+E501V+Y504T; T65A+S105P+Q327W; T65A+S105P+Q327F; T65A+Q327W+S364P; T65A+Q327F+S364P; T65A+S103N+Q327F; P2N+P4S+P11F+K34Y+T65A+Q327F; P2N+P4S+P11F+T65A+Q327F+D445N+V447S; P2N+P4S+P11F+T65A+I172V+Q327F; P2N+P4S+P11F+T65A+Q327F+N502*; P2N+P4S+P11F+T65A+Q327F+N502T+P563S+K571E; P2N+P4S+P11F+R31S+K33V+T65A+Q327F+N564D+K571S; P2N+P4S+P11F+T65A+Q327F+S377T; P2N+P4S+P11F+T65A+V325T+Q327W; P2N+P4S+P11F+T65A+Q327F+D445N+V447S+E501V+Y504T; P2N+P4S+P11F+T65A+I172V+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+Q327F+S377T+E501V+Y504T; P2N+P4S+P11F+D26N+K34Y+T65A+Q327F; P2N+P4S+P11F+T65A+Q327F+I375A+E501V+Y504T; P2N+P4S+P11F+T65A+K218A+K221D+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+S103N+Q327F+E501V+Y504T; P2N+P4S+T10D+T65A+Q327F+E501V+Y504T; P2N+P4S+F12Y+T65A+Q327F+E501V+Y504T; K5A+P11F+T65A+Q327F+E501V+Y504T; P2N+P4S+T10E+E18N+T65A+Q327F+E501V+Y504T; P2N+T10E+E18N+T65A+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+Q327F+E501V+Y504T+T568N; P2N+P4S+P11F+T65A+Q327F+E501V+Y504T+K524T+G526A; P2N+P4S+P11F+K34Y+T65A+Q327F+D445N+V447S+E501V+Y504T; P2N+P4S+P11F+R31S+K33V+T65A+Q327F+D445N+V447S+E501V+Y504T; P2N+P4S+P11F+D26N+K34Y+T65A+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+F80*+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+K112S+

Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+Q327F+E501V+Y504T+T516P+K524T+G526A; P2N+P4S+P11F+T65A+Q327F+E501V+N502T+Y504*; P2N+P4S+P11F+T65A+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+S103N+Q327F+E501V+Y504T; K5A+P11F+T65A+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+Q327F+E501V+Y504T+T516P+K524T+G526A; P2N+P4S+P11F+T65A+V79A+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+V79G+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+V79I+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+V79L+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+V79S+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+L72V+Q327F+E501V+Y504T; S255N+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+E74N+V79K+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+G220N+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+Y245N+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+Q253N+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+D279N+Q327F+E501V+Y504T; P2N+P4S+P11F+T65A+Q327F+S359N+E501V+Y504T; P2N+P4S+P11F+T65A+Q327F+D370N+E501V+Y504T; P2N+P4S+P11F+T65A+Q327F+V460S+E501V+Y504T; P2N+P4S+P11F+T65A+Q327F+V460T+P468T+E501V+Y504T; P2N+P4S+P11F+T65A+Q327F+T463N+E501V+Y504T; P2N+P4S+P11F+T65A+Q327F+S465N+E501V+Y504T; and P2N+P4S+P11F+T65A+Q327F+T477N+E501V+Y504T.

In one embodiment, the *Penicillium oxalicum* glucoamylase variant has a K79V substitution (using SEQ ID NO: 2 of WO2011/127802 for numbering), corresponding to the PE001 variant, and further comprises one of the following substitutions or combinations of substitutions:

P11F+T65A+Q327F;
P2N+P4S+P11F+T65A+Q327F;
P11F+D26C+K33C+T65A+Q327F;
P2N+P4S+P11F+T65A+Q327W+E501V+Y504T;
P2N+P4S+P11F+T65A+Q327F+E501V+Y504T; and
P11F+T65A+Q327W+E501V+Y504T.

Additional glucoamylases contemplated for use with the present invention can be found in WO2011/153516 (the content of which is incorporated herein).

Additional polynucleotides encoding suitable glucoamylases may be obtained from microorganisms of any genus, including those readily available within the UniProtKB database (www.uniprot.org).

The glucoamylase coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding glucoamylases from strains of different genera or species, as described supra.

The polynucleotides encoding glucoamylases may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc.) as described supra.

Techniques used to isolate or clone polynucleotides encoding glucoamylases are described supra.

In one embodiment, the glucoamylase has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any one of the glucoamylases described or referenced herein (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230). In another embodiment, the glucoamylase has a mature polypeptide sequence that is a fragment of the any one of the glucoamylases described or referenced herein (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230). In

one embodiment, the number of amino acid residues in the fragment is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of amino acid residues in referenced full length glucoamylase (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230). In other embodiments, the glucoamylase may comprise the catalytic domain of any glucoamylase described or referenced herein (e.g., the catalytic domain of any one of SEQ ID NOs: 8, 102-113, 229 and 230).

The glucoamylase may be a variant of any one of the glucoamylases described supra (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230). In one embodiment, the glucoamylase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to any one of the glucoamylases described supra (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230).

In one embodiment, the glucoamylase has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of the glucoamylases described supra (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230). In one embodiment, the glucoamylase has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) of amino acid sequence of any one of the glucoamylases described supra (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230). In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

In some embodiments, the glucoamylase has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the glucoamylase activity of any glucoamylase described or referenced herein (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230) under the same conditions.

In one embodiment, the glucoamylase coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any glucoamylase described or referenced herein (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230). In one embodiment, the glucoamylase coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any glucoamylase described or referenced herein (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230).

In one embodiment, the glucoamylase comprises the coding sequence of any glucoamylase described or referenced herein (any one of SEQ ID NOs: 8, 102-113, 229 and 230). In one embodiment, the glucoamylase comprises a coding sequence that is a subsequence of the coding sequence from any glucoamylase described or referenced herein, wherein the subsequence encodes a polypeptide having glucoamylase activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

The referenced coding sequence of any related aspect or embodiment described herein can be the native coding

sequence or a degenerate sequence, such as a codon-optimized coding sequence designed for use in a particular host cell (e.g., optimized for expression in *Saccharomyces cerevisiae*).

The glucoamylase can also include fused polypeptides or cleavable fusion polypeptides, as described supra.

Proteases

The expressed and/or exogenous protease can be any protease that is suitable for the fermenting organisms and/or their methods of use described herein, such as a naturally occurring protease or a variant thereof that retains protease activity. Any protease contemplated for expression by a fermenting organism described below is also contemplated for aspects of the invention involving exogenous addition of a protease.

Proteases are classified on the basis of their catalytic mechanism into the following groups: Serine proteases (S), Cysteine proteases (C), Aspartic proteases (A), Metallo proteases (M), and Unknown, or as yet unclassified, proteases (U), see Handbook of Proteolytic Enzymes, A. J. Barrett, N. D. Rawlings, J. F. Woessner (eds), Academic Press (1998), in particular the general introduction part.

Protease activity can be measured using any suitable assay, in which a substrate is employed, that includes peptide bonds relevant for the specificity of the protease in question. Assay-pH and assay-temperature are likewise to be adapted to the protease in question. Examples of assay-pH-values are pH 6, 7, 8, 9, 10, or 11. Examples of assay-temperatures are 30, 35, 37, 40, 45, 50, 55, 60, 65, 70 or 80° C.

In some aspects, the fermenting organism comprising a heterologous polynucleotide encoding a protease has an increased level of protease activity compared to the fermenting organism without the heterologous polynucleotide encoding the protease, when cultivated under the same conditions. In some aspects, the fermenting organism has an increased level of protease activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the fermenting organism without the heterologous polynucleotide encoding the protease, when cultivated under the same conditions.

Exemplary proteases that may be expressed with the fermenting organisms and used with the methods described herein include, but are not limited to, proteases shown in Table 3 (or derivatives thereof).

TABLE 3

Donor Organism (catalytic domain)	SEQ ID NO: (mature polypeptide)	Family
<i>Aspergillus niger</i>	9	A1
<i>Trichoderma reesei</i>	10	
<i>Thermoascus</i>	11	M35
<i>aurantiacus</i>		
<i>Dichomitus squaleus</i>	12	S53
<i>Nocardioptis prasina</i>	13	S1
<i>Penicillium</i>	14	S10
<i>simplicissimum</i>		
<i>Aspergillus niger</i>	15	
<i>Meriphilus giganteus</i>	16	S53
<i>Lecanicillium</i> sp.	17	S53
WMM742		
<i>Talaromyces</i>	18	S53
<i>proteolyticus</i>		
<i>Penicillium</i>	19	A1A
<i>ranomafanaense</i>		
<i>Aspergillus oryzae</i>	20	S53
<i>Talaromyces liani</i>	21	S10

TABLE 3-continued

Donor Organism (catalytic domain)	SEQ ID NO: (mature polypeptide)	Family
5 <i>Thermoascus</i>	22	S53
<i>thermophilus</i>		
<i>Pyrococcus furiosus</i>	23	
<i>Trichoderma reesei</i>	24	
<i>Rhizomucor miehei</i>	25	
<i>Lenzites betulinus</i>	26	S53
10 <i>Neolentinus lepideus</i>	27	S53
<i>Thermococcus</i> sp.	28	S8
<i>Thermococcus</i> sp.	29	S8
<i>Thermomyces</i>	30	S53
<i>lanuginosus</i>		
<i>Thermococcus</i>	31	S53
15 <i>thioreducens</i>		
<i>Polyporus arcularius</i>	32	S53
<i>Ganoderma lucidum</i>	33	S53
<i>Ganoderma lucidum</i>	34	S53
<i>Ganoderma lucidum</i>	35	S53
<i>Trametes</i> sp. AH28-2	36	S53
<i>Cinereomyces lindbladii</i>	37	S53
20 <i>Trametes versicolor</i>	38	S53
O82DDP		
<i>Paecilomyces hepiali</i>	39	S53
<i>Isaria tenuipes</i>	40	S53
<i>Aspergillus tamarii</i>	41	S53
<i>Aspergillus brasiliensis</i>	42	S53
25 <i>Aspergillus iizukae</i>	43	S53
<i>Penicillium</i> sp-72364	44	S10
<i>Aspergillus denticulatus</i>	45	S10
<i>Hamigera</i> sp. t184-6	46	S10
<i>Penicillium janthinellum</i>	47	S10
<i>Penicillium vasconiae</i>	48	S10
30 <i>Hamigera paravellanea</i>	49	S10
<i>Talaromyces variabilis</i>	50	S10
<i>Penicillium arenicola</i>	51	S10
<i>Nocardioptis</i>	52	S1
<i>kunsanensis</i>		
<i>Streptomyces parvulus</i>	53	S1
35 <i>Saccharopolyspora</i>	54	S1
<i>endophytica</i>		
<i>luteus</i> cellwall	55	S1
enrichments K		
<i>Saccharothrix</i>	56	S1
<i>australiensis</i>		
<i>Nocardioptis</i>	57	S1
40 <i>baichengensis</i>		
<i>Streptomyces</i> sp. SM15	58	S1
<i>Actinoalloteichus</i>	59	S1
<i>spitiensis</i>		
<i>Byssoschlamys</i>	60	M35
<i>verrucosa</i>		
45 <i>Hamigera terricola</i>	61	M35
<i>Aspergillus tamarii</i>	62	M35
<i>Aspergillus niveus</i>	63	M35
<i>Penicillium sclerotiorum</i>	64	A1
<i>Penicillium bilaiae</i>	65	A1
<i>Penicillium antarcticum</i>	66	A1
50 <i>Penicillium sumatrense</i>	67	A1
<i>Trichoderma lixii</i>	68	A1
<i>Trichoderma</i>	69	A1
<i>brevicompactum</i>		
<i>Penicillium</i>	70	A1
<i>cinnamopurpureum</i>		
55 <i>Bacillus licheniformis</i>	71	S8
<i>Bacillus subtilis</i>	72	S8
<i>Trametes cf versicol</i>	73	S53

Additional polynucleotides encoding suitable proteases may be derived from microorganisms of any suitable genus, including those readily available within the UniProtKB database (www.uniprot.org).

In one embodiment, the protease is derived from *Aspergillus*, such as the *Aspergillus niger* protease of SEQ ID NO: 9, the *Aspergillus tamarii* protease of SEQ ID NO: 41, or the *Aspergillus denticulatus* protease of SEQ ID NO: 45. In one embodiment, the protease is derived from *Dichomitus*, such

as the *Dichomitus squalens* protease of SEQ ID NO: 12. In one embodiment, the protease is derived from *Penicillium*, such as the *Penicillium simplicissimum* protease of SEQ ID NO: 14, the *Penicillium antarcticum* protease of SEQ ID NO: 66, or the *Penicillium sumatrense* protease of SEQ ID NO: 67. In one aspect, the protease is derived from *Meriphilus*, such as the *Meriphilus giganteus* protease of SEQ ID NO: 16. In one aspect, the protease is derived from *Talaromyces*, such as the *Talaromyces liani* protease of SEQ ID NO: 21. In one aspect, the protease is derived from *Thermoascus*, such as the *Thermoascus thermophilus* protease of SEQ ID NO: 22. In one aspect, the protease is derived from *Ganoderma*, such as the *Ganoderma lucidum* protease of SEQ ID NO: 33. In one aspect, the protease is derived from *Hamigera*, such as the *Hamigera terricola* protease of SEQ ID NO: 61. In one aspect, the protease is derived from *Trichoderma*, such as the *Trichoderma brevicompactum* protease of SEQ ID NO: 69.

The protease coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding proteases from strains of different genera or species, as described supra.

The polynucleotides encoding proteases may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc.) as described supra.

Techniques used to isolate or clone polynucleotides encoding proteases are described supra.

In one embodiment, the protease has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any one of SEQ ID NOs: 9-73 (e.g., any one of SEQ ID NOs: 9, 14, 16, 21, 22, 33, 41, 45, 61, 62, 66, 67, and 69; such as any one of SEQ NOs: 9, 14, 16, and 69). In another embodiment, the protease has a mature polypeptide sequence that is a fragment of the protease of any one of SEQ ID NOs: 9-73 (e.g., wherein the fragment has protease activity). In one embodiment, the number of amino acid residues in the fragment is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of amino acid residues in referenced full length protease (e.g. any one of SEQ ID NOs: 9-73). In other embodiments, the protease may comprise the catalytic domain of any protease described or referenced herein (e.g., the catalytic domain of any one of SEQ ID NOs: 9-73).

The protease may be a variant of any one of the proteases described supra (e.g., any one of SEQ ID NOs: 9-73. In one embodiment, the protease has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to any one of the proteases described supra (e.g., any one of SEQ ID NOs: 9-73).

In one embodiment, the protease has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of the proteases described supra (e.g., any one of SEQ ID NOs: 9-73). In one embodiment, the protease has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) of amino acid sequence of any one of the proteases described supra (e.g., any one of SEQ ID NOs: 9-73). In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

In one embodiment, the protease coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any protease described or referenced herein (e.g., any one of SEQ ID NOs: 9-73). In one embodiment, the protease coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any protease described or referenced herein (e.g., any one of SEQ ID NOs: 9-73).

In one embodiment, the protease comprises the coding sequence of any protease described or referenced herein (any one of SEQ ID NOs: 9-73). In one embodiment, the protease comprises a coding sequence that is a subsequence of the coding sequence from any protease described or referenced herein, wherein the subsequence encodes a polypeptide having protease activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

The referenced coding sequence of any related aspect or embodiment described herein can be the native coding sequence or a degenerate sequence, such as a codon-optimized coding sequence designed for use in a particular host cell (e.g., optimized for expression in *Saccharomyces cerevisiae*).

The protease can also include fused polypeptides or cleavable fusion polypeptides, as described supra.

In one embodiment, the protease used according to a process described herein is a Serine proteases. In one particular embodiment, the protease is a serine protease belonging to the family 53, e.g., an endo-protease, such as S53 protease from *Meripilus giganteus*, *Dichomitus squalens*, *Trametes versicolor*, *Polyporus arcularius*, *Lezites betulinus*, *Ganoderma lucidum*, *Neolentinus lepideus*, or *Bacillus* sp. 19138, in a process for producing ethanol from a starch-containing material, the ethanol yield was improved, when the S53 protease was present/or added during saccharification and/or fermentation of either gelatinized or un-gelatinized starch. In one embodiment, the proteases is selected from: (a) proteases belonging to the EC 3.4.21 enzyme group; and/or (b) proteases belonging to the EC 3.4.14 enzyme group; and/or (c) Serine proteases of the peptidase family S53 that comprises two different types of peptidases: tripeptidyl aminopeptidases (exo-type) and endo-peptidases; as described in 1993, *Biochem. J.* 290:205-218 and in MEROPS protease database, release, 9.4 (31 Jan. 2011) (www.merops.ac.uk). The database is described in Rawlings, N. D., Barrett, A. J. and Bateman, A., 2010, "MEROPS: the peptidase database", *Nucl. Acids Res.* 38: D227-D233.

For determining whether a given protease is a Serine protease, and a family S53 protease, reference is made to the above Handbook and the principles indicated therein. Such determination can be carried out for all types of proteases, be it naturally occurring or wild-type proteases; or genetically engineered or synthetic proteases.

Peptidase family S53 contains acid-acting endopeptidases and tripeptidyl-peptidases. The residues of the catalytic triad are Glu, Asp, Ser, and there is an additional acidic residue, Asp, in the oxyanion hole. The order of the residues is Glu, Asp, Asp, Ser. The Ser residue is the nucleophile equivalent

to Ser in the Asp, His, Ser triad of subtilisin, and the Glu of the triad is a substitute for the general base, His, in subtilisin.

The peptidases of the S53 family tend to be most active at acidic pH (unlike the homologous subtilisins), and this can be attributed to the functional importance of carboxylic residues, notably Asp in the oxyanion hole. The amino acid sequences are not closely similar to those in family S8 (i.e. serine endopeptidase subtilisins and homologues), and this, taken together with the quite different active site residues and the resulting lower pH for maximal activity, provides for a substantial difference to that family. Protein folding of the peptidase unit for members of this family resembles that of subtilisin, having the clan type SB.

In one embodiment, the protease used according to a process described herein is a Cysteine proteases.

In one embodiment, the protease used according to a process described herein is a Aspartic proteases. Aspartic acid proteases are described in, for example, Hand-book of Proteolytic Enzymes, Edited by A. J. Barrett, N. D. Rawlings and J. F. Woessner, Academic Press, San Diego, 1998, Chapter 270). Suitable examples of aspartic acid protease include, e.g., those disclosed in R. M. Berka et al. Gene, 96, 313 (1990)); (R. M. Berka et al. Gene, 125, 195-198 (1993)); and Gomi et al. Biosci. Biotech. Biochem. 57, 1095-1100 (1993), which are hereby incorporated by reference.

The protease also may be a metalloprotease, which is defined as a protease selected from the group consisting of:

- (a) proteases belonging to EC 3.4.24 (metalloendopeptidases); preferably EC 3.4.24.39 (acid metalloproteinases);
- (b) metalloproteases belonging to the M group of the above Handbook;
- (c) metalloproteases not yet assigned to clans (designation: Clan MX), or belonging to either one of clans MA, MB, MC, MD, ME, MF, MG, MH (as defined at pp. 989-991 of the above Handbook);
- (d) other families of metalloproteases (as defined at pp. 1448-1452 of the above Handbook);
- (e) metalloproteases with a HEXXH motif;
- (f) metalloproteases with an HEFTH motif;
- (g) metalloproteases belonging to either one of families M3, M26, M27, M32, M34, M35, M36, M41, M43, or M47 (as defined at pp. 1448-1452 of the above Handbook);
- (h) metalloproteases belonging to the M28E family; and
- (i) metalloproteases belonging to family M35 (as defined at pp. 1492-1495 of the above Handbook).

In other particular embodiments, metalloproteases are hydrolases in which the nucleophilic attack on a peptide bond is mediated by a water molecule, which is activated by a divalent metal cation. Examples of divalent cations are zinc, cobalt or manganese. The metal ion may be held in place by amino acid ligands. The number of ligands may be five, four, three, two, one or zero. In a particular embodiment the number is two or three, preferably three.

There are no limitations on the origin of the metalloprotease used in a process of the invention. In an embodiment the metalloprotease is classified as EC 3.4.24, preferably EC 3.4.24.39. In one embodiment, the metalloprotease is an acid-stable metalloprotease, e.g., a fungal acid-stable metalloprotease, such as a metalloprotease derived from a strain of the genus *Thermoascus*, preferably a strain of *Thermoascus aurantiacus*, especially *Thermoascus aurantiacus* CGMCC No. 0670 (classified as EC 3.4.24.39). In another embodiment, the metalloprotease is derived from a strain of the genus *Aspergillus*, preferably a strain of *Aspergillus oryzae*.

In one embodiment the metalloprotease has a degree of sequence identity to amino acids -178 to 177, -159 to 177, or preferably amino acids 1 to 177 (the mature polypeptide) of SEQ ID NO: 1 of WO2010/008841 (a *Thermoascus aurantiacus* metalloprotease) of at least 80%, at least 82%, at least 85%, at least 90%, at least 95%, or at least 97%; and which have metalloprotease activity. In particular embodiments, the metalloprotease consists of an amino acid sequence with a degree of identity to SEQ ID NO: 1 as mentioned above.

The *Thermoascus aurantiacus* metalloprotease is a preferred example of a metalloprotease suitable for use in a process of the invention. Another metalloprotease is derived from *Aspergillus oryzae* and comprises the sequence of SEQ ID NO: 11 disclosed in WO2003/048353, or amino acids -23-353; -23-374; -23-397; 1-353; 1-374; 1-397; 177-353; 177-374; or 177-397 thereof, and SEQ ID NO: 10 disclosed in WO2003/048353.

Another metalloprotease suitable for use in a process of the invention is the *Aspergillus oryzae* metalloprotease comprising SEQ ID NO: 5 of WO2010/008841, or a metalloprotease is an isolated polypeptide which has a degree of identity to SEQ ID NO: 5 of at least about 80%, at least 82%, at least 85%, at least 90%, at least 95%, or at least 97%; and which have metalloprotease activity. In particular embodiments, the metalloprotease consists of the amino acid sequence of SEQ ID NO: 5 of WO2010/008841.

In a particular embodiment, a metalloprotease has an amino acid sequence that differs by forty, thirty-five, thirty, twenty-five, twenty, or by fifteen amino acids from amino acids -178 to 177, -159 to 177, or +1 to 177 of the amino acid sequences of the *Thermoascus aurantiacus* or *Aspergillus oryzae* metalloprotease.

In another embodiment, a metalloprotease has an amino acid sequence that differs by ten, or by nine, or by eight, or by seven, or by six, or by five amino acids from amino acids -178 to 177, -159 to 177, or +1 to 177 of the amino acid sequences of these metalloproteases, e.g., by four, by three, by two, or by one amino acid.

In particular embodiments, the metalloprotease a) comprises or b) consists of

- i) the amino acid sequence of amino acids -178 to 177, -159 to 177, or +1 to 177 of SEQ ID NO:1 of WO2010/008841;
- ii) the amino acid sequence of amino acids -23-353, -23-374, -23-397, 1-353, 1-374, 1-397, 177-353, 177-374, or 177-397 of SEQ ID NO: 3 of WO2010/008841;
- iii) the amino acid sequence of SEQ ID NO: 5 of WO2010/008841; or allelic variants, or fragments, of the sequences of i), ii), and iii) that have protease activity.

A fragment of amino acids -178 to 177, -159 to 177, or +1 to 177 of SEQ ID NO: 1 of WO2010/008841 or of amino acids -23-353, -23-374, -23-397, 1-353, 1-374, 1-397, 177-353, 177-374, or 177-397 of SEQ ID NO: 3 of WO2010/008841; is a polypeptide having one or more amino acids deleted from the amino and/or carboxyl terminus of these amino acid sequences. In one embodiment a fragment contains at least 75 amino acid residues, or at least 100 amino acid residues, or at least 125 amino acid residues, or at least 150 amino acid residues, or at least 160 amino acid residues, or at least 165 amino acid residues, or at least 170 amino acid residues, or at least 175 amino acid residues.

To determine whether a given protease is a metalloprotease or not, reference is made to the above "Handbook of Proteolytic Enzymes" and the principles indicated therein. Such determination can be carried out for all types

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of proteases, be it naturally occurring or wild-type proteases; or genetically engineered or synthetic proteases.

The protease may be a variant of, e.g., a wild-type protease, having thermostability properties defined herein. In one embodiment, the thermostable protease is a variant of a metallo protease. In one embodiment, the thermostable protease used in a process described herein is of fungal origin, such as a fungal metallo protease, such as a fungal metallo protease derived from a strain of the genus *Thermoascus*, preferably a strain of *Thermoascus aurantiacus*, especially *Thermoascus aurantiacus* CGMCC No. 0670 (classified as EC 3.4.24.39).

In one embodiment, the thermostable protease is a variant of the mature part of the metallo protease shown in SEQ ID NO: 2 disclosed in WO2003/048353 or the mature part of SEQ ID NO: 1 in WO2010/008841 further with one of the following substitutions or combinations of substitutions:

S5*+D79L+S87P+A112P+D142 L;
D79L+S87P+A112P+T124V+D142 L;
S5*+N26R+D79L+S87P+A112P+D142L;
N26R+T46R+D79L+S87P+A112P+D142L;
T46R+D79L+S87P+T116V+D142L;
D79L+P81R+S87P+A112P+D142L;
A27K+D79L+S87P+A112P+T124V+D142L;
D79L+Y82F+S87P+A112P+T124V+D142L;
D79L+Y82F+S87P+A112P+T124V+D142L;
D79L+S87P+A112P+T124V+A126V+D142L;
D79L+S87P+A112P+D142L;
D79L+Y82F+S87P+A112P+D142L;
S38T+D79L+S87P+A112P+A126V+D142L;
D79L+Y82F+S87P+A112P+A126V+D142L;
A27K+D79L+S87P+A112P+A126V+D142L;
D79L+S87P+N98C+A112P+G135C+D142L;
D79L+S87P+A112P+D142L+T141C+M161C;
S36P+D79L+S87P+A112P+D142L;
A37P+D79L+S87P+A112P+D142L;
S49P+D79L+S87P+A112P+D142L;
S50P+D79L+S87P+A112P+D142L;
D79L+S87P+D104P+A112P+D142L;
D79L+Y82F+S87G+A112P+D142L;
S70V+D79L+Y82F+S87G+Y97W+A112P+D142L;
D79L+Y82F+S87G+Y97W+D104P+A112P+D142L;
S70V+D79L+Y82F+S87G+A112P+D142L;
D79L+Y82F+S87G+D104P+A112P+D142L;
D79L+Y82F+S87G+A112P+A126V+D142L;
Y82F+S87G+S70V+D79L+D104P+A112P+D142L;
Y82F+S87G+D79L+D104P+A112P+A126V+D142L;
A27K+D79L+Y82F+S87G+D104P+A112P+A126V+D142L;
A27K+Y82F+S87G+D104P+A112P+A126V+D142L;
A27K+D79L+Y82F+D104P+A112P+A126V+D142L;
A27K+Y82F+D104P+A112P+A126V+D142L;
A27K+D79L+S87P+A112P+D142L; and
D79L+S87P+D142L.

In one embodiment, the thermostable protease is a variant of the metallo protease disclosed as the mature part of SEQ ID NO: 2 disclosed in WO2003/048353 or the mature part of SEQ ID NO: 1 in WO2010/008841 with one of the following substitutions or combinations of substitutions:

D79L+S87P+A112P+D142L;
D79L+S87P+D142L; and
A27K+D79L+Y82F+S87G+D104P+A112P+A126V+D142L.

In one embodiment, the protease variant has at least 75% identity preferably at least 80%, more preferably at least 85%, more preferably at least 90%, more preferably at least 91%, more preferably at least 92%, even more preferably at

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least 93%, most preferably at least 94%, and even most preferably at least 95%, such as even at least 96%, at least 97%, at least 98%, at least 99%, but less than 100% identity to the mature part of the polypeptide of SEQ ID NO: 2 disclosed in WO2003/048353 or the mature part of SEQ ID NO: 1 in WO2010/008841.

The thermostable protease may also be derived from any bacterium as long as the protease has the thermostability properties.

In one embodiment, the thermostable protease is derived from a strain of the bacterium *Pyrococcus*, such as a strain of *Pyrococcus furiosus* (pfu protease).

In one embodiment, the protease is one shown as SEQ ID NO: 1 in U.S. Pat. No. 6,358,726-B1 (Takara Shuzo Company).

In one embodiment, the thermostable protease is a protease having a mature polypeptide sequence of at least 80% identity, such as at least 85%, such as at least 90%, such as at least 95%, such as at least 96%, such as at least 97%, such as at least 98%, such as at least 99% identity to SEQ ID NO: 1 in U.S. Pat. No. 6,358,726-B1. The *Pyrococcus furiosus* protease can be purchased from Takara Bio, Japan.

The *Pyrococcus furiosus* protease may be a thermostable protease as described in SEQ ID NO: 13 of WO2018/098381. This protease (PfuS) was found to have a thermostability of 110% (80° C./70° C.) and 103% (90° C./70° C.) at pH 4.5 determined.

In one embodiment a thermostable protease used in a process described herein has a thermostability value of more than 20% determined as Relative Activity at 80° C./70° C. determined as described in Example 2 of WO2018/098381.

In one embodiment, the protease has a thermostability of more than 30%, more than 40%, more than 50%, more than 60%, more than 70%, more than 80%, more than 90%, more than 100%, such as more than 105%, such as more than 110%, such as more than 115%, such as more than 120% determined as Relative Activity at 80° C./70° C.

In one embodiment, protease has a thermostability of between 20 and 50%, such as between 20 and 40%, such as 20 and 30% determined as Relative Activity at 80° C./70° C. In one embodiment, the protease has a thermostability between 50 and 115%, such as between 50 and 70%, such as between 50 and 60%, such as between 100 and 120%, such as between 105 and 115% determined as Relative Activity at 80° C./70° C.

In one embodiment, the protease has a thermostability value of more than 10% determined as Relative Activity at 85° C./70° C. determined as described in Example 2 of WO2018/098381.

In one embodiment, the protease has a thermostability of more than 10%, such as more than 12%, more than 14%, more than 16%, more than 18%, more than 20%, more than 30%, more than 40%, more than 50%, more than 60%, more than 70%, more than 80%, more than 90%, more than 100%, more than 110% determined as Relative Activity at 85° C./70° C.

In one embodiment, the protease has a thermostability of between 10% and 50%, such as between 10% and 30%, such as between 10% and 25% determined as Relative Activity at 85° C./70° C.

In one embodiment, the protease has more than 20%, more than 30%, more than 40%, more than 50%, more than 60%, more than 70%, more than 80%, more than 90% determined as Remaining Activity at 80° C.; and/or the protease has more than 20%, more than 30%, more than

40%, more than 50%, more than 60%, more than 70%, more than 80%, more than 90% determined as Remaining Activity at 84° C.

Determination of "Relative Activity" and "Remaining Activity" is done as described in Example 2 of WO2018/098381.

In one embodiment, the protease may have a thermostability for above 90, such as above 100 at 85° C. as determined using the Zein-BCA assay as disclosed in Example 3 of WO2018/098381.

In one embodiment, the protease has a thermostability above 60%, such as above 90%, such as above 100%, such as above 110% at 85° C. as determined using the Zein-BCA assay of WO2018/098381.

In one embodiment, protease has a thermostability between 60-120, such as between 70-120%, such as between 80-120%, such as between 90-120%, such as between 100-120%, such as 110-120% at 85° C. as determined using the Zein-BCA assay of WO2018/098381.

In one embodiment, the thermostable protease has at least 20%, such as at least 30%, such as at least 40%, such as at least 50%, such as at least 60%, such as at least 70%, such as at least 80%, such as at least 90%, such as at least 95%, such as at least 100% of the activity of the JTP196 protease variant or Protease Pfu determined by the AZCL-casein assay of WO2018/098381, and described herein.

In one embodiment, the thermostable protease has at least 20%, such as at least 30%, such as at least 40%, such as at least 50%, such as at least 60%, such as at least 70%, such as at least 80%, such as at least 90%, such as at least 95%, such as at least 100% of the protease activity of the Protease 196 variant or Protease Pfu determined by the AZCL-casein assay of WO2018/098381, and described herein.

Pullulanases

In some embodiments, a pullulanase is present and/or added in liquefaction step and/or saccharification step, or simultaneous saccharification and fermentation (SSF).

Pullulanases (E.C. 3.2.1.41, pullulan 6-glucano-hydrolase), are debranching enzymes characterized by their ability to hydrolyze the alpha-1,6-glycosidic bonds in, for example, amylopectin and pullulan.

In some embodiments, the fermenting organism comprises a heterologous polynucleotide encoding a pullulanase. Any pullulanase described or referenced herein is contemplated for expression in the fermenting organism.

The pullulanase may be any pullulanase that is suitable for the host cells and/or the methods described herein, such as a naturally occurring pullulanase or a variant thereof that retains pullulanase activity.

In some embodiments, the fermenting organism comprising a heterologous polynucleotide encoding a pullulanase has an increased level of pullulanase activity compared to the host cells without the heterologous polynucleotide encoding the pullulanase, when cultivated under the same conditions. In some embodiments, the fermenting organism has an increased level of pullulanase activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the fermenting organism without the heterologous polynucleotide encoding the pullulanase, when cultivated under the same conditions.

Exemplary pullulanases that can be used with the host cells and/or the methods described herein include bacterial, yeast, or filamentous fungal pullulanases, e.g., obtained from any of the microorganisms described or referenced herein, as described supra under the sections related to alpha-amylases.

Contemplated pullulanases include the pullulanases from *Bacillus amyloclaviformis* disclosed in U.S. Pat. No. 4,560,651 (hereby incorporated by reference), the pullulanase disclosed as SEQ ID NO: 2 in WO01/151620 (hereby incorporated by reference), the *Bacillus deramificans* disclosed as SEQ ID NO: 4 in WO01/151620 (hereby incorporated by reference), and the pullulanase from *Bacillus acidopullulyticus* disclosed as SEQ ID NO: 6 in WO01/151620 (hereby incorporated by reference) and also described in FEMS Mic. Let. (1994) 115, 97-106.

Additional pullulanases contemplated include the pullulanases from *Pyrococcus woesei*, specifically from *Pyrococcus woesei* DSM No. 3773 disclosed in WO92/02614.

In one embodiment, the pullulanase is a family GH57 pullulanase. In one embodiment, the pullulanase includes an X47 domain as disclosed in U.S. 61/289,040 published as WO2011/087836 (which are hereby incorporated by reference). More specifically the pullulanase may be derived from a strain of the genus *Thermococcus*, including *Thermococcus litoralis* and *Thermococcus hydrothermalis*, such as the *Thermococcus hydrothermalis* pullulanase truncated at site X4 right after the X47 domain (i.e., amino acids 1-782). The pullulanase may also be a hybrid of the *Thermococcus litoralis* and *Thermococcus hydrothermalis* pullulanases or a *T. hydrothermalis*/*T. litoralis* hybrid enzyme with truncation site X4 disclosed in U.S. 61/289,040 published as WO2011/087836 (which is hereby incorporated by reference).

In another embodiment, the pullulanase is one comprising an X46 domain disclosed in WO2011/076123 (Novozymes).

The pullulanase may be added in an effective amount which include the preferred amount of about 0.0001-10 mg enzyme protein per gram DS, preferably 0.0001-0.10 mg enzyme protein per gram DS, more preferably 0.0001-0.010 mg enzyme protein per gram DS. Pullulanase activity may be determined as NPUN. An Assay for determination of NPUN is described in WO2018/098381.

Suitable commercially available pullulanase products include PROMOZYME D, PROMOZYME™ D2 (Novozymes A/S, Denmark), OPTIMAX L-300 (DuPont-Danisco, USA), and AMANO 8 (Amano, Japan).

In one embodiment, the pullulanase is derived from the *Bacillus subtilis* pullulanase of SEQ ID NO: 114. In one embodiment, the pullulanase is derived from the *Bacillus licheniformis* pullulanase of SEQ ID NO: 115. In one embodiment, the pullulanase is derived from the *Oryza sativa* pullulanase of SEQ ID NO: 116. In one embodiment, the pullulanase is derived from the *Triticum aestivum* pullulanase of SEQ ID NO: 117. In one embodiment, the pullulanase is derived from the *Clostridium phytofermentans* pullulanase of SEQ ID NO: 118. In one embodiment, the pullulanase is derived from the *Streptomyces avermitilis* pullulanase of SEQ ID NO: 119. In one embodiment, the pullulanase is derived from the *Klebsiella pneumoniae* pullulanase of SEQ ID NO: 120.

Additional pullulanases contemplated for use with the present invention can be found in WO2011/153516 (the content of which is incorporated herein).

Additional polynucleotides encoding suitable pullulanases may be obtained from microorganisms of any genus, including those readily available within the UniProtKB database (www.uniprot.org).

The pullulanase coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding pullulanases from strains of different genera or species, as described supra.

The polynucleotides encoding pullulanases may also be identified and obtained from other sources including micro-organisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc.) as described supra.

Techniques used to isolate or clone polynucleotides encoding pullulanases are described supra.

In one embodiment, the pullulanase has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any one of the pullulanases described or referenced herein (e.g., any one of SEQ ID NOs: 114-120). In another embodiment, the pullulanase has a mature polypeptide sequence that is a fragment of the any one of the pullulanases described or referenced herein (e.g., any one of SEQ ID NOs: 114-120). In one embodiment, the number of amino acid residues in the fragment is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of amino acid residues in referenced full length pullulanase. In other embodiments, the pullulanase may comprise the catalytic domain of any pullulanase described or referenced herein (e.g., any one of SEQ ID NOs: 114-120).

The pullulanase may be a variant of any one of the pullulanases described supra (e.g., any one of SEQ ID NOs: 114-120). In one embodiment, the pullulanase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to any one of the pullulanases described supra (e.g., any one of SEQ ID NOs: 114-120).

In one embodiment, the pullulanase has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of the pullulanases described supra (e.g., any one of SEQ ID NOs: 114-120). In one embodiment, the pullulanase has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) of amino acid sequence of any one of the pullulanases described supra (e.g., any one of SEQ ID NOs: 114-120). In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

In some embodiments, the pullulanase has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the pullulanase activity of any pullulanase described or referenced herein under the same conditions (e.g., any one of SEQ ID NOs: 114-120).

In one embodiment, the pullulanase coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any pullulanase described or referenced herein (e.g., any one of SEQ ID NOs: 114-120). In one embodiment, the pullulanase coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any pullulanase described or referenced herein (e.g., any one of SEQ ID NOs: 114-120).

In one embodiment, the pullulanase comprises the coding sequence of any pullulanase described or referenced herein (e.g., any one of SEQ ID NOs: 114-120). In one embodi-

ment, the pullulanase comprises a coding sequence that is a subsequence of the coding sequence from any pullulanase described or referenced herein, wherein the subsequence encodes a polypeptide having pullulanase activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

The referenced coding sequence of any related aspect or embodiment described herein can be the native coding sequence or a degenerate sequence, such as a codon-optimized coding sequence designed for use in a particular host cell (e.g., optimized for expression in *Saccharomyces cerevisiae*).

The pullulanase can also include fused polypeptides or cleavable fusion polypeptides, as described supra.

Methods Using a Cellulosic-Containing Material

In some aspects, the methods described herein produce a fermentation product from a cellulosic-containing material. The predominant polysaccharide in the primary cell wall of biomass is cellulose, the second most abundant is hemicellulose, and the third is pectin. The secondary cell wall, produced after the cell has stopped growing, also contains polysaccharides and is strengthened by polymeric lignin covalently cross-linked to hemicellulose. Cellulose is a homopolymer of anhydrocellobiose and thus a linear beta-(1-4)-D-glucan, while hemicelluloses include a variety of compounds, such as xylans, xyloglucans, arabinoxylans, and mannans in complex branched structures with a spectrum of substituents. Although generally polymorphous, cellulose is found in plant tissue primarily as an insoluble crystalline matrix of parallel glucan chains. Hemicelluloses usually hydrogen bond to cellulose, as well as to other hemicelluloses, which help stabilize the cell wall matrix.

Cellulose is generally found, for example, in the stems, leaves, hulls, husks, and cobs of plants or leaves, branches, and wood of trees. The cellulosic-containing material can be, but is not limited to, agricultural residue, herbaceous material (including energy crops), municipal solid waste, pulp and paper mill residue, waste paper, and wood (including forestry residue) (see, for example, Wiseloge et al., 1995, in *Handbook on Bioethanol* (Charles E. Wyman, editor), pp. 105-118, Taylor & Francis, Washington D.C.; Wyman, 1994, *Bioresource Technology* 50: 3-16; Lynd, 1990, *Applied Biochemistry and Biotechnology* 24/25: 695-719; Mosier et al., 1999, Recent Progress in Bioconversion of Lignocellulosics, in *Advances in Biochemical Engineering/Biotechnology*, T. Scheper, managing editor, Volume 65, pp. 23-40, Springer-Verlag, New York). It is understood herein that the cellulose may be in the form of lignocellulose, a plant cell wall material containing lignin, cellulose, and hemicellulose in a mixed matrix. In one embodiment, the cellulosic-containing material is any biomass material. In another embodiment, the cellulosic-containing material is lignocellulose, which comprises cellulose, hemicelluloses, and lignin.

In one embodiment, the cellulosic-containing material is agricultural residue, herbaceous material (including energy crops), municipal solid waste, pulp and paper mill residue, waste paper, or wood (including forestry residue).

In another embodiment, the cellulosic-containing material is arundo, bagasse, bamboo, corn cob, corn fiber, corn stover, miscanthus, rice straw, switchgrass, or wheat straw.

In another embodiment, the cellulosic-containing material is aspen, eucalyptus, fir, pine, poplar, spruce, or willow.

In another embodiment, the cellulosic-containing material is algal cellulose, bacterial cellulose, cotton linter, filter paper, microcrystalline cellulose (e.g., AVICEL®), or phosphoric-acid treated cellulose.

In another embodiment, the cellulosic-containing material is an aquatic biomass. As used herein the term "aquatic biomass" means biomass produced in an aquatic environment by a photosynthesis process. The aquatic biomass can be algae, emergent plants, floating-leaf plants, or submerged plants.

The cellulosic-containing material may be used as is or may be subjected to pretreatment, using conventional methods known in the art, as described herein. In a preferred embodiment, the cellulosic-containing material is pretreated.

The methods of using cellulosic-containing material can be accomplished using methods conventional in the art. Moreover, the methods of can be implemented using any conventional biomass processing apparatus configured to carry out the processes.

Cellulosic Pretreatment

In one embodiment the cellulosic-containing material is pretreated before saccharification.

In practicing the processes described herein, any pretreatment process known in the art can be used to disrupt plant cell wall components of the cellulosic-containing material (Chandra et al., 2007, *Adv. Biochem. Engin./Biotechnol.* 108: 67-93; Galbe and Zacchi, 2007, *Adv. Biochem. Engin./Biotechnol.* 108: 41-65; Hendriks and Zeeman, 2009, *Biore-source Technology* 100: 10-18; Mosier et al., 2005, *Biore-source Technology* 96: 673-686; Taherzadeh and Karimi, 2008, *Int. J. Mol. Sci.* 9: 1621-1651; Yang and Wyman, 2008, *Biofuels Bioproducts and Biorefining-Biofr.* 2: 26-40).

The cellulosic-containing material can also be subjected to particle size reduction, sieving, pre-soaking, wetting, washing, and/or conditioning prior to pretreatment using methods known in the art.

Conventional pretreatments include, but are not limited to, steam pretreatment (with or without explosion), dilute acid pretreatment, hot water pretreatment, alkaline pretreatment, lime pretreatment, wet oxidation, wet explosion, ammonia fiber explosion, organosolv pretreatment, and biological pretreatment. Additional pretreatments include ammonia percolation, ultrasound, electroporation, microwave, supercritical CO₂, supercritical H₂O, ozone, ionic liquid, and gamma irradiation pretreatments.

In a one embodiment, the cellulosic-containing material is pretreated before saccharification (i.e., hydrolysis) and/or fermentation. Pretreatment is preferably performed prior to the hydrolysis. Alternatively, the pretreatment can be carried out simultaneously with enzyme hydrolysis to release fermentable sugars, such as glucose, xylose, and/or cellobiose. In most cases the pretreatment step itself results in some conversion of biomass to fermentable sugars (even in absence of enzymes).

In one embodiment, the cellulosic-containing material is pretreated with steam. In steam pretreatment, the cellulosic-containing material is heated to disrupt the plant cell wall components, including lignin, hemicellulose, and cellulose to make the cellulose and other fractions, e.g., hemicellulose, accessible to enzymes. The cellulosic-containing material is passed to or through a reaction vessel where steam is injected to increase the temperature to the required temperature and pressure and is retained therein for the desired reaction time. Steam pretreatment is preferably performed at 140-250° C., e.g., 160-200° C. or 170-190° C., where the

optimal temperature range depends on optional addition of a chemical catalyst. Residence time for the steam pretreatment is preferably 1-60 minutes, e.g., 1-30 minutes, 1-20 minutes, 3-12 minutes, or 4-10 minutes, where the optimal residence time depends on the temperature and optional addition of a chemical catalyst. Steam pretreatment allows for relatively high solids loadings, so that the cellulosic-containing material is generally only moist during the pretreatment. The steam pretreatment is often combined with an explosive discharge of the material after the pretreatment, which is known as steam explosion, that is, rapid flashing to atmospheric pressure and turbulent flow of the material to increase the accessible surface area by fragmentation (Duff and Murray, 1996, *Biore-source Technology* 855: 1-33; Galbe and Zacchi, 2002, *Appl. Microbiol. Biotechnol.* 59: 618-628; U.S. Patent Application No. 2002/0164730). During steam pretreatment, hemicellulose acetyl groups are cleaved and the resulting acid autocatalyzes partial hydrolysis of the hemicellulose to monosaccharides and oligosaccharides. Lignin is removed to only a limited extent.

In one embodiment, the cellulosic-containing material is subjected to a chemical pretreatment. The term "chemical treatment" refers to any chemical pretreatment that promotes the separation and/or release of cellulose, hemicellulose, and/or lignin. Such a pretreatment can convert crystalline cellulose to amorphous cellulose. Examples of suitable chemical pretreatment processes include, for example, dilute acid pretreatment, lime pretreatment, wet oxidation, ammonia fiber/freeze expansion (AFEX), ammonia percolation (APR), ionic liquid, and organosolv pretreatments.

A chemical catalyst such as H₂SO₄ or SO₂ (typically 0.3 to 5% w/w) is sometimes added prior to steam pretreatment, which decreases the time and temperature, increases the recovery, and improves enzymatic hydrolysis (Ballesteros et al., 2006, *Appl. Biochem. Biotechnol.* 129-132: 496-508; Varga et al., 2004, *Appl. Biochem. Biotechnol.* 113-116: 509-523; Sassner et al., 2006, *Enzyme Microb. Technol.* 39: 756-762). In dilute acid pretreatment, the cellulosic-containing material is mixed with dilute acid, typically H₂SO₄, and water to form a slurry, heated by steam to the desired temperature, and after a residence time flashed to atmospheric pressure. The dilute acid pretreatment can be performed with a number of reactor designs, e.g., plug-flow reactors, counter-current reactors, or continuous counter-current shrinking bed reactors (Duff and Murray, 1996, *Biore-source Technology* 855: 1-33; Schell et al., 2004, *Biore-source Technology* 91: 179-188; Lee et al., 1999, *Adv. Biochem. Eng. Biotechnol.* 65: 93-115). In a specific embodiment the dilute acid pretreatment of cellulosic-containing material is carried out using 4% w/w sulfuric acid at 180° C. for 5 minutes.

Several methods of pretreatment under alkaline conditions can also be used. These alkaline pretreatments include, but are not limited to, sodium hydroxide, lime, wet oxidation, ammonia percolation (APR), and ammonia fiber/freeze expansion (AFEX) pretreatment. Lime pretreatment is performed with calcium oxide or calcium hydroxide at temperatures of 85-150° C. and residence times from 1 hour to several days (Wyman et al., 2005, *Biore-source Technology* 96: 1959-1966; Mosier et al., 2005, *Biore-source Technology* 96: 673-686). WO2006/110891, WO2006/110899, WO2006/110900, and WO2006/110901 disclose pretreatment methods using ammonia.

Wet oxidation is a thermal pretreatment performed typically at 180-200° C. for 5-15 minutes with addition of an oxidative agent such as hydrogen peroxide or over-pressure of oxygen (Schmidt and Thomsen, 1998, *Biore-source Tech-*

nology 64: 139-151; Palonen et al., 2004, *Appl. Biochem. Biotechnol.* 117: 1-17; Varga et al., 2004, *Biotechnol. Bioeng.* 88: 567-574; Martin et al., 2006, *J. Chem. Technol. Biotechnol.* 81: 1669-1677). The pretreatment is performed preferably at 1-40% dry matter, e.g., 2-30% dry matter or 5-20% dry matter, and often the initial pH is increased by the addition of alkali such as sodium carbonate.

A modification of the wet oxidation pretreatment method, known as wet explosion (combination of wet oxidation and steam explosion) can handle dry matter up to 30%. In wet explosion, the oxidizing agent is introduced during pretreatment after a certain residence time. The pretreatment is then ended by flashing to atmospheric pressure (WO2006/032282).

Ammonia fiber expansion (AFEX) involves treating the cellulosic-containing material with liquid or gaseous ammonia at moderate temperatures such as 90-150° C. and high pressure such as 17-20 bar for 5-10 minutes, where the dry matter content can be as high as 60% (Gollapalli et al., 2002, *Appl. Biochem. Biotechnol.* 98: 23-35; Chundawat et al., 2007, *Biotechnol. Bioeng.* 96: 219-231; Alizadeh et al., 2005, *Appl. Biochem. Biotechnol.* 121: 1133-1141; Teymouri et al., 2005, *Bioresource Technology* 96: 2014-2018). During AFEX pretreatment cellulose and hemicelluloses remain relatively intact. Lignin-carbohydrate complexes are cleaved.

Organosolv pretreatment delignifies the cellulosic-containing material by extraction using aqueous ethanol (40-60% ethanol) at 160-200° C. for 30-60 minutes (Pan et al., 2005, *Biotechnol. Bioeng.* 90: 473-481; Pan et al., 2006, *Biotechnol. Bioeng.* 94: 851-861; Kurabi et al., 2005, *Appl. Biochem. Biotechnol.* 121: 219-230). Sulphuric acid is usually added as a catalyst. In organosolv pretreatment, the majority of hemicellulose and lignin is removed.

Other examples of suitable pretreatment methods are described by Schell et al., 2003, *Appl. Biochem. Biotechnol.* 105-108: 69-85, and Mosier et al., 2005, *Bioresource Technology* 96: 673-686, and U.S. Published Application 2002/0164730.

In one embodiment, the chemical pretreatment is carried out as a dilute acid treatment, and more preferably as a continuous dilute acid treatment. The acid is typically sulfuric acid, but other acids can also be used, such as acetic acid, citric acid, nitric acid, phosphoric acid, tartaric acid, succinic acid, hydrogen chloride, or mixtures thereof. Mild acid treatment is conducted in the pH range of preferably 1-5, e.g., 1-4 or 1-2.5. In one aspect, the acid concentration is in the range from preferably 0.01 to 10 wt. % acid, e.g., 0.05 to 5 wt. % acid or 0.1 to 2 wt. % acid. The acid is contacted with the cellulosic-containing material and held at a temperature in the range of preferably 140-200° C., e.g., 165-190° C., for periods ranging from 1 to 60 minutes.

In another embodiment, pretreatment takes place in an aqueous slurry. In preferred aspects, the cellulosic-containing material is present during pretreatment in amounts preferably between 10-80 wt. %, e.g., 20-70 wt. % or 30-60 wt. %, such as around 40 wt. %. The pretreated cellulosic-containing material can be unwashed or washed using any method known in the art, e.g., washed with water.

In one embodiment, the cellulosic-containing material is subjected to mechanical or physical pretreatment. The term "mechanical pretreatment" or "physical pretreatment" refers to any pretreatment that promotes size reduction of particles. For example, such pretreatment can involve various types of grinding or milling (e.g., dry milling, wet milling, or vibratory ball milling).

The cellulosic-containing material can be pretreated both physically (mechanically) and chemically. Mechanical or physical pretreatment can be coupled with steaming/steam explosion, hydrothermolysis, dilute or mild acid treatment, high temperature, high pressure treatment, irradiation (e.g., microwave irradiation), or combinations thereof. In one aspect, high pressure means pressure in the range of preferably about 100 to about 400 psi, e.g., about 150 to about 250 psi. In another aspect, high temperature means temperature in the range of about 100 to about 300° C., e.g., about 140 to about 200° C. In a preferred aspect, mechanical or physical pretreatment is performed in a batch-process using a steam gun hydrolyzer system that uses high pressure and high temperature as defined above, e.g., a Sunds Hydrolyzer available from Sunds Defibrator AB, Sweden. The physical and chemical pretreatments can be carried out sequentially or simultaneously, as desired.

Accordingly, in one embodiment, the cellulosic-containing material is subjected to physical (mechanical) or chemical pretreatment, or any combination thereof, to promote the separation and/or release of cellulose, hemicellulose, and/or lignin.

In one embodiment, the cellulosic-containing material is subjected to a biological pretreatment. The term "biological pretreatment" refers to any biological pretreatment that promotes the separation and/or release of cellulose, hemicellulose, and/or lignin from the cellulosic-containing material. Biological pretreatment techniques can involve applying lignin-solubilizing microorganisms and/or enzymes (see, for example, Hsu, T.-A., 1996, Pretreatment of biomass, in *Handbook on Bioethanol: Production and Utilization*, Wyman, C. E., ed., Taylor & Francis, Washington, DC, 179-212; Ghosh and Singh, 1993, *Adv. Appl. Microbiol.* 39: 295-333; McMillan, J. D., 1994, Pretreating lignocellulosic biomass: a review, in *Enzymatic Conversion of Biomass for Fuels Production*, Himmel, M. E., Baker, J. O., and Overend, R. P., eds., ACS Symposium Series 566, American Chemical Society, Washington, DC, chapter 15; Gong, C. S., Cao, N. J., Du, J., and Tsao, G. T., 1999, Ethanol production from renewable resources, in *Advances in Biochemical Engineering/Biotechnology*, Scheper, T., ed., Springer-Verlag Berlin Heidelberg, Germany, 65: 207-241; Olsson and Hahn-Hagerdal, 1996, *Enz. Microb. Tech.* 18: 312-331; and Vallander and Eriksson, 1990, *Adv. Biochem. Eng./Biotechnol.* 42: 63-95).

Saccharification and Fermentation of Cellulosic-Containing Material

Saccharification (i.e., hydrolysis) and fermentation, separate or simultaneous, include, but are not limited to, separate hydrolysis and fermentation (SHF); simultaneous saccharification and fermentation (SSF); simultaneous saccharification and co-fermentation (SSCF); hybrid hydrolysis and fermentation (HHF); separate hydrolysis and co-fermentation (SHCF); hybrid hydrolysis and co-fermentation (HHCF).

SHF uses separate process steps to first enzymatically hydrolyze the cellulosic-containing material to fermentable sugars, e.g., glucose, cellobiose, and pentose monomers, and then ferment the fermentable sugars to ethanol. In SSF, the enzymatic hydrolysis of the cellulosic-containing material and the fermentation of sugars to ethanol are combined in one step (Philippidis, G. P., 1996, Cellulose bioconversion technology, in *Handbook on Bioethanol: Production and Utilization*, Wyman, C. E., ed., Taylor & Francis, Washington, DC, 179-212). SSCF involves the co-fermentation of multiple sugars (Sheehan and Himmel, 1999, *Biotechnol. Prog.* 15: 817-827). HHF involves a separate hydrolysis

step, and in addition a simultaneous saccharification and hydrolysis step, which can be carried out in the same reactor. The steps in an HHF process can be carried out at different temperatures, i.e., high temperature enzymatic saccharification followed by SSF at a lower temperature that the fermentation organism can tolerate. It is understood herein that any method known in the art comprising pretreatment, enzymatic hydrolysis (saccharification), fermentation, or a combination thereof, can be used in the practicing the processes described herein.

A conventional apparatus can include a fed-batch stirred reactor, a batch stirred reactor, a continuous flow stirred reactor with ultrafiltration, and/or a continuous plug-flow column reactor (de Castilhos Corazza et al., 2003, *Acta Scientiarum. Technology* 25: 33-38; Gusakov and Sinitsyn, 1985, *Enz. Microb. Technol.* 7: 346-352), an attrition reactor (Ryu and Lee, 1983, *Biotechnol. Bioeng.* 25: 53-65). Additional reactor types include fluidized bed, upflow blanket, immobilized, and extruder type reactors for hydrolysis and/or fermentation.

In the saccharification step (i.e., hydrolysis step), the cellulosic and/or starch-containing material, e.g., pretreated, is hydrolyzed to break down cellulose, hemicellulose, and/or starch to fermentable sugars, such as glucose, cellobiose, xylose, xylulose, arabinose, mannose, galactose, and/or soluble oligosaccharides. The hydrolysis is performed enzymatically e.g., by a cellulolytic enzyme composition. The enzymes of the compositions can be added simultaneously or sequentially.

Enzymatic hydrolysis may be carried out in a suitable aqueous environment under conditions that can be readily determined by one skilled in the art. In one aspect, hydrolysis is performed under conditions suitable for the activity of the enzymes(s), i.e., optimal for the enzyme(s). The hydrolysis can be carried out as a fed batch or continuous process where the cellulosic and/or starch-containing material is fed gradually to, for example, an enzyme containing hydrolysis solution.

The saccharification is generally performed in stirred-tank reactors or fermentors under controlled pH, temperature, and mixing conditions. Suitable process time, temperature and pH conditions can readily be determined by one skilled in the art. For example, the saccharification can last up to 200 hours, but is typically performed for preferably about 12 to about 120 hours, e.g., about 16 to about 72 hours or about 24 to about 48 hours. The temperature is in the range of preferably about 25° C. to about 70° C., e.g., about 30° C. to about 65° C., about 40° C. to about 60° C., or about 50° C. to about 55° C. The pH is in the range of preferably about 3 to about 8, e.g., about 3.5 to about 7, about 4 to about 6, or about 4.5 to about 5.5. The dry solids content is in the range of preferably about 5 to about 50 wt. %, e.g., about 10 to about 40 wt. % or about 20 to about 30 wt. %.

Saccharification may be carried out using a cellulolytic enzyme composition. Such enzyme compositions are described below in the "Cellulolytic Enzyme Composition"-section below. The cellulolytic enzyme compositions can comprise any protein useful in degrading the cellulosic-containing material. In one aspect, the cellulolytic enzyme composition comprises or further comprises one or more (e.g., several) proteins selected from the group consisting of a cellulase, an AA9 (GH61) polypeptide, a hemicellulase, an esterase, an expansin, a ligninolytic enzyme, an oxidoreductase, a pectinase, a protease, and a swollenin.

In another embodiment, the cellulase is preferably one or more (e.g., several) enzymes selected from the group consisting of an endoglucanase, a cellobiohydrolase, and a beta-glucosidase.

In another embodiment, the hemicellulase is preferably one or more (e.g., several) enzymes selected from the group consisting of an acetylmannan esterase, an acetylxyylan esterase, an arabinanase, an arabinofuranosidase, a coumaric acid esterase, a feruloyl esterase, a galactosidase, a glucuronidase, a glucuronoyl esterase, a mannanase, a mannosidase, a xylanase, and a xylosidase. In another embodiment, the oxidoreductase is one or more (e.g., several) enzymes selected from the group consisting of a catalase, a laccase, and a peroxidase. The enzymes or enzyme compositions used in a processes of the present invention may be in any form suitable for use, such as, for example, a fermentation broth formulation or a cell composition, a cell lysate with or without cellular debris, a semi-purified or purified enzyme preparation, or a host cell as a source of the enzymes. The enzyme composition may be a dry powder or granulate, a non-dusting granulate, a liquid, a stabilized liquid, or a stabilized protected enzyme. Liquid enzyme preparations may, for instance, be stabilized by adding stabilizers such as a sugar, a sugar alcohol or another polyol, and/or lactic acid or another organic acid according to established processes.

In one embodiment, an effective amount of cellulolytic or hemicellulolytic enzyme composition to the cellulosic-containing material is about 0.5 to about 50 mg, e.g., about 0.5 to about 40 mg, about 0.5 to about 25 mg, about 0.75 to about 20 mg, about 0.75 to about 15 mg, about 0.5 to about 10 mg, or about 2.5 to about 10 mg per g of the cellulosic-containing material.

In one embodiment, such a compound is added at a molar ratio of the compound to glucosyl units of cellulose of about 10^{-6} to about 10, e.g., about 10^{-6} to about 7.5, about 10^{-6} to about 5, about 10^{-6} to about 2.5, about 10^{-6} to about 1, about 10^{-5} to about 1, about 10^{-5} to about 10^{-1} , about 10^{-4} to about 10^{-1} , about 10^{-3} to about 10^{-1} , or about 10^{-3} to about 10^{-2} . In another aspect, an effective amount of such a compound is about 0.1 μ M to about 1 M, e.g., about 0.5 μ M to about 0.75 M, about 0.75 μ M to about 0.5 M, about 1 μ M to about 0.25 M, about 1 μ M to about 0.1 M, about 5 μ M to about 50 mM, about 10 μ M to about 25 mM, about 50 μ M to about 25 mM, about 10 μ M to about 10 mM, about 5 μ M to about 5 mM, or about 0.1 mM to about 1 mM.

The term "liquor" means the solution phase, either aqueous, organic, or a combination thereof, arising from treatment of a lignocellulose and/or hemicellulose material in a slurry, or monosaccharides thereof, e.g., xylose, arabinose, mannose, etc., under conditions as described in WO2012/021401, and the soluble contents thereof. A liquor for cellulolytic enhancement of an AA9 polypeptide (GH61 polypeptide) can be produced by treating a lignocellulose or hemicellulose material (or feedstock) by applying heat and/or pressure, optionally in the presence of a catalyst, e.g., acid, optionally in the presence of an organic solvent, and optionally in combination with physical disruption of the material, and then separating the solution from the residual solids. Such conditions determine the degree of cellulolytic enhancement obtainable through the combination of liquor and an AA9 polypeptide during hydrolysis of a cellulosic substrate by a cellulolytic enzyme preparation. The liquor can be separated from the treated material using a method standard in the art, such as filtration, sedimentation, or centrifugation.

In one embodiment, an effective amount of the liquor to cellulose is about 10^{-6} to about 10 g per g of cellulose, e.g., about 10^{-6} to about 7.5 g, about 10^{-6} to about 5 g, about 10^{-6} to about 2.5 g, about 10^{-6} to about 1 g, about 10^{-5} to about 1 g, about 10^{-5} to about 10^{-1} g, about 10^{-4} to about 10^{-1} g, about 10^{-3} to about 10^{-1} g, or about 10^{-3} to about 10^{-2} g per g of cellulose.

In the fermentation step, sugars, released from the cellulosic-containing material, e.g., as a result of the pretreatment and enzymatic hydrolysis steps, are fermented to ethanol, by a fermenting organism, such as yeast described herein. Hydrolysis (saccharification) and fermentation can be separate or simultaneous.

Any suitable hydrolyzed cellulosic-containing material can be used in the fermentation step in practicing the processes described herein. Such feedstocks include, but are not limited to carbohydrates (e.g., lignocellulose, xylans, cellulose, starch, etc.). The material is generally selected based on economics, i.e., costs per equivalent sugar potential, and recalcitrance to enzymatic conversion.

Production of ethanol by a fermenting organism using cellulosic-containing material results from the metabolism of sugars (monosaccharides). The sugar composition of the hydrolyzed cellulosic-containing material and the ability of the fermenting organism to utilize the different sugars has a direct impact in process yields. Prior to Applicant's disclosure herein, strains known in the art utilize glucose efficiently but do not (or very limitedly) metabolize pentoses like xylose, a monosaccharide commonly found in hydrolyzed material.

Compositions of the fermentation media and fermentation conditions depend on the fermenting organism and can easily be determined by one skilled in the art. Typically, the fermentation takes place under conditions known to be suitable for generating the fermentation product. In some embodiments, the fermentation process is carried out under aerobic or microaerophilic (i.e., where the concentration of oxygen is less than that in air), or anaerobic conditions. In some embodiments, fermentation is conducted under anaerobic conditions (i.e., no detectable oxygen), or less than about 5, about 2.5, or about 1 mmol/L/h oxygen. In the absence of oxygen, the NADH produced in glycolysis cannot be oxidized by oxidative phosphorylation. Under anaerobic conditions, pyruvate or a derivative thereof may be utilized by the host cell as an electron and hydrogen acceptor in order to generate NAD⁺.

The fermentation process is typically run at a temperature that is optimal for the recombinant fungal cell. For example, in some embodiments, the fermentation process is performed at a temperature in the range of from about 25° C. to about 42° C. Typically the process is carried out at a temperature that is less than about 38° C., less than about 35° C., less than about 33° C., or less than about 38° C., but at least about 20° C., 22° C., or 25° C.

A fermentation stimulator can be used in a process described herein to further improve the fermentation, and in particular, the performance of the fermenting organism, such as, rate enhancement and product yield (e.g., ethanol yield). A "fermentation stimulator" refers to stimulators for growth of the fermenting organisms, in particular, yeast. Preferred fermentation stimulators for growth include vitamins and minerals. Examples of vitamins include multivitamins, biotin, pantothenate, nicotinic acid, meso-inositol, thiamine, pyridoxine, para-aminobenzoic acid, folic acid, riboflavin, and Vitamins A, B, C, D, and E. See, for example, Alfenore et al., Improving ethanol production and viability of *Saccharomyces cerevisiae* by a vitamin feeding strategy during

fed-batch process, Springer-Verlag (2002), which is hereby incorporated by reference. Examples of minerals include minerals and mineral salts that can supply nutrients comprising P, K, Mg, S, Ca, Fe, Zn, Mn, and Cu.

Cellulolytic Enzymes and Compositions

A cellulolytic enzyme or cellulolytic enzyme composition may be present and/or added during saccharification. A cellulolytic enzyme composition is an enzyme preparation containing one or more (e.g., several) enzymes that hydrolyze cellulosic-containing material. Such enzymes include endoglucanase, cellobiohydrolase, beta-glucosidase, and/or combinations thereof.

In some embodiments, the fermenting organism comprises one or more (e.g., several) heterologous polynucleotides encoding enzymes that hydrolyze cellulosic-containing material (e.g., an endoglucanase, cellobiohydrolase, beta-glucosidase or combinations thereof). Any enzyme described or referenced herein that hydrolyzes cellulosic-containing material is contemplated for expression in the fermenting organism.

The cellulolytic enzyme may be any cellulolytic enzyme that is suitable for the host cells and/or the methods described herein (e.g., an endoglucanase, cellobiohydrolase, beta-glucosidase), such as a naturally occurring cellulolytic enzyme or a variant thereof that retains cellulolytic enzyme activity.

In some embodiments, the fermenting organism comprising a heterologous polynucleotide encoding a cellulolytic enzyme has an increased level of cellulolytic enzyme activity (e.g., increased endoglucanase, cellobiohydrolase, and/or beta-glucosidase) compared to the host cells without the heterologous polynucleotide encoding the cellulolytic enzyme, when cultivated under the same conditions. In some embodiments, the fermenting organism has an increased level of cellulolytic enzyme activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the fermenting organism without the heterologous polynucleotide encoding the cellulolytic enzyme, when cultivated under the same conditions.

Exemplary cellulolytic enzymes that can be used with the host cells and/or the methods described herein include bacterial, yeast, or filamentous fungal cellulolytic enzymes, e.g., obtained from any of the microorganisms described or referenced herein, as described supra under the sections related to proteases.

The cellulolytic enzyme may be of any origin. In an embodiment the cellulolytic enzyme is derived from a strain of *Trichoderma*, such as a strain of *Trichoderma reesei*; a strain of *Humicola*, such as a strain of *Humicola insolens*, and/or a strain of *Chrysosporium*, such as a strain of *Chrysosporium lucknowense*. In a preferred embodiment the cellulolytic enzyme is derived from a strain of *Trichoderma reesei*.

The cellulolytic enzyme composition may further comprise one or more of the following polypeptides, such as enzymes: AA9 polypeptide (GH61 polypeptide) having cellulolytic enhancing activity, beta-glucosidase, xylanase, beta-xylosidase, CBH I, CBH II, or a mixture of two, three, four, five or six thereof.

The further polypeptide(s) (e.g., AA9 polypeptide) and/or enzyme(s) (e.g., beta-glucosidase, xylanase, beta-xylosidase, CBH I and/or CBH II) may be foreign to the cellulolytic enzyme composition producing organism (e.g., *Trichoderma reesei*).

In an embodiment the cellulolytic enzyme composition comprises an AA9 polypeptide having cellulolytic enhancing activity and a beta-glucosidase.

In another embodiment the cellulolytic enzyme composition comprises an AA9 polypeptide having cellulolytic enhancing activity, a beta-glucosidase, and a CBH I.

In another embodiment the cellulolytic enzyme composition comprises an AA9 polypeptide having cellulolytic enhancing activity, a beta-glucosidase, a CBH I and a CBH II.

Other enzymes, such as endoglucanases, may also be comprised in the cellulolytic enzyme composition.

As mentioned above the cellulolytic enzyme composition may comprise a number of difference polypeptides, including enzymes.

In one embodiment, the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Thermoascus aurantiacus* AA9 (GH61A) polypeptide having cellulolytic enhancing activity (e.g., WO2005/074656), and *Aspergillus oryzae* beta-glucosidase fusion protein (e.g., one disclosed in WO2008/057637, in particular shown as SEQ ID NOs: 59 and 60).

In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Thermoascus aurantiacus* AA9 (GH61A) polypeptide having cellulolytic enhancing activity (e.g., SEQ ID NO: 2 in WO2005/074656), and *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO2005/047499).

In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Penicillium emersonii* AA9 (GH61A) polypeptide having cellulolytic enhancing activity, in particular the one disclosed in WO2011/041397, and *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO2005/047499).

In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Penicillium emersonii* AA9 (GH61A) polypeptide having cellulolytic enhancing activity, in particular the one disclosed in WO2011/041397, and *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO2005/047499) or a variant disclosed in WO2012/044915 (hereby incorporated by reference), in particular one comprising one or more such as all of the following substitutions: F100D, S283G, N456E, F512Y.

In an embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic composition, further comprising an AA9 (GH61A) polypeptide having cellulolytic enhancing activity, in particular the one derived from a strain of *Penicillium emersonii* (e.g., SEQ ID NO: 2 in WO2011/041397), *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 in WO2005/047499) variant with one or more, in particular all of the following substitutions: F100D, S283G, N456E, F512Y and disclosed in WO2012/044915; *Aspergillus fumigatus* Cel7A CBH1, e.g., the one disclosed as SEQ ID NO: 6 in WO2011/057140 and *Aspergillus fumigatus* CBH II, e.g., the one disclosed as SEQ ID NO: 18 in WO2011/057140.

In a preferred embodiment the cellulolytic enzyme composition is a *Trichoderma reesei*, cellulolytic enzyme composition, further comprising a hemicellulase or hemicellulolytic enzyme composition, such as an *Aspergillus fumigatus* xylanase and *Aspergillus fumigatus* beta-xylosidase.

In an embodiment the cellulolytic enzyme composition also comprises a xylanase (e.g., derived from a strain of the

genus *Aspergillus*, in particular *Aspergillus aculeatus* or *Aspergillus fumigatus*; or a strain of the genus *Talaromyces*, in particular *Talaromyces leycettanus*) and/or a beta-xylosidase (e.g., derived from *Aspergillus*, in particular *Aspergillus fumigatus*, or a strain of *Talaromyces*, in particular *Talaromyces emersonii*).

In an embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Thermoascus aurantiacus* AA9 (GH61A) polypeptide having cellulolytic enhancing activity (e.g., WO2005/074656), *Aspergillus oryzae* beta-glucosidase fusion protein (e.g., one disclosed in WO2008/057637, in particular as SEQ ID NOs: 59 and 60), and *Aspergillus aculeatus* xylanase (e.g., Xyl II in WO94/21785).

In another embodiment the cellulolytic enzyme composition comprises a *Trichoderma reesei* cellulolytic preparation, further comprising *Thermoascus aurantiacus* GH61A polypeptide having cellulolytic enhancing activity (e.g., SEQ ID NO: 2 in WO2005/074656), *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO2005/047499) and *Aspergillus aculeatus* xylanase (Xyl II disclosed in WO94/21785).

In another embodiment the cellulolytic enzyme composition comprises a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Thermoascus aurantiacus* AA9 (GH61A) polypeptide having cellulolytic enhancing activity (e.g., SEQ ID NO: 2 in WO2005/074656), *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO2005/047499) and *Aspergillus aculeatus* xylanase (e.g., Xyl II disclosed in WO94/21785).

In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Penicillium emersonii* AA9 (GH61A) polypeptide having cellulolytic enhancing activity, in particular the one disclosed in WO2011/041397, *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO2005/047499) and *Aspergillus fumigatus* xylanase (e.g., Xyl III in WO2006/078256).

In another embodiment the cellulolytic enzyme composition comprises a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Penicillium emersonii* AA9 (GH61A) polypeptide having cellulolytic enhancing activity, in particular the one disclosed in WO2011/041397, *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO2005/047499), *Aspergillus fumigatus* xylanase (e.g., Xyl III in WO2006/078256), and CBH I from *Aspergillus fumigatus*, in particular Cel7A CBH1 disclosed as SEQ ID NO: 2 in WO2011/057140.

In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Penicillium emersonii* AA9 (GH61A) polypeptide having cellulolytic enhancing activity, in particular the one disclosed in WO2011/041397, *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO2005/047499), *Aspergillus fumigatus* xylanase (e.g., Xyl III in WO2006/078256), CBH I from *Aspergillus fumigatus*, in particular Cel7A CBH1 disclosed as SEQ ID NO: 2 in WO2011/057140, and CBH II derived from *Aspergillus fumigatus* in particular the one disclosed as SEQ ID NO: 4 in WO2013/028928.

In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Penicillium emersonii* AA9 (GH61A) polypeptide having cellulolytic enhancing activity, in particular the one disclosed in WO2011/041397, *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO2005/047499) or variant thereof with one or more, in

particular all, of the following substitutions: F100D, S283G, N456E, F512Y; *Aspergillus fumigatus* xylanase (e.g., Xyl III in WO2006/078256), CBH I from *Aspergillus fumigatus*, in particular Cel7A CBH I disclosed as SEQ ID NO: 2 in WO2011/057140, and CBH II derived from *Aspergillus*

fumigatus, in particular the one disclosed in WO2013/028928.

In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition comprising the CBH I (GENSEQP Accession No. AZY49536 (WO2012/103293); a CBH II (GENSEQP Accession No. AZY49446 (WO2012/103288); a beta-glucosidase variant (GENSEQP Accession No. AZU67153 (WO2012/44915)), in particular with one or more, in particular all, of the following substitutions: F100D, S283G, N456E, F512Y; and AA9 (GH61 polypeptide) (GENSEQP Accession No. BAL61510 (WO2013/028912)).

In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition comprising a CBH I (GENSEQP Accession No. AZY49536 (WO2012/103293)); a CBH II (GENSEQP Accession No. AZY49446 (WO2012/103288); a GH10 xylanase (GENSEQP Accession No. BAK46118 (WO2013/019827)); and a beta-xylosidase (GENSEQP Accession No. AZI04896 (WO2011/057140)).

In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition comprising a CBH I (GENSEQP Accession No. AZY49536 (WO2012/103293)); a CBH II (GENSEQP Accession No. AZY49446 (WO2012/103288)); and an AA9 (GH61 polypeptide; GENSEQP Accession No. BAL61510 (WO2013/028912)).

In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition comprising a CBH I (GENSEQP Accession No. AZY49536 (WO2012/103293)); a CBH II (GENSEQP Accession No. AZY49446 (WO2012/103288)), an AA9 (GH61 polypeptide; GENSEQP Accession No. BAL61510 (WO2013/028912)), and a catalase (GENSEQP Accession No. BAC11005 (WO2012/130120)).

In an embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition comprising a CBH I (GENSEQP Accession No. AZY49446 (WO2012/103288); a CBH II (GENSEQP Accession No. AZY49446 (WO2012/103288)), a beta-glucosidase variant (GENSEQP Accession No. AZU67153 (WO2012/44915)), with one or more, in particular all, of the following substitutions: F100D, S283G, N456E, F512Y; an AA9 (GH61 polypeptide; GENSEQP Accession No. BAL61510 (WO2013/028912)), a GH10 xylanase (GENSEQP Accession No. BAK46118 (WO2013/019827)), and a beta-xylosidase (GENSEQP Accession No. AZI04896 (WO2011/057140)).

In an embodiment the cellulolytic composition is a *Trichoderma reesei* cellulolytic enzyme preparation comprising an EG I (Swissprot Accession No. P07981), EG II (EMBL Accession No. M19373), CBH I (supra); CBH II (supra); beta-glucosidase variant (supra) with the following substitutions: F100D, S283G, N456E, F512Y; an AA9 (GH61 polypeptide; supra), GH10 xylanase (supra); and beta-xylosidase (supra).

All cellulolytic enzyme compositions disclosed in WO2013/028928 are also contemplated and hereby incorporated by reference.

The cellulolytic enzyme composition comprises or may further comprise one or more (several) proteins selected from the group consisting of a cellulase, a AA9 (i.e., GH61)

polypeptide having cellulolytic enhancing activity, a hemi-cellulase, an expansin, an esterase, a laccase, a ligninolytic enzyme, a pectinase, a peroxidase, a protease, and a swollenin.

In one embodiment the cellulolytic enzyme composition is a commercial cellulolytic enzyme composition. Examples of commercial cellulolytic enzyme compositions suitable for use in a process of the invention include: CELLIC® CTec (Novozymes A/S), CELLIC® CTec2 (Novozymes A/S), CELLIC® CTec3 (Novozymes A/S), CELLULAST™ (Novozymes A/S), SPEZYME™ CP (Genencor Int.), ACCELLERASE™ 1000, ACCELLERASE 1500, ACCELLERASE™ TRIO (DuPont), FILTRASE® NL (DSM); METHAPLUS® S/L 100 (DSM), ROHAMENT™ 7069 W (Rohm GmbH), or ALTERNAFUEL® CMAX3™ (Dyadic International, Inc.). The cellulolytic enzyme composition may be added in an amount effective from about 0.001 to about 5.0 wt. % of solids, e.g., about 0.025 to about 4.0 wt. % of solids or about 0.005 to about 2.0 wt. % of solids.

Additional enzymes, and compositions thereof can be found in WO2011/153516 and WO2016/045569 (the contents of which are incorporated herein).

Additional polynucleotides encoding suitable cellulolytic enzymes may be obtained from microorganisms of any genus, including those readily available within the UniProtKB database (www.uniprot.org).

The cellulolytic enzyme coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding cellulolytic enzymes from strains of different genera or species, as described supra.

The polynucleotides encoding cellulolytic enzymes may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc.) as described supra.

Techniques used to isolate or clone polynucleotides encoding cellulolytic enzymes are described supra.

In one embodiment, the cellulolytic enzyme has a mature polypeptide sequence of at least 60%, e.g., at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to any cellulolytic enzyme described or referenced herein (e.g., any endoglucanase, cellobiohydrolase, or beta-glucosidase). In one aspect, the cellulolytic enzyme has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from any cellulolytic enzyme described or referenced herein. In one embodiment, the cellulolytic enzyme has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any cellulolytic enzyme described or referenced herein, allelic variant, or a fragment thereof having cellulolytic enzyme activity. In one embodiment, the cellulolytic enzyme has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) amino acids. In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

In some embodiments, the cellulolytic enzyme has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the cellulolytic enzyme activity of any cellulolytic enzyme

described or referenced herein (e.g., any endoglucanase, cellobiohydrolase, or beta-glucosidase) under the same conditions.

In one embodiment, the cellulolytic enzyme coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any cellulolytic enzyme described or referenced herein (e.g., any endoglucanase, cellobiohydrolase, or beta-glucosidase). In one embodiment, the cellulolytic enzyme coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any cellulolytic enzyme described or referenced herein.

In one embodiment, the polynucleotide encoding the cellulolytic enzyme comprises the coding sequence of any cellulolytic enzyme described or referenced herein (e.g., any endoglucanase, cellobiohydrolase, or beta-glucosidase). In one embodiment, the polynucleotide encoding the cellulolytic enzyme comprises a subsequence of the coding sequence from any cellulolytic enzyme described or referenced herein, wherein the subsequence encodes a polypeptide having cellulolytic enzyme activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

The cellulolytic enzyme can also include fused polypeptides or cleavable fusion polypeptides, as described supra.

Xylose Metabolism

In one aspect, the fermenting organism (e.g., yeast cell) further comprises a heterologous polynucleotide encoding a xylose isomerase (XI). The xylose isomerase may be any xylose isomerase that is suitable for the host cells and the methods described herein, such as a naturally occurring xylose isomerase or a variant thereof that retains xylose isomerase activity. In one embodiment, the xylose isomerase is present in the cytosol of the host cells.

In some embodiments, the fermenting organism comprising a heterologous polynucleotide encoding a xylose isomerase has an increased level of xylose isomerase activity compared to the host cells without the heterologous polynucleotide encoding the xylose isomerase, when cultivated under the same conditions. In some embodiments, the fermenting organisms have an increased level of xylose isomerase activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the host cells without the heterologous polynucleotide encoding the xylose isomerase, when cultivated under the same conditions.

Exemplary xylose isomerases that can be used with the host cells and methods of use described herein include, but are not limited to, XIs from the fungus *Piromyces* sp. (WO2003/062430) or other sources (Madhavan et al., 2009, *Appl Microbiol Biotechnol.* 82(6), 1067-1078) have been expressed in *S. cerevisiae* host cells. Still other XIs suitable for expression in yeast have been described in US 2012/0184020 (an XI from *Ruminococcus flavefaciens*), WO2011/078262 (several XIs from *Reticulitermes speratus* and *Mastotermes darwiniensis*) and WO2012/009272 (constructs and fungal cells containing an XI from *Abiotrophia defectiva*). U.S. Pat. No. 8,586,336 describes a *S. cerevisiae*

host cell expressing an XI obtained by bovine rumen fluid (shown herein as SEQ ID NO: 74).

Additional polynucleotides encoding suitable xylose isomerases may be obtained from microorganisms of any genus, including those readily available within the UniProtKB database (www.uniprot.org). In one embodiment, the xylose isomerases is a bacterial, a yeast, or a filamentous fungal xylose isomerase, e.g., obtained from any of the microorganisms described or referenced herein, as described supra.

The xylose isomerase coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding xylose isomerases from strains of different genera or species, as described supra.

The polynucleotides encoding xylose isomerases may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc.) as described supra.

Techniques used to isolate or clone polynucleotides encoding xylose isomerases are described supra.

In one embodiment, the xylose isomerase has a mature polypeptide sequence of having at least 60%, e.g., at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to any xylose isomerase described or referenced herein (e.g., the xylose isomerase of SEQ ID NO: 74). In one aspect, the xylose isomerase has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from any xylose isomerase described or referenced herein (e.g., the xylose isomerase of SEQ ID NO: 74). In one embodiment, the xylose isomerase has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any xylose isomerase described or referenced herein (e.g., the xylose isomerase of SEQ ID NO: 74), allelic variant, or a fragment thereof having xylose isomerase activity. In one embodiment, the xylose isomerase has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) amino acids. In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

In some embodiments, the xylose isomerase has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the xylose isomerase activity of any xylose isomerase described or referenced herein (e.g., the xylose isomerase of SEQ ID NO: 74) under the same conditions.

In one embodiment, the xylose isomerase coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any xylose isomerase described or referenced herein (e.g., the xylose isomerase of SEQ ID NO: 74). In one embodiment, the xylose isomerase coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the

coding sequence from any xylose isomerase described or referenced herein (e.g., the xylose isomerase of SEQ ID NO: 74).

In one embodiment, the heterologous polynucleotide encoding the xylose isomerase comprises the coding sequence of any xylose isomerase described or referenced herein (e.g., the xylose isomerase of SEQ ID NO: 74). In one embodiment, the heterologous polynucleotide encoding the xylose isomerase comprises a subsequence of the coding sequence from any xylose isomerase described or referenced herein, wherein the subsequence encodes a polypeptide having xylose isomerase activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

The xylose isomerases can also include fused polypeptides or cleavable fusion polypeptides, as described supra.

In one aspect, the fermenting organism (e.g., yeast cell) further comprises a heterologous polynucleotide encoding a xylulokinase (XK). A xylulokinase, as used herein, provides enzymatic activity for converting D-xylulose to xylulose 5-phosphate. The xylulokinase may be any xylulokinase that is suitable for the host cells and the methods described herein, such as a naturally occurring xylulokinase or a variant thereof that retains xylulokinase activity. In one embodiment, the xylulokinase is present in the cytosol of the host cells.

In some embodiments, the fermenting organisms comprising a heterologous polynucleotide encoding a xylulokinase have an increased level of xylulokinase activity compared to the host cells without the heterologous polynucleotide encoding the xylulokinase, when cultivated under the same conditions. In some embodiments, the host cells have an increased level of xylose isomerase activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the host cells without the heterologous polynucleotide encoding the xylulokinase, when cultivated under the same conditions.

Exemplary xylulokinases that can be used with the fermenting organisms and methods of use described herein include, but are not limited to, the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75. Additional polynucleotides encoding suitable xylulokinases may be obtained from microorganisms of any genus, including those readily available within the UniProtKB database (www.uniprot.org). In one embodiment, the xylulokinases is a bacterial, a yeast, or a filamentous fungal xylulokinase, e.g., obtained from any of the microorganisms described or referenced herein, as described supra.

The xylulokinase coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding xylulokinases from strains of different genera or species, as described supra.

The polynucleotides encoding xylulokinases may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc.) as described supra.

Techniques used to isolate or clone polynucleotides encoding xylulokinases are described supra.

In one embodiment, the xylulokinase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to any xylulokinase described or

referenced herein (e.g., the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75). In one embodiment, the xylulokinase has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from any xylulokinase described or referenced herein (e.g., the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75). In one embodiment, the xylulokinase has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any xylulokinase described or referenced herein (e.g., the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75), allelic variant, or a fragment thereof having xylulokinase activity. In one embodiment, the xylulokinase has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) amino acids. In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

In some embodiments, the xylulokinase has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the xylulokinase activity of any xylulokinase described or referenced herein (e.g., the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75) under the same conditions.

In one embodiment, the xylulokinase coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any xylulokinase described or referenced herein (e.g., the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75). In one embodiment, the xylulokinase coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any xylulokinase described or referenced herein (e.g., the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75).

In one embodiment, the heterologous polynucleotide encoding the xylulokinase comprises the coding sequence of any xylulokinase described or referenced herein (e.g., the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75). In one embodiment, the heterologous polynucleotide encoding the xylulokinase comprises a subsequence of the coding sequence from any xylulokinase described or referenced herein, wherein the subsequence encodes a polypeptide having xylulokinase activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

The xylulokinases can also include fused polypeptides or cleavable fusion polypeptides, as described supra.

In one aspect, the fermenting organism (e.g., yeast cell) further comprises a heterologous polynucleotide encoding a ribulose 5 phosphate 3-epimerase (RPE1). A ribulose 5 phosphate 3-epimerase, as used herein, provides enzymatic activity for converting L-ribulose 5-phosphate to L-xylulose 5-phosphate (EC 5.1.3.22). The RPE1 may be any RPE1 that is suitable for the host cells and the methods described herein, such as a naturally occurring RPE1 or a variant thereof that retains RPE1 activity. In one embodiment, the RPE1 is present in the cytosol of the host cells. In one

embodiment, the recombinant cell comprises a heterologous polynucleotide encoding a ribulose 5 phosphate 3-epimerase (RPE1), wherein the RPE1 is *Saccharomyces cerevisiae* RPE1, or an RPE1 having at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to a *Saccharomyces cerevisiae* RPE1.

In one aspect, the fermenting organism (e.g., yeast cell) further comprises a heterologous polynucleotide encoding a ribulose 5 phosphate isomerase (RKI1). A ribulose 5 phosphate isomerase, as used herein, provides enzymatic activity for converting ribose-5-phosphate to ribulose 5-phosphate. The RKI1 may be any RKI1 that is suitable for the host cells and the methods described herein, such as a naturally occurring RKI1 or a variant thereof that retains RKI1 activity. In one embodiment, the RKI1 is present in the cytosol of the host cells.

In one embodiment, the fermenting organism comprises a heterologous polynucleotide encoding a ribulose 5 phosphate isomerase (RKI1), wherein the RKI1 is a *Saccharomyces cerevisiae* RKI1, or an RKI1 having a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to a *Saccharomyces cerevisiae* RKI1.

In one aspect, the fermenting organism (e.g., yeast cell) further comprises a heterologous polynucleotide encoding a transketolase (TKL1). The TKL1 may be any TKL1 that is suitable for the host cells and the methods described herein, such as a naturally occurring TKL1 or a variant thereof that retains TKL1 activity. In one embodiment, the TKL1 is present in the cytosol of the host cells.

In one embodiment, the fermenting organism comprises a heterologous polynucleotide encoding a transketolase (TKL1), wherein the TKL1 is a *Saccharomyces cerevisiae* TKL1, or a TKL1 having a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to a *Saccharomyces cerevisiae* TKL1.

In one aspect, the fermenting organism (e.g., yeast cell) further comprises a heterologous polynucleotide encoding a transaldolase (TAL1). The TAL1 may be any TAL1 that is suitable for the host cells and the methods described herein, such as a naturally occurring TAL1 or a variant thereof that retains TAL1 activity. In one embodiment, the TAL1 is present in the cytosol of the host cells.

In one embodiment, the fermenting organism comprises a heterologous polynucleotide encoding a transketolase (TAL1), wherein the TAL1 is a *Saccharomyces cerevisiae* TAL1, or a TAL1 having a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to a *Saccharomyces cerevisiae* TAL1.

Fermentation Products

A fermentation product can be any substance derived from the fermentation. The fermentation product can be, without limitation, an alcohol (e.g., arabinitol, n-butanol, isobutanol, ethanol, glycerol, methanol, ethylene glycol, 1,3-propanediol [propylene glycol], butanediol, glycerin, sorbitol, and xylitol); an alkane (e.g., pentane, hexane, heptane, octane, nonane, decane, undecane, and dodecane), a cycloalkane (e.g., cyclopentane, cyclohexane, cycloheptane, and cyclooctane), an alkene (e.g., pentene, hexene, heptene, and octene); an amino acid (e.g., aspartic acid, glutamic acid, glycine, lysine, serine, and threonine); a gas (e.g., methane, hydrogen (H₂), carbon dioxide (CO₂), and carbon monoxide (CO)); isoprene; a ketone (e.g., acetone); an organic acid (e.g., acetic acid, acetic acid, adipic acid, ascorbic acid, citric acid, 2,5-diketo-D-gluconic acid, formic

acid, fumaric acid, glucaric acid, gluconic acid, glucuronic acid, glutaric acid, 3-hydroxypropionic acid, itaconic acid, lactic acid, malic acid, malonic acid, oxalic acid, oxaloacetic acid, propionic acid, succinic acid, and xylonic acid); and polyketide.

In one aspect, the fermentation product is an alcohol. The term "alcohol" encompasses a substance that contains one or more hydroxyl moieties. The alcohol can be, but is not limited to, n-butanol, isobutanol, ethanol, methanol, arabinitol, butanediol, ethylene glycol, glycerin, glycerol, 1,3-propanediol, sorbitol, xylitol. See, for example, Gong et al., 1999, Ethanol production from renewable resources, in *Advances in Biochemical Engineering/Biotechnology*, Scheper, T., ed., Springer-Verlag Berlin Heidelberg, Germany, 65: 207-241; Silveira and Jonas, 2002, *Appl. Microbiol. Biotechnol.* 59: 400-408; Nigam and Singh, 1995, *Process Biochemistry* 30(2): 117-124; Ezeji et al., 2003, *World Journal of Microbiology and Biotechnology* 19(6): 595-603. In one embodiment, the fermentation product is ethanol.

In another aspect, the fermentation product is an alkane. The alkane may be an unbranched or a branched alkane. The alkane can be, but is not limited to, pentane, hexane, heptane, octane, nonane, decane, undecane, or dodecane.

In another aspect, the fermentation product is a cycloalkane. The cycloalkane can be, but is not limited to, cyclopentane, cyclohexane, cycloheptane, or cyclooctane.

In another aspect, the fermentation product is an alkene. The alkene may be an unbranched or a branched alkene. The alkene can be, but is not limited to, pentene, hexene, heptene, or octene. In another aspect, the fermentation product is an amino acid. The organic acid can be, but is not limited to, aspartic acid, glutamic acid, glycine, lysine, serine, or threonine. See, for example, Richard and Margarithis, 2004, *Biotechnology and Bioengineering* 87(4): 501-515.

In another aspect, the fermentation product is a gas. The gas can be, but is not limited to, methane, H₂, CO₂, or CO. See, for example, Kataoka et al., 1997, *Water Science and Technology* 36(6-7): 41-47; and Gunaseelan, 1997, *Biomass and Bioenergy* 13(1-2): 83-114.

In another aspect, the fermentation product is isoprene.

In another aspect, the fermentation product is a ketone. The term "ketone" encompasses a substance that contains one or more ketone moieties. The ketone can be, but is not limited to, acetone.

In another aspect, the fermentation product is an organic acid. The organic acid can be, but is not limited to, acetic acid, acetic acid, adipic acid, ascorbic acid, citric acid, 2,5-diketo-D-gluconic acid, formic acid, fumaric acid, glucaric acid, gluconic acid, glucuronic acid, glutaric acid, 3-hydroxypropionic acid, itaconic acid, lactic acid, malic acid, malonic acid, oxalic acid, propionic acid, succinic acid, or xylonic acid. See, for example, Chen and Lee, 1997, *Appl. Biochem. Biotechnol.* 63-65: 435-448.

In another aspect, the fermentation product is polyketide. Recovery

The fermentation product, e.g., ethanol, can optionally be recovered from the fermentation medium using any method known in the art including, but not limited to, chromatography, electrophoretic procedures, differential solubility, distillation, or extraction. For example, alcohol is separated from the fermented cellulosic material and purified by conventional methods of distillation. Ethanol with a purity of up to about 96 vol. % can be obtained, which can be used as, for example, fuel ethanol, drinking ethanol, i.e., potable neutral spirits, or industrial ethanol.

In some aspects of the methods, the fermentation product after being recovered is substantially pure. With respect to the methods herein, "substantially pure" intends a recovered preparation that contains no more than 15% impurity, wherein impurity intends compounds other than the fermentation product (e.g., ethanol). In one variation, a substantially pure preparation is provided wherein the preparation contains no more than 25% impurity, or no more than 20% impurity, or no more than 10% impurity, or no more than 5% impurity, or no more than 3% impurity, or no more than 1% impurity, or no more than 0.5% impurity.

Suitable assays to test for the production of ethanol and contaminants, and sugar consumption can be performed using methods known in the art. For example, ethanol product, as well as other organic compounds, can be analyzed by methods such as HPLC (High Performance Liquid Chromatography), GC-MS (Gas Chromatography Mass Spectroscopy) and LC-MS (Liquid Chromatography-Mass Spectroscopy) or other suitable analytical methods using routine procedures well known in the art. The release of ethanol in the fermentation broth can also be tested with the culture supernatant. Byproducts and residual sugar in the fermentation medium (e.g., glucose or xylose) can be quantified by HPLC using, for example, a refractive index detector for glucose and alcohols, and a UV detector for organic acids (Lin et al., *Biotechnol. Bioeng.* 90:775-779 (2005)), or using other suitable assay and detection methods well known in the art.

The invention may further be described in the following numbered paragraphs:

Paragraph [1]. A method of producing a fermentation product from a starch-containing or cellulosic-containing material comprising:

(a) saccharifying the starch-containing or cellulosic-containing material; and

(b) fermenting the saccharified material of step (a) with a fermenting organism;

wherein the fermenting organism comprises a heterologous polynucleotide encoding an alpha-amylase or a heterologous polynucleotide encoding a trehalase.

Paragraph [2]. The method of paragraph [1], wherein the alpha-amylase has a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of any one of SEQ ID NOs: 76-101, 121-174 and 231.

Paragraph [3]. The method of paragraph [1] or [2], wherein the heterologous polynucleotide encodes an alpha-amylase having a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of SEQ ID NOs: 76-101, 121-174 and 231.

Paragraph [4]. The method of any one of paragraphs [1]-[3], wherein the heterologous polynucleotide encodes an alpha-amylase having a mature polypeptide sequence comprising or consisting of the amino acid sequence of any one of SEQ ID NOs: SEQ ID NOs: 76-101, 121-174 and 231.

Paragraph [5]. The method of any one of paragraphs [1]-[4], wherein the trehalase has mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of any one of SEQ ID NOs: 175-226.

Paragraph [6]. The method any one of paragraphs [1]-[5], wherein the heterologous polynucleotide encodes a trehalase having a mature polypeptide sequence that differs by no

more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of SEQ ID NOs: 175-226.

Paragraph [7]. The method of any one of paragraphs [1]-[6], wherein the heterologous polynucleotide encodes a trehalase having a mature polypeptide sequence comprising or consisting of the amino acid sequence of any one of SEQ ID NOs: SEQ ID NOs: 175-226.

Paragraph [8]. The method of any one of paragraphs [1]-[7], wherein saccharification of step (a) occurs on a starch-containing material, and wherein the starch-containing material is either gelatinized or ungelatinized starch.

Paragraph [9]. The method of paragraph [8], comprising liquefying the starch-containing material by contacting the material with an alpha-amylase prior to saccharification.

Paragraph [10]. The method of paragraph [9], wherein liquefying the starch-containing material and/or saccharifying the starch-containing material is conducted in presence of exogenously added protease.

Paragraph [11]. The method of any one of paragraphs [1]-[10], wherein fermentation is performed under reduced nitrogen conditions (e.g., less than 1000 ppm urea or ammonium hydroxide, such as less than 750 ppm, less than 500 ppm, less than 400 ppm, less than 300 ppm, less than 250 ppm, less than 200 ppm, less than 150 ppm, less than 100 ppm, less than 75 ppm, less than 50 ppm, less than 25 ppm, or less than 10 ppm).

Paragraph [12]. The method of any one of paragraphs [1]-[11], wherein fermentation and saccharification are performed simultaneously in a simultaneous saccharification and fermentation (SSF).

Paragraph [13]. The method of any one of paragraphs [1]-[11], wherein fermentation and saccharification are performed sequentially (SHF).

Paragraph [14]. The method of any one of paragraphs paragraph [1]-[13], comprising recovering the fermentation product from the from the fermentation.

Paragraph [15]. The method of paragraph [14], wherein recovering the fermentation product from the from the fermentation comprises distillation.

Paragraph [16]. The method of any one of paragraphs [1]-[15], wherein the fermentation product is ethanol.

Paragraph [17]. The method of any one of paragraphs [1]-[16], wherein the fermenting organism comprises a heterologous polynucleotide encoding a glucoamylase.

Paragraph [18]. The method of paragraph [17], wherein the glucoamylase has a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of a *Pycnoporus glycoamylase* (e.g., a *Pycnoporus sanguineus* glucoamylase of SEQ ID NO: 229), a *Gloeophyllum* glucoamylase (e.g. a *Gloeophyllum sepiarium* of SEQ ID NO: 8), or a glucoamylase of any one of SEQ ID NOs: 102-113 (e.g., a *Saccharomycopsis fibuligera* glucoamylase of SEQ ID NO: 103 or 104, or a *Trichoderma reesei* glucoamylase of SEQ ID NO: 230).

Paragraph [19]. The method of any one of paragraphs [1]-[18], wherein the fermenting organism comprises a heterologous polynucleotide encoding a protease.

Paragraph [20]. The method of paragraph [19], wherein the protease has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence of any one of SEQ ID NOs: 9-73 (e.g., any one of

SEQ ID NOs: 9, 14, 16, 21, 22, 33, 41, 45, 61, 62, 66, 67, and 69; such as any one of SEQ NOs: 9, 14, 16, and 69). Paragraph [21]. The method of any one of paragraphs [1]-[20], wherein saccharification of step (a) occurs on a cellulosic-containing material, and wherein the cellulosic-

containing material is pretreated. Paragraph [22]. The method of paragraph [21], wherein the pretreatment is a dilute acid pretreatment.

Paragraph [23]. The method of any one of paragraphs [1]-[20], wherein saccharification occurs on a cellulosic-containing material, and wherein the enzyme composition comprises one or more enzymes selected from a cellulase, an AA9 polypeptide, a hemicellulase, a CIP, an esterase, an expansin, a ligninolytic enzyme, an oxidoreductase, a pectinase, a protease, and a swollenin.

Paragraph [24]. The method of paragraph [23], wherein the cellulase is one or more enzymes selected from an endoglucanase, a cellobiohydrolase, and a beta-glucosidase.

Paragraph [25]. The method of paragraph or [24], wherein the hemicellulase is one or more enzymes selected a xylanase, an acetylxylan esterase, a feruloyl esterase, an arabinofuranosidase, a xylosidase, and a glucuronidase.

Paragraph [26]. The method of any one of paragraphs [1]-[25], wherein the fermenting organism is a *Saccharomyces*, *Rhodotorula*, *Schizosaccharomyces*, *Kluyveromyces*, *Pichia*, *Hansenula*, *Rhodospiridium*, *Candida*, *Yarrowia*, *Lipomyces*, *Cryptococcus*, or *Dekkera* sp. cell.

Paragraph [27]. The method of any one of paragraphs [1]-[26], wherein the fermenting organism is a *Saccharomyces cerevisiae* cell.

Paragraph [28]. A recombinant yeast cell comprising a heterologous polynucleotide encoding an alpha-amylase or a heterologous polynucleotide encoding a trehalase.

Paragraph [29]. The recombinant yeast cell of paragraph [28], wherein the alpha-amylase has a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of any one of SEQ ID NOs: 76-101, 121-174 and 231.

Paragraph [30]. The recombinant yeast cell of paragraph or [29], wherein the heterologous polynucleotide encodes an alpha-amylase having a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of SEQ ID NOs: 76-101, 121-174 and 231.

Paragraph [31]. The recombinant yeast cell of any one of paragraphs [28]-[30], wherein the heterologous polynucleotide encodes an alpha-amylase having a mature polypeptide sequence comprising or consisting of the amino acid sequence of any one of SEQ ID NOs: 76-101, 121-174 and 231.

Paragraph [32]. The recombinant yeast cell of any one of paragraphs [28]-[31], wherein the trehalase has mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of any one of SEQ ID NOs: 175-226.

Paragraph [33]. The recombinant yeast cell of any one of paragraphs [28]-[32], wherein the heterologous polynucleotide encodes a trehalase having a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of SEQ ID NOs: 175-226.

Paragraph [34]. The recombinant yeast cell of any one of paragraphs [28]-[33], wherein the heterologous polynucleotide encodes a trehalase having a mature polypeptide sequence comprising or consisting of the amino acid sequence of any one of SEQ ID NOs: 175-226. Paragraph [35]. The recombinant yeast cell of any one of paragraphs [28]-[34], wherein the fermenting organism comprises a heterologous polynucleotide encoding a glucoamylase.

Paragraph [36]. The recombinant yeast cell of paragraph [35], wherein the glucoamylase has a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of a *Pycnoporus glycoamylase* (e.g., a *Pycnoporus sanguineus* glucoamylase of SEQ ID NO: 229), a *Gloeophyllum glucoamylase* (e.g. a *Gloeophyllum sepiarium* of SEQ ID NO: 8), or a *glucoamylase* of any one of SEQ ID NOs: 102-113 (e.g., a *Saccharomycopsis fibuligera* glucoamylase of SEQ ID NO: 103 or 104, or a *Trichoderma reesei* glucoamylase of SEQ ID NO: 230).

Paragraph [37]. The recombinant yeast cell of any one of paragraphs [28]-[36], wherein the fermenting organism comprises a heterologous polynucleotide encoding a protease.

Paragraph [38]. The recombinant yeast cell of paragraph [37], wherein the protease has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence of any one of SEQ ID NOs: 9-73 (e.g., any one of SEQ ID NOs: 9, 14, 16, 21, 22, 33, 41, 45, 61, 62, 66, 67, and 69; such as any one of SEQ NOs: 9, 14, 16, and 69).

Paragraph [39]. The recombinant yeast of any one of paragraphs [28]-[38], wherein the cell is a *Saccharomyces*, *Rhodotorula*, *Schizosaccharomyces*, *Kluyveromyces*, *Pichia*, *Hansenula*, *Rhodospiridium*, *Candida*, *Yarrowia*, *Lipomyces*, *Cryptococcus*, or *Dekkera* sp. cell.

Paragraph [40]. The recombinant yeast of paragraph [39], wherein the cell is a *Saccharomyces cerevisiae* cell.

The invention described and claimed herein is not to be limited in scope by the specific aspects herein disclosed, since these aspects are intended as illustrations of several aspects of the invention. Any equivalent aspects are intended to be within the scope of this invention. Indeed, various modifications of the invention in addition to those shown and described herein will become apparent to those skilled in the art from the foregoing description. Such modifications are also intended to fall within the scope of the appended claims. In the case of conflict, the present disclosure including definitions will control. All references are specifically incorporated by reference for that which is described.

The following examples are offered to illustrate certain aspects of the present invention, but not in any way intended to limit the scope of the invention as claimed.

EXAMPLES

Materials and Methods

Chemicals used as buffers and substrates were commercial products of at least reagent grade.

ETHANOL RED® ("ER"): *Saccharomyces cerevisiae* yeast available from Fermentis/Lesaffre, USA.

YPD+clonNAT plates were composed of 10 g of yeast extract, 20 g of peptone, 20 g bacto agar, and deionized water to 960 ml, followed by autoclave treatment. 40 ml

sterile 50% glucose and 1 ml clonNAT stock solution was added, followed by mixing and pouring.

clonNAT stock solution was composed of 2 g nourseo-thricin sulfate and deionized water to 20 ml.

Example 1: Construction of Yeast Strains Expressing a Heterologous Alpha-Amylase

This example describes the construction of yeast cells containing a heterologous alpha-amylase under control of an *S. cerevisiae* TDH3 promoter (SEQ ID NO: 1) or ADH1 promoter (SEQ ID NO: 5). Three pieces of DNA containing the promoter, gene and terminator were designed to allow for homologous recombination between the 3 DNA fragments and into the X-3 locus of the yeast yMHCT484 (PCT/US2018/035596). The resulting strain has one promoter containing fragment (left), one gene containing fragment (middle) and one ENO2 terminator (SEQ ID NO: 228) fragment (right) integrated into the *S. cerevisiae* genome at the X-3 locus.

Construction of the Promoter Containing Fragments (Left Fragments)

Synthetic linear uncloned DNA containing 300 bp homology to the X-3 site, *S. cerevisiae* promoter ADH1 (SEQ ID NO: 5) or THD3 (SEQ ID NO: 1) and *S. cerevisiae* EXG1 signal sequence (SEQ ID NO: 227) were synthesized by Thermo Fisher Scientific. The 2 linear DNAs were designated 17ABCK4P and 17ABCK3P for each promoter listed above, respectively. To generate additional linear DNA for

transformation into yeast, the DNA containing the left cassette was PCR amplified from 17ABCK4P and 17ABCK3P.

Construction of the Alpha-Amylase-Containing Fragments (Middle Fragments)

Synthetic linear uncloned DNA containing *S. cerevisiae* EXG1 signal peptide coding sequence (encoding the signal of SEQ ID NO: 227), a codon-optimized alpha-amylase gene and 50 bp of ENO2 terminator (SEQ ID NO: 228), were synthesized by Thermo Fisher Scientific.

To generate linear DNA for transformation into yeast, the DNA containing the alpha-amylase cassette was PCR amplified from the synthetic DNA with primers 1222985 (5'-ATGAT GAAAA AATAA GCAGA AAAGA CTAAT AATTC TTAGT TAAAA GC-3'; SEQ ID NO: 235) and 1222984 (5'-ATGCT TTCGC TTTAA ACGTT ACTGT G-3'; SEQ ID NO: 236). Fifty pmoles each of forward and reverse primer was used in a PCR reaction containing 5 ng of plasmid DNA as template, 0.1 mM each dATP, dGTP, dCTP, dTTP, 1× Phusion HF Buffer (Thermo Fisher Scientific), and 2 units Phusion Hot Start DNA polymerase in a final volume of 50 µL. The PCR was performed in a T100™ Thermal Cycler (Bio-Rad Laboratories, Inc.) programmed for one cycle at 98° C. for 1 minute followed by 32 cycles each at 98° C. for 10 seconds, 55° C. for 30 seconds, and 72° C. for 2 minutes with a final extension at 72° C. for 10 minutes. Following thermocycling, the PCR reaction products gel isolated and cleaned up using the NucleoSpin Gel and PCR clean-up kit (Machery-Nagel). The resulting linear DNAs were designated as indicated in Table 4.

TABLE 4

Alpha-amylase DNA product names and associated enzyme					
Product Number	DNA format	Signal peptide	Donor Organism (catalytic domain)	SEQ ID NO: (mature polypeptide)	Terminator Fragment
17ABDQYP	linear	EXG1	<i>Rhizomucor pusillus</i>	121	ENO2
17ABDQXP	linear	EXG1	<i>Bacillus licheniformis</i>	122	ENO2
17ABDQWP	linear	EXG1	<i>Aspergillus niger</i>	123	ENO2
17ABDQVP	linear	EXG1	<i>Aspergillus tamaritii</i>	124	ENO2
17ABDQUP	linear	EXG1	<i>Acidomyces richmondensis</i>	125	ENO2
17ABDQTP	linear	EXG1	<i>Aspergillus bombycis</i>	126	ENO2
17ABDQSP	linear	EXG1	<i>Alternaria</i> sp	127	ENO2
17ABDQRP	linear	EXG1	<i>Rhizopus microsporus</i>	128	ENO2
17ABDQQP	linear	EXG1	<i>Syncephalastrum racemosum</i>	129	ENO2
17ABDQPP	linear	EXG1	<i>Rhizomucor pusillus</i>	130	ENO2
17ABDQOP	linear	EXG1	<i>Dichotomocladium hesseltinei</i>	131	ENO2
17ABDQNP	linear	EXG1	<i>Lichtheimia ramosa</i>	132	ENO2
17ABDQMP	linear	EXG1	<i>Penicillium aethiopicum</i>	133	ENO2
17ABDQLP	linear	EXG1	<i>Subulisporea</i> sp	134	ENO2
17ABDQKP	linear	EXG1	<i>Trichoderma paraviridescens</i>	135	ENO2
17ABDQJP	linear	EXG1	<i>Byssosascus striatosporus</i>	136	ENO2
17ABDQIP	linear	EXG1	<i>Aspergillus brasiliensis</i>	137	ENO2
17ABDQHP	linear	EXG1	<i>Penicillium subspinosum</i>	138	ENO2
17ABDQGP	linear	EXG1	<i>Penicillium antarcticum</i>	139	ENO2
17ABDQFP	linear	EXG1	<i>Penicillium coprophilum</i>	140	ENO2
17ABDQEP	linear	EXG1	<i>Penicillium olsonii</i>	141	ENO2
17ABDQDP	linear	EXG1	<i>Penicillium vasconiae</i>	142	ENO2
17ABDQCP	linear	EXG1	<i>Penicillium</i> sp	143	ENO2
17ABDQBP	linear	EXG1	<i>Heterocephalum aurantiacum</i>	144	ENO2
17ABDQAP	linear	EXG1	<i>Neosartorya massa</i>	145	ENO2
17ABDP7P	linear	EXG1	<i>Penicillium janthinellum</i>	146	ENO2
17ABDP6P	linear	EXG1	<i>Aspergillus brasiliensis</i>	147	ENO2
17ABDP5P	linear	EXG1	<i>Aspergillus westerdijkiae</i>	148	ENO2
17ABDP4P	linear	EXG1	<i>Hamigera avellanea</i>	149	ENO2
17ABDP3P	linear	EXG1	<i>Hamigera avellanea</i>	150	ENO2
17ABDP2P	linear	EXG1	<i>Meripilus giganteus</i>	151	ENO2

TABLE 4-continued

Alpha-amylase DNA product names and associated enzyme					
Product Number	DNA format	Signal peptide	Donor Organism (catalytic domain)	SEQ ID NO: (mature polypeptide)	Terminator Fragment
17ABDPZP	linear	EXG1	<i>Cerrena unicolor</i>	152	ENO2
17ABDPYP	linear	EXG1	<i>Physalacria cryptomeriae</i>	153	ENO2
17ABDPXP	linear	EXG1	<i>Lenzites betulinus</i>	154	ENO2
17ABDPWP	linear	EXG1	<i>Trametes lujbarskyi</i>	155	ENO2
17ABDPVP	linear	EXG1	<i>Bacillus subtilis</i>	156	ENO2
17ABDPUP	linear	EXG1	<i>Bacillus subtilis</i> subsp. <i>subtilis</i>	157	ENO2
17ABDPTP	linear	EXG1	<i>Schwanniomyces occidentalis</i>	158	ENO2
17ABDPSP	linear	EXG1	<i>Rhizomucor pusillus</i>	159	ENO2
17ABDPRP	linear	EXG1	<i>Aspergillus niger</i>	160	ENO2
17ABDPQP	linear	EXG1	<i>Bacillus stearothermophilus</i>	161	ENO2
17ABDPPP	linear	EXG1	<i>Bacillus halmapalus</i>	162	ENO2
17ABDPOP	linear	EXG1	<i>Aspergillus oryzae</i>	163	ENO2
17ABDPNP	linear	EXG1	<i>Bacillus amyloliquefaciens</i>	164	ENO2
17ABDPMP	linear	EXG1	<i>Rhizomucor pusillus</i>	165	ENO2
17ABDPLP	linear	EXG1	<i>Kionochaeta ivoriensis</i>	166	ENO2
17ABDPKP	linear	EXG1	<i>Aspergillus niger</i>	167	ENO2
17ABDPJP	linear	EXG1	<i>Aspergillus oryzae</i>	168	ENO2
17ABDPIP	linear	EXG1	<i>Penicillium canescens</i>	169	ENO2
17ABDPHP	linear	EXG1	<i>Acidomyces acidothermus</i>	170	ENO2
17ABDQ4P	linear	EXG1	<i>Kionochaeta ivoriensis</i>	171	ENO2
17ABDQ3P	linear	EXG1	<i>Aspergillus terreus</i>	172	ENO2
17ABDQ2P	linear	EXG1	<i>Thamnidium elegans</i>	173	ENO2
17ABDQZP	linear	EXG1	<i>Meripilus giganteus</i>	174	ENO2

Integration of the Left, Middle and Right-Hand Fragments to Generate Yeast Strains with a Heterologous Alpha-Amylase

The yeast yMHCT484 (PCT/US2018/035596) was transformed with the left, middle and right integration fragments described above. In each transformation pool a fixed left fragment and right fragment were used. The middle fragment consisted of a pool of 19-21 middle fragments containing the alpha-amylase gene with 100-600 ng of each fragment (1500 ng total). To aid homologous recombination of the left, middle and right fragments at the genomic X-3 sites a plasmid containing Cas9 and guide RNA specific to X-3 (pMcTs442) was also used in the transformation. These four components were transformed into the into *S. cerevisiae*

strain yMHCT484 following a yeast electroporation protocol (See, Thompson et al. Yeast. 1998 Apr. 30; 14(6):565-71). Transformants were selected on YPD+cloNAT to select for transformants that contain the CRISPR/Cas9 plasmid pMcTs442 (FIG. 3). Transformants were picked using a Q-pix Colony Picking System (Molecular Devices) to inoculate 1 well of 96-well plate containing YPD+clonNAT media. The plates were grown for 2 days then glycerol was added to 20% final concentration and the plates were stored at -80° C. until needed. Integration of specific alpha-amylase construct was verified by PCR with locus specific primers and subsequent sequencing. The strains generated in this example are shown in Table 5.

TABLE 5

Alpha-amylase expressing <i>S. cerevisiae</i> strains.					
left piece	middle piece	Promoter	Signal peptide	Alpha-amylase gene donor (catalytic domain)	SEQ ID NO: (mature polypeptide) strain ID
17ABCK4P	17ABDQJP	ADH1	EXG1	<i>Byssosascus striatosporus</i>	136 P110-A08
17ABCK4P	17ABDQHP	ADH1	EXG1	<i>Penicillium subspinosum</i>	138 P110-A09
17ABCK4P	17ABDQUP	ADH1	EXG1	<i>Acidomyces richmondensis</i>	125 P110-B01
17ABCK4P	17ABDQKP	ADH1	EXG1	<i>Trichoderma paraviridescens</i>	135 P110-B04
17ABCK4P	17ABDQXP	ADH1	EXG1	<i>Bacillus licheniformis</i>	122 P110-B05
17ABCK4P	17ABDQMP	ADH1	EXG1	<i>Penicillium aethiopicum</i>	133 P110-B08
17ABCK4P	17ABDQSP	ADH1	EXG1	<i>Alternaria sp</i>	127 P110-C05
17ABCK4P	17ABDQOP	ADH1	EXG1	<i>Dichotomocladium hesseltinei</i>	131 P110-D01
17ABCK4P	17ABDQTP	ADH1	EXG1	<i>Aspergillus bombycis</i>	126 P110-D02
17ABCK4P	17ABDQIP	ADH1	EXG1	<i>Aspergillus brasiliensis</i>	137 P110-D10
17ABCK4P	17ABDQVP	ADH1	EXG1	<i>Aspergillus tamarii</i>	124 P110-F02
17ABCK4P	17ABDQTP	ADH1	EXG1	<i>Aspergillus bombycis</i>	126 P110-F07
17ABCK4P	17ABDQHP	ADH1	EXG1	<i>Penicillium subspinosum</i>	138 P110-G03
17ABCK4P	17ABDQUP	ADH1	EXG1	<i>Acidomyces richmondensis</i>	125 P110-G04
17ABCK4P	17ABDQSP	ADH1	EXG1	<i>Alternaria sp</i>	127 P110-G06
17ABCK4P	17ABDQLP	ADH1	EXG1	<i>Subulispora sp</i>	134 P110-H02

TABLE 5-continued

Alpha-amylase expressing <i>S. cerevisiae</i> strains.						
left piece	middle piece	Promoter	Signal peptide	Alpha-amylase gene donor (catalytic domain)	SEQ ID NO: (mature polypeptide)	strain ID
17ABCK4P	17ABDQHP	ADH1	EXG1	<i>Penicillium subspinosum</i>	138	P110-H05
17ABCK4P	17ABDQGP	ADH1	EXG1	<i>Penicillium antarcticum</i>	139	P110-H07
17ABCK4P	17ABDPZP	ADH1	EXG1	<i>Cerrena unicolor</i>	152	P111-C03
17ABCK4P	17ABDQAP	ADH1	EXG1	<i>Neosartorya massa</i>	145	P111-D10
17ABCK4P	17ABDQDP	ADH1	EXG1	<i>Penicillium vasconiae</i>	142	P111-F01
17ABCK4P	17ABDQCP	ADH1	EXG1	<i>Penicillium</i> sp	143	P111-H08
17ABCK4P	17ABDPJP	ADH1	EXG1	<i>Aspergillus oryzae</i>	168	P112-A03
17ABCK4P	17ABDQ3P	ADH1	EXG1	<i>Aspergillus terreus</i>	172	P112-A07
17ABCK4P	17ABDQ3P	ADH1	EXG1	<i>Aspergillus terreus</i>	172	P112-B11
17ABCK4P	17ABDQ2P	ADH1	EXG1	<i>Thamnidium elegans</i>	173	P112-C09
17ABCK4P	17ABDPPP	ADH1	EXG1	<i>Bacillus halmapalus</i>	162	P112-D05
17ABCK4P	17ABDPJP	ADH1	EXG1	<i>Aspergillus oryzae</i>	168	P112-D06
17ABCK4P	17ABDPMP	ADH1	EXG1	<i>Rhizomucor pusillus</i>	165	P112-H03
17ABCK3P	17ABDQIP	TDH3	EXG1	<i>Aspergillus brasiliensis</i>	137	P113-A03
17ABCK3P	17ABDQYP	TDH3	EXG1	<i>Rhizomucor pusillus</i>	121	P113-B05
17ABCK3P	17ABDQXP	TDH3	EXG1	<i>Bacillus licheniformis</i>	122	P113-B06
17ABCK3P	17ABDQTP	TDH3	EXG1	<i>Aspergillus bombycis</i>	126	P113-C03
17ABCK3P	17ABDQNP	TDH3	EXG1	<i>Lichtheimia ramosa</i>	132	P113-C06
17ABCK3P	17ABDQVP	TDH3	EXG1	<i>Aspergillus tamarii</i>	124	P113-C09
17ABCK3P	17ABDQYP	TDH3	EXG1	<i>Rhizomucor pusillus</i>	121	P113-C10
17ABCK3P	17ABDQRP	TDH3	EXG1	<i>Rhizopus microsporus</i>	128	P113-D07
17ABCK3P	17ABDQVP	TDH3	EXG1	<i>Aspergillus tamarii</i>	124	P113-D08
17ABCK3P	17ABDQSP	TDH3	EXG1	<i>Alternaria</i> sp	127	P113-D10
17ABCK3P	17ABDQNP	TDH3	EXG1	<i>Lichtheimia ramosa</i>	132	P113-F02
17ABCK3P	17ABDQQP	TDH3	EXG1	<i>Syncephalastrum racemosum</i>	129	P113-F05
17ABCK3P	17ABDQJP	TDH3	EXG1	<i>Byssosascus striatosporus</i>	136	P113-G04
17ABCK3P	17ABDQTP	TDH3	EXG1	<i>Aspergillus bombycis</i>	126	P113-G09
17ABCK3P	17ABDPSP	TDH3	EXG1	<i>Rhizomucor pusillus</i>	159	P114-A04
17ABCK3P	17ABDP4P	TDH3	EXG1	<i>Hamigera avellanea</i>	149	P114-B02
17ABCK3P	17ABDPUP	TDH3	EXG1	<i>Bacillus subtilis</i> subsp. <i>subtilis</i>	157	P114-B08
17ABCK3P	17ABDPUP	TDH3	EXG1	<i>Bacillus subtilis</i> subsp. <i>subtilis</i>	157	P114-C01
17ABCK3P	17ABDP2P	TDH3	EXG1	<i>Meripilus giganteus</i>	151	P114-C04
17ABCK3P	17ABDPVP	TDH3	EXG1	<i>Bacillus subtilis</i>	156	P114-C05
17ABCK3P	17ABDQAP	TDH3	EXG1	<i>Neosartorya massa</i>	145	P114-C06
17ABCK3P	17ABDQEP	TDH3	EXG1	<i>Penicillium olsonii</i>	141	P114-C07
17ABCK3P	17ABDPTP	TDH3	EXG1	<i>Schwanniomyces occidentalis</i>	158	P114-D02
17ABCK3P	17ABDPRP	TDH3	EXG1	<i>Aspergillus niger</i>	160	P114-D07
17ABCK3P	17ABDP3P	TDH3	EXG1	<i>Hamigera avellanea</i>	150	P114-F06
17ABCK3P	17ABDP6P	TDH3	EXG1	<i>Aspergillus brasiliensis</i>	147	P114-F07
17ABCK3P	17ABDPUP	TDH3	EXG1	<i>Bacillus subtilis</i> subsp. <i>subtilis</i>	157	P114-F08
17ABCK3P	17ABDP2P	TDH3	EXG1	<i>Meripilus giganteus</i>	151	P114-H02
17ABCK3P	17ABDQAP	TDH3	EXG1	<i>Neosartorya massa</i>	145	P114-H03
17ABCK3P	17ABDPZP	TDH3	EXG1	<i>Cerrena unicolor</i>	152	P114-H07
17ABCK3P	17ABDQAP	TDH3	EXG1	<i>Neosartorya massa</i>	145	P114-H08
17ABCK3P	17ABDPKP	TDH3	EXG1	<i>Aspergillus niger</i>	167	P115-B03
17ABCK3P	17ABDPMP	TDH3	EXG1	<i>Rhizomucor pusillus</i>	165	P115-C11
17ABCK3P	17ABDPMP	TDH3	EXG1	<i>Rhizomucor pusillus</i>	165	P115-D09
17ABCK3P	17ABDQ3P	TDH3	EXG1	<i>Aspergillus terreus</i>	172	P115-F06
17ABCK3P	17ABDPJP	TDH3	EXG1	<i>Aspergillus oryzae</i>	168	P115-G04

Example 2: Activity Assay of Yeast Strains Expressing Alpha-Amylase

Yeast strains from Example 1 were cultivated overnight in standard YPD media containing 2% or 6% glucose. The cultured yeast medium was subjected to centrifugation at 3500 rpm for 10 min to harvest the supernatant. The culture supernatant is used for enzyme activity assays, as described below. Yeast may also be cultivated using other cultivation media such as minimal YNB media or clarified and filtered industrial liquefied corn mash.

Glucoamylase Activity Assay

Glucoamylase activity was measured using maltose as substrate. Enzyme-catalyzed hydrolysis of maltose yields glucose as the reaction product which can be detected and

quantified using commercially available assay kits such as Wako Diagnostics AUTOKIT GLUCOSE C2. Reagents provided in the assay kits react with glucose resulted in a color change with maximal absorbance at 505 nm. The absorbance intensity measured spectrophotometrically is proportional to glucoamylase activity. The absorbance at 505 nm can be fit to standard curve generated using a purified glucoamylase enzyme to estimate enzyme activity. Reaction conditions and color development are described in Table 5 and Table 6, respectively. Glucoamylase units (AGU) for glucoamylase activity is defined as the amount of enzyme required to hydrolyze one micromole maltose per minute under the reaction conditions.

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TABLE 5

Glucoamylase reaction conditions	
Appropriate amount of yeast supernatant	10-200 μ l
Substrate	maltose, 10 mM
Buffer	acetate, 0.1M
pH	5.0 $\pm$ 0.05
Incubation temperature	32° C.
Reaction time	5-20 min
Glucoamylase assay range	0.001-0.036 AGU/ml

TABLE 6

Color development	
Reaction mixture	10 μ l
AUTOKIT GLUCOSE C2	200 μ l
developing reagent	
Incubation temperature	room temperature or 37° C.
Reaction time	10-25 min
Wavelength	505 nm

Alpha Amylase Activity Assay

Alpha amylase activity was measured using blocked-p-nitrophenyl-maltoheptaoside (BPNPG7) as substrate, which is included as the amylase HR reagent from Megazymes. Enzyme hydrolysis of the alpha-bond of BPNPG7 releases a blocked maltosaccharide oligomer and a p-nitrophenyl maltosaccharide oligomer. The p-nitrophenyl maltosaccharide will react with a glucoamylase from Megazymes yielding p-nitrophenol which may be detected using commercially available assay kits such as MEGAZYMES R-AMHR4. Reagents provided in the assay kits will specifically react with p-nitrophenol resulting in color formation. The color intensity measured using a spectrophotometer or microplate reader is proportional to alpha-amylase activity. Reaction conditions and color development are described in Table 7 and Table 8, respectively.

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TABLE 7

Alpha-amylase reaction conditions	
Appropriate amount of yeast supernatant	10-200 μ l
Substrate	blocked-p-nitrophenyl-maltoheptaoside (BPNPG7), 10 mM
Buffer	acetate, 0.1M
pH	5.0 $\pm$ 0.05
Incubation temperature	32° C.
Reaction time	20 min
Alpha amylase assay range	5-200 ng/ml

TABLE 8

Color development	
Reaction mixture	20 μ l
blocked-p-nitrophenyl-maltoheptaoside (BPNPG7)	80 μ l
Stop solution (4% Tris)	100 μ l
Incubation temperature	Room temperature - 32° C.
Reaction time	10-25 min
Wavelength	400 nm

Results

The absorbance at 505 nm increases as the amount of purified glucoamylase added to hydrolyze maltose to glucose increases. The absorbance at 400 nm increases as the amount of purified alpha-amylase added increases. Specifically, the alpha-amylase hydrolyzes blocked-p-nitrophenyl-maltoheptaoside (BPNPG7) releasing a blocked maltosaccharide oligomer and a p-nitrophenyl maltosaccharide oligomer. The p-nitrophenyl maltosaccharide reacts with a glucoamylase from Megazymes yielding p-nitrophenol which absorbs at 400 nm. A purified glucoamylase and alpha amylase standard curve was generated and used to estimate glucoamylase and alpha-amylase activity in yeast supernatants.

Results for alpha-amylase activity and glucoamylase activity are shown Table 9. A graphical representation of comparative alpha-amylase activity is shown in FIG. 1.

TABLE 9

Alpha-amylase (AA) and glucoamylase (GA) activity and estimated secretion							
Yeast strain no.	Promoter for alpha-amylase expression	SEQ ID NO: (mature polypeptide)	AA gene donor (catalytic domain)	Glucoamylase		Alpha-amylase	
				activity	Conc. (ug/mL)	activity	Conc. (ug/mL)
1	Background strain with glucoamylase gene, without alpha amylase gene			1.03	10.6	N/A	N/A
1	Background strain with glucoamylase gene, without alpha amylase gene			0.98	9.8	N/A	N/A
1	Background strain with glucoamylase gene, without alpha amylase gene			1.00	10.0	N/A	N/A
2	PADH1	125	<i>Acidomyces richmondensis</i>	0.86	8.1	0.1753	0.04
3	pADH1	125	<i>Acidomyces richmondensis</i>	1.03	10.6	0.5529	0.16
4	PADH1	136	<i>Byssosascus striatosporus</i>	0.76	6.6	0.084	0.01
5	pADH1	162	<i>Bacillus halmapalus</i>	1.58	18.5	0.4314	0.12
6	pADH1	172	<i>Aspergillus terreus</i>	0.91	8.8	1.4154	0.44
7	pTDH3	129	<i>Syncephalastrum racemosum</i>	0.87	8.2	1.3655	0.43
8	pTDH3	151	<i>Meripilus</i>	0.60	4.3	1.6636	0.52
9	pTDH3	141	<i>Penicillium olsonii</i>	0.99	10.0	0.0846	0.01
11	pADH1	131	<i>Dichotomocladium hesseltinei</i>	0.75	6.5	0.084	0.01

TABLE 9-continued

Alpha-amylase (AA) and glucoamylase (GA) activity and estimated secretion							
Yeast	Promoter for alpha- amylase	SEQ ID NO:	AA gene donor	Glucoamylase		Alpha-amylase	
strain no.	expression	(mature polypeptide)	(catalytic domain)	activity	Conc. (ug/mL)	activity	Conc. (ug/mL)
12	pADH1	122	<i>Bacillus licheniformis</i>	0.93	9.1	0.1147	0.02
13	pADH1	133	<i>Penicillium aethiopicum</i>	1.03	10.5	0.0842	0.01
14	pADH1	143	<i>Penicillium</i> sp	0.88	8.4	0.2047	0.05
15	pADH1	168	<i>Aspergillus oryzae</i>	0.90	8.7	1.561	0.49
16	pTDH3	122	<i>Bacillus licheniformis</i>	0.81	7.3	0.1219	0.02
17	pTDH3	126	<i>Aspergillus bombycis</i>	0.98	9.8	0.1628	0.03
18	pTDH3	145	<i>Neosartorya massa</i>	0.92	8.9	1.902	0.60
19	pTDH3	160	<i>Aspergillus niger</i>	0.95	9.3	0.0899	0.01
20	pTDH3	167	<i>Aspergillus niger</i>	0.98	9.8	1.2357	0.38
21	pADH1	126	<i>Aspergillus bombycis</i>	0.95	9.3	0.3891	0.11
22	pADH1	127	<i>Alternaria</i> sp	0.93	9.1	0.0866	0.01
23	pADH1	138	<i>Penicillium subspinulosum</i>	0.97	9.7	0.0848	0.01
24	pADH1	145	<i>Neosartorya massa</i>	0.93	9.1	0.8786	0.27
25	pTDH3	132	<i>Lichtheimia ramosa</i>	0.89	8.5	1.1903	0.37
26	pTDH3	132	<i>Lichtheimia ramosa</i>	0.88	8.3	1.7498	0.55
27	pTDH3	121	<i>Rhizomucor pusillus</i>	0.78	7.0	1.334	0.42
28	pTDH3	159	<i>Aspergillus niger</i>	0.90	8.7	1.9582	0.62
29	pTDH3	147	<i>Aspergillus brasiliensis</i>	1.03	10.6	0.3469	0.09
30	pTDH3	168	<i>Aspergillus oryzae</i>	0.92	9.0	1.5655	0.49
31	pADH1	124	<i>Aspergillus tamarii</i>	0.87	8.3	1.1583	0.36
32	pADH1	138	<i>Penicillium subspinulosum</i>	0.90	8.7	0.0848	0.01
34	pADH1	172	<i>Aspergillus terreus</i>	0.88	8.4	1.3645	0.43
35	pTDH3	137	<i>Aspergillus brasiliensis</i>	0.92	8.9	1.2183	0.38
36	pTDH3	169	<i>Penicillium canescens</i>	0.83	7.7	1.8212	0.57
37	pTDH3	127	<i>Alternaria</i> sp	0.95	9.4	0.0877	0.01
38	pTDH3	151	<i>Meripilus</i>	0.56	3.8	1.3665	0.43
39	pTDH3	152	<i>Cerreia unicolor</i>	0.98	9.8	0.0831	0.01
40	pTDH3	172	<i>Aspergillus terreus</i>	0.98	9.9	0.8774	0.27
41	pADH1	134	<i>Subulisporea</i> sp	0.72	6.1	0.0916	0.01
42	pADH1	127	<i>Alternaria</i> sp	0.80	7.3	0.0938	0.01
43	pADH1	137	<i>Aspergillus brasiliensis</i>	0.94	9.2	1.1772	0.36
44	pADH1	168	<i>Aspergillus oryzae</i>	0.92	9.0	1.5347	0.48
46	pTDH3	126	<i>Aspergillus bombycis</i>	1.03	10.5	0.1955	0.05
47	pTDH3	128	<i>Rhizopus microsporus</i>	0.65	5.1	0.9639	0.30
48	pTDH3	157	<i>Bacillus subtilis subsp. subtilis</i>	0.94	9.2	1.2992	0.40
49	pTDH3	156	<i>Bacillus subtilis</i>	0.93	9.1	1.1427	0.35
50	pTDH3	157	<i>Bacillus subtilis subsp. subtilis</i>	0.95	9.4	1.3374	0.42
51	pADH1	140	<i>Penicillium coprophilum</i>	0.90	8.6	0.7588	0.23
52	pADH1	136	<i>Byssosascus striatosporus</i>	0.94	9.2	0.085	0.01
53	pADH1	146	<i>Penicillium janthinellum</i>	0.79	7.0	0.3325	0.09
54	pADH1	173	<i>Thamnidium elegans</i>	0.91	8.8	1.1844	0.37
55	pADH1	163	<i>Aspergillus oryzae</i>	0.88	8.4	1.7175	0.54
56	pTDH3	137	<i>Aspergillus brasiliensis</i>	0.93	9.0	0.9012	0.27
57	pTDH3	123	<i>Aspergillus niger</i>	0.94	9.2	1.2994	0.40
58	pTDH3	150	<i>Hamigera avellanea</i>	0.87	8.3	0.7698	0.23
59	pTDH3	149	<i>Hamigera avellanea</i>	0.89	8.5	0.8048	0.24
60	pTDH3	165	<i>Rhizomucor pusillus</i>	0.94	9.3	1.9117	0.60
61	PADH1	138	<i>Penicillium subspinulosum</i>	0.76	6.7	0.1233	0.02
62	PADH1	126	<i>Aspergillus bombycis</i>	0.94	9.3	0.3626	0.10
63	pADH1	142	<i>Penicillium vasconiae</i>	0.74	6.4	0.1007	0.01
64	pADH1	165	<i>Rhizomucor pusillus</i>	0.94	9.3	1.7183	0.54
66	pTDH3	136	<i>Byssosascus striatosporus</i>	0.96	9.6	0.084	0.01
67	pTDH3	124	<i>Aspergillus tamarii</i>	0.90	8.6	1.0263	0.32
68	pTDH3	149	<i>Hamigera avellanea</i>	1.04	10.7	0.4173	0.12
69	pTDH3	145	<i>Neosartorya massa</i>	0.96	9.5	0.2249	0.05

TABLE 9-continued

Alpha-amylase (AA) and glucoamylase (GA) activity and estimated secretion							
Yeast strain no.	Promoter for alpha-amylase expression	SEQ ID NO: (mature polypeptide)	AA gene donor (catalytic domain)	Glucoamylase activity	Conc. (ug/mL)	Alpha-amylase activity	Conc. (ug/mL)
70	pTDH3	157	<i>Bacillus subtilis</i> subsp. <i>subtilis</i>	0.99	9.9	1.4	0.44
71	pTDH3	165	<i>Rhizomucor pusillus</i>	0.89	8.4	2.2181	0.70
72	pADH1	135	<i>Trichoderma paraviridescens</i>	0.92	9.0	0.0873	0.01
73	pADH1	139	<i>Penicillium antarcticum</i>	0.78	6.9	0.0926	0.01
74	pADH1	152	<i>Cerrena unicolor</i>	0.93	9.1	0.0858	0.01
76	pADH1	173	<i>Thamnidium elegans</i>	0.89	8.6	1.2458	0.39
77	pTDH3	121	<i>Rhizomucor pusillus</i>	0.80	7.2	1.6326	0.51
78	pTDH3	124	<i>Aspergillus tamaris</i>	0.80	7.3	1.2964	0.40
79	pTDH3	158	<i>Schwannomyces occidentalis</i>	1.00	10.0	0.2679	0.07
80	pTDH3	150		0.83	7.7	0.7079	0.21
81	pTDH3	145	<i>Neosartorya massa</i>	0.90	8.7	0.1957	0.05
82	pTDH3	165	<i>Rhizomucor pusillus</i>	0.81	7.3	2.1273	0.67

Example 3: Simultaneous Saccharification and Fermentation (SSF) of Yeast Strains Expressing Alpha-Amylase

Yeast strains were cultivated overnight in standard YPD media containing 2% glucose. The cultured yeast medium was centrifuged at 3000 rpm for 10 min to collect the supernatant. The supernatant was used for enzyme activity assay, as described below.

Alpha Amylase Activity Assay

Alpha-amylase activity was detected by measuring the amount of starch degraded through enzymatic hydrolysis of starch. Potassium iodide and iodine reagent was used to measure the residual starch based on the color development from application of the reagent. The color intensity measured on a spectrophotometer or microplate reader is inversely proportional to alpha-amylase activity. Reaction conditions and color development were described in Table 11 and Table 12, respectively.

TABLE 11

Alpha-amylase reaction condition	
Amount of yeast supernatant	20 μ l
Amount of substrate	130 μ l
Substrate	2 mM starch
Buffer	Sodium acetate, 0.1M, 0.01% Triton 100
pH	5.0 $\pm$ 0.05
Incubation temperature	20° C.
Reaction time	2-3 hr

TABLE 12

Color development	
Reaction mixture	150 μ l
Amount of reagent	50 μ l
Reagent	14.5 mM potassium iodide, 0.9 mM iodine

TABLE 12-continued

Color development	
Incubation temperature	20° C.
Reaction time	10-15 min
Wavelength	595 nm

Simultaneous saccharification and fermentation (SSF) was performed via mini-scale fermentations using industrial corn mash (Avantec® Amp, Novozymes, A/S) using conditions shown in Table 13. Yeast strains were cultivated overnight in YPD media with 2% glucose for 24 hours at 30° C. and 300 rpm. The corn mash was supplemented with 250 ppm of urea. Approximately 0.6 mg of corn mash was dispensed per well to 96 well microtiter plates, followed by the addition of approximately 10⁸ yeast cells/g of corn mash from the overnight culture. Plates were incubated at 32° C. without shaking. Duplicates of each strain were analyzed after 48 hour fermentations. Fermentation was stopped by the addition of 100 μ L of 8% H₂SO₄, followed by centrifugation at 3000 rpm for 10 min. The supernatant was analyzed for ethanol using H PLC.

TABLE 13

Microtiter plate fermentation reaction conditions	
Substrate	Avantec® Amp corn mash
Yeast pitch	10 ⁸ cells/g corn mash
Supplementary urea	250 ppm
pH	5.0 $\pm$ 0.05
Incubation temperature	32° C.
Reaction time	48 hours

As shown in Table 14, higher ethanol was obtained from yeast expressing a heterologous alpha-amylase compared to yeast lacking heterologous alpha-amylase expression. “Mean (residual starch)” column shows the results from the YPD based alpha-amylase activity assay where the residual starch is inversely proportional to alpha-amylase activity, while “Mean (normalized ethanol)” column shows the ethanol at the 48 hour timepoint from two different simultaneous and saccharification fermentation (SSF) experiments, normalized to that of the strain without heterologous alpha-amylase expression (yMHCT484).

TABLE 14

Strain IDs and normalized ethanol and alpha-amylase activity data.						
Promoter	Signal peptide	Alpha-amylase gene donor (catalytic domain)	SEQ ID NO: (mature polypeptide)	Mean (normalized ethanol) (exp 1)	Mean (normalized ethanol) (exp 2)	Mean (residual starch)
		Background strain with glucoamylase gene, without alpha amylase gene		1.00	1.00	0.74
ADH1	EXG1	<i>Byssosascus striatosporus</i>	136	1.18	0.69	1.007
ADH1	EXG1	<i>Penicillium subspinosum</i>	138	1.27	0.97	1.029
ADH1	EXG1	<i>Acidomyces richmondensis</i>	125	1.10	0.88	0.414
ADH1	EXG1	<i>Trichoderma paraviridescens</i>	135	1.14	0.78	0.564
ADH1	EXG1	<i>Bacillus licheniformis</i>	122	1.28	0.85	0.453
ADH1	EXG1	<i>Penicillium aethiopicum</i>	133	1.19	0.97	0.954
ADH1	EXG1	<i>Alternaria</i> sp	127	1.17	0.97	0.460
ADH1	EXG1	<i>Dichotomocladium hesseltinei</i>	131	1.15	0.74	0.980
ADH1	EXG1	<i>Aspergillus bombycis</i>	126	1.13	0.98	0.423
ADH1	EXG1	<i>Aspergillus brasiliensis</i>	137	1.16	1.06	0.507
ADH1	EXG1	<i>Aspergillus tamarii</i>	124	1.22	1.04	0.440
ADH1	EXG1	<i>Aspergillus bombycis</i>	126	1.14	0.97	0.479
ADH1	EXG1	<i>Acidomyces richmondensis</i>	125	1.11	1.00	0.464
ADH1	EXG1	<i>Alternaria</i> sp	127	1.13	0.77	0.411
ADH1	EXG1	<i>Subulispora</i> sp	134	1.03	0.64	0.452
ADH1	EXG1	<i>Penicillium subspinosum</i>	138	1.11	0.82	0.607
ADH1	EXG1	<i>Penicillium antarcticum</i>	139	1.17	0.71	0.476
ADH1	EXG1	<i>Cerrena unicolor</i>	152	1.07	0.95	0.533
ADH1	EXG1	<i>Neosartorya massa</i>	145	1.05	1.13	0.422
ADH1	EXG1	<i>Penicillium vasconiae</i>	142	1.15	0.72	0.394
ADH1	EXG1	<i>Penicillium</i> sp	143	1.09	0.88	0.412
ADH1	EXG1	<i>Aspergillus oryzae</i>	168	1.21	1.16	0.038
ADH1	EXG1	<i>Aspergillus terreus</i>	172	1.15	1.22	0.038
ADH1	EXG1	<i>Aspergillus terreus</i>	172	1.17	1.22	0.037
ADH1	EXG1	<i>Thamnidium elegans</i>	173	1.25	1.13	0.037
ADH1	EXG1	<i>Aspergillus oryzae</i>	168	1.10	1.16	0.044
ADH1	EXG1	<i>Rhizomucor pusillus</i>	165	1.23	1.23	0.036
TDH3	EXG1	<i>Aspergillus brasiliensis</i>	137	1.16	1.07	0.042
TDH3	EXG1	<i>Rhizomucor pusillus</i>	121	1.27	1.20	0.040
TDH3	EXG1	<i>Bacillus licheniformis</i>	122	0.91	0.93	0.046
TDH3	EXG1	<i>Aspergillus bombycis</i>	126	1.20	0.97	0.039
TDH3	EXG1	<i>Lichtheimia ramosa</i>	132	1.19	1.12	0.051
TDH3	EXG1	<i>Aspergillus tamarii</i>	124	1.21	1.11	0.048
TDH3	EXG1	<i>Rhizopus microsporus</i>	128	1.02	1.01	0.041
TDH3	EXG1	<i>Aspergillus tamarii</i>	124	1.18	1.04	0.040
TDH3	EXG1	<i>Syncephalastrum racemosum</i>	129	1.17	1.24	0.042
TDH3	EXG1	<i>Byssosascus striatosporus</i>	136	1.04	0.96	0.541
TDH3	EXG1	<i>Aspergillus bombycis</i>	126	1.05	0.97	0.042
TDH3	EXG1	<i>Rhizomucor pusillus</i>	159		1.25	0.141
TDH3	EXG1	<i>Hamigera avellanea</i>	149	1.11	1.03	0.055
TDH3	EXG1	<i>Bacillus subtilis</i> subsp. <i>subtilis</i>	157	1.19	1.08	0.043
TDH3	EXG1	<i>Bacillus subtilis</i> subsp. <i>subtilis</i>	157	1.25	1.11	0.044
TDH3	EXG1	<i>Meripilus giganteus</i>	151	1.20	1.14	0.045
TDH3	EXG1	<i>Bacillus subtilis</i>	156	1.35	1.27	0.048
TDH3	EXG1	<i>Neosartorya massa</i>	145	1.15	1.03	0.048
TDH3	EXG1	<i>Penicillium olsonii</i>	141	0.96	0.98	0.052
TDH3	EXG1	<i>Aspergillus niger</i>	160	1.06	0.98	0.053
TDH3	EXG1	<i>Hamigera avellanea</i>	150	1.11	1.04	0.046
TDH3	EXG1	<i>Bacillus subtilis</i> subsp. <i>subtilis</i>	157	1.16	1.13	0.045

TABLE 14-continued

Strain IDs and normalized ethanol and alpha-amylase activity data.						
Promoter	Signal peptide	Alpha-amylase gene donor (catalytic domain)	SEQ ID NO: (mature polypeptide)	Mean (normalized ethanol) (exp 1)	Mean (normalized ethanol) (exp 2)	Mean (residual starch)
TDH3	EXG1	<i>Meripilus giganteus</i>	151	1.11	1.14	0.039
TDH3	EXG1	<i>Cerrena unicolor</i>	152	1.02	0.96	0.546
TDH3	EXG1	<i>Neosartorya massa</i>	145	1.25	1.04	0.053
TDH3	EXG1	<i>Aspergillus niger</i>	167	1.08	1.05	0.040
TDH3	EXG1	<i>Rhizomucor pusillus</i>	165	1.30	1.23	0.039
TDH3	EXG1	<i>Aspergillus terreus</i>	172	1.13	1.11	0.043
TDH3	EXG1	<i>Aspergillus oryzae</i>	168	1.13	1.17	0.037

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Example 4: Construction of Yeast Strains Expressing a Heterologous Trehalase

This example describes the construction of yeast cells containing a heterologous trehalase under control of an *S. cerevisiae* CCW12 promoter (SEQ ID NO: 232) or PGK1 promoter (SEQ ID NO: 4). Three pieces of DNA containing the promoter, gene and terminator were designed to allow for homologous recombination between the 3 DNA fragments and into the X-3 locus of the yeast yMHCT484 (PCT/US2018/035596). The resulting strain has one promoter containing fragment (left), one gene containing fragment (middle) and one TEF1 terminator (SEQ ID NO: 233) fragment (right) integrated into the *S. cerevisiae* genome at the X-3 locus.

Construction of the Promoter Containing Fragments (Left Fragments)

Synthetic linear uncloned DNA containing 300 bp homology to the X-3 site, *S. cerevisiae* promoter CCW12 (SEQ ID NO: 232) or PGK1 (SEQ ID NO: 4) and *S. cerevisiae* AGA2 signal sequence (SEQ ID NO: 234) were synthesized by Thermo Fisher Scientific. The 2 linear DNAs were designated 17ABCK6P and 17ABCK7P for each promoter listed above, respectively.

To generate additional linear DNA for transformation into yeast, the DNA containing the left cassette was PCR amplified from 17ABCK6P and 17ABCK7P. Fifty pmoles each of forward and reverse primer was used in a PCR reaction containing 12.5 ng of linear DNA as template, 0.1 mM each dATP, dGTP, dCTP, dTTP, 1× Phusion HF Buffer (Thermo

Fisher Scientific), and 2 units Phusion Hot Start DNA polymerase in a final volume of 50 µL. The PCR was performed in a T100™ Thermal Cycler (Bio-Rad Laboratories, Inc.) programmed for one cycle at 98° C. for 1 minute followed by 32 cycles each at 98° C. for 10 seconds, 60° C. for 30 seconds, and 72° C. for 1 minute with a final extension at 72° C. for 10 minutes. Following thermocycling, the PCR reaction products gel isolated and cleaned up using the NucleoSpin Gel and PCR clean-up kit (Machery-Nagel).

Construction of the Trehalase Containing Fragments (Middle Fragments)

Synthetic linear uncloned DNA containing *S. cerevisiae* AGA2 signal peptide coding sequence (encoding the signal of SEQ ID NO: 234), a codon-optimized trehalase gene and 50 bp of TEF1 terminator (SEQ ID NO: 233), were synthesized by Thermo Fisher Scientific. The resulting linear DNAs were designated as indicated in Table 15.

A subset of the trehalase containing fragments were ordered as cloned synthetic plasmid DNA rather than linear uncloned DNA. Synthetic plasmid DNA containing *S. cerevisiae* AGA2 signal coding sequence, a codon-optimized trehalase gene and 50 bp of TEF1 terminator, were synthesized by Thermo Fisher Scientific. The resulting DNAs were designated as indicated in Table 15.

Construction of the Terminator Contain Fragment (Right Fragment)

Synthetic linear uncloned DNA containing *S. cerevisiae* TEF1 terminator and 300 bp homology to the X-3 site, were synthesized by Thermo Fisher Scientific.

TABLE 15

Trehalase DNA product names and associated enzyme					
Product number	DNA format	Signal Peptide	Donor Organism (catalytic domain)	SEQ ID NO: (mature polypeptide)	Terminator Fragment
17ABFBKP	linear	AGA2	<i>Chaetomium megalocarpum</i>	175	TEF1
17ABFBJP	linear	AGA2	<i>Lecanicillium psalliotae</i>	176	TEF1
17ABFBIP	linear	AGA2	<i>Doratomyces</i> sp	177	TEF1
17ABFBHP	linear	AGA2	<i>Mucor moelleri</i>	178	TEF1
17ABFBGP	linear	AGA2	<i>Phialophora cyclaminis</i>	179	TEF1
17ABFBFP	linear	AGA2	<i>Thielavia arenaria</i>	180	TEF1
17ABFBEP	linear	AGA2	<i>Thielavia antarctica</i>	181	TEF1
17ABFBDP	linear	AGA2	<i>Chaetomium</i> sp	182	TEF1
17ABFBGP	linear	AGA2	<i>Chaetomium nigricolor</i>	183	TEF1
17ABFBBP	linear	AGA2	<i>Chaetomium jodhpurens</i>	184	TEF1
17ABFBAP	linear	AGA2	<i>Chaetomium piluliferum</i>	185	TEF1
17ABFA7P	linear	AGA2	<i>Myceliophthora himpalea</i>	186	TEF1
17ABFA6P	linear	AGA2	<i>Chloridium virescens</i>	187	TEF1
17ABFA5P	linear	AGA2	<i>Gelasinospora cratophora</i>	188	TEF1
17ABFA4P	linear	AGA2	<i>Acidobacteriaceae bacterium</i>	189	TEF1

TABLE 15-continued

Trehalase DNA product names and associated enzyme					
Product number	DNA format	Signal Peptide	Donor Organism (catalytic domain)	SEQ ID NO: (mature polypeptide)	Terminator Fragment
17ABFA3P	linear	AGA2	<i>Acidobacterium capsulatum</i>	190	TEF1
17ABFA2P	linear	AGA2	<i>Acidovorax wautersii</i>	191	TEF1
17ABFAZP	linear	AGA2	<i>Xanthomonas arboricola</i>	192	TEF1
17ABFAYP	linear	AGA2	<i>Kosakonia sacchari</i>	193	TEF1
17ABFAXP	linear	AGA2	<i>Enterobacter</i> sp	194	TEF1
17ABFAWP	linear	AGA2	<i>Saitozyma flava</i>	195	TEF1
17ABFAVP	linear	AGA2	<i>Phaeotremella skinneri</i>	196	TEF1
17ABFAUP	linear	AGA2	<i>Trichoderma asperellum</i>	197	TEF1
17ABFATP	linear	AGA2	<i>Corynascus sepedonium</i>	198	TEF1
17ABFASP	linear	AGA2	<i>Myceliophthora thermophila</i>	199	TEF1
17ABFARP	linear	AGA2	<i>Trichoderma reesei</i> GH37	200	TEF1
17ABFAQP	linear	AGA2	<i>Chaetomium virescens</i>	201	TEF1
17ABFAPP	linear	AGA2	<i>Rhodothermus marinus</i>	202	TEF1
17ABFAOP	linear	AGA2	<i>Myceliophthora sepedonium</i>	203	TEF1
17ABFANP	linear	AGA2	<i>Moelleriella libera</i>	204	TEF1
17ABFAMP	linear	AGA2	<i>Acremonium dichromosporum</i>	205	TEF1
17ABFALP	linear	AGA2	<i>Fusarium sambucinum</i>	206	TEF1
17ABFAKP	linear	AGA2	<i>Phoma</i> sp	207	TEF1
17ABFAJP	linear	AGA2	<i>Lentinus similis</i>	208	TEF1
17ABFAIP	linear	AGA2	<i>Diaporthe nobilis</i>	209	TEF1
17ABFAHP	linear	AGA2	<i>Solicoccozyma terricola</i>	210	TEF1
17ABFAGP	linear	AGA2	<i>Dioszegia cryoxerica</i>	211	TEF1
17ABFO6P	plasmid	AGA2	<i>Talaromyces finiculus</i>	212	TEF1
17ABFO5P	plasmid	AGA2	<i>Hamigera avellanea</i>	213	TEF1
17ABFO4P	plasmid	AGA2	<i>Talaromyces ruber</i>	214	TEF1
17ABFO3P	plasmid	AGA2	<i>Trichoderma lixii</i>	215	TEF1
17ABFO2P	plasmid	AGA2	<i>Aspergillus cervinus</i>	216	TEF1
17ABFOZP	plasmid	AGA2	<i>Rasamsonia brevistipitata</i>	217	TEF1
17ABFOYP	plasmid	AGA2	<i>Acremonium curvulum</i>	218	TEF1
17ABFOXP	plasmid	AGA2	<i>Talaromyces piceae</i>	219	TEF1
17ABFOWP	plasmid	AGA2	<i>Penicillium</i> sp	220	TEF1
17ABFOVP	plasmid	AGA2	<i>Talaromyces aurantiacus</i>	221	TEF1
17ABFOUP	plasmid	AGA2	<i>Talaromyces pinophilus</i>	222	TEF1
17ABFOTP	plasmid	AGA2	<i>Talaromyces leycettanus</i>	223	TEF1
17ABFOSP	plasmid	AGA2	<i>Talaromyces variabilis</i>	224	TEF1
17ABFORP	plasmid	AGA2	<i>Aspergillus niger</i>	225	TEF1
17ABFOQP	plasmid	AGA2	<i>Trichoderma reesei</i> GH65	226	TEF1

Integration of the Left, Middle and Right-Hand Fragments to Generate Yeast Strains with a Heterologous Trehalase

The yeast yMHCT484 (PCT/US2018/035596) was transformed with the left, middle and right integration fragments described above. In each transformation pool a fixed left fragment and right fragment were used. The middle fragment consisted of a pool of 13-21 middle fragments containing the trehalase gene with 100-600 ng of each fragment (1000 ng total). To aid homologous recombination of the left, middle and right fragments at the genomic X-3 sites a plasmid containing Cas9 and guide RNA specific to X-3 (pMcTs442) was also used in the transformation. These four components were transformed into the *S. cerevisiae* strain yMHCT484 following a yeast electroporation protocol (See, Thompson et al. Yeast. 1998 Apr. 30; 14(6):565-71). Transformants were selected on YPD+clonNAT to select for transformants that contain the CRISPR/Cas9 plasmid pMcTs442. Transformants were picked using a Q-pix Colony Picking System (Molecular Devices) to inoculate 1 well of 96-well plate containing YPD+clonNAT media. The plates were grown for 2 days then glycerol was added to 20% final concentration and the plates were stored at -80° C. until needed. Integration of specific trehalase construct was verified by PCR with locus specific primers and subsequent sequencing.

Example 5: Activity Assay of Yeast Strain Expressing Trehalase

Yeast expressing a trehalase gene from *Corynascus sepedonium* driven by the promoter ADH1 (supra) was cultivated overnight in standard YPD media containing 2% glucose. The cultured yeast medium was centrifuged at 3000 rpm for 10 min to collect the supernatant. The supernatant was used for enzyme activity assay, as described below.

Trehalase activity was detected by measuring the amount of glucose released through enzymatic hydrolysis of trehalose. Glucose oxidase reagent was used to measure the glucose based on the color development from application of the reagent. The color intensity measured on a spectrophotometer or microplate reader is proportional to trehalase activity. Reaction conditions and color development are described in Table 16 and Table 17, respectively.

The Trehalase Novozymes Unit (TNU(A)) for trehalase assay standard is measured relative to an enzyme standard of declared activity.

TABLE 16

Trehalase reaction condition	
Amount of yeast supernatant	20 μ l
Amount of substrate	100 μ l

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TABLE 16-continued

Trehalase reaction condition	
Substrate	Trehalose, 60 mM
Buffer	Sodium acetate, 0.1M, 0.01% Triton 100
pH	5.0 ± 0.05
Incubation temperature	20° C.
Reaction time	2-3 hr
Trehalase assay range	0.004-0.017 TNU(A)/ml

TABLE 17

Color development	
Reaction mixture	20 µl
Glucose oxidase reagent	200 µl
Incubation temperature	20° C.
Reaction time	10-15 min
Wavelength	490 nm

Assay results showed that trehalase expression proportionally increased the glucose released, measured as the optical density at 490 nm (0.10 for background strain lacking the trehalase gene compared to 0.88 for the

Example 6: Activity Assay of Yeast Strains Expressing Trehalase

Yeast strains from Example 4 were cultivated overnight in standard YPD media containing 2% or 6% glucose. The cultured yeast medium was subjected to centrifugation at 3500 rpm for 10 min to harvest the supernatant. The culture supernatant is used for the described enzyme activity assays.

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Yeast may also be cultivated using other cultivation media such as minimal YNB media or clarified and filtered industrial liquefied corn mash.

Glucoamylase activity was measured using maltose as substrate as described supra.

Trehalase activity was measured using trehalose as substrate. Enzyme hydrolysis of trehalose will release glucose as reaction product which may be detected using commercially available assay kits such as Wako Diagnostics AUTOKIT GLUCOSE C2. Reagents provided in the assay kits will specifically react with glucose resulted in color formation. The color intensity measured on spectrophotometer or microplate reader, is proportional to trehalase activity. Reaction conditions are described in Table 18. The Trehalase Unit (TNU) for standard trehalase is defined as the amount of enzyme, which hydrolyzes one micromole trehalose per minute under the standard conditions.

TABLE 18

Trehalase reaction conditions	
Appropriate amount of yeast supernatant	10-200 µl
Substrate	trehalose, 10 mM
Buffer	acetate, 0.1M
pH	5.0 ± 0.05
Incubation temperature	32° C.
Reaction time	5-20 min
Trehalase assay range	0.002-0.036 TNU/ml

The absorbance at 505 nm increases as the amount of purified glucoamylase or trehalase added to hydrolyze maltose or trehalose, respectively, to glucose increases. A purified glucoamylase and trehalase standard curve was generated and used to estimate glucoamylase and trehalase activity in yeast supernatants.

Results for trehalase activity and glucoamylase activity are shown Table 19. A graphical representation of comparative trehalase activity is shown in FIG. 2.

TABLE 19

Trehalase and glucoamylase (GA) activity, and estimated enzyme secretion.							
Yeast	Promoter for	SEQ ID NO:	Donor Organism	Glucoamylase		Trehalase	
strain no.	trehalase expression	(mature polypeptide)	(catalytic domain)	activity	Conc. (ug/mL)	activity	Conc. (ug/mL)
1	Background strain with glucoamylase gene, without trehalase gene			0.352	10.7	N/A	N/A
1	Background strain with glucoamylase gene, without trehalase gene			0.312	8.9	N/A	N/A
1	Background strain with glucoamylase gene, without trehalase gene			0.308	8.7	N/A	N/A
2	pPGK1	189	<i>Acidobacteriaceae bacterium</i>	0.357	10.9	0.05	1.04
3	pCCW12	191	<i>Acidovorax wautersii</i>	0.301	9.1	1.8	22.79
4	pPGK1	218	<i>Acremonium curvulum</i>	0.287	7.8	0.07	1.08
5	pCCW12	218	<i>Acremonium curvulum</i>	0.261	6.7	0.08	1.09
6	pCCW12	205	<i>Acremonium dichromosporum</i>	0.304	8.6	0.78	3.78
7	pCCW12	205	<i>Acremonium dichromosporum</i>	0.309	8.8	0.89	4.63
8	pCCW12	216	<i>Aspergillus cervinus</i>	0.299	8.4	0.12	1.18
9	pPGK1	216	<i>Aspergillus cervinus</i>	0.309	8.8	0.13	1.2
10	pPGK1	225	<i>Aspergillus niger</i>	0.308	8.8	0.48	2.22
11	pCCW12	225	<i>Aspergillus niger</i>	0.307	8.7	0.67	3.11
12	pPGK1	184	<i>Chaetomium jodhpurens</i>	0.327	9.6	0.15	1.26

TABLE 19-continued

Trehalase and glucoamylase (GA) activity, and estimated enzyme secretion.							
Yeast	Promoter for	SEQ ID NO:	Donor Organism	Glucoamylase		Trehalase	
strain no.	trehalase expression	(mature polypeptide)	(catalytic domain)	activity	Conc. (ug/mL)	activity	Conc. (ug/mL)
13	pCCW12	184	<i>Chaetomium jodhpurens</i>	0.302	8.5	1.77	21.85
14	pCCW12	175	<i>Chaetomium megalocarpum</i>	0.302	8.5	1.77	21.85
15	pCCW12	175	<i>Chaetomium megalocarpum</i>	0.303	8.5	0.4	1.95
16	pPGK1	183	<i>Chaetomium nigricolor</i>	0.381	11.9	0.36	1.8
17	pCCW12	183	<i>Chaetomium nigricolor</i>	0.315	9.1	0.66	3.05
18	pCCW12	183	<i>Chaetomium nigricolor</i>	0.314	9	0.63	2.9
19	pPGK1	182	<i>Chaetomium</i> sp	0.329	9.7	0.14	1.22
20	pCCW12	182	<i>Chaetomium</i> sp	0.295	8.2	0.27	1.54
21	pCCW12	182	<i>Chaetomium</i> sp	0.301	8.4	0.12	1.18
22	pCCW12	201	<i>Chaetomium virescens</i>	0.341	10.2	0.53	2.43
23	pPGK1	201	<i>Chaetomium virescens</i>	0.318	9.2	0.23	1.43
24	pPGK1	187	<i>Chloridium virescens</i>	0.317	9.2	0.24	1.47
25	pPGK1	187	<i>Chloridium virescens</i>	0.349	10.5	0.15	1.25
26	pCCW12	211	<i>Dioszegia cryoxerica</i>	0.299	8.4	0.07	1.07
27	pPGK1	211	<i>Dioszegia cryoxerica</i>	0.384	12.1	0.44	2.07
28	pPGK1	177	<i>Doratomyces</i> sp	0.314	9	1.08	6.44
29	pCCW12	177	<i>Doratomyces</i> sp	0.315	8.9	1.87	25.75
30	pPGK1	194	<i>Enterobacter</i> sp	0.321	9.3	0.9	4.71
31	pCCW12	194	<i>Enterobacter</i> sp	0.312	8.9	1.87	25.75
32	pPGK1	206	<i>Fusarium sambucinum</i>	0.323	9.4	0.05	1.05
33	pCCW12	188	<i>Gelasinospora cratophora</i>	0.315	9.1	0.66	3.05
34	pCCW12	188	<i>Gelasinospora cratophora</i>	0.33	9.7	0.8	3.91
35	pPGK1	213	<i>Hamigera avellanea</i>	0.302	8.5	0.63	2.89
36	pCCW12	213	<i>Hamigera avellanea</i>	0.324	9.4	0.98	5.38
37	pCCW12	213	<i>Hamigera avellanea</i>	0.322	9.4	1.13	6.97
38	pPGK1	213	<i>Hamigera avellanea</i>	0.326	9.5	0.1	1.15
39	pPGK1	193	<i>Kosakonia sacchari</i>	0.356	10.8	0.12	1.19
40	pCCW12	193	<i>Kosakonia sacchari</i>	0.323	9.4	0.06	1.07
41	pPGK1	176	<i>Lecanicillium psalliotae</i>	0.349	10.5	0.15	1.25
42	pCCW12	176	<i>Lecanicillium psalliotae</i>	0.333	9.9	0.07	1.08
43	pPGK1	176	<i>Lecanicillium psalliotae</i>	0.446	14.8	0.06	1.06
44	pCCW12	208	<i>Lentinus similis</i>	0.314	9	0.05	1.05
45	pPGK1	208	<i>Lentinus similis</i>	0.328	9.6	0.05	1.04
46	pPGK1	204	<i>Moelleriella libera</i>	0.304	8.6	0.13	1.21
47	pPGK1	198	<i>Corynascus sepedonium</i>	0.322	9.4	0.58	2.66
48	pCCW12	198	<i>Corynascus sepedonium</i>	0.316	9.1	1.13	6.98
49	pCCW12	198	<i>Corynascus sepedonium</i>	0.31	8.8	1.2	7.9
50	pCCW12	178	<i>Mucor moelleri</i>	0.316	9.1	0.34	1.75
51	pPGK1	178	<i>Mucor moelleri</i>	0.305	8.7	0.78	3.77
52	pCCW12	186	<i>Myceliophthora himnulea</i>	0.299	8.4	0.99	5.51
53	pPGK1	186	<i>Myceliophthora himnulea</i>	0.348	10.5	0.38	1.85
54	pPGK1	203	<i>Myceliophthora sepedonium</i>	0.328	9.6	0.43	2.04

TABLE 19-continued

Trehalase and glucoamylase (GA) activity, and estimated enzyme secretion.							
Yeast	Promoter for	SEQ ID NO:	Donor Organism	Glucoamylase		Trehalase	
strain no.	trehalase expression	(mature polypeptide)	(catalytic domain)	activity	Conc. (ug/mL)	activity	Conc. (ug/mL)
55	pPGK1	203	<i>Myceliophthora sepedonium</i>	0.328	9.6	0.41	1.97
56	pCCW12	203	<i>Myceliophthora sepedonium</i>	0.31	8.8	1.2	7.9
57	pPGK1	199	<i>Myceliophthora thermophila</i>	0.329	9.7	0.3	1.63
58	pCCW12	199	<i>Myceliophthora thermophila</i>	0.315	9.1	0.74	3.54
59	pPGK1	220	<i>Penicillium</i> sp	0.334	9.9	0.05	1.04
60	pCCW12	220	<i>Penicillium</i> sp	0.326	9.6	0.05	1.04
61	pPGK1	179	<i>Phialophora cyclaminis</i>	0.345	10.4	0.2	1.36
62	pCCW12	179	<i>Phialophora cyclaminis</i>	0.315	9.1	0.66	3.05
63	pPGK1	207	<i>Phoma</i> sp	0.312	9	0.07	1.09
64	pCCW12	207	<i>Phoma</i> sp	0.283	7.7	0.09	1.12
65	pCCW12	217	<i>Rasamsonia brevistipitata</i>	0.29	8	0.61	2.79
66	pPGK1	217	<i>Rasamsonia brevistipitata</i>	0.321	9.3	0.54	2.46
67	pCCW12	202	<i>Rhodothermus marinus</i>	0.319	9.3	0.3	1.61
68	pPGK1	202	<i>Rhodothermus marinus</i>	0.318	9.2	0.12	1.18
69	pPGK1	195	<i>Saitozyma flava</i>	0.312	8.9	0.25	1.48
70	pCCW12	195	<i>Saitozyma flava</i>	0.309	8.8	0.39	1.9
71	pCCW12	221	<i>Talaromyces aurantiacus</i>	0.332	9.8	0.6	2.77
72	pPGK1	221	<i>Talaromyces aurantiacus</i>	0.345	10.4	0.57	2.63
73	pCCW12	223	<i>Talaromyces leycectanus</i>	0.293	8.1	1.13	7.02
74	pPGK1	223	<i>Talaromyces leycectanus</i>	0.311	8.9	0.71	3.35
75	pPGK1	219	<i>Talaromyces piceae</i>	0.308	8.8	0.45	2.11
76	pCCW12	219	<i>Talaromyces piceae</i>	0.313	9	0.73	3.48
77	pCCW12	222	<i>Talaromyces pinophilus</i>	0.368	11.4	1.29	9.26
78	pCCW12	222	<i>Talaromyces pinophilus</i>	0.302	8.5	0.95	5.13
79	pPGK1	222	<i>Talaromyces pinophilus</i>	0.365	11.3	0.62	2.87
80	pCCW12	224	<i>Talaromyces variabilis</i>	0.296	8.2	0.06	1.06
81	pPGK1	181	<i>Thielavia antarctica</i>	0.338	10.1	0.08	1.1
82	pCCW12	180	<i>Thielavia arenaria</i>	0.302	8.5	0.25	1.5
83	pCCW12	215	<i>Trichoderma lixii</i>	0.306	8.7	0.06	1.06
84	pPGK1	215	<i>Trichoderma lixii</i>	0.326	9.5	0.1	1.15
85	pPGK1	200	<i>Trichoderma reesei</i> GH37	0.312	8.9	0.05	1.04
86	pCCW12	226	<i>Trichoderma reesei</i> GH65	0.301	8.4	0.18	1.3
87	pCCW12	192	<i>Xanthomonas arboricola</i>	0.306	8.7	0.46	2.17
88	pPGK1	192	<i>Xanthomonas arboricola</i>	0.33	9.7	0.09	1.12

Example 7: Simultaneous Saccharification and Fermentation (SSF) of Yeast Strains Expressing Trehalase

Simultaneous saccharification and fermentation (SSF) was performed via mini-scale fermentations for various trehalase-expressing strains described supra using industrial corn mash (Avantec® Amp, Novozymes, A/S). Yeast strains were cultivated overnight in YPD media with 2% glucose

for 24 hours at 30° C. and 300 rpm. The corn mash was supplemented with 250 ppm of urea, dosed with 0.15 AGU/g-DS of an exogenous glucoamylase enzyme product (Spirizyme® Excel, Novozymes, A/S), and 30 mM trehalose. Approximately 0.6 mg of corn mash was dispensed per well to 96 well microtiter plates, followed by the addition of approximately 10⁸ yeast cells/g of corn mash from the overnight culture. Plates were incubated at 32° C. without shaking. Duplicates of each strain were analyzed after 48

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hour fermentations. Fermentation was conducted using the conditions shown in Table 20, and then stopped by the addition of 100 μ L of 8% H₂SO₄, followed by centrifugation at 3000 rpm for 10 min. The supernatant was analyzed for trehalose using HPLC.

TABLE 20

Microtiter plate fermentation reaction conditions	
Substrate	Avantec $\otimes$ Amp corn mash
Yeast pitch	10 ⁸ cells/g corn mash
Supplementary urea	250 ppm
Supplementary trehalose	30 mM
Exogenous glucoamylase product dose	0.15 AGU/g-DS
PH	5.0 $\pm$ 0.05
Incubation temperature	32° C.
Reaction time	48 hours

As shown in Table 21, higher reduction in trehalose was obtained from yeast expressing a heterologous trehalase compared to yeast lacking heterologous trehalase expression at 48 hours of simultaneous and saccharification fermentation (SSF).

TABLE 21

Promoter	SEQ ID NO: (mature polypeptide)	Trehalase Donor Organism	Mean (trehalose % w/v)
		Background strain with glucoamylase gene, without trehalase gene	0.96
pCCW12	191	<i>Acidovorax wautersii</i>	0.23
pCCW12	218	<i>Acremonium curvulum</i>	0.29
pCCW12	205	<i>Acremonium dichromosporum</i>	0.12
pCCW12	216	<i>Aspergillus cervinus</i>	0.23
pCCW12	225	<i>Aspergillus niger</i>	0.22
pCCW12	184	<i>Chaetomium jodhpurens</i>	0.09
pCCW12	175	<i>Chaetomium megalocarpum</i>	0.14
pCCW12	183	<i>Chaetomium nigricolor</i>	0.12
pCCW12	182	<i>Chaetomium sp</i>	0.19
pCCW12	201	<i>Chaetomium virescens</i>	0.13
pCCW12	21	<i>Dioszegia cryoxerica</i>	0.25
pCCW12	177	<i>Doratomyces sp</i>	0.15
pCCW12	194	<i>Enterobacter sp</i>	0.06
pCCW12	188	<i>Gelasinospora cratophora</i>	0.12
pCCW12	213	<i>Hamigera avellanea</i>	0.10
pCCW12	193	<i>Kosakonia sacchari</i>	0.23
pCCW12	176	<i>Lecanicillium psalliotae</i>	0.35
pCCW12	208	<i>Lentinus similis</i>	0.43
pCCW12	198	<i>Corynascus sepedonium</i>	0.13
pCCW12	178	<i>Mucor moelleri</i>	0.13
pCCW12	186	<i>Myceliophthora himmplea</i>	0.14
pCCW12	203	<i>Myceliophthora thermophila</i>	0.13
pCCW12	199	<i>Myceliophthora thermophila</i>	0.11

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TABLE 21-continued

	SEQ ID NO: (mature polypeptide)	Trehalase Donor Organism	Mean (trehalose % w/v)
5	Promoter		
	pCCW12	220 <i>Penicillium sp</i>	0.95
	pCCW12	179 <i>Phialophora cyclaminis</i>	0.09
	pCCW12	207 <i>Phoma sp</i>	0.33
	pCCW12	217 <i>Rasamsonia brevistipitata</i>	0.16
10	pCCW12	202 <i>Rhodothermus marinus</i>	0.42
	pCCW12	195 <i>Saitozyma flava</i>	0.19
	pCCW12	221 <i>Talaromyces aurantiacus</i>	0.15
	pCCW12	223 <i>Talaromyces leycettianus</i>	0.19
	pCCW12	219 <i>Talaromyces piceae</i>	0.19
	pCCW12	222 <i>Talaromyces pinophilus</i>	0.16
15	pCCW12	224 <i>Talaromyces variabilis</i>	0.99
	pCCW12	180 <i>Thielavia arenaria</i>	0.11
	pCCW12	215 <i>Trichoderma lixii</i>	0.88
	pCCW12	226 <i>Trichoderma reesei</i> GH65	0.27
	pCCW12	192 <i>Xanthomonas arboricola</i>	0.36
	pPGK1	189 <i>Acidobacteriaceae bacterium</i>	0.30
20	pPGK1	218 <i>Acremonium curvulum</i>	0.40
	pPGK1	216 <i>Aspergillus cervinus</i>	0.29
	pPGK1	225 <i>Aspergillus niger</i>	0.11
	pPGK1	184 <i>Chaetomium jodhpurens</i>	0.17
	pPGK1	183 <i>Chaetomium nigricolor</i>	0.39
	pPGK1	182 <i>Chaetomium sp</i>	0.16
	pPGK1	201 <i>Chaetomium virescens</i>	0.17
25	pPGK1	187 <i>Chloridium virescens</i>	0.20
	pPGK1	211 <i>Dioszegia cryoxerica</i>	0.15
	pPGK1	177 <i>Doratomyces sp</i>	0.19
	pPGK1	194 <i>Enterobacter sp</i>	0.18
	pPGK1	206 <i>Fusarium sambucinum</i>	0.26
	pPGK1	213 <i>Hamigera avellanea</i>	0.26
30	pPGK1	193 <i>Kosakonia sacchari</i>	0.25
	pPGK1	176 <i>Lecanicillium psalliotae</i>	0.24
	pPGK1	208 <i>Lentinus similis</i>	0.15
	pPGK1	204 <i>Moelleriella libera</i>	0.18
	pPGK1	198 <i>Corynascus sepedonium</i>	0.18
	pPGK1	178 <i>Mucor moelleri</i>	0.20
35	pPGK1	186 <i>Myceliophthora himmplea</i>	0.17
	pPGK1	203 <i>Myceliophthora sepedonium</i>	0.16
	pPGK1	199 <i>Myceliophthora thermophila</i>	0.16
	pPGK1	220 <i>Penicillium sp</i>	1.19
	pPGK1	179 <i>Phialophora cyclaminis</i>	0.20
	pPGK1	207 <i>Phoma sp</i>	0.32
40	pPGK1	217 <i>Rasamsonia brevistipitata</i>	0.11
	pPGK1	202 <i>Rhodothermus marinus</i>	0.44
	pPGK1	195 <i>Saitozyma flava</i>	0.19
	pPGK1	221 <i>Talaromyces aurantiacus</i>	0.16
	pPGK1	223 <i>Talaromyces leycettianus</i>	0.13
	pPGK1	219 <i>Talaromyces piceae</i>	0.20
45	pPGK1	222 <i>Talaromyces pinophilus</i>	0.11
	pPGK1	181 <i>Thielavia antarctica</i>	0.25
	pPGK1	215 <i>Trichoderma lixii</i>	0.35
	pPGK1	200 <i>Trichoderma reesei</i> GH37	0.42
	pPGK1	192 <i>Xanthomonas arboricola</i>	0.42

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SVHLESLEK LPHGWKAET PSPSSQIVLQ VALTQONIDQ LESRLAAVST PTSSTYGKYL 60
DVDEINSIFA PSDASSSAVE SWLQSHGVT S YTKQGSSIWF QTNISTANAM LSTNPHYTSYD 120
LTGAKKVRTL KYSIPESLIG HVDLISPTTY FGTTKAMRKL KSSGVSPAAD ALAARQEPSS 180
CKGTLVFEGE TFNVFQPDCL RTEYSVDGYT PSVKSGSRIG FGSFLNESAS FADQALFEKH 240
FNIPSQNFVS VLINGGTDLP QPPSDANDGE ANLDAQTILT IAHPLPITEF ITAGSPPYFP 300
DPVEPAGTFN ENEPYLQYYE FLLSKSNAEI PQVITNSYGD EEQTVPRSYA VRVCNLIGLL 360
GLRGISVLHS SGDEGVGASC VATNSTTPQF NPIFPATCPY VTSVGGTVSF NPEVWAGSS 420
GGFSYFYSRP WYQQEAVGTY LEKYVSAETK KYYGPYVDFS GRGFPDVAAH SVSPDYVPVQ 480
GGELTPSGGT SAASPVVAI VALLNDARLR EGKPTLGFLN PLIYLHASKG FTDITSQOSE 540
GCNGNNTQTG SPLPGAGFIA GAHWNATKGW DPTTGFGVFN LKKLLALVRF 590

SEQ ID NO: 11     moltype = AA length = 511
FEATURE          Location/Qualifiers
source           1..511
                 mol_type = protein
                 organism = Thermoascus aurantiacus

SEQUENCE: 11
VPVEVAGSAQ GLDVTLSQVG NTRIKAVVKN TGSEDTVTFVH LNFFKDAAPV QKVSFLRNAT 60
EVQFQGIQQR LITEGLSDDA LTTLPAGATI EDEFDIASTS DLSEGGTITI NSNGLVPITT 120

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DNKVTGYIPF TSNELSIDVD AAEASVTQA VKILERRTRI SSCSGSRQSA LTTALRNAAS 180
LANKAADAAQ SGSASKFSEY FKTTSSSTRQ TVAARLRAVA REASSSSSSGA TTYICLDPFG 240
YCSGNVLAYT LPSYNI IANC PIFYTYLPPL TSTCHAQDQA TTVLHEFTHA PGVYSPGTL 300
LAYGYQAAMG LSSSQAVMNA DTYALYANAI YLGCTRISSC SGSRQSALT ALRNAASLAN 360
AAADAAQSGS ASKFSEYFKT TSSSTRQTVA ARLRAVAREA SSSSSGATTY YCDDPYGYCS 420
SNVLAYTLPS YNIIANC DIF YTYLPALTST CHAQDQATTA LHEFTHAPGV YSPGTDLLAY 480
GYQAAMGLSS SQAVMNADTY ALYANAIYLG C 511

```

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SEQ ID NO: 12      moltype = AA length = 550
FEATURE           Location/Qualifiers
source            1..550
                  mol_type = protein
                  organism = Dichomitus squalens

```

```

SEQUENCE: 12
KPTARNLKLH ESRSPAPNGF SLVGSADSNR TLKLRLLALAE SNFSELERKL YDVSTPKSAN 60
YGKHLASKAEV QQLVAPGQDS IDAVNAWLKE NDITAKTISS TGEWISFEVP VSKANDLFDA 120
DFSVFKHDDT GVEAIRLSY SIPAELQGH DLVHPTVTFP NPYSHLPVFQ SPVKKTAEIQ 180
NFTAGAISS CSSTITPAQL QAIYNIPTTA ATSSNQLGV TGFIDQYANK KDLKTFLLKY 240
RTDISSSTTF TLQTLDDGSN SQTGSKAGVE ANLDIQYTVG VATGVPTFI SVGDDFQDGD 300
LEGFLDVINA LLEDAPPVSV LTTSYGQDES TISRALAVKL CNAYAQLGAR GVSILFASGD 360
GGVSGSQSAS CSKFVPTFPS GCPYMTSVGA TQGVNPTAA DFSSGGFSNY WGVDPYQSDA 420
VSTYLSALGK TNSGKYNASG RGFDPVSTQG VSFEVVVDGS VEAVDGTSCA SPTFASIISL 480
VNDKLVAAGK SPLGLNPFPL YSDGVAALND ITSGSNPGCN TNGFPAKKGW DPVTGLGTPD 540
FKKLLTAVGL 550

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SEQ ID NO: 13      moltype = AA length = 353
FEATURE           Location/Qualifiers
source            1..353
                  mol_type = protein
                  organism = Nocardiosis prasina

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```

SEQUENCE: 13
ATGALPQSPT PEADAVSMQE ALQRDLDLTS AEAEELLAAQ DTAFEVDEAA AEAAGDAYGG 60
SVFDTESLEL TVLVTDAAAV EAVEATGAGT ELVSYGIDGL DEIVQELNAA DAVPGVVGWY 120
PDVAGDTTVL EVLEGSGADV SGLLDAGVD ASAVEVTTSD QPELYADIIG GLAYTMGGRC 180
SVGFAATNAA GQPGFVTAGH CGRVGTQVTI GNGRGVFEQS VFPGNDAAFV RGTSNFTLTN 240
LVSRNTTGGY ATVAGHNQAP IGSSVCRSGS TTGWHCCTIQ ARGQSVSYPE GTVTNMTRTT 300
VCAEPGDSGG SYISGTQAQG VTSGGSGNCR TGGTTFYQEV TPMVNSWGV RLT 353

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SEQ ID NO: 14      moltype = AA length = 456
FEATURE           Location/Qualifiers
source            1..456
                  mol_type = protein
                  organism = Penicillium simplicissimum

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```

SEQUENCE: 14
APASTAKDSV SSVVKNVVKY TVFEHAATGA KMEFVKNSGI CETTPGVNQY SGYLSVGSNM 60
NMWFWFWEAR NNPQAPLAA WFNNGGPGCS MIGLFQENGP CHFVNGDSTP SLNEYSWNNY 120
ANMLYVDQPI GVGFSYGTDD VTSTVTAAPY VWKLLQAFYA QFPEYESRDF AIFTESYGGH 180
YGPEFASYIQ EQNSAIKTS ISGENINLVA LGVNNGWIDS TIQEKAYIDF SYNNSYQQLI 240
DDSQRTSLLS AYNSQCLPAI QKCTKSGSNS DCQNADSVCY NKIEGPISSS GDWDVVDIRE 300
PSNDPYPPST YSTYLSNADV VKAIGAQSSY QECPNGPYNK FASTGDNPRS FLSTLSSVVK 360
SGINVLVWAG DADWICNLG NYEVANAVDF SGHTEFSAKD LAPYTVNGTE KGMFKNVANF 420
SFLKVGAGH EVPPYQPDAT LQVFEQVLQN KPIFST 456

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SEQ ID NO: 15      moltype = AA length = 502
FEATURE           Location/Qualifiers
source            1..502
                  mol_type = protein
                  organism = Aspergillus niger

```

```

SEQUENCE: 15
LQNPERRAVP PPLSHRSVAS RSVPPERRT DFEYLTNKTA RFLVNGTSIP EVDFDVGESY 60
AGLLPNTPTG NSSLFFWFFP SQNPEASDEI TIWLNCGPGC SSLDGLLQEN GPFLWQPGTY 120
KVPNPNYSWT NLTNVYIDQ PAGTGFSPPG STVNNEEDVA AQFNSWFKHF VDTFDLHGRK 180
VYITGESYAG MYVPYIADAM LNEEDTTYFN LKGIQINDPS INSDSVMMYS PAVRHLNHYN 240
NIFQLNSTFL SYINAKADK GYNALDKAI TYPPSPFPET APEITEDCQV WDEVVMAAYD 300
INPCFNYYHL IDFCPYLWDV LGFPLASGP NNYFNRSDVQ KILHVPPTDY SVCSETVIFA 360
NGDGDSPSSW GPLPSVIERT NNTIIGHGWL DYLLFLNGSL ATIQNMWTNG KQGFQRPVPE 420
PLFVPYHYGL AELYWGDPEP PYNLDAGAGY LGTAHTERGL TFSVYLSGH EIPQYVPGAA 480
YRQLEFLGR ISSLSAKGNY TS 502

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SEQ ID NO: 16      moltype = AA length = 547
FEATURE           Location/Qualifiers
source            1..547
                  mol_type = protein
                  organism = Meriphilus giganteus

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```

SEQUENCE: 16
TPTGRNLKLH EAREDLPAGF SLRGAASPD TTKLRALVQ NNFAELEDKL YDVSTPSSAN 60
YGNHLSKEEV EQYIAPAPES VKAVNAWLTE NGLDAHTISP AGDWLAFEPV VSKANELFDA 120
DFSVFTHDES GLEAIRLAY SIPAELQGH DLVHPTVTFP NPNALPVVR STQPIRNLTG 180
RAIPASCAST ITPACLOAIY GIPTTKATQS SNKLAVSGFI DQFANKADLK SFLAQFRKDI 240

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SSSTTFSLQT LDGGENDQSP SEAGIEANLD IQYTVGLATG VPTTFISVGD DFQDGNLEGF 300
LDIINFLLGE SNPPQVLTTT YGQNTENTISA KLANQLCNAY AQLGARGTSI LFASGDGGVS 360
GSQSAHCNSF VPTFPSCGPF MTSVGATQGV SPETAAAFSS GGFENVFGIP SYQASAVSGY 420
LSALGSTNSG KFNRSRGFP DVSTQGVDFQ IVSGGQTIGV DGTSCASPTF ASVISLVNDR 480
LIAAGKSPLG FLNPFLYSSA GKAALNDVTS GSNPGCSTNG FPAKAGWDPV TGLGTPNFAK 540
LLTAVGL 547

```

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SEQ ID NO: 17      moltype = AA  length = 541
FEATURE           Location/Qualifiers
source            1..541
                  mol_type = protein
                  organism = Lecanicillium sp.

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SEQUENCE: 17
APAPHGLPVK FGEITKLPSK WIATGAADSD AVIKAQIGIK QNNIKGLQDK LADIADPNSP 60
NYGQWLSKEE VDKYSAPAAA DVAAVKAWLA SSGITDVTMP TNDWIEFSVP VSKMESLLGS 120
KYEWFVHLET GEKVPRTKQF SVPQNLHDLI DVVTPPTVLY HNMGPFAHAS PQAADASGLT 180
SPASIKSAYN VDYKGTGNTL VGTGTGFLGVG ASHQDYANFA RQFSPGLTDF KDVSINGGSN 240
SGDGSALERN LDTQYCGALA APNPSEYLAH APEGSDGSSF NDAMLAFGNY LNANSNPSPA 300
VSTSYGGEED GTDPNYMDRI CNEFMKAGSR GVSIFPSSGD NGVGGNGESS CYNGYYPLWP 360
ASCPYVTTVG GTEFDGSGRE VVANFEQYNK NVKSPGGGFS NHFPAPSYNK NVTAYANSL 420
SAAQKQRLNP NRGFPDIAL VSVKYQNVN GQISQVLGTS ASSPSMAGLV GLLNDYRKTQ 480
GKPNLGFINP LLYSDKVKPA LRDVTSGANK GCDSSGLPAK TGWDAASGLG SFDPAKLRTL 540
V 541

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SEQ ID NO: 18      moltype = AA  length = 633
FEATURE           Location/Qualifiers
source            1..633
                  mol_type = protein
                  organism = Talaromyces proteolyticus

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```

SEQUENCE: 18
VPAPSKRHVV HERRDALPHS WSEPRRDVGR TQLPVRIGLT QSNIDESHDM LMDIASPSSP 60
NYRKYMTVHE VNELFAPAGE AVSAVRDWLE SAGIAAERV TQANKQWLQF DGDAAEVESL 120
LGAEYIYTH DTNGRSHMGC EKHYVPEHIS HHIDYIIPGV KSLEVRFPQ AELEKRTFGF 180
RKPQPPLFKA LPESLETIIN SILGGLDLCL STVITPSCI TLYNITEGTT ATKNELGIF 240
EDLDGYYSQT DLDLFFTLFY SQIPAGTGPT LKGIDGAQAP TQTLTQAGPE SLDLQVVSYP 300
IIWPQNSILF QTDDANYEAN YTFNGFLNNF LDAIDGSYCT YSAFGIDGNT ADDPPYPDPA 360
SNGYKGLQC GVYEPTNVIS ISYGGDEAGL SVNYQKRQCN EYKKLGLQGV SVVSSGDSG 420
VAGADGCLGG GKIFNPDPFA GCPYITTVGA TYLPSGASST SDSEVAVSRF PSGGGFSNIY 480
SQPSYQSDAV NTYLTQHTPP YPAYETSDNS SVGANGGIYN KAGRGYPDVA AVGDNIIVFN 540
AGAPTLIGGT SASAPIFASI LTRINEVLLA KKGTTVGFVN PTLYANPDFA HDITSGDNP 600
CSTNGFSTAP GWDPVTGLGT PNYPALLKVF LGE 633

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SEQ ID NO: 19      moltype = AA  length = 371
FEATURE           Location/Qualifiers
source            1..371
                  mol_type = protein
                  organism = Penicillium ranomafanaense

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SEQUENCE: 19
VPTGKKSFV VQVAVSATK TQNFANNYAR ALAKYGAKVP THVQAAQQS GSATTTPESD 60
DEEYLTPVMV GGTTLNLDFD TGSADLWVFS SELPASEQTG HSLYKPNNGT KLSGYTWSIS 120
YGDGSSASGD VYRDTVSGG VKATGQAVEA ASTISQQFTQ DQNDGLGL AFSSINTVKP 180
KSQTTFFDTV KSTLASPLFA VSLKHNPAGS YDFGFIDKSK YTGSLTYTDV DSSQGFWGFT 240
ADSYKIGSTT GSSIKGIADT GTTLNLLDDE VVSAYYQVKS GAASDSSAGG YTFDCSSTLP 300
DFTVSISGYD AVVPGSLINY TPVSQSSSKC LGGIQSNISGL GFSIFGDIFL KSQYVVFDSN 360
GPRLGFAAQS S 371

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SEQ ID NO: 20      moltype = AA  length = 578
FEATURE           Location/Qualifiers
source            1..578
                  mol_type = protein
                  organism = Aspergillus oryzae

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```

SEQUENCE: 20
EAFEKLSAVP KGWHYSSTPK GNTEVCLKIA LAQKDAAGFE KTVLEMSDPD HPSYQGHTT 60
HDEMKRMLLP RDDTVDAVRQ WLENGGVTD F TQDADWINFC TTVDTANKLL NAQFKWYVSD 120
VKHIRRLRTL QYDVPESVTP HINTIQPTTR FGKISPKKAV THSKPSQLDV TALAAAVVAK 180
NISHCDSIIT PTCLKELYNI GDYQADANSK SKIAFASYLE EYARYADLEN FENYLAPWAK 240
GQNFVSTTFN GGLNDQNSSS DSGEANLDLQ YILGVSAPLP VTEFSTGGRG PLVPDLTQPD 300
PNSNSNEPYL EFFQNVLKLD QKDLPQVIST SYGENEQEIP EKYARTVCNL IAQLGSRGVS 360
VLFSSGDSGV GEGCMTNDGT NRTHFPQFP AACPWVTSVG ATFKTTPERG TYFSSGGFSD 420
YWPRPEWQDE AVSSYLETIG DTFKGLYNSS GRAFPDVAQ GMNFAVYDKG TLGEFDGTS 480
SAPAFSAVIA LNDARLRAG KPTLGLPLNP LYKTGRQGLQ DITLGASIG TGRARFGGAP 540
DGGPVVPYAS WNATQGWDPV TGLGTPDFAE LKKLALGN 578

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SEQ ID NO: 21      moltype = AA  length = 456
FEATURE           Location/Qualifiers
source            1..456
                  mol_type = protein
                  organism = Talaromyces liani

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SEQUENCE: 21

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APASTTKDNV	SSVVKNGVTY	TVFEHAATGA	KMEFVKNSGI	CETTPGVNQY	SGYLSVGNM	60
NMWFVFFEAR	NNPQTAPLAA	WFNGGPGCSS	MIGLFQENGP	CHFVNGASTP	SLNEYSWNNY	120
ANMLYVDQPI	GVGFSYGTDD	VTSTVTAAPY	VWKLQAFYA	QFPEYQSRDF	AIFTESYGGH	180
YGPEFAAYIQ	EQNSGIAAGS	VSGENINLIA	LGVNNGWIDP	AIQEKAYIDF	SYNNSYQQLI	240
DDSQRTNLLS	DYNDQCLPAI	QQCAQTGRNS	DCQNADNVCY	DTIEGPISSS	GNWDVYDIRE	300
PSNDPYPPST	YSSYLSNSRV	VKAIGAQTSY	QECPNGPYNK	FASTGDNPRS	FLSTLSSVVQ	360
SGIHLVWAG	DADWICNWL	NYRVANAVDF	PGHAEFSAKA	LAPYTVNGTE	KGMFKNVDF	420
SFLKVYGAGH	EVPYYPQATA	LQVFEQILQN	KSITST			456

SEQ ID NO: 22 moltype = AA length = 589
 FEATURE Location/Qualifiers
 source 1..589
 mol_type = protein
 organism = Thermoascus thermophilus

SEQUENCE: 22

EVFERLRVAVP	EGWRFSATPS	DDQPIRLQIA	LQQHDVEGFE	RAVLDMSTPS	SPNYGKHFQS	60
HDEMKRMLLP	SDDAVDAVLD	WLQSAGITDI	EEDADWINFR	TTVGVANELL	DTQFQWFVSE	120
TSSHVRRLRA	LEYSIPESVT	PHIHMVQPTT	RFGQIGRHHT	TSREKPIVSG	ADIHASIAGA	180
NNQTTGTDCN	TEITPKCLQD	LYKFGGYKAS	ANSGSKVGFC	SYLEEYARYD	DLALFEEALA	240
PYAAGQNFSV	ITYNGGLNDQ	HSSSDSGEAN	LDLQYIVGVS	APLPVTEFST	GGRGELVPDL	300
DQPNPADNSN	EPYLDLFQNV	LKLDQKDLPO	VISTSYGENE	QSVPEKYARS	VCNLFMQLGS	360
RGVSIVFSSG	DSGVGSACLT	NDGKNQTRFM	PQFPASCPWV	TSVGSTQHIA	PEEATYFSSG	420
GFSDLWPMPP	YQKSAVGEYL	DRLGSKWAGL	YNPQGRGPPD	VAAQGVNPNV	YDKGSLKRFD	480
GTSCSAPTFA	GVIALLNDAR	LRARQPPMGF	LNPWLYGAGK	GGLNDIVNGG	STGCDGNARF	540
GGAPNGSPVV	PFASWNATQG	WDPVSGLGTP	DPSRLKLAV	PSRVGGRLA		589

SEQ ID NO: 23 moltype = AA length = 413
 FEATURE Location/Qualifiers
 source 1..413
 mol_type = protein
 organism = Pyrococcus furiosus

SEQUENCE: 23

AEEGLDES	AQVMATYVWN	LGVDGSGITI	GIIDTGIDAS	HPDLQGVIG	WVDFVNGRSY	60
PYDDHGHGTH	VASIAAGTGA	ASNGKYKGMA	PGAKLAGIKV	LGADGSGSIS	TIKGVAV	120
DNKDKYGIKV	INLSLGSSQS	SDGTDALSQA	VNAAWDAGLV	VVVAAGNSGP	NKYTIGSPAA	180
ASKVITVGAV	DKYDVITSFS	SRGPTADGRL	KPEVVAPGNW	IIAARASGTS	MGQPINDYYT	240
AAPGTSMATP	HVAGIAALLL	QAHPSTWTPDK	VKTALITETAD	IVKPDEIADI	AYGAGRVNAY	300
KAINYDNYAK	LVFTGYVANK	GSQTHQFVIS	GASFVTATLY	WDNANSDDL	LYDPNGNQV	360
DYSYTAYYDF	EKVGYYNPTD	GTWTIKVVS	SGSANYQVDV	VSDGSLSQPG	SSP	413

SEQ ID NO: 24 moltype = AA length = 387
 FEATURE Location/Qualifiers
 source 1..387
 mol_type = protein
 organism = Trichoderma reesei

SEQUENCE: 24

LPTEGQKTAS	VEVQYNKNYV	PHGPTALFKA	KRKYGAPISD	NLKSIVAARQ	AKQALAKRQT	60
GSAPNHPSDS	ADSEYITSVS	IGTPAQVLPL	DFDTGSSDLW	VFSSETPKSS	ATGHAIYTPS	120
KSSTSKKVS	ASWSISYGDG	SSSSGDVYTD	KVTIGGFSVN	TQGVESATRV	STEFVQDTVI	180
SGLVGLAFDS	GNQVRPHPOK	TWFSNAASSL	AEPLFTADLR	HGQNGSYNFG	YIDTSVAKGP	240
VAYTPVDNSQ	GFWEFTASGY	SVGGGKLNRN	SIDGIADTGT	TLLLLDDNV	DAYYANVQSA	300
QYDNQQEGVV	FDCDEDLPSF	SFGVGSSTIT	IPGDLNLTP	LEEGSSSTCFG	GLQSSSGIGI	360
NIFGDVALKA	ALVVFDLGNE	RLGWAQK				387

SEQ ID NO: 25 moltype = AA length = 408
 FEATURE Location/Qualifiers
 source 1..408
 mol_type = protein
 organism = Rhizomucor miehei

SEQUENCE: 25

RPVSKQSESK	DKLLALPLTS	VSRKFSQTKF	GQQQLAEKLA	GLKPFSEAAA	DGSVDTPGYY	60
DFDLEEYAI	VSIGTPGQDF	LLLFDTGSSD	TWVPHKGCTK	SEGCVGSRFF	DPSASSTFKA	120
TNYNLNITYG	TGGANGLYFE	DSIAIGDITV	TKQILAYVDN	VRGPTAEQSP	NADIFLDGLF	180
GAAYPDNTAM	EAEGYSTYNT	VHVNLYKQGL	ISSPLFSVYM	NTNSGTGEVV	FGGVNNTLLG	240
GDIAYTDVMS	RYGGYFWD	PVTGITVDGS	AAVRFSRPQA	FTIDTGTNFF	IMPSSAASKI	300
VKAALPDATE	TQQGWVPCA	SYQNSKSTIS	IVMQSGSSS	DTIEISVPVS	KMLLPVDQSN	360
ETCMFIILPD	GGNQYIVGNL	FLRFFVNVYD	FGNNRIGFAP	LASAYENE		408

SEQ ID NO: 26 moltype = AA length = 548
 FEATURE Location/Qualifiers
 source 1..548
 mol_type = protein
 organism = Lenzites betulinus

SEQUENCE: 26

KPMGRNLKVH	EAREEIPDGF	SLQGSAAAPDT	TLKLRIALVQ	SNFALEQKL	YDVSTPSSPN	60
YGAHLSKEEV	EQLVAPSADS	VDAVNAWLKE	NDLSAQTISP	AGDWLAPEVP	VSKANELFDA	120
DFSVFTHDQT	GLEAIRMSY	SIPAELOQHL	DLVHPTVTFP	NPYSHLPVVR	SPIKASQNL	180
SRATIPASCA	STITPAQLQD	IYGIPTTKAT	QSSNKLAVSG	FIDQFANSAD	LATFLKKFRT	240
DISSTTTFAL	QTLDDGSSNSQ	SGSQAGVEAN	LDIQYTVGLA	SGVPVTFISV	GDNFQDGDLE	300

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GFLDIINFLL AESAPPQVLT TSYGQNTNTI SVKLANQLCN AYAQLGARGT SILFASGDGG 360
VSGSQSSSSCS KFPVTFPSGC PFMTSVGATQ GVNPEAADF SSGGFSNYFG IPSYQATAVK 420
TYLTALGTTN SGKFNTSGRA FPDVSTQGV DFEIVVDGRTE GVDGTSCASP TFAAIIISLVN 480
DKLIAAGKSP LGFLNPFLYS TGASAFDIT SGSNPGCNTK GPPAKAGWDP VTGLGTPNFA 540
KLLAAAGV 548

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SEQ ID NO: 27      moltype = AA length = 547
FEATURE           Location/Qualifiers
source            1..547
                  mol_type = protein
                  organism = Neolentinus lepideus

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SEQUENCE: 27
GPAPRNVLVH ESRDGVPEGF VKSSTASPDIT TLKLRIALVQ GDMASLEKAL YDVSVPSSPL 60
YQHLKSKQEV EEEYKPTQES VDAVNQWLSS EGITANTISP AGDWLQFSVP VSKANEMFDA 120
DFSVFTHTES GQQAIRLSY SIPKELVGH LVLVHPTITFP NPYSHLPVVS SPAPRNLTID 180
ASVPSSCGST ITPTCLQDLY GIPPTAATQS SNKLAVSGFI DQYANKADLK SFLTTRYRKDI 240
SSSTTFTLET IDGGENPDQD SDAGVEANLD TQYTVGLATG VPTYFISVGD DYQDGDLEGF 300
LDIVNYLLSM DQPPQVLTTS YGQNTNTMSR SLANNLCNAY MQLGARGTSI LFASGDGGVS 360
GSQSGSCGSK FVPTFPSCGK YLTSVGATTG INPEVAASF SSGGFSNYWGV PSYQQSVVSS 420
YISGLGSTNK GKYNSSGRGF PDVSAQGENV EIVVDGSTEG VDGTCSSPI FASIVSLND 480
ELIAAGKSPL GFLNPFLYSD GASAFNDITS GDNPGCNTNG FSAKSGWDPV TGLGTPNYAK 540
LRTAVGF 547

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SEQ ID NO: 28      moltype = AA length = 399
FEATURE           Location/Qualifiers
source            1..399
                  mol_type = protein
                  organism = Thermococcus sp.

```

```

SEQUENCE: 28
VSAEKVRVLI TIDKDFNENS VFALGGNVVA RGKVPFIVIA ELSPRVERL KNAKGVVRVE 60
YDAEVQVLKG KSPGAGKPKP SQPAQTIPWG IERIKAPDVW SITDGSSSGV IEVAILDTGI 120
DYDHPDLAAN LAWGVSVLRG KVKTKPKDYK DQNGHGTHVA GTVAALNNDI GVVGVAPAVE 180
IYAVRVLDAS GRGSYSIIL GIEQALLGPD GVLDSGDGI IVGDPDDDA EVISMSLGLL 240
SDVQAFHDAI IEAYNYGVVI VAASGNEGAS SPSYPAAYPE VIAVGATDVN DQVPWWSNRG 300
VEVSAPGVDV LSTYPPDSYE TLTSGTSMATP HVSQGVVALIQ AAYYNKYGSV LPVGTFFDNT 360
MSTVRGILHI TADDLGSSGW DADYGYGIVR ADLAVQAVN 399

```

```

SEQ ID NO: 29      moltype = AA length = 396
FEATURE           Location/Qualifiers
source            1..396
                  mol_type = protein
                  organism = Thermococcus sp.

```

```

SEQUENCE: 29
EKVRVITIID KDFNENSVFA LGGNVVARGK VPPIVIAELS PRAVERLKNA KGVVRVEYDA 60
EVQVLKKGKSP GAGKPKPSQP AQTPWPWIER IKAPDVWSIT DGSSSGVIEV AILDGTGIDYD 120
HPDLAANLAW GVSVLRGKVS TKPKDYKDQD GHGTHVAGTV AALNNDIGVV GVAPAVEIYA 180
VRVLDASGRG SYSIILGIE QALLGPDGVL DSDGDGIIVG DPDDDAAEVI SMSLGLSDV 240
QAFHDAIIEA YNYGVVIVAA SGNEGASSPS YPAAYPEVIA VGATDVNDQV PWWSNRGVEV 300
SAPGVDVLST YPDDSYETLS GTSMATPHVS GVVALIQAA YNKYGSVLPV GTFDDNTMST 360
VRGILHITAD DLGSSGWDAD YGYGIVRADL AVQAVN 396

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SEQ ID NO: 30      moltype = AA length = 572
FEATURE           Location/Qualifiers
source            1..572
                  mol_type = protein
                  organism = Thermomyces lanuginosus

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SEQUENCE: 30
APFQVVERLS APPDGWIKKE KAAPSAQIQF RLGLPQONSE QLEQLALNIA TPGHELYRKH 60
LKRDEIKALV RPLASVSEKV LAWLRLDEGVP EDRIHDDGAW IKFTVPVSTA EKLNTTEFFV 120
FHNERTGAEQ IRTLEYSVPQ DIHSLVKFIQ PTHFSSSLGP QVRRVVPLDV LPKLRLITLED 180
CNKKITPDCL KQLYKIGDYV APEDPRNRIG ISGYLEQFAR YADFEFPLES YAPDRTDANF 240
TVVSINGGRN DQNSTLDSTE ASLDIDYAVT LSYKTQAVYV TTAGRGPLVP DESQDPDPNEV 300
SNEPYMEQLQ FLDDLPEEL PTVLTTSYGE NEQSLPGSYA DETCNMPRL LGMRGVSVIFS 360
SGDWGTGIVC KANDGSERIK FDPVYPASCP YVTSVGGTTG VNPERAVEFS SGGFSDRFPF 420
PKYQDEAVRS YLTKLGDHWK GLYNESGRAP PDVAAQADNF VVRDQGGQWVS VGGTSASAPV 480
FAAIIANVNA ELLKAGKPL GFLNPWLYGL KGRGFTDVVH GGSTGCPGTV PWTGLPAGHV 540
PYASWNATEG WDPVTGLGTP LYDELVKAAAL GK 572

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SEQ ID NO: 31      moltype = AA length = 397
FEATURE           Location/Qualifiers
source            1..397
                  mol_type = protein
                  organism = Thermococcus thioreducens

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SEQUENCE: 31
EKPELVVRVIV HVDRGHFNDA DVATIGGHVV YQFKLIDAVV VEVPTAVGR LKKLPGVKMW 60
EPDCHKARILA GPPSWLGGGQ PSQQIPWGIS RVRAPDVWGI TDGSGGVIEV AVLDTGVDYD 120
HPDLAGNIAW CVSTLRGRVT TNPAQCKDQN GHGTHVIGTI AALNNDIGVV GVAPGVEIYS 180
IRVLDASGSG SYSDIAGIE QALLGPDGIL DKDGDGIIVG DPDDDAAEVI SMSLGGPTDD 240
QYLHDMIITA YNYGVVIVAA SGNEGASSPS YPAAYPEVIA VGASDVNDQI ASWSNRQPEV 300

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SAPGVDILST	YPDDTYETLS	GTSMATPHVS	GVVALIQAAV	YNKYGKVLVP	GTFDDMGTNT	360
VRGILHVTAD	DLGDAGWDIY	YGVGIVRADL	AVQAAIG			397

SEQ ID NO: 32	moltype = AA	length = 549
FEATURE	Location/Qualifiers	
source	1..549	
	mol_type =	protein
	organism =	Polyporus arcularius

SEQUENCE: 32						
KPMARSMKLH	ESREGIPEGF	SLRGAAQPEQ	TIKLRLALVQ	SNFAELERKL	MDVSTPSSAN	60
YKGHLKSKEV	QQLVAPTQDS	VDAVKSWLKE	NDISAKTISA	TGDWLSFEVP	VSKANELFDA	120
DFSIVTHDET	GTEAVRTLST	SIPAEHQGHL	DLVHPTVTFP	NPRGLPPVFT	APIKAEQNL	180
TSRATIPSSC	ARTITPAQLQ	AIYNIPSTPA	TESSNKLAVT	GFIEQFANKA	DLKTFLTRFR	240
TDISSSTSFT	LQTLDDGGSNP	QSSSEAGVEA	NLDIQTIVGV	ATGVPTVFIS	VGEDFQDGD	300
EGFLDVVNSL	LDEDTPPFVM	TTSYGQNT	ISRNLANNLC	NAYAQLGARG	VSILFASGDG	360
GVAGSQSASC	SKFVPTFPSPG	CPMNTSVGAT	QGFSPETAAD	FSSGGFSNYF	AIPDYQTSV	420
SGYIKALGNT	NSGKYNATGR	GFPDIATQGV	NFEVVVGQGS	GTVEGTSCSS	PTLASIISLL	480
NDRLIAAGKS	PLGFLNPFLY	STGTSALNDI	TSGSNPGCNT	NGFPAKAGWD	PVTGLGTPDF	540
NKLLSAVGL						549

SEQ ID NO: 33	moltype = AA	length = 548
FEATURE	Location/Qualifiers	
source	1..548	
	mol_type =	protein
	organism =	Ganoderma lucidum

SEQUENCE: 33						
KSTRNLKLH	ETRQAGPSGF	SHTGSADPNQ	TLKLRLALVQ	GNTAELEERKL	YDVSTPSSAN	60
YKGHLKSKEV	RQLVAPAQGS	VDAVNAWLRE	NGITAKSTSA	AGDWLSFEVP	VSKANELFDA	120
DFSVPKHDDT	GVKAVRTLST	SIPAEHQGHL	DLVHPTVTFP	NPNGHMPVFQ	APVKDTEAVQ	180
NFSARAVPSS	CSNTITPAQL	QALYNIPSDA	ATQSSNKLAV	TGFIEQYANQ	VDLAVFLKQY	240
RADISSNTTF	ALQTLDDGGSN	SQTNVPGVEA	NLDIQTIVGI	ATGVPTVFIS	VGQYQDGD	300
EGFLDVINFL	LDEDTPPYVV	TTSYGQDEHT	ISRKLAQNLC	NAYAQLGARG	VSILFASGDG	360
GVAGSRSSNC	SKFVPTFPSPG	CPYMTSVGAT	QGVPEAADF	SSGGFSNYFG	TPDYQASAVK	420
SYLSTLGSTN	RGKFNASGRG	FDPVATQGVN	FEVIDGVEVE	GVSGTSAASP	MFAAIVALLN	480
DKLIAAGKSP	LGFLNPFLYS	KGVEALNDIT	TGSNPGCGTI	GFPAGEGWDP	VTGLGTPDFQ	540
KLASAAGL						548

SEQ ID NO: 34	moltype = AA	length = 548
FEATURE	Location/Qualifiers	
source	1..548	
	mol_type =	protein
	organism =	Ganoderma lucidum

SEQUENCE: 34						
KTATRNKLH	ETSQAGPSGF	SLTGSADPDQ	TLKLRLALVQ	GNVAELERRL	YDVSTPSSPN	60
YKGHLKSKEV	QQLVAPAQDS	IDAINAWLKE	NGISAKTTSA	TGDWLSFEVP	VSKANELFDA	120
DFSVPKHDDT	GMEVVRTLSY	SIPAEHQAH	DLVHPTVTFP	NPKGHPPVFQ	APAMITNDVQ	180
NFSAGAVPSS	CSSRITPAQL	QALYNIPSDP	ATQPSNKLAV	TGYIEQYANQ	DDLAVFLKEY	240
RADMSNTTF	TLQTLDDGVN	SQTDEAGIEA	NLDVQYTVGI	ATGVPTVFIS	VGQYQDGNL	300
EGFLDVVNSL	LDEDTPPYVM	TTSYGQDEHT	MSRKLQNLC	NAYAQLGARG	VSILFASGDG	360
GVAGSRSSSC	SKFVPTFPSPG	CPYMTSVGAT	QGVPEAADF	SSGGFSNYFG	IPDYQASAVS	420
GYSALGHTN	KGKYNASGRG	FDPVSTQGVN	FEVMVDGALE	GVSGTSAASP	TFAAVVALLN	480
DRLIAAGKSP	LGFLNPFLYS	KGVSALNDIT	SGSNPGCRTN	GFPAGEGWDP	VTGLGTPDFQ	540
KLASAAGL						548

SEQ ID NO: 35	moltype = AA	length = 541
FEATURE	Location/Qualifiers	
source	1..541	
	mol_type =	protein
	organism =	Ganoderma lucidum

SEQUENCE: 35						
KPTARNLRLH	ETRQAGPSGF	SLTGSADPNQ	TVRLRLALVQ	GNTGELEERKL	YDVSTPSSAN	60
YKGHLKSKEV	QQLVAPAQGS	IDAVNAWLKE	NDITAKTISA	TGDWLSFEVP	VNKANELFDA	120
DFSVPKHDDT	GMEAVRTLST	SIPAEHQGHL	DLVHPTVTFP	NPKGNLPLFQ	TPIKSKRDVP	180
ADCSNNITPA	CLQALYNIPS	DAATQSSNTL	AVTGIEQYQA	NQQDLTSFLG	QFRPDISSNT	240
TFALQITIDG	SNSQSGSDAG	GEANLDIQT	VGLATGVPTV	FISVGEQYQD	GDLGGLLDVI	300
NFVLAEDAPP	NVITTSYQGN	ENTISLKLAQ	NLCNAYAQLG	ARGVSILFAS	GDGGVAGSQS	360
DNCTQFVPTF	PSGCPYMTSV	GATQGVPEA	ADFSTGGFSN	LFSVPDYQAA	AVQSYLSALG	420
GTYQGLFNAS	GRAFPDVSTQ	GVNFETVVDG	SVSGASGTS	ASPTFAAIVA	LLNDRLVAAG	480
KSPLGFLNPF	LYSTGASALN	DIATGSNPGC	GTNGFSAQKG	WDPVTGLGTP	DFQKLAAAAG	540
L						541

SEQ ID NO: 36	moltype = AA	length = 547
FEATURE	Location/Qualifiers	
source	1..547	
	mol_type =	protein
	organism =	Trametes sp.

SEQUENCE: 36						
TPTGRNLKLH	EAREDIPTGY	SLRGAASPD	TLKLRLALVQ	NNFALEDK	YDVSTPSSAN	60
YGNHLSKEEV	BOYIAPAPES	VKAVNAWLTE	NGLDAHTISP	AGDWLAFEV	VSKANELFDA	120

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DFSVFTHDES  GLEAIRTLAY  SIPAELQGH  DLVHPTVTFP  NPNAHLPVVR  STKPIQNLTG  180
RAIPASCAST  ITPACLQAIY  GIPTTKATQS  SNKLAVSGFI  DQFANSADLK  SFLSTFRKDI  240
SSSTTFALQT  LDGGQNNQSP  SQAGIEANLD  IQYTVGLATG  VPVTFISVGD  NFQDGDLEGF  300
LDIINFLLE  SNPPQVLTS  YQONENTISA  KLANQLCNAY  AQLGARGTSI  LFASGDGGVA  360
GSQSSSCRNF  VPTFPSCGPF  MTSVGATQGV  SPETAADFSS  GGFSNVFGIP  SYQTSVAVGY  420
LSALGNTNSG  KFNRSRGFP  DVATQGVNFQ  IVSGDGTGGV  DGTSCASPTF  ASVISLINDR  480
LIAAGKSPLG  FLNPFLYSAA  GKAALNDVTS  GSNPGCNTNG  FPAKAGWDPV  TGLGTPNFAK  540
LLTAVGL  547

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SEQ ID NO: 37      moltype = AA  length = 553
FEATURE            Location/Qualifiers
source             1..553
                  mol_type = protein
                  organism = Cinereomyces lindbladii

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SEQUENCE: 37
KPTARNLLVH  ESLDGVPTGF  QLVGPASPD  VLMSRIALVQ  SDPAGLEAAL  YDVSTPSSAS  60
YGNHLSKAEV  EKFSVPTSES  VQAVNAWLTE  NDLTATQLSP  AGDWLGFEVP  VSKAEDLFGT  120
QFSVFTHEAT  GMQTVRTLST  SIPSELQGH  DLVFPPTINF  DPANALPVFR  HASKKREVT  180
LNANLTSDAV  PSSCADTITP  ACLQALYGIP  TTPATSSTNQ  LGVSGFIDQF  ANQADLKFTL  240
QNFRDIDISS  TTFSLDTLDG  GSNSQNRGDA  GVEANLDTQY  TVGLATDVPT  VFISVGEDNQ  300
DGSLLGGFLDI  INFLLDQDSP  PQVLTTSYGQ  NENTVSRAVA  NNLCNAYAQL  GARGTSILFA  360
SGDGGVSGSQ  SASCRTFVPT  FPGSCPMTS  VGATTGINPE  TAATFSAGGF  SNYFGTPSYQ  420
ASAVSSYLAA  LGSTNSGKFN  TSGRGYPDVS  TQGENFEIVV  SGEIEGVDTG  SCASPTFASI  480
ISLVNDRLIA  AGKPPLGLFN  PFLYSTGASA  FTDITTDGND  GCNTNGFFAK  SGWDPVTGLG  540
TPNFSKLLTA  VGL  553

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SEQ ID NO: 38      moltype = AA  length = 559
FEATURE            Location/Qualifiers
source             1..559
                  mol_type = protein
                  organism = Trametes versicolor

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SEQUENCE: 38
AVASTLQLHE  ARKGIPAGFS  LHGAASPD  LNLRLMALVQ  NFAGLEERLY  DVSTPSSANY  60
GKHLKSAEVE  QYVAPRQSI  TAVKAWLAAN  GLSGTSSISA  GDWIAAKVPV  SKANKLLGAQ  120
FSVFNNDATG  RQIIRTLAYS  IPAELKGHL  LVHPTITPAD  IKPLVPVVS  RRESRVLVDS  180
DLVANTIPAS  CNAAITPA  QDLYGIPSTP  ATQSSNQLGV  SGFIDQFANQ  ADLATFLTEF  240
RPDVSNSTTF  TLQTLDDGGQ  PQDPSDAGVE  ANLDTQYTVG  VATNVPTTF  SVGDDTKDGI  300
FGFLDLISFL  LAAAAPPQVL  TTSYGADGG  LSAANLVRNLC  QAYAQLGARG  TSILFSSGDG  360
GVSGSAEGC  VDFVPTFP  CPPLTSVGAT  QLTTASGLTV  ETAAGFSSGG  FSNYPPTPY  420
QQAQVDAIY  KTLVNGTVNE  GLFNASGRA  PDVSAVGVDY  LIVVGGGTDI  VSGTSASSPL  480
FASVIALIND  RRLAAGKPPL  GFLNPFLYSQ  AGASALNDVT  VGSNPGCASP  GFPAAGWDP  540
VTGLGTPNFA  KLLAAAL  559

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SEQ ID NO: 39      moltype = AA  length = 541
FEATURE            Location/Qualifiers
source             1..541
                  mol_type = protein
                  organism = Paecilomyces hepialid

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SEQUENCE: 39
APAPHGLPVK  FGEIRKLPSK  WVATGAADAN  AVIKQIGIK  QNNIQGLQAK  LADIADPN  60
NYGQWLSKEE  VDKYSAPAAA  DVAAVKAWLA  SSGITDVTMP  TNDWIEFSVP  VSKMESLLGS  120
KYEWFVHLET  GEKVPRTKEF  SVQNLHDLI  DVVPTTVLY  HNINPHTSS  PQAAGAAGLT  180
SPASIKSAYN  VDYKGTGNTL  VGTGFLGVG  ASHTDYANFG  QQFSPGLKDF  QDVSVNGGSN  240
SGDGSALGN  LDTQYCGALA  APNPSEYLAH  APEGSDNNSF  NDAMLAFGNY  LNSARNPPSA  300
VSTSYGGEED  GVDAGYLDRI  CNEFMKAGSR  GVSIFSSGD  NGVGGNGESS  CQNGYPLWP  360
ATCPYVTTVG  GTEFDNSGRE  VVANFEQYNK  NIKSPGGGYS  NHFAAPSYNK  AVTTSYANGL  420
AAPQQRQLNP  NGRGYPDISL  VSVKYQVNVN  NQISQVLGTS  ASSPSIAGLV  GLLNDYRK  480
GKPNLGFINP  LLYSDKVKPA  LRDVTSGSNK  GCDSVGLPAK  TGWDAASGLG  SFDGKLRTL  540
V  541

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SEQ ID NO: 40      moltype = AA  length = 541
FEATURE            Location/Qualifiers
source             1..541
                  mol_type = protein
                  organism = Isaria tenuipes

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SEQUENCE: 40
APAPHGLPVK  FGELKKLP  WVATGAANGD  AVIKAQIGIK  QNNIKGLQDK  LAEISDPNSP  60
SYGQWLSKEE  VAKYAPADA  DVAAVKAWLS  SAGITEVTMP  TNDWIEFSVP  VSKMESLLGS  120
KYEWFVHLET  GEKAPRTKEF  SVQNLHGII  DVVPTTVLY  HNINPNHSHG  ELSASASGLT  180
SPASIKSAYN  VDYKGTGNTL  VATTGFLGVG  ASHNDYLA  HQFSPGLKDF  QDVSVNGGSN  240
SGDGSALGN  LDTQYCGALA  SPNPSQYLAN  SPEGSDNNSF  NDAMTAFGNY  LNSASNP  300
VSTSYGGEED  GVDAGYLDRI  CNEFMKAGSR  GISVFFSSGD  NGVGGNGEPS  CQNGYPLWP  360
ATCPYVTTVG  GTEFDNSGRE  VVANFEQYNK  NVKSPGGGYS  NHFPAPDYNK  NVTAYANSL  420
SAAQQRQLNP  NGRGYPDISL  VSVKYQVSLN  GQTKQVLGTS  ASSPSVAGLV  GLLNDYRK  480
GKSNLGLFNP  LLYSGKVNA  LRDVTSGSNK  GCDSVGLPAK  SGWDAASGLG  SFDFAKL  540
I  541

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SEQ ID NO: 41      moltype = AA  length = 578
FEATURE            Location/Qualifiers

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source          1..578
                 mol_type = protein
                 organism = Aspergillus tamarii

SEQUENCE: 41
EAFEKLSAVP KGWHSSTPE GSTSVCLKIA LAQKDAAGFE KRVYEMSDPD HPNYGQHFTT 60
HEEMKRMLLP RDDTVDAVRQ WLENGGVTDV RQSDSWINFC TTVDTANKLL NAQFKWYVSD 120
VKHIRRLRTL QYDVPGSVAS HVNTIQPTTR FGKITPKKAV THSKPSQLDV TALAAAVVAK 180
NISHCDSIIT PTCLKELYNI GDYQADANSK SKIAFASYLE EYARYADLEN FENYLAPWAK 240
GQNFVSVITYN GGLNDQNSSS DSGEANLDLQ YILGVSAPLP VTEFSTGGRG PLVPDLTQPD 300
PNANSNEPYL EFFQNVLLKD QEQLPQVIST SYGENEQEIP EKYARTVCNL IAQLGSRGVS 360
VLFSSGDSGV GEGCMTNDGT NRTHFPQPFP AACPWVTSVG ATYKTTPERA TYFSSGGFSD 420
YWARPEWQEE AVSSYLETIG DAFKGLYNAS GRAFPDVAAQ GMNFAVYDKG TLGEFDGTS 480
SAPAFSAIIA LLNDARLRAG KPTLGFLNPW LYKTGRQGLQ DITLGASTGC TGRARFGGAP 540
DGGPVVPFAS WNATQGWDPV TGLGTPDFAE LKKLALAN 578

SEQ ID NO: 42      moltype = AA length = 587
FEATURE           Location/Qualifiers
source            1..587
                 mol_type = protein
                 organism = Aspergillus brasiliensis

SEQUENCE: 42
EIFEKLQGVN NGWRYANNPQ GNEVIRLQIA LQQHDVTGFE QAVMDMSTPG HADYGKHFR 60
HEEMKRMLLP SDTAVDSVRD WLESAGVHNI QVDADWIKPH TTVTKANALL DADFKWYVSE 120
ARHIRRLRTL QYSIPDALVS HINMIQPTTR FGQIQPNRAT MRSKPKHADE TFLTAATLAQ 180
NTSHCDSIIT PSCCLKQLVNI GDYQADPKSG SKIGFASYLE EYARYADLEK FEQHLAPNAI 240
GQNFTVVQFN GGLNDQLSTK DSGEANLDLQ YILGVSAPLP VTEYSTGGRG ELVPDLSSPD 300
PNDNSNEPYL DFLQNLKLN NSDLQPQVIST SYGEDEQTIP VPYARAVCNL YAQLGSRGVS 360
VIFSSGDSGV GAACLTNDGT NRTHFPQPFP ASCPWVTSVG ATSKTSPEQA VSFSSGGFSD 420
LWPRPSYQHA AVQTYLTHEL GNKFSGLFNA SGRAPPDVSA QGVNNAVYDK GILGQFDGTS 480
CSAPTFSGVI ALLNDARLRA GLPVMGFLNP FLYGAGSKLG GLNDIVTGGG VGCDGRNRF 540
GTPNGSPVVP PASWNATTGW DPVSGLGTPD FAKLKVVVALG ESEGDE 587

SEQ ID NO: 43      moltype = AA length = 582
FEATURE           Location/Qualifiers
source            1..582
                 mol_type = protein
                 organism = Aspergillus iizukae

SEQUENCE: 43
EVFDTLAAVP KGWHSYHTPR ADQPISLKIA LKQHNVEGFE QAVLDMSTPG HEHYGKHFR 60
HDEMKRMLLP SDATVDAVKD WLLAADVTDY EVDADWINLH TTVQANELL DTEFAWYVSD 120
VRAVRRLRTL RYVSPDAVAP HINMVQPTTR FGQIHPDRAT FRAGSTHFGA HILSAMSAGV 180
DVSSANVTD DVIPTPLCKE LYKVDGYRAE AEHGSKIAPA SYLEEYARYD DMVRFQEKLA 240
PYAKGENFSV ILYNGGVDDQ QSTSDSGEAN LDLQTIMGLS APLPITEYIT GGRGLIPDL 300
SQPDPNDNSN EPYLEWIQNV LKHSPEELPQ VISTSYGEDE QTIPRGYAES VCNLLAQLGS 360
RGVSVIFSSG DSGVGSACQT NDGTNTTHFP PQFPASCPWV TSVGATSKTH PEEAVYFSSG 420
GFSDLWARPA WQDDAVSTYI ESIGGKFAGL YNASGRAFPD VSAQQQNYAI FDKGRLGKMD 480
GTSCSAPAPA GIVSLLNDAR LRANRPVLGF LNPWLYGTAR EGLNDIVHGG SKGCDGRDRP 540
GGKPNPSPVVP PYASWNATPG WDPVSGLGTP NFATLVQVAL HD 582

SEQ ID NO: 44      moltype = AA length = 456
FEATURE           Location/Qualifiers
source            1..456
                 mol_type = protein
                 organism = Penicillium sp.

SEQUENCE: 44
APASTAKDSV SSVVKNQVVKY TVFEHAATGA KMEFVKNSGI CETTPGVNQY SGYLSVGSNM 60
NMWFWFPEAR NNPQAPLAA WFNNGPGCSS MIGLFQENGP CHFVNGDSTP SLNEYSWNNY 120
ANMLYVDQPI GVGFSYGTDD VTSTVTAAPY VWKLLQAFYA QFPEYESRDF AIFTESYGGH 180
YGPEFASYIQ DQNAAIKAGS VSGENINLVA LGVNNGWIDS TIQEKAYIDF SYNNSYKQLI 240
DDSQRTSLLS AYNDQCLPAI QKCTSSGSNS DCKNADSVCY NQIEGPISSS GDWDVYDIRE 300
PSNDPYPPST YSTYLSNADY VKAIGAQSSY QECPNPGPYNK FTSTGDNPRS FLSTLSSVVK 360
SGINVLVWAG DADWICNWLG NYEVANAVDF SGHTDFSAGD LAPYTVNGTE KGLFKNVDNF 420
SFRLRVYGAGH EVPPYQPDAT LQVFQQLQK KPIFST 456

SEQ ID NO: 45      moltype = AA length = 456
FEATURE           Location/Qualifiers
source            1..456
                 mol_type = protein
                 organism = Aspergillus denticulatus

SEQUENCE: 45
STASAAKDSV SSIVKNQVVKY TVFEHAATGA KMEFVKNSGI CETTPGVNQY SGYLSVGDNM 60
NMWFWFPEAR NNPQAPLAA WFNNGPGCSS MIGLFQEHGP CHFVNGEDTP SLNEYSWNNY 120
ANMLYVDQPI GVGFSYGTDD VTSTVTAAPY VWKLLQAFYA QFPEYESRDF AVFTESYGGH 180
YGPEFASYIQ DQNAAIKAGT VSGENINLIA LGVNNGWIDS AIQEKAYIDF SYNNTYKQLI 240
SSSDRTRLLS VYNSQCLPAI QKCTSTGTGA ACRNADSVCY NNIEGPISSS GDWDVYDIRE 300
PANDPYPPAT YSTYLADPDV VKAIGAQTSY QECPNPGPYNK FASTGDNPRS FLSTLSNVVK 360
SGINVLVWAG DADWICNWLG NYEVANAVDY PGQSEFEAKD LAPYTVNGAE KGMFKNVDNF 420
SFRLRVYGAGH EVPPYQPETA LQVFQQLQK KPIFST 456

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-continued

SEQ ID NO: 46 moltype = AA length = 456
 FEATURE Location/Qualifiers
 source 1..456
 mol_type = protein
 organism = Hamigera sp.

SEQUENCE: 46

APASTAKDTL	SSIVKNGVTY	NVFEHADSGA	KIEFVKNSGI	CETTPGVNQY	SGYLSVGDNM	60
NMWFWFPEAR	NNPQKAPLAA	WFNGGPGCSS	MIGLPQENGP	CHFVNGENTP	SLNEYSWNNY	120
ANMLYVDQPI	GVGFSYGTDD	VDSTVTAAPY	VWKLLQAFYA	QFPEYESRDF	AIFTESYGGH	180
YGPEFAHYIQ	QQNAAIKSGS	VKGENINLIG	LGVNNGWIDS	AIQEKAYIDF	SYNNSYKQLI	240
DFSQRTSLMR	AYKNQCLPAI	QKCYQTGTNA	DCTDASSVCY	NNIEGPISSS	GDWDVVDIRE	300
PSNDPYPPKT	YSSYLSDPKV	VKAIGARTNY	KECPNGPYNK	FSTTGDNPRS	FLSTLSDVVK	360
SGINVLWAG	DADWICNWL	GYGVANAVDY	PGHAQFRKA	LAPYTVNGTE	KGQFKTVDNF	420
QFLKVYGAGH	EVPYYQPETA	LQVFEQILQK	KPIHST			456

SEQ ID NO: 47 moltype = AA length = 456
 FEATURE Location/Qualifiers
 source 1..456
 mol_type = protein
 organism = Penicillium janthinellum

SEQUENCE: 47

APASTAKDTV	SSVVKDGVTY	TVFEHAATGA	KMEFVKNSGI	CETTPGVNQY	SGYLSVGSNM	60
NMWFWFPEAR	NNPQQAPLAA	WFNGGPGCSS	MIGLPQENGP	CHFVNGESTP	SLNENSWNNY	120
ANMLYVDQPI	GVGFSYGTDR	VTSTVTAAPY	VWKLLQAFYA	QFPEYESRDF	AIFTESYGGH	180
YGPEFASYIE	QQNAAIKAGS	VTGQNVNIVA	LGVNNGWIDA	TIQEKAYIDF	SYNNSYQQII	240
DSSTRDSLDD	AYNNQCLPAL	QQCAQSGSNS	DCTNADSVCY	QNIEGPISSS	GDFDVVDIRE	300
PSNDPYPPKT	YSTYLSDPV	VKAIGARTNY	QECPNGPYNK	FASTGDNPRS	FLSTLSSVVQ	360
SGINVLWAG	DADWICNWL	NYAVANAVDF	PGNAQFSAMD	LAPYTVNGVE	KGQFKTVDNF	420
SFLKVYGAGH	EVPYYQPDTA	LQVFKQILQK	KPISST			456

SEQ ID NO: 48 moltype = AA length = 456
 FEATURE Location/Qualifiers
 source 1..456
 mol_type = protein
 organism = Penicillium vasconiae

SEQUENCE: 48

APASTAKDSV	SSVVKNGVKY	TVFEHAATGA	KMEFVKNSGI	CETTPGVNQY	SGYLSVGSNM	60
NMWFWFPEAR	NNPQQAPLAA	WFNGGPGCSS	MIGLPQENGP	CHFVNGDSTP	SLNEYSWNNY	120
ANMLYVDQPI	GVGFSYGTDD	VTSTVTAAPY	VWKLLQAFYA	QFPEYESRDF	AIFTESYGGH	180
YGPEFASYIQ	EQNAAITAGS	VSGQKINLIA	LGVNNGWIDS	TIQEKAYIDF	SYNNSYQQLI	240
DDSQRTSLLS	AYNKQCLPAI	QKCTQTGSNS	ACQNAANVCY	NNIEGPISSS	GDWDVVDIRE	300
PSNDPYPPST	YSTYLSADV	VKAIGAQSSY	QECPNGPYNK	FASTGDNPRS	FLSTLSSVVK	360
SGINVLWAG	DADWICNWL	NYEVANAVDF	SGHAEFSAKD	LAPYTVNGAE	KGMFKNVDNF	420
SFLKVYGAGH	EVPYYQPETA	LQVFEQILQK	KPISST			456

SEQ ID NO: 49 moltype = AA length = 454
 FEATURE Location/Qualifiers
 source 1..454
 mol_type = protein
 organism = Hamigera paravellanea

SEQUENCE: 49

APSLRDKRSF	VERDGVITYV	FEHAATGAKM	EFVQNSGICE	TTPGVNQYSG	YLSVGDNMNM	60
WFWFPEARNN	PTAAPLAAWF	NGGPGCSSMI	GLFQENGPCP	FVNGESTPSL	NEYSFNYYAN	120
VLYVDQPIGT	GFSYGTDDVT	STVTAAPYVW	KLLQAFYAQF	PEYESRDFGI	FTESYGGHYG	180
PEFASYIQEQ	NAAIKAGSVS	GDNINLVALG	INNGWFDAGI	QEKAYIDFSY	NNSYRQIISS	240
SQRSSYLDAY	NHDCPLAIES	CASSGTNSAC	KNAESVCYNG	IEGPISAAAD	FDVDVVRQPS	300
NDPYPPATYS	TYLQASAVRK	AIGARTKYQE	CPNGPYNKFE	TTGDNRSFL	STLSDVVNTG	360
ITVLVWAGDA	DWICNWVGGH	AVADAVTFAR	QKTFQAKPLE	PYTVNGTEKG	RFKTVDNFTF	420
LRVYEAGHEV	PYYQPETALQ	VFVQTMQKKA	IFST			454

SEQ ID NO: 50 moltype = AA length = 453
 FEATURE Location/Qualifiers
 source 1..453
 mol_type = protein
 organism = Talaromyces variabilis

SEQUENCE: 50

AAVPQDKRSI	VKRDGVITYV	FEHAATGAKM	EFVKNSGICE	TTPGVNQYSG	YLSVGDNMNM	60
WFWFPESRNN	ASGAPLAAWF	NGGPGCSSMI	GLFQENGPCP	FVNGEKKPSL	NKYSFNEYAN	120
VLYVDQPIGV	GFSYGTDDVT	STESAAPYVW	KLLQAFYAQF	PQYESRDFGI	FTESYGGHYG	180
PEFAHYLQQQ	NEGVKNSVD	GENINLVALG	INNGWFDTQL	QEGAYIDYAY	SNNYKKIIDS	240
SQRSSLEDSL	KSDCLPAVKQ	CLSSGSDSDC	ENASDTCGQI	ESSIQQAADF	DVYDVREPSN	300
DPYPPSTYS	YLADSSVKA	IGAKSTYKEC	PNGPYKFS	TGDNTRSFLS	ELSSVVQSGI	360
QVLVWAGDAD	WICNYMGVQR	VADAVEFDGS	SQFSNATLKP	YTVNGTKKGE	YKNVDNFSYL	420
RVYAGAGHEV	YYQPAVALQV	FKQTMQQQAI	KST			453

SEQ ID NO: 51 moltype = AA length = 456
 FEATURE Location/Qualifiers
 source 1..456
 mol_type = protein

-continued

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organism = Penicillium arenicola

SEQUENCE: 51
APATHLQDKR SIVERDGVNY TVFEHAATGA KLEFVTNSGI CETTSGVNQY SGYLSVGTNM 60
NMWFWFFESR NSPSTAPLAA WFNGGPCSS MIGLFQENGP CQFYDGASTP SLNPYSFNEY 120
ANMIYIDQPI GVGFSYGTDD VTSTVTAAPY VWKLIQAFYA SFPAYESREF GLFTESYGGH 180
YGPEPAYIYQ QQNAAIASGT VTGDTIDIVA LGINNGWIDS ALQEKAYIEY SYNNSYKQII 240
TSSQRTSYLS TYTNDCLPAI NKCTTGGSNS ACSNAADVCY NDIESPIMSD ADFDVYDIRQ 300
PSNDAYPPET YVTYLQTSV VKAIGASSTY QECPDAPYNK FATTDGNDRS FLATLSTVVQ 360
SGITVLLWAG DADWICNVG NQYVADAVTW SGQSSFAAQ TPTPTVNGSE VGTFKTLNLD 420
SFLRVYEAGH EVPPYQDATA LQAFIQTMQK KALSST 456

SEQ ID NO: 52      moltype = AA length = 354
FEATURE           Location/Qualifiers
source            1..354
                  mol_type = protein
                  organism = Nocardiosis kunsanensis

SEQUENCE: 52
APAPQNPTEP AEATTMAEAL ERDLGLNEAE ATDLIDAEQES ALDVAEATE AAGEHYGGSL 60
FDTETHDLTV LVTDSAAVPG VEAAGAEAAV VEHGVEGLDD LISDLSAGA QEGVVGWYPE 120
VENDTVVIET LEGADADVDA LLSSAGVDPD DVRVETDEA PEVYANIVGG DAYTIGGSSR 180
CSVGFPASDS YGQPGFVTAG HCGTTGSSVS IGNNGSVFSQ SVFPGNDAAF VRGTSNFSLT 240
NLVNRVNSGS DVAVSGSTQA PIGSQVCRSG STTGWHCGTI QARGQTVSYP QGTVRDLTRT 300
SVCAEPGDSG GSFISGSQAQ GVTSGGSGNC SWGGTTYQYE VNPMNSWNL NLST 354

SEQ ID NO: 53      moltype = AA length = 425
FEATURE           Location/Qualifiers
source            1..425
                  mol_type = protein
                  organism = Streptomyces parvulus

SEQUENCE: 53
GTAPSPAAPT AAESLRADAA PPALLRAMER DLGLGREQAE RRLGNEAEAG AVAGRLRADL 60
GGDFAGAWVR GAESGTLTVA TTDADVPAI EARGAVAEV RHLADLGA KSRLDRAAAH 120
RDTAEAPVRY VDVRTNTVTV QAVRPSAARA LLAAAGVDAG LARVETSAER PRPLYDLRGG 180
EAYYINNSGR CSVGFPVTKG TQGGFATAGH CGRAGASTSG ANRVAQGTFO GSVFPGRDMA 240
WVAANSQWTA TPYVSGAGGQ NVQVAGSTQA PVGASVCRSG STTGWHCGTI QQHDTSVTYP 300
EGTITGVTRT TVCAEPGDSG GSYISGSQAQ GVTSGGSGNC GSGGTTFFQP INPLLQNYGL 360
TLKTTGGGGE DPGEPGEPGG TWAAGTVYRP GDTVTYGGAT YRCLQGHQAQ RGWEPANVPA 420
LWQRV 425

SEQ ID NO: 54      moltype = AA length = 350
FEATURE           Location/Qualifiers
source            1..350
                  mol_type = protein
                  organism = Saccharopolyspora endophytica

SEQUENCE: 54
LTATIADPAG PPVSPDLVTA MQRDLGLTAD QAVARLQGEA VAARADSALR DALAGSYGGS 60
YFDANLGKLV VGTDDAKSD EVRAAGAEPR QVDASERQLD GIVEALNRRG AQVPAAVTGW 120
YADVRENADV VTTQPGTAEQ ATGFVRDAQV PQESVRVWES PAQPETYADV VGGYAYYTAS 180
GARCSMGFAV QGGFVTAGHC GAPGESTTQP TGYFAGSFP GNDYAFVNTG TDDTGYPVLY 240
NYSSGYVRVS GSAPLPLGSS ICRSGSTTGW HCGTVLAKNQ SVRYQEGTVS GLTRTNVCAE 300
PGDSGGSFIS GNQAQGMSTG GWGDCRTGGE TTYQPVREAL SAYGLTLLTQ 350

SEQ ID NO: 55      moltype = AA length = 355
FEATURE           Location/Qualifiers
source            1..355
                  mol_type = protein
                  organism = Luteus cellwall

SEQUENCE: 55
ASGPLPQSPS PDSVATTMA EALERDLNLT STEAQELLTA QEAAFEADEA AAQAAGDAYG 60
GSVFDTEITD LTVMTDAAA VQAVEATGAK ADVVSYGIDG LDTIIDDLNE ADAPEGVVGW 120
YPDIDSDTVV LEVLEGSGAD VDALLAEAGV DASAVKVEST TEQPELYADI IGGLAYYMGG 180
RCSVGFAATN ASGQPGFVTA GHCGRVGTQV TIGNRGVFE RSVFPGNDAA FVRGTSNFTL 240
TNLVSRVNSG GYATVSGSSV APIGSSVCRS GSTTGWRCTG IQARGQVTY PQGTIYNMTR 300
TSACAEPGDS GGSFISGTQA QGVTSGGSGN CSWGGTTFYQ EVNPMNSWN LRLRT 355

SEQ ID NO: 56      moltype = AA length = 406
FEATURE           Location/Qualifiers
source            1..406
                  mol_type = protein
                  organism = Saccharothrix australiensis

SEQUENCE: 56
GPPTTHQEEs GLIAAMARDF KITPDQARAR LVREKAATT EQSLKSLRGG HYAGAWLNEG 60
ATELVAVVTD AAQAKVVEDA GATPKVVQRS QIQLDELKAK LDANKNAPKD VPAWYVDVKT 120
NSVVVLARNT ASAKAFARAS GLSEADVRIE QSTEDPRPLI DVIGNAYYM GSGGRCSVGF 180
SVNGGFVTAG HCGRVGTTTT QPSGTFAGST FPGRDYAWVR VSSGNTMRGL VNRYPGTVPV 240
KGSNESSVGA SVCRSGSTTG WHCGTIQQKN TSVTYPEGTI SGVTRTNACA EPGDSGGSWL 300
TGDQAQGVTS GGSNGCSSGG TTYFQPVNPI LQAYGLQLVI EGGPTGTTGP TTTSSNPGGT 360
TWQPGVAYTA GTTVTYEGVG YECLQGHTSQ IGWEPSAVPA LWERVG 406

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SEQ ID NO: 57 moltype = AA length = 346
 FEATURE Location/Qualifiers
 source 1..346
 mol_type = protein
 organism = *Nocardioopsis baichengensis*

SEQUENCE: 57

DAFPEGTEPL	AEAIEDLGV	ASGQADELLT	AESARSLEK	EAEKAAGEAF	AGAVFDTETH	60
ELTVSVADPS	AVEAVEATGA	ETRVVEASQD	ELDAAMADLD	AASEDGVSEE	VTGWHVDLES	120
NTVVVEALEG	SEDAAEDLIA	DAGLDSAPVV	VEKADAQPET	FGAIVGGDAY	YPGNSRCSIG	180
FSVRGGFVTA	GHCSTGTSTV	SGSAGESGRV	AGSVFPGRDM	GYVRANSQWT	PSPYVNNYRG	240
GRVAVRGSNE	ASVGASVCRS	GSTTGWHCGT	IQAKNQTVNY	PQGTVRGLTR	TTACAEFGDS	300
GGSWLSGNQA	QGVTSGGSGN	CSWGGTTFQ	PVNPILSQWG	LSLT		346

SEQ ID NO: 58 moltype = AA length = 353
 FEATURE Location/Qualifiers
 source 1..353
 mol_type = protein
 organism = *Streptomyces* sp.

SEQUENCE: 58

NDTLTERADA	AVAELPAGVL	DAMERDLGLS	EQEAGLQLVA	QYDASLLGET	LSADLDAYAG	60
SWLADGTDLV	VATTDRAEAA	QITEAGAKVE	IVDHTLLELE	SVKAALDEAA	ESYDITDAPV	120
WYVDITTNV	VLLTSDTAEA	KGFVEAAGVD	AGAVSIQTS	EQQAFYDLV	GGDAYVMGGG	180
RCSVGFSVTQ	GSTPGFATAG	HCGTVGTSTT	GFNQAAQGT	EESFFPGDDM	AWVSVNSNWN	240
TTPTVNDGAV	TVSGSTQGA	GASICRSGST	TGWHCGTIEQ	HNTSVTYPEP	TITGVTRTSV	300
CAEPGDSGGS	YISGSAQGV	TSGSGNCTS	GGTTYHQPIN	PLLSAYGLDL	VTG	353

SEQ ID NO: 59 moltype = AA length = 353
 FEATURE Location/Qualifiers
 source 1..353
 mol_type = protein
 organism = *Actinoalloteichus spitiensis*

SEQUENCE: 59

DTPSPDGADA	TVASPEMLSA	MQRDLGLTEQ	EALTRVAVEA	TAVETEDELR	ASLGPAFGGA	60
HPDGTNTLV	VGVTSAAKAD	EVRAAGATPE	VVAFSADTLD	GVVSTLNETS	EVPDGVGTGWY	120
VDTADNTVVV	TTALGSGEAA	ADFVAESGVN	ADAVTVVEST	EQPRTLYDII	GGDAYYFGGS	180
RCSVGFSVSV	GYVTAGHCGG	VGTATQGYNR	VSSGQVAGSV	PPGSDMGYVR	TNANWTPRPL	240
VNRYSGGATV	TVSGSNEAAV	GASICRSGST	TGWRCGTVQA	KNQTVFYAQ	AVSGLTRTNA	300
CAEGGDSGGS	WLSGSAQGV	TSGSGNCTW	GGTTYFQPLN	PILSRWGLSL	TRG	353

SEQ ID NO: 60 moltype = AA length = 338
 FEATURE Location/Qualifiers
 source 1..338
 mol_type = protein
 organism = *Byssoschlamys verrucosa*

SEQUENCE: 60

FPAAVDVKRA	PSSLGITLSQ	VSNTLIKAVV	QNTGRGEVSF	IHLNFFKDDA	PVKKVAVYRN	60
GSEVQFEGIQ	RRYKSTGLTR	DAFTTLAPGK	TAEDVFDIAS	TCDLISGGPV	TIRSEGVVPY	120
ATANGIDTAG	YIPYSSNELT	IDVDGAIAS	VSKAIAPLNR	RTNISSCSGS	EQSTLTMAK	180
NAASLAHAAA	DAAESGSASK	FSEYFKTTAS	STRKTVAARL	RAVAQEASSS	SSGSTTYYC	240
DAYGYCTTNV	LAYTLPSHNT	IATCDLYYTN	LSALTRTCHA	QDQATTSLHE	FTHAPGVYSP	300
GTDDLAYGYA	SSTSLSSSQ	VMNADSYALY	ANAIYVGC			338

SEQ ID NO: 61 moltype = AA length = 339
 FEATURE Location/Qualifiers
 source 1..339
 mol_type = protein
 organism = *Hamigera terricola*

SEQUENCE: 61

SPVNVNVGRE	ELPALDVTLS	QIGNTQIKAV	VKNTGSEDVT	FMHLNFFTDS	APVKKVSFVQ	60
NNTEVEFQGI	LRRVKYTDVS	TDSVTTLAPG	ASIEDVFDIA	TTTDLASGGA	VTVKTDGFVP	120
ILASAENKVT	GYARYTSNEL	HLDVDGPSAA	TVSKAIAPLD	RRTRLSSCSG	SRSSALQTAL	180
RNTVSLANAA	ANAARSGSAS	KFSEYFKTTS	SSVRSTVAAR	LSAVASEASS	TSSGSTTYYC	240
NDPYGYCSTD	VLAYTLPSYN	IIANCDIYYS	YLPALTGSCH	AQDQATTTLH	EFTHAPGVYS	300
PGTEDYGYGY	NAATSLSSSQ	AVLNADSYAL	YANAIYLGC			339

SEQ ID NO: 62 moltype = AA length = 334
 FEATURE Location/Qualifiers
 source 1..334
 mol_type = protein
 organism = *Aspergillus tamarii*

SEQUENCE: 62

IPVEVPASAP	GLDVTLSQVG	NTRIKAVVKN	TGSEEVTFVH	LNFFKDAAPV	QKVSLEFRNAT	60
EVQFQGIKQR	LITEGLSDEA	LTTLAPGATI	EDEFDIAS	DLSEGGTITI	NSNGLVPITT	120
ENKVTGYIPF	ASNELSIDV	AEEAATVSQA	VKILDRRTKV	TSCSGSRSSA	LQTALRNTVS	180
LARAAASAAQ	SGSSSRFQEY	FKTTSSSTRS	TVAARLNAVA	NEAASTASGS	TTYCSDVYV	240
YCSSNVLAYT	LPAYNIIANC	DLYYSYLPAL	TSTCHAQDQA	TTTLHEPHTA	PGVYSPGTTD	300
LGYGYSAAAT	LSASQALLNA	DTYALFANAV	NLNC			334

SEQ ID NO: 63 moltype = AA length = 334

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FEATURE
source Location/Qualifiers
1..334
mol_type = protein
organism = *Aspergillus niveus*

SEQUENCE: 63

LPAKTGEQLQ	KLDVALSQVD	NTLIKAVVKN	TGSEDTFVH	LNFFRDTPV	KKVSLFRNTT	60
EVPPHGIKQR	LRSDGLSADA	LTVLAPGESI	EDEFDIAATS	DLSEGGSIIT	SADGFVPIAS	120
GNKITGYVPF	SSNELSVEVD	AAQAASVASA	VKPLDRRTKV	ASCSGSRSSA	LTQALRNTVS	180
LANAASAAQ	SGSSTRFQEQ	FKTTSSSVRS	SVAARFRAVA	SEASSTSAGS	TTYCTDVYG	240
YCSSNVLAYT	LPAYNIIANC	DIYYTYLPAL	TSTCHAQDQA	TTTLHEFTHA	PGVYSPGTDD	300
LGGYDAATA	LSSSQALNNA	DSYALFANAV	NLNC			334

SEQ ID NO: 64 moltype = AA length = 372

FEATURE
source Location/Qualifiers
1..372
mol_type = protein
organism = *Penicillium sclerotiorum*

SEQUENCE: 64

IPTGGKKSSF	SVDQVAIPAT	KTKNFADTYA	RAISKFGGNV	PSHVRAAAQQ	SGAATTTPEA	60
NDEEYLTPVN	VGGTTLNLDF	DTGSADLWVF	SEQLPSSSEQ	GHSVYKPMNG	TKLSGATWSI	120
SYGDGSSASG	DVYKDTVSVG	PVKATGQAVE	AASKISAQFT	RDSNNDGLLG	LAFSSINTVK	180
PKAQTTFPDT	VKSSLASPLF	AVTLKHNAAG	TYDFGFVDSS	KYTGSLAYTD	VDNSQGFWEF	240
TADSYKVGSG	SGSSIKGIAD	TGTTLLLLDD	EVVSAYYKQV	SGASDSQSAG	GYTFDCSADL	300
PDFTVTISGY	DAVVPGLIN	YAPVSDGSST	CLGGIQSNSG	IGFSIFGDIF	LKSQYVVFDS	360
NGPRLGFAPQ	SS					372

SEQ ID NO: 65 moltype = AA length = 371

FEATURE
source Location/Qualifiers
1..371
mol_type = protein
organism = *Penicillium bilaiae*

SEQUENCE: 65

VPTGGKKSFS	INQVAIPATK	TKNFAGNYAH	AIKYGGNVP	SHVEAAAQSS	GAATTTPESN	60
DEEYLTPVNV	GGTTLNLDFD	TGSADLWVFS	AELPSAEQSG	HALYKPSNGT	KLGSASWSIS	120
YGDGSSASGD	VYKDTVSVGS	VKATGQAVEA	ASKISAQFTK	DKNNDGLLGL	AFSSINTVKP	180
KAQTTFPFDV	KSSLASPLFA	VTLKHNAAGT	YDFGFIDKSK	YTGSLAYADV	DNSQGFWEFT	240
ADSYSVGSSK	GSSIKGIADT	GTTLLLLDDE	VVSAYYKQVQ	GAQQDSSAGG	YTFDCSSKLP	300
DFTVTISGYD	AVVPGDLINF	APASEGSSTC	LGGIQSNSGI	GFSIFGDIFL	KSQYVVFDSN	360
GPRLGFAPQS	S					371

SEQ ID NO: 66 moltype = AA length = 373

FEATURE
source Location/Qualifiers
1..373
mol_type = protein
organism = *Penicillium antarticum*

SEQUENCE: 66

SPLVTPRKGF	TINQETRAVT	KSKTVNLPV	YAQALSKYGA	TVPQHVHAAA	VSGSAVTTPPE	60
ESDVEYLTPV	NVGTTNLNLD	FDTGSADLWV	FSSELTSSQQ	SGHSIYKPSS	SATKLSGSSW	120
SISYGDGSSA	SGDVYKDTVT	VGGVKATGQA	VEAASKISSA	FLQDVNNNDGL	LGLAFSSINT	180
VSPRAQTTFP	DTVKSQDLSP	LFAVTLKHNA	PGSYDFGYID	KSKYTGSLTY	ANVDDSQGFW	240
SFTASSYKIG	TTTGGSIITGI	ADTGTTLTLL	PDSVVSAYYK	KVSGSQNSNY	YGGYVFPSCA	300
TLPDFVTVTIN	GYNAVVPGNL	INFAQATTGS	STCYGGIQSN	SGIGFSIFGD	IFLKSQYVVF	360
DSEGPRLGFA	AQA					373

SEQ ID NO: 67 moltype = AA length = 370

FEATURE
source Location/Qualifiers
1..370
mol_type = protein
organism = *Penicillium sumatrense*

SEQUENCE: 67

VPTNNVASKF	SVNQVSRPAT	KTTNFAANYG	RALSKYGAGV	PSHVEAAAAA	SGSAVTTPES	60
NDVEYLTPVS	IGGTTNLNLD	DTGSADLWVF	STELSSSEQS	GHSVYNPSKS	GKKISGASWD	120
ISYGDGSGAS	GDVYTDVTV	GGVTASKQAV	EAAKQISSQF	QQDNDNDGLL	GLAFSSINTV	180
SPTPQKTFFD	NVKSSLSQPL	FAVALKHNA	GVYDFGFIDS	SKHTGSIAYT	SVDSSQGFWS	240
FTVDGYKVG	KSGAGFDGIA	DTGTTLLLLD	DSVVSAYYSQ	VSGAKNDMNA	GGYVFDSCAD	300
LPDFSVTIGS	YTATVPGSLI	NYGDSGDN	IGGIQNSNGI	GFSIFGDIFL	KSQYVVFNAN	360
GPKLGFAPQA						370

SEQ ID NO: 68 moltype = AA length = 384

FEATURE
source Location/Qualifiers
1..384
mol_type = protein
organism = *Trichoderma lixii*

SEQUENCE: 68

LPTEGQKTAS	IEVTYNKNYV	AHGPTALFKA	KRKYGAPISD	NLRAAVALAKH	SLTKRQTGSA	60
NTNPSDSADD	EYITSVSI	PAQVLPLDFD	TGSSDLWVFS	SETPKSSASG	HVTYSPSKSS	120
TAKKLSGSTW	SITYGDHSSS	SGDVYTDVVS	IGGFSVKTQA	IESATKVSTQ	FVQDTVISGL	180
VGLGPDVGN	KPRAQKTWF	SNAASSLAEP	LFTADLRHQE	TGSYNFGFID	NSLAKGTIGY	240
TPADGSEGYW	GFTATGYSVG	GAKLGRSSIT	GIADTGTTL	LLPDNVVDAY	YNNVESAQYD	300

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DSQEGVVVDC	SEDLPSPSFG	VGGQTITISG	DLNLTPIEE	GSSTCFGGLQ	SSADIGINIF	360
GDVALKAALV	VFDLGNRLG	FAQK				384

SEQ ID NO: 69	moltype = AA	length = 384
FEATURE	Location/Qualifiers	
source	1..384	
	mol_type = protein	
	organism = Trichoderma brevicompactum	

SEQUENCE: 69						
LPTEGQKTAS	VEVTYNQNYA	AHGPTQLYKA	KRKYGAPISD	NLKAIVANRK	ALIKRQTGSA	60
PNHPSDSADD	EYITNVSIGT	PAQVLPLDFD	TGSSDLWVFS	SETPKSSASG	HTIYTPSKSS	120
TSKKLSGATW	SIEYGDKSTS	SGDVYTDKVT	VGGFSVSTQA	VESATKVSQA	FVQDTANGL	180
LGLAFDSINT	VSPRQQTWF	SNAANSLAQ	LFTANLNHQA	TGSYNFGFID	TSLASGPINY	240
VPVDNSQGF	GFTASGSYVG	GKLNRSLS	GIADTGTLL	LLPDAVVNAY	YANVESAEYD	300
DEQEGVVVDC	SEDLPSPSFG	VGGQTITIPG	DLNLTPIDS	SGQTCYGGGLQ	SSADIGINIF	360
GDVALKAALV	VFDLGNRLG	WAQK				384

SEQ ID NO: 70	moltype = AA	length = 379
FEATURE	Location/Qualifiers	
source	1..379	
	mol_type = protein	
	organism = Penicillium cinnamopurpureum	

SEQUENCE: 70						
IPTEGPNRKG	FTVNQVRPV	TNGTKSKTLN	LPAIYANALS	KYGVAVPANI	KAAAESGTAT	60
TPEDNDIEY	LTPVDVGTT	LNLDFTGSA	DLWVFSSELP	SSESSGHSIY	KPSQSGKKLD	120
GYSWKISYGD	SSASGDVYT	DTVTVGVT	DGQAVEAAK	ISQQFVQDKN	NDGLLGLAFS	180
SINTVQPKAQ	TTFPDTVKDQ	LDSPLFAVTL	KHNAPGSYDF	GFIDKSKYTG	SLTYADVDSK	240
DGFWARTADG	YSVGSSSSS	SRIKGIADTG	TLLLLIDDEI	VSAYYKQVDG	AQESYSVGGY	300
TFDCSTKLDP	FNKIGIDYTA	VIPGDVINYA	PVQQGSSTCF	GGIQSNSGLP	FSIFGDIPLK	360
SQYVVDANG	PRLGFAAQA					379

SEQ ID NO: 71	moltype = AA	length = 350
FEATURE	Location/Qualifiers	
source	1..350	
	mol_type = protein	
	organism = Bacillus licheniformis	

SEQUENCE: 71						
AQPAKNVEKD	YIVGFGSGVK	TASVKKDIK	ESGGKVDKQF	RIINAAKAKL	DKEALKEVKN	60
DPDVAYVEED	HVAHALAQTV	PYGIPLIKAD	KVQAQGFKA	NVKVAVLDTG	IQASHPDLNV	120
VGGASFVAGE	AYNTDGNHGH	THVAGTVAAL	DNTTGVLGVA	PSVSLYAVKV	LNSSGSGSYS	180
GIVSGIEWAT	TNGMDVINMS	LGGASGSTAM	KQAVDNAYAR	GVVVVAAAGN	SGSSGNTNTI	240
GYPKYDSVI	AVGAVDSNSN	RASFSSVGA	LEVMAPGAGV	YSTYPTNTYA	TLNGTSMAS	300
HVAGAAALIL	SKHPNLSASQ	VRNRLSSTAT	YLGSSFFYKG	GLINVEAAAQ		350

SEQ ID NO: 72	moltype = AA	length = 360
FEATURE	Location/Qualifiers	
source	1..360	
	mol_type = protein	
	organism = Bacillus subtilis	

SEQUENCE: 72						
FSNMSAQAG	KSSTEKKYIV	GFKQTSAMS	SAKKKDVI	KGKVKQKQF	YVNAAAATLD	60
EKAVKELKED	PSVAYVEEDH	IAHEYAQSV	YGISQIKAPA	LHSQGYTGSN	VKVAVIDSGI	120
DSSHDPDLNVR	GGASFVPSET	NPYQDGSSHG	THVAGTIAAL	NNSIGVLGVA	PSASLYAVKV	180
LDSTGSGQYS	WIINGIEWAI	SNMMDVINMS	LGGPTGSTAL	KTVVDKAVSS	GIVVAAAAGN	240
EGSSGSTSTV	GYPKYPSSTI	AVGAVNSNQ	RASFSSAGSE	LDVMAPGVSI	QSTLPGGTG	300
AYNGTSMATP	HVAGAAALIL	SKHPTWTNAQ	VRDRLESTAT	YLGSSFFYKG	GLINVQAAAQ	360

SEQ ID NO: 73	moltype = AA	length = 548
FEATURE	Location/Qualifiers	
source	1..548	
	mol_type = protein	
	organism = Trametes cf versicol	

SEQUENCE: 73						
TPTARNLKLH	ESREEIPAGF	SLSGAASPD	TLKLRLALVQ	SNFAELEDKL	YDVSTPSSAN	60
YQHLKSKEEV	EQLVAPSAES	VNAVNAWLTE	NGLTAQTISP	AGDWLAFVFP	VSKANELFDA	120
DFSVPTHDES	GLKAVRTLAY	SIPAEQGH	DLVHPTITFP	PNNSHLPVVR	SPVKPVQNL	180
SRAVPASCAS	TITPAQLAL	YGIPTTKATQ	SSNKLAVSGF	IDQFANSADL	KTFPLGKFRD	240
ISSSTTFTLQ	TLDGGSNSQS	SSQAGVEANL	DIQYTVGLAS	AVPTIFISVG	DDFQDGLLEG	300
FLDIINFLN	ESAPPQVLT	SYQONENTIS	AKLANQLCNA	YAQLGARGTS	ILFASGDGGV	360
SGSQSSSSCK	FVPTFPSCGP	FMTSVGATQG	INPETAADFS	SGGFSNVFAR	PSYQSTAVSS	420
YLTALGSTNS	GKFNTSGRAF	PDIAQGVDF	EIVVSGRTEG	VDGTSASPT	LAAIISLLND	480
RLIAAGKSPL	GFLNPFLYSA	AGTAALTDIT	SGSNPGCNTN	GFPKAGWDP	VTGLGTPNFA	540
KLTAVAL						548

SEQ ID NO: 74	moltype = AA	length = 439
FEATURE	Location/Qualifiers	
source	1..439	
	mol_type = protein	
	organism = Bos taurus	

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SEQUENCE: 74
 MAKEYFPPTG KIPFEGKDSK NVMAFHYEY EKVVMGKKMK DWLKFAMAW HTLGGASADQ 60
 FGGQTRSYEW DKAECVPQRA KDKMDAGFEI MDKLGIEYFC FHDVDLVEEA PTIAEYEERM 120
 KAITDYAQEK MKQFPNIKLL WGTANVFGNK RYANGASTNP DFDVVARAIV QIKNSIDATI 180
 KLGGTNYVFW GREGYMSLL NTDQKREKEH MATMLGMARD YARAKGPKGT FLIEPKPMEP 240
 SKHQYDVDTF TVIGFLKAHG LDKDFKVNIE VNHAFLAGHT FEHELACAVD AGMLGSIDAN 300
 RGDQNGWDT DQFPIDNFEL TQAMLEIIRN GGLGNGGTNF DAKIRRNSTD LEDLFIAHIS 360
 GMDAMARALM NAADILENSE LPAMKKARYA SFDSGIGKDF EDGKLTFEQV YEYGKKVEEP 420
 KQTSKGQEKY ETIVALHCK 439

SEQ ID NO: 75 moltype = AA length = 591
 FEATURE Location/Qualifiers
 source 1..591
 mol_type = protein
 organism = *Saccharomyces cerevisiae*

SEQUENCE: 75
 MLCSVIQRT REVSNMTSLD SYILGFDLST QQLKCLAINQ DLKIVHSETV EFEKDLPHYH 60
 TKKGVYIHGD TIECPVAMWL GALDLVLISKY REAKFPLNKV MAVSGSCQOH GSVWSSQAE 120
 SLLEQLNKKP EKDLLHYVSS VAFARQTAPN WQDHSSTAKQC QEFECEIGGP EKMAQLTGSR 180
 AHFRFTGPQI LKIAQLEPEA YEKTKTISLV SNFLTSILVG HLVELEEADA CGMNLVDIRE 240
 RKFMYYELLHL IDSSSKDKTI RQKLMRAPMK NLIAGTICKY FIEKYGFNTN CKVSPMTGDN 300
 LATICSLPLR KNDVLVSLGT STTVLLVTDK YHPSPNYHLF IHPTLPNHYM GMICYCNGSL 360
 ARERIRDELN KERENNYEKT NDWTLFNQAV LDDSESSENE LGVYFPLGEI VPSVKAINKR 420
 VIFNPKTGMI BREVAKFKDK RHDANKIVES QALSCRVRIS PLLSDSNASS QORLNEDTIV 480
 KFDYDESPLR DYLNKRPERT FFVGGASKND AIVKKFAQVI GATKGNFRLE TPNSCALGGC 540
 YKAMWSLLYD SNKIAVPFDK FLNDNFPWHV MESISDVND NWAIIIPRLS P 591

SEQ ID NO: 76 moltype = AA length = 444
 FEATURE Location/Qualifiers
 source 1..444
 mol_type = protein
 organism = *Bacillus subtilis*

SEQUENCE: 76
 ETANKSNELT APSIKSGTIL HAWNWSFNTL KHNMKDIHDA GYTAIQTSPI NQVKEGNQGD 60
 KSMNSWYWL YQTSYQIGNR YLGTEQEFKE MCAAEEYGI KVIIDAVINH TTFDYAAISN 120
 EVKSPNWNH GNTQIKNWSL RWDVTQNSLL GLYDWNQNT QVQSYLKRFL ERALNDGADG 180
 FRFDAKHIE LPDDGSYGSQ FWPNTINTSA EFQYGEILQD SASRDAAYAN YMDVTASNYG 240
 HSIRSALKNR NLGVSNNISHY AYDVSADKLW TWVESHDTYA NDDEESTWMS DDDIRLGWAV 300
 IASRSGSTPL FFSRPEGGSN GVRFPKGSQI GDRGSAFED QSITAVNRFH NVMAQPEEL 360
 SNPNNGNNQIF MNQRGSHGVV LANAGSSSVS INTPTKLPDG RYDNKAGAGS FQVNDGKLTG 420
 TINARSAVL YPDDIEIRCN TFFQ 444

SEQ ID NO: 77 moltype = AA length = 476
 FEATURE Location/Qualifiers
 source 1..476
 mol_type = protein
 organism = *Saccharomycopsis fibuligera*

SEQUENCE: 77
 QPVTLFKRET NADKWSQSI YQIVTDRFAR TDGDTASASN TEDRLYCGGS FQGIKKLDY 60
 IKDMGFTAIW ISPVENIPD NTAYGYAYHG YWMKNIYKIN ENFGTADDLK SLAQELHHRD 120
 MLLMVDIVTN HYSDGSGSDS IDYSEYTPFN DQKYFHNLYC ISNYDDQAV QSCWEGDSSV 180
 ALPDLRTEDS DVASVFNWSV KDFVGNYSID GLRIDSARKH DQGFPPDFVS ASGVYSVGEV 240
 FQGDPAITCP YQNYIPGVSN YPLYPTTRF FKTIDSSSSE LTQMISSVAS SCSPTLLTN 300
 FVENHDNERF ASMTSDQSLI SNAIAFVLLG DGIPVIYYGQ EQGLSGKSDP NNREALWLSG 360
 YNKESDYKLF IAKANAARNA AVYQDSSYAT SQLSVIFSND HVIATKRGSV VSVFNNLGSS 420
 GSSDVTISNT GYSSGEDLVE VLTCTVSGS SDLQVSIQGG QPQIFVPAKY ASDICS 476

SEQ ID NO: 78 moltype = AA length = 487
 FEATURE Location/Qualifiers
 source 1..487
 mol_type = protein
 organism = *Debaryomyces occidentalis*

SEQUENCE: 78
 QPIIFDKRDV GSSADKWKDQ SIYQIVTDRF ARSDGSTTAD CLVSDRKYCG GSYKGIIDKL 60
 DYIQMGFTA IWISPVVEQI PDNTAYGYAY HGYWMKNIDE LNTNFGTAD LKQLASELHS 120
 RSMLLMVDVV YNHAWNGDG SSVDYSSFTP FNQSYFHDY CLITNYNDQT NVEDCWEGDT 180
 EVSLPDLSTE DNEVIGVFT WVSDFVQNS IDGLRIDSAR HVDTASLTKF EDASGVYNLG 240
 EYVQGDPTYT CPYQSYMKG V TNYPLYPVY RFFSDTSATS SELTSMISTL QSSCDVSL 300
 GNFIENHDQV RFPVTSSTS LIKNAMAFII LGDGIPIIY GQEQGLNGGS DPANREALWL 360
 SGYNTDSEYY ELISKLNQIR NQAIKKDSAY STYKSSVSS SDHYIATRKG SDANQLISIF 420
 NNLGNSGSD ITVSTNTGYSS GDKVIDIISC NSVSAGDFGS LSVSISGGM QVYAPSSVLS 480
 GSGICNQ 487

SEQ ID NO: 79 moltype = AA length = 487
 FEATURE Location/Qualifiers
 source 1..487
 mol_type = protein
 organism = *Debaryomyces occidentalis*

SEQUENCE: 79

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KPIFLSKRDA GSSAAAAWRS ESIYQLVTDR FARTDGSTSA TCNTGDRVYC GGTFQGIIDK 60
LDYIQMGFT AIWISPVVEQ IPDDTGYGYA YHGWMKDIY AINSNFGTAD DLKNSNELH 120
KRNMKLMVDI VTNHYAWNGA GSSVAYSNNY PFNQQSYFHD YCLITNYDDQ TNVEDCWEGD 180
NTVSLPDLRT EDSDVSSIFN LWVAELVSNY SIDGLRDSA KHVESFYPS FQSAAGVYLL 240
GEVYDGDPAI TCPYQNYMSG VTNYPYLYPM LRFFQGTSNS VDELNAMISS LESDCKDITL 300
LGNFIENHDQ PRLPSYTSDS ALIKNAIAFN LMSDGIPIIY YGQEQGYSGS SDPNNREALW 360
LSGYSTSNFY YKLISVNNQI RNQAIYKDSK YTTYWSDVLY ASGHVIALQR GADDQRIVSV 420
FNNLGSSSQS TVTFSTKYSK GEKVVDVLTG QTSYANSNST LTVSISGGAP RIYAPASLIA 480
NSGICNF 487

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SEQ ID NO: 80      moltype = AA  length = 570
FEATURE           Location/Qualifiers
source            1..570
                  mol_type = protein
                  organism = Lipomyces kononenkoae

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SEQUENCE: 80
MCGSTLSASL YVYNDYDKI VTLYYLTSSG TTGSTLALIL PVWSNNWELW TLSAIAAGAV 60
EITGASYVDS DTSVYTTSL DLPLTTTSAS VPTGTAANWR GRSIYQVVD RFARTDGSIT 120
YSCDVTDRVY CGGSYRGIIIN MLDYIQMGFT TAIWISPIVE NIPDDTGYGY AYHGWMKDI 180
FALNTNFGGA DDLIALATEL HNRGMYLMVD IVVNHFAFSG NHADVDYSEY FPYSSQDYFH 240
SPCWITDYSN QTNVEECWLG DDSVPLVDVN TQLDVTKSEY QSWVKQLIAN YSIDGLRIDT 300
VKHVQMDFWA PFQEAAGIYT VGEVFDGDP YTCPYQENLD GVLNYPVYYP VVSAPQRVGG 360
SISSLVDMID TLKSECIDTT LLGSFLENQD NPRFPSYTS ESLIKNAIAF TILSDGIPII 420
YYGQEQGLNG GNDPYNREAL WPTGYSTTST FYEYIASLNQ IRNHAIYIDD TYLTYQNWVI 480
YSDSTTIAMR KGFTGNQIIT VLSNLGSSGS SYTLTSLNTG YTASSVVEI LTCTAVTVDL 540
SGNLAVPMMSG GLPRVFYFES QLVGSGICSM 570

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SEQ ID NO: 81      moltype = AA  length = 476
FEATURE           Location/Qualifiers
source            1..476
                  mol_type = protein
                  organism = Lipomyces kononenkoae

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SEQUENCE: 81
KTAAEWKELS IYQVITDRFA TTNLTPDCW IRAYCGGTWK GLERKLDYIQ NMGFDAVWIS 60
PVIHNIEVNT TWGFAPHGYY GDDPYRLNEH FGTAADLKS LSDLHARGMS LMDVVVINHL 120
ASYTLPOVDV YSLYPAPFNT SSFAHQPCPI DFSNQSSIED CWLVTEPAPA LVDLKNEDQV 180
ILDALINSVY DLVETDYDIDG IRLDTARHVP KPSLAKFQEK VGVFVTGEAL NQSVPYVAQY 240
QGPLNSAINY PLWYALVDSF MGRRTTFDYLE SVVKSEQATF SDAHALTNFL DNQDQPRFAS 300
YLGDNQGGDV LRDENAATFL FVSGIPVIY YGFEQRFDGG FDPVNRPEMW TSGYNTSTPL 360
YNYLARLNAI RKYAASITGT QVYFSDDTVF LSGGVSHMAM QRGPLVIVLT NVGQHIIDNT 420
GYTVTGSQFS AGDSLTDLVS CTKVKVVGAN GTFTSPSNGG KARIWIKSKY AGKFCS 476

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SEQ ID NO: 82      moltype = AA  length = 626
FEATURE           Location/Qualifiers
source            1..626
                  mol_type = protein
                  organism = Bacillus subtilis

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SEQUENCE: 82
ETANKSNELT APSIKSGTIL HAWNWSFNTL KHNMKDIHDA GYTAIQTSP I NQVKEGNQGD 60
KSMNSWYWL Y QPSTYQIGNR YLGTEQEFKE MCAAEEYGI KVIIDAVINH TTSYAAISN 120
EVKSI PNWTH GNTQIKWSD RWDVTQNSLL GLYDWTQNT QVQSYLKRFL DRALNDGADG 180
FRFDAKHIE LPDDGSYGSQ FWPNTINTSA EFQYGEILQD SASRDAAYAN YMDVTASNYG 240
HSIRSALKNR NLGVSNSH Y ASDVSADKL TWVESHDTYA NDDEESTWMS DDIRLGLWAV 300
IASRSGSTPL FFSRPEGGN GVRFPKGSQI GDRGSALFED QAITAVNRFH NVMAQPEEL 360
SNPNGNNGIF MNQRGSHGVV LANAGSSSVS INTATKLPDG RYDNKAGAGS FQVNDGKLTG 420
TINARSVAVL YPDDIAKAPH VPLENYKTGV THSFNDQLTI TLRADANTTK AVYQINNGPE 480
TAFKDGQFT IGKGDPPGKT YTIMLKGTNS DGVTRTEKYS FVKRDPASAK TIGYQNPNNH 540
SQVNAYIYKH DGSRIELTG SWPGKPMKN ADGIYTLTLP ADTDTTNAKV IFNNGSAQVP 600
GQNQPGFDYV LNGLYNDSSL SGSLPH 626

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SEQ ID NO: 83      moltype = AA  length = 483
FEATURE           Location/Qualifiers
source            1..483
                  mol_type = protein
                  organism = Bacillus subtilis

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SEQUENCE: 83
VNGTLMQYFE WYTPNDGQHW KRLQNDAEHL SDIGITAVWI PPAYKGLSQS DNGYGPYDLY 60
DLGEFQQKGT VRTKYGKSE LQDAIGSLHS RNVQVYGDV LNHKAGADAT EDVTAVEVNP 120
ANRNQETSEE YQIKAWTDFR FPGRGNTYSD FKWHWHYFDG ADWDESRKIS RIFKFRGEGK 180
AWDWEVSSN GNYDLYMYAD VDYDNPVVA ETKKWGNWYA NELSLDGFR I DAAKHIFKSF 240
LRDWVQAVRQ ATGKEMFTVA EYWNQNAAGL ENYLNKTSFN QSVFDVPLHF NLQAASSQGG 300
GYDMRRLLDG TVVSRHPEKA VTFVENHDTQ PQQSLESTVQ TWFKPLAYAF ILTRESGYPO 360
VFGDMYGTGK GTSPEIPSL KDNIEPILKA RKEYAYGPQH DYIDHPDVIG WTREGDSAA 420
KSGLAALITD GPGGSKRMYA FLWPNAGETWY DITGNRSDTV KIGSDGWGEF HVNDGSVSIY 480
VQK 483

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SEQ ID NO: 84      moltype = AA  length = 589
FEATURE           Location/Qualifiers
source            1..589

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mol_type = protein
organism = Bacillus subtilis

SEQUENCE: 84
MMEYAAIHQ PFSTDAYSYD GRTVHIKIRT KKGADAHIRF IWGDPYEYND GKWSANEQPM 60
RKIAATEMHD YWFAEVVPPF RRLQYAFVVT DDHEDIFFGS SGVCPYNEKT LETIHYYKFK 120
PFVHEADTFQ APEWVKSTVW YQIFPERFAN GREDELSPKNA LPWGSKDPGV NDFFGDLQGG 180
IVDKLDYLED LGVNGIYLT PFSAPSNNHY DTLDYFSIDP HFGDPEIFRT LVSQHLQGRM 240
RIMLDAVFNH IGSASPQWQD VVKNGDQSR YKDWFIHSFP VTDDNYDRFA FTADMPKLNT 300
ANPEVQKYLL DIALYWIREF DIDGWRLDVA NEVDHVFVWKT FRQAVSTEKP DVYILGEIWH 360
SAEPWLGRDE PHAAMNYPFT EPMIEYFADQ TISASRMAHR VNAHLMNGMK QANEVMFNLL 420
DSHDTKRLLT RCRNDEKKAR ALLAFMFAQT GSPCIYYGTE IGLDGENDPL CRKCMVWEKE 480
KQNDMLQFM KRLIALRKQE NTLLETEGHLE WNLDDKDNF ISFSRTLDEK ILIYFFNQGN 540
VVQHISLREL NIDRNNKICD AWTEQPLHYH DVIAPQGEF LILSAAAPV 589

SEQ ID NO: 85      moltype = AA length = 483
FEATURE           Location/Qualifiers
source            1..483
                  mol_type = protein
                  organism = Bacillus lichenformis

SEQUENCE: 85
ANLNGTLMQY FEWYMPNDGQ HWKRLQNDQA YLAEHGITAV WIPPAYKGT S QADVGYGAYD 60
LYDLGEFHKQ GTVTRTKYGTK GELQSAIKSL HSRDINVYGD VVINHKGGAD ATEDVTAVEV 120
DPADRNRVIS GEHLIKAWTH FHFPGRGSTY SDFKWHWYHF DGTDWDESRL LNRIYKFQGG 180
AWDWEVSNEN GNYDYLMYAD IDYDHPDVA EIKRWGTWYA NELQLDGFRL DAVKHIFKSF 240
LRDWNHVRE KTGKEMFTVA EYQNDLGLAL ENYLNKTNFN HSVFDVPLHY QFHAASTQGG 300
GYDMRKLLNG TVVSKHPLKS VTFVDNHDTQ PGQSLESTVQ TWFKPLAYAF ILTRESGYQP 360
VPYGDYMYTK GDSQREIPAL KHKIEPILKA RKQYAYGAQH DYFDHHDIVG WTREGDSSVA 420
NSGLAALITD GPGGAKRMYV GRQNAGETWH DITGNRSEPV VINSEGWGEF HVNGGSVSIY 480
VQR 483

SEQ ID NO: 86      moltype = AA length = 478
FEATURE           Location/Qualifiers
source            1..478
                  mol_type = protein
                  organism = Aspergillus niger

SEQUENCE: 86
ATPADWRSQS IYFLLTDRFA RTDGSTTATC NTADQKYCGG TWQGIIDKLD YIQGMGFTAI 60
WITPVTAQLP QTTAYGDAYH GYWQDDIYSL NENYGTADDL KALSSALHER GMYLMVDVVA 120
NHMGYDGAGS SVDYSVFKPF SSQDYFHPFC FIQNYEDQTQ VEDCWLGDNT VSLPDLDTTK 180
DVVKNEWYDW VGSLSVNSYI DGLRIDTVKH VQKDFWPGYN KAAGVYCIGE VLDGDPAYTC 240
PYQNVMDGVL NYPIIYPLLN AFKSTSGSMD DLYNMINTVK SDCPDSTLLG TFVENHDNPR 300
FASYTNDIAL AKNVAAPIIL NDGIPIIYAG QEQHYAGGND PANREATWLS GYPTDSELYK 360
LIASRNAIRN YAIKSDTGFV TYKNWPIYKD DTTIPMRKGT DGSQIVTILS NKGASGDSYT 420
LSLSGAGYTA GQQLTEVIGC TTVTVGSDGN VPVPMAGGLP RVLYPTEKLA GSKICSSS 478

SEQ ID NO: 87      moltype = AA length = 477
FEATURE           Location/Qualifiers
source            1..477
                  mol_type = protein
                  organism = Aspergillus niger

SEQUENCE: 87
ATPADWRSQS IYFLLTDRFA RTDGSTTATC NTADQKYCGG TWQGIIDKLD YIQGMGFTAI 60
WITPVTAQLP QTTAYGDAYH GYWQDDIYSL NENYGTADDL KALSSALHER GMYLMVDVVA 120
NHMGYDGAGS SVDYSVFKPF SSQDYFHPFC FIQNYEDQTQ VEDCWLGDNT VSLPDLDTTK 180
DVVKNEWYDW VGSLSVNSYI DGLRIDTVKH VQKDFWPGYN KAAGVYCIGE VLDGDPAYTC 240
PYQNVMDGVL NYPIIYPLLN AFKSTSGSMD DLYNMINTVK SDCPDSTLLG TFVENHDNPR 300
FASYTNDIAL AKNVAAPIIL NDGIPIIYAG QEQHYAGGND PANREATWLS GYPTDSELYK 360
LIASRNAIRN YAIKSDTGFV TYKNWPIYKD DTTIPMRKGT DGSQIVTILS NKGASGDSYT 420
LSLSGAGYTA GQQLTEVIGC TTVTVGSDGN VPVPMAGGLP RVLYPTEKLA GSKICYG 477

SEQ ID NO: 88      moltype = AA length = 431
FEATURE           Location/Qualifiers
source            1..431
                  mol_type = protein
                  organism = Streptomyces avermitilis

SEQUENCE: 88
SPPGTDKDTA VLFWEKFDV ARECTNTLGP AGYGYVQVSP PAEHIQGSQW WTSYQPVSYK 60
IAGRLGDATA FQNMINTCHT AGVKVVVDTV VNHMSAGSGT GTGGSAYTKY NYPGLYSSYD 120
MDDCTATITD YTNRANVQNC ELVGLADLDT GEEYVRKTIA GYMNTLLGYG ADGFRVDAVK 180
HIPAADLANI KSLRTNPSVY WKQEVIIYASG EAVQPTTEYTG NGDVQEPFRA YDLKRVFNNE 240
NLAYLKNYGE GSWGLNSSVA GVFPVDNHDTE RRGSTLNYKD GANYTLANVF MLAYPYGAD 300
INSGYEWSDA DAGPPGGGTV NACWQDGWKC QHAWPEIKAM VAFRNATRGE SVTNWWDNNG 360
DAIAFGRGAK GYVAINHESG SLRTRYQTSL TAGTYCNVQN NTGVTVDSSG RFTATLGANT 420
ALALYSGKST C 431

SEQ ID NO: 89      moltype = AA length = 643
FEATURE           Location/Qualifiers
source            1..643
                  mol_type = protein

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organism = Clostridium phytofermentans

SEQUENCE: 89
MYTLKSKLRD LYRHPVGYDV INKILLQAGL SKGLIENPVI GALPLSFLNR IAGKKLGNF 60
FDALLALLNQ SDRRLDPYSS KDKKITPTWW KEAVFYQIYP RSFMDGNGDG VGDLPGLISK 120
LDYLKELGVD ALWLSPIYDS PGDDNGYDIR DYQKIDSQFG TMEDFDLLLT ELHARNMRLV 180
MDLVVNHTSD EHHWFKEALK SSESTYRDY FLRKEPNWNT SFFSGSAWNH YPEEDLWGLH 240
LFSKKQMDLN WENPKLRQDI YQMIRWWLEK GVDGFRDLVI NYISKETGLP DGDSFIGNLM 300
GFTGIEHYFY GPKLHNHLQE IQKEAFTPYQ AFSVGETPGI GMKMGKLLTD DSRGELNMMF 360
SFDHLETS GH ARFDQYEDL NYYKSYIMDW MENFADTSWM SLFYDNHNDP RMLSKVDHTH 420
THRQELAKML AMIQMTLKG PFLYQGGELG MINKDFHEIS NFRDVSINK YKELCEKMPK 480
EEAFLQILAG SRDHARTPMQ WSAKPGCGFS NAVPWIDSDG DELVCNAEIQ MQDSESVLSF 540
YRDLIALRRK TPALITYGDI FTHKKRKDIL IYTRYLEGET YLIICNLSND EQKLPGNVVP 600
SESLEGLESL SASADERKGL VLCNYPKVM KSLRAYEGRV YRI 643

SEQ ID NO: 90      moltype = AA length = 1016
FEATURE            Location/Qualifiers
source              1..1016
                    mol_type = protein
                    organism = Clostridium phytofermentans

SEQUENCE: 90
ATDTITIHYH RDDGDYKWN LWLWAEKDG AAYYFDGEDA FGPYVSVSLD KSADRIGFIV 60
RTDSWEKDV EDRFIDTSLG DEIWISSGES TFSYEAPEGY EKEVSIESFQ LKLNLYRYDE 120
EYTDISFRLT PEDGTTDFLT KEHMRIENGI LKAEKEVKYG KKITLDVLKN GLEEDYQGV 180
FSTAKIDEES KLEMYWMQGT GTISPKADFI KRSKEIESAL ITSMKEITVK LSVPCRVDI 240
KQDGGFKLSPK LAVSKVEATS TRDSEYKTIK EGYADTFIIT MEEPLDMSKK YALSKTDYGS 300
RNLTLDGLY TSEEFEAAYT YEGNDLGATY SKEKTVPKVV SPSAESISVL FYPHGEAKDG 360
EKPEITYPMK QTGAGVQAE IEGDLKNKY VYQVTDGKT KLVVDPYAKA AGVNGERGMV 420
IDLSETDPDG FREHSSPEFK NPVDAYIEI HVRDLSMNEV SGIEKNGKFL GFTETGTNS 480
AGLSTGLDHM KELGVTHVHL LPSFDYKTID ESKLGENKFN WGYDPQNYNL PEGSYTDPY 540
QGEVRVREYK EMVQALHENG LHVVMDEVYN HTYTAGDSNF TSLVPGYYR TDTINGNFTNG 600
SGCGNETASE RAMVRKFIV SVVYWATEYK VDGFRFDLMG LHDITMNMV REALDKIDPS 660
ILLYGEGWTG GSTPLPDSKQ AIKNNAVELN ERIACFSDDI RDAIKGSVFD ASDTGFINSG 720
KKNVSNRDES IKFGIVASVS HPQVNLGVP YSSRFWANEP SQTINYASAH DNLTLDWKL 780
ETNKMASKEE LVQMNKLSAA IVLTSQGI PFQAGEEMART KKGNDNSYQS PDSINMLNWD 840
NKTEYKDLFE YYKGLIALRK TYDAFRMQTA EEIQQKLEFV DSDSSVIAYR IHDVKGRE 900
IALIFNGTLE EKEVVLSANA WDLVNVQDTA GTDVIETITG GTIKVPAKST LVLLENKDAV 960
IKGDKDAVKG DEIQELPTNM QEVAEKESGN ABLWVGIIATV CVLAGGVLPW ILKRKR 1016

SEQ ID NO: 91      moltype = AA length = 554
FEATURE            Location/Qualifiers
source              1..554
                    mol_type = protein
                    organism = Clostridium phytofermentans

SEQUENCE: 91
MKNTNTLHPW WESAAAYQIY PRSFMDSDNGD GVGDLQGIIS RLPYLSELGF DLIWICPIY 60
SPNDDNGYDI SDYQNIQKEY GTMEDFEELL HKAHERGIRV IMDLVVNHTS SSHPWIESR 120
SSKDNPKRDW YIWKDGKDNV EPNWESIFG GSTWEYDEKS GQYFLHVFVK TMPDINWENT 180
QVKKAIFDMI CWLWDKIDG FRVDAISHIK KPDNDMPNP KNERVVSFD KHMNQSGLD 240
LLNELKENAF SKYDIFTVAE ANGVRIEIE EWSSEKGIF NSLFQFDHLN LWNVGSEEGK 300
ISIKKLKNAI TKWQKAAPMD GNVALVMENH DLVRSISRFG SEDKYWKESA KCLALMYMQ 360
KGVPPIYQGG EIGMLNADYE SHLDFRDPT LFAYQDRINN GMSPAESLQV LKSSRDNSR 420
TPMQWDASPH AGFTTGTFWM KVNQNVHWN AEVQKEDEDS ILNFKYKLIK IKKETTGLIY 480
GDYKLLMEES ESIYAYTREY EEKNYLVCN LSEELSELQI DLDITKGEIL ISNYEDRNSK 540
EMLKPYECR LYSL 554

SEQ ID NO: 92      moltype = AA length = 538
FEATURE            Location/Qualifiers
source              1..538
                    mol_type = protein
                    organism = Clostridium phytofermentans

SEQUENCE: 92
MVKKWWHSSV VYQIYPRSFN DSNMGIGIDL KGIIKLDYL KNLGIDVIWL SPVFKSPNDD 60
NGYDISDYED IMDEFGTLED MELLKEANN RGIKILMDLV ANHTSDEHKW FIESRKSND 120
AYRDYIWRD PVDGHEPNLD GSTFSGSAWE WDEATGQYYL HLFKKQPDN NWNPIVREE 180
VWKSMMFWID KGIGGFRMDV IELLGKIPDE KIISNGPMLH EYIREMNRNS FGDKDLTGV 240
ECWGATPEIA KMSYNDGSE LSMVQFEHI GLDQIPGDK WDLQPLNLID LKNVPHKWQ 300
CFHDDGWNLS FWNHNDTPRI VSRWGNKVY KIESAKMLAT LLHGLKGTYP IQGSEELGMA 360
NKFKDINQY KDIELTNMYK DRLNKGKHE DIMESIYAKG RDNARTPMQW SDEIDGGFTT 420
GTPWIEVNP FTEINAKQV SNPNSIYNY KKLIEIRKN EVIVYGFEM LLPEDKNIFA 480
YVRTLKDSKI VVVCNFYENE VEYNIPKEYE EKKEVLISNY GLSLTGLRFP FEAIMYRV 538

SEQ ID NO: 93      moltype = AA length = 555
FEATURE            Location/Qualifiers
source              1..555
                    mol_type = protein
                    organism = Clostridium phytofermentans

SEQUENCE: 93
CKKADVQNP SELNQDESQK EKEENDDEGT PEVSQDETGA VIPYDYVQNL NIIDDNYRNF 60
YEIVVYSFYD SNGDGIGDIN GVISKLDYIN DGNATDSDL GFNGIWLMPI MPSTTYHKYD 120

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VTDYVYNIDPQ YGTLEDFKNL VSECHKRGIH LIIDFVFNHT SAKHPWFLEA VSYLESKEG 180
EEPDLEKCPY VGYHYFTKDY NGSKTYKAG TSNWYEGVF WDQMPDLALE NENVRKEIED 240
IAKYWLDLGV DGFRLDAAKE YFSGEKERNI EVLKWFSYV KSVKEDADIV AEWVDEEGTI 300
AAYYESGIPS LFNFPLSQHN GLITNTARKL GTSSGKNFAK TLLRLDEKYK EGNPKYIDAP 360
FISNHDTTTRI SAQCVDNEDQ MKMSAGMLLT MNGSPYVYVG EEIGMNSKGT KDENKRLPMQ 420
WSATDTTGTI TPPANADSV E QKFPVDEQM KDPLSLYNY KRAVRIRNEN PEIARGDMSV 480
IEELCTKDIS AIKKVYQGSE IVILYNINTE SANILLKDAG LTELNIRGYL SVDGNAVMTS 540
DGVVSMPKYS IVILK 555

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SEQ ID NO: 94      moltype = AA length = 583
FEATURE           Location/Qualifiers
source            1..583
                  mol_type = protein
                  organism = Clostridium phytofermentans

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SEQUENCE: 94
MKFEAIYHRT SDNYCYPLNE EDLIINIKTG HDIERVFIYY GDPFEGGILG GNWTWNGVEE 60
ELIYKKNLTH HIWWTITVKP KFKRCKYYPK LVANDTSYYY FEDGFYTEAE MNHQDKNLVY 120
FTFPWMNSID INKTPDWVND TVWYQIFPER FNNQDKENDP KNVKAWGFHT VSNDEFYGGD 180
LQGINRLDY LADIGISGIY LTPIFEANTS HKYDTKDYMK IDPHFGDEKV FKNLVDTAHE 240
KGIRIMLDGV FNHCGNQFAP WLDVLKNGPD SKYFNWFMIN KWPFNKEDHN TNDGSFYSA 300
FTSRMPKLNT NNPEVIKYL DVVEYVWKNF DIDGIRLDVA NEISHRFCKD LRKLTKELKP 360
DFYILGELWH DAITWLHGDE FDGVMNYPLA TSLADYVWVP EKTNYDFECA INHNFTMYMQ 420
QTNDVLFNLL DSHDNTNRLID KVKDIDIFYQ QLAVLFTMPG SPCIIYGTETI AMEGSYDPDC 480
RRCMPWEDID AGLFKDRIEI IKALIHRLKT NNAFKSRHYH FIEDKNNNRV IHYIKTDEDH 540
KQVEVILNCS KDSIVVQRKG NELFSLNED TILKPKGVFI QQI 583

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SEQ ID NO: 95      moltype = AA length = 575
FEATURE           Location/Qualifiers
source            1..575
                  mol_type = protein
                  organism = Clostridium thermocellum

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SEQUENCE: 95
MKLEAIYHKP YSEFAFPVAP DTLVIRLRTA KNDVNTCILI YHEKYDTSQR GKVKMDKVAS 60
DGMFDYEEVE LNVGIKRIKY MFYLEDNYSI KWSYSSDGFED YMPQWGHFTY SYICKDDIFH 120
EVEWFRNSTI YQIFPDRFAK FPPDTENSGK RTIHGGNIKG IIDRFDHLVK LGVDVVYVLP 180
IFKSESYHRY DVVDYIEIDP MFGSKEELRE LMDLCHKNGI KVIIDGPFVNH SGDKFFAFRD 240
VVEKGEKSKY ANWYFINSFP VQGYPRENPE CFSFYGGMPK LNTGNPETAK YFLDVVKYWT 300
VEFGVDGWRL DAADVEDRKF WRKLRDMLKD LNKDVVLIGE IFDEASSWLW GDQFDSVINY 360
PLKAMINDLF AYRSIDVETP RNRISGYIMK FNKKVLSLV NIISTHDTTP FLTLNCGDEK 420
RFEMAVVFQF TFPGVPLIYY DEIGMEGEG DPDCRRPMIW DEAKWNKKT LELYKFLIGLR 480
KRFDALRTGE YGELPVTGCN GILAYRRGRG ENGIIVAMNT LDRKENVVVE TGDSPFTVKA 540
FESLKDEERL NVDDKKRINIC LNPFEWRIYK ACCEL 575

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SEQ ID NO: 96      moltype = AA length = 655
FEATURE           Location/Qualifiers
source            1..655
                  mol_type = protein
                  organism = Thermobifida fusca

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SEQUENCE: 96
MIGRFPILDV SPVVDIGTAK AVVGETFPVR ATVFREGHEA LGAGVVLYTP EGQRQPLVPL 60
REIAPGTRDY EAEVTVTSEG LWHFAIEAWS DPYATWCHDA RIKIPAGQDV ELMLEEGARL 120
LERAARRVPR PALAEIAAA MRDGSRSAHE RLDLALSILV RDELAERPLR ELVTRSORFP 180
VMVSRRRALF GSWYEFFPRS EGAVLDTEDG EPRSGTFATA ARRLPAIADM GFDVVYIPPI 240
HPVGYSFRKG RNNSTVAQPG DPGSVWAIGS HEGGHDAIHP DLGTIDDFDA FVARARELGL 300
ETAMDALQA SPDPHPVKEH PEWFTVRADG SIAYAENPPK KYQDIYPINF DKDPEGIFTE 360
VRRIVRYWMS HGVRIFRVDN PHTKPVAFWE RLLADIAATD PDVIFLSEAF TRPAMMHTLA 420
KIGFHQSYTY FTWRNTQQL EEYLTELTEG AAAYMRPNFF VNTPDILHAY LQHGGRPAFE 480
VRAILAATLS PTWGMYSGYE LCENRALKPG SEEYLDSEKY QYKPRDWEAA EAAGITITPL 540
IRKLNSLRSS HPALQELRNL RFHYADQPEI ICYSKRLAGA NHGADDTILV VANLDPHHTR 600
EATVWLDMPA LGFAPGDHIT VTDQLSGHSY HWVEANYVRL DPHVQTAHIF TVAPA 655

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SEQ ID NO: 97      moltype = AA length = 572
FEATURE           Location/Qualifiers
source            1..572
                  mol_type = protein
                  organism = Thermobifida fusca

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SEQUENCE: 97
APSGNRDVIV HLFQWRWXS ADECRITLGP HGFQAVQVSP PQEHVVLPAE DYPWWQDYQP 60
VGYKLDQTRR GSRADFIDMV NTCREAGVKI YVDAVINHMT GTGSAGAGPG SAGSSYSKYD 120
YPGIYQSQDF NDCRRDITNW NDKWEVQHCE LVGLADLKTS SPYVQDRIAA YLNLIDLGV 180
AGFRIDAARK IPEGDLQAIL SRLKNVHPAW GGGKPYIFQE VIADSTISTG SYTHLGSVTE 240
FOYHRDISHA FANGNIAHLT GLSGSLTPSD KAVVFPVNH D TQRYEPILTH TDGARYDLAQ 300
KFMLAHPYGT PKVMSSYTWS GDDKAGPPMH SDGTTTPTDC SADRWLCEHR AVAGMVGPHN 360
AVAGQGIGSA VTDGNGRLAF ARGSAGYAAF NATNTAWTRT FTTSLPDGVY CDVANGTFVD 420
GVCDGPSYQV SGGKFTATVP ANGAVALHVE APGSCGPDGC GTPPGGGDDC TTVTARFHAT 480
VTTWYQGEVA VVGSIPELGS WQPAQGVRLR TDSGTYPVWS GAVDLPAVG FEYKYVKLNP 540
DGTVEWEQGG NRIATVDDSG GGCSQNFYDS WR 572

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SEQ ID NO: 98      moltype = AA length = 825

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-continued

FEATURE
source

Location/Qualifiers
1..825
mol_type = protein
organism = Anaerocellum thermophilum

SEQUENCE: 98

MLVRAYIDDF	NEIVVVLSQL	VHSVKKEDFK	VFLNEEEIDI	EKIDKIIPHS	DNPAAEATRG	60
YEICEQKGKI	RFVLKEGHFD	YHRKPYKKPV	FVIGEMNDWQ	ISPEWEMTYS	KLRGRYELIK	120
DLKEIKIGQK	FKFAEGASQK	LWYPPGFGND	IVITEYFDRE	TAFTNMIRII	PSNRLLPNLK	180
YKVVKSEHI	WARPREILTR	PEFFYPGELG	IKYEPYGTYP	KLWAPTAYKV	KVKVPDESEN	240
FRFEKEMARS	ENGWNIIYLT	DLKNHYLY	EVWHYNYDED	BGFIVYVDP	PYSKASSNS	300
QKSFIIDPAD	TLIEGWQQDE	FVKTIEKQD	AIIYEMHVRD	FTIDKNSGVD	EKFRGKFLGL	360
CQKSPYKEKF	STGLLHLKEL	GITHIHLPI	SDFGSVDDKN	PDKKYNWGYD	PVLYQCPEYW	420
YSTKSGGIEA	LKELKTMKT	LHQNGIGVVM	DVFNHTYHT	KGGKFSIFDK	IVPGYFYRID	480
DYGDYSNATG	CGNEIATEKP	MVRKFILDTI	IYWTEDFHID	GFRFDLMGLI	DTLTMRMIK	540
EVKRKNPYAL	IYEGGWVMD	SMCLLEERAT	IESTAHHGYS	IGLFNDRIRD	SIRGDLGDFK	600
TGYMHGNLSD	IERLKQGIRA	AIDDFAKEPD	ECVNVYSCHD	NLTFLDKAQK	TMVGEDIFWI	660
DRVCRLANAI	ILTSQGIPFL	HGGVEFNRSK	GGHPTYNAG	DNINKIDWSL	KEKFYDTFKF	720
YCDLKLKRKE	HVAFRMRSSG	EIRKYLKFLP	APDGIVAFLI	SYPYDAWKKI	IVAYNPFKEK	780
KVITLPEGVW	KIKANDGIIF	SENELEAIG	SFEISPVSLF	IAYQK		825

SEQ ID NO: 99

FEATURE
source

moltype = AA length = 1104
Location/Qualifiers
1..1104
mol_type = protein
organism = Anaerocellum thermophilum

SEQUENCE: 99

DEKTLIIHY	YRYNEDYQGW	NLWIWPVEPV	GAEGKAYEFT	SKDDFGVKAV	VELPGKVTKV	60
GIIVRKGWNE	AKDVAVDRFI	SGISGSKEVW	LIEGEEQIYT	SQPKTPKMT	AFIDGLNTIV	120
VKLAKKADIL	SNNRTQGPKV	TAFYEEVPIK	KVEPVLPKIN	KNFKPEEAGY	ELIDGGTKVK	180
FILKPGAGDF	KFTDTSGLKD	VYVSGTMNDW	GGTASSEGGY	KPLPAWKMTW	NAEKGYVELV	240
KELGKDGVI	GAKPKFTSWD	GTSAKWYDPG	MGNDKVIEL	YTGNEKITKV	DTFKITTEDE	300
LEPQVPYVVS	KDSFKPTVAQ	ARNILDNPKY	YYKGNLGGCT	YTKAYSAPRL	WAPTAIGVIL	360
RLYDDYKTK	YKEYMQQS	NGTWYKING	DLKGKYQYE	VWHSNSITD	DTIRKVVVPD	420
PYSRATSANS	ERTLIFDPKD	TNPVGWEKDT	FVTLKNQEDA	IYETHVRDF	TIDASSGVRP	480
EPRGKYLGT	QTGAKPGNGV	KTGIDHLKEL	GITHVHLPT	YDFGSIDETN	PDKGYNWGYD	540
PVLYQNVES	YATNPNTIVR	IKYKQMVMA	LHKAGIGIIQ	DVFNHTFQI	GDAKFSIFDK	600
IVPGYFYRKD	KDGNYSNAG	CGNEIATEKP	MVRKFIIIDL	TYLTKEYHID	GFRFDLMAAI	660
DRVMTAKAQE	EVRKINPSAV	IYEGEWLAGS	TPLDSSLRME	IGSFNQAGLH	IGLFNDRIRE	720
AIRGNLDNES	KGFMQGNYSF	RLEDLKRGIQ	GGLGDFAADP	DECINYVSAH	DNLTLDWDLQ	780
KSPVNEPDYI	KDKMGRLANA	IVLTAQGVPF	LHGGVEFNRT	KYMNHNSYNA	GDKINKYNWN	840
LKVKWYNTFK	YYQGLIALRK	AHPAFRMTTA	EDIQKYLTFI	QTPKGTGLGR	LYPKDTWND	900
IIVVYNSTKK	VQEVTLPEGN	WVVVANGDEV	GTTPIKNLTN	FVAGKALVAP	ISMVYAYKSN	960
EPFQGTQKVT	GKDPVSLSS	STVTPKVYV	NGNIEVTFKV	KVPHGTDGDD	IYLAGSFGKA	1020
GLSDWNPBGD	DGAIELVRLQ	DGYTIVTVKL	NAGETFEYKY	TRGSWTTVEK	GANKKEIENR	1080
KLTVKDEGGG	KMIVSDTVLN	WADK				1104

SEQ ID NO: 100

FEATURE
source

moltype = AA length = 611
Location/Qualifiers
1..611
mol_type = protein
organism = Anaerocellum thermophilum

SEQUENCE: 100

MRKPHIEAI	IGNTKVLGQL	DSNGILQRFY	WPAVDYYQQL	KLFLAAVFLD	GLVFPEDENF	60
KTKSGFVDDF	VYFPEYKIAD	KTIFQLDFVD	FETDSLVLRLW	ETGFEDPYVF	LEPMINSSSL	120
FNAAKVDKEN	BIVYAYFKGT	YIGLAFENKI	KSFTVKNGID	DANDNQLEGW	NEATNPQIAV	180
KLKNTGKVV	FLAFGNSKDE	IYQKLSYLKQ	KGYDEVYRQN	KAFWEKKFSK	VKLICTQDPK	240
DMQLQKRSAY	VFYVLQNSKT	GGILAAASEVD	EKFPHCGGYG	FVWGRDAAFI	VSAMDELGLS	300
REVEKFFGFK	FSCQKEGFW	DQRYTDTGSL	APSWGQIDE	TASVVWGFL	HCEKQNSLHL	360
IDLHKEQLKK	ALLFLIAAVD	SEKGVIFRSF	DLWEEREGIH	LYSNASIYAA	LKKAKKYFPE	420
LESEIEKKLK	AIKNQMATRF	YSPKLSRYVR	STDVRIPHEE	FLKLPEENRY	MQKDERYEIT	480
YFVKQDEVV	DISMLGIYYP	FEMVDSDDKA	FKATILAIER	ECQNSIVGGY	KRYSDDRYIG	540
GNPWILTTLW	LAIIYKKTGQ	IDRAEKLFEW	AKAHSPLNGL	PPEQVDRITG	KPAWVPLAW	600
SHAMVVLIIY	E					611

SEQ ID NO: 101

FEATURE
source

moltype = AA length = 529
Location/Qualifiers
1..529
mol_type = protein
organism = Streptomyces avermitilis

SEQUENCE: 101

MTSFRPAPAW	LADAVFYQIY	PQSFADSDGD	GIGDFNGIVQ	RLDHLVWLV	TAVWLNPCFV	60
SPFRDAGYDV	SDYLNVAAPRY	GSADDLAELV	DEAGRRGIRV	LLDLVAGHTS	DEHPWFTASA	120
NDPDDHRYIW	APGGRPDGTV	TSPGTRPGAY	LPNFFDTQPA	LNFGYGRKNP	AEPWRQPVDA	180
AGPRANREAL	RTIMDHWLGL	GLAGFRVMDA	ASLVKDDPGR	TETARIWTEL	RHWLDTAHPD	240
AVLLSEWGEP	EVSVFAGFHT	DFFLQFGGAT	DGLPLRSLWS	NGDGTVNEAW	DPLDCFFDAS	300
GKGSPPPFVE	AWRKASDAVG	ATGFVSLPTA	NHDFSRLNCG	PRTAEQLPAA	FAFQLTWPTL	360
PAIYYGDEIG	MRYVGLPLDK	EGSVLGPFRY	RAGSRTPMQW	DDGPGAGFST	APADRLLPL	420
DSPDRPTVA	AQRADDGSL	HLVRLVALR	ASTPALGSGG	SVEVLHTGYP	FVYVRGGRYL	480
VVVPQRNEV	RCPYDATREA	RALEASGVRV	GNGTIEAEGF	SYGVFDLGR		529

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SEQ ID NO: 102 moltype = AA length = 431
 FEATURE Location/Qualifiers
 source 1..431
 mol_type = protein
 organism = *Streptomyces avermitilis*

SEQUENCE: 102

SPPGTKDVTA	VLFWEKFDVS	ARECTNTLGP	AGYGYVQVSP	PAEHIQGSQW	WTSYQPVSYSK	60
IAGRLGDATA	FQNMINTCHT	AGVKVVVDTV	VNHMSAGSGT	GTGGSAYTKY	NYPGLYSSYD	120
MDDCTATITD	YTNRAVQNC	ELVGLADLDT	GEEYVRKTIA	GYMNTLLGYG	ADGFRVDAVK	180
HIPAADLANI	KSRLTNPSVY	WKQEVIIYASG	EAVQPTTEYTG	NGDVQEFRYA	YDLKRVFNNE	240
NLAYLKNYGE	GWGYLNSVA	GVFVDNHDTE	RNGSTLNYKD	GANYTLANVF	MLAYPYGAPD	300
INSGYEWSDA	DAGPPGGGTV	NACWQDGWKC	QHAWPEIKAM	VAFRNATRGE	SVTNWWDNGG	360
DAIAFGRGAK	GYVAINHESG	SLRTRYQTSL	TAGTYCNVQN	NTGVTVDSSG	RFTATLGANT	420
ALALYSGKST	C					431

SEQ ID NO: 103 moltype = AA length = 503
 FEATURE Location/Qualifiers
 source 1..503
 mol_type = protein
 organism = *Saccharomycopsis fibuligera*

SEQUENCE: 103

LPLQEGPLNK	RAYPSFEAYS	NYKVDRTDLE	TFLDKQKDVS	LYYLLQNIAY	PEGQFNDGVP	60
GTVIASPTST	NPDIYYQWTR	DSAITFLTIVL	SELEDNNFNT	TLAKAVEYYI	NTSYNLQRTS	120
NPSGSFDDEN	HKGLGEPKFN	TDGSAYTGAW	GRPQNDGPAL	RAYAISRYLN	DVNSLNKGKL	180
VLTDSGDINF	SSTEDIYKNI	IKPDLEYVIG	YWDSTGFDLW	EENQGRHFFT	SLVQQKALAY	240
AVDIAKSPDD	GDFANTLSST	ASTLESYLSG	SDGGFVNTDV	NHIVENPDLL	QQNSRQGLDS	300
ATYIGPLLLH	DIGESSSTPF	DVDNEVVLQS	YVLLLEDNKG	RYSVNSAYSA	GAAIGRYPED	360
VYNGDGSSEG	NPWFLATAYA	AQVPYKLVYD	AKSASNDITI	NKINYDPFNK	YIVDLSTINS	420
GYQSSDSVTI	KSGSDEFNTV	ADNLVTFGDS	FLQVILDHIN	DDGSLNEQLN	RNTGYSTSAY	480
SLTWSSGALL	BAIRLRNKVK	ALA				503

SEQ ID NO: 104 moltype = AA length = 497
 FEATURE Location/Qualifiers
 source 1..497
 mol_type = protein
 organism = *Saccharomycopsis fibuligera*

SEQUENCE: 104

VPVELDKRNT	GHFQAYSQYT	VARSNFTQWI	HEQPAVSWYI	LLQNIIDYPEG	QFKSAKPGVV	60
VASPTSEPD	YFYQWTRDTA	ITFLSLIAEV	EDHSFSNTTL	AKVVEYYISN	TYTLQRVSNP	120
SGNFDSPNHD	GLGEPKFNVD	DTAYTASWGR	PQNDGPALRA	YAIRSYLNAV	AKHNNKLLL	180
AQONGIPYSS	ASDIYWKIIK	PDLQHVSTHW	STSGFDLWEE	NQGTTHFFAL	VQLKALSYGI	240
PLSKTYNDPG	FTSWLEKQKD	ALNSYINSSG	FVNSGKKHIV	ESPQLSSRGG	LDSATYIAAL	300
ITHDIGDDDT	YTPFNVDSY	VLNSLYLLV	DNKNRYKING	NYKAGAAVGR	YPEDVYNGVG	360
TSEGNPWQLA	TAYAGQTFYT	LAYNSLKNKK	NLVIEKLYND	LYNSFIADLS	KIDSSYASKD	420
SLTLTYGSDN	YKNVIKSLQ	FGDSFLKVL	DHIDDNGQLT	BEINRYTGFG	AGAVSLTWSS	480
GSLLSANRAR	NKLIELL					497

SEQ ID NO: 105 moltype = AA length = 747
 FEATURE Location/Qualifiers
 source 1..747
 mol_type = protein
 organism = *Saccharomyces cerevisiae*

SEQUENCE: 105

FPTALVPRGS	SSSNITSSGP	SSTPFSSATE	SFSTGTTVTP	SSSKYPGSKT	ETSVSSTTET	60
TIVPTTTTTT	VITPSTTTIT	TTVCSTGTNS	AGETTSGCSP	KTITTTVPCS	TSPSETASES	120
TTTSPTTPVT	TVVSTTVVTT	EYASTSTKQG	GEITTTFTVK	NIPTTYLTTI	APTSSVTTVT	180
NFTPTTITTT	VCSTGTNSAG	ETTSKSPKT	VTTTVPCTSG	TGEYTTATA	PVTTAVTTTV	240
VTTESSTGTN	SAGKTTTSYT	TKSVPTTYVF	DFGKGILDQS	CGGVFSNNGS	SQVQLRDVVL	300
MNGTVVYDSN	GAWDSSPLEE	WLQRQKKVSI	ERIFENIGPS	AVYPSILPGV	VIASPSQTHP	360
DYFYQWIRDS	ALTINSIVSH	SADPAIETLL	QYLNVSFHLQ	RTNNTLGAGI	GYTNDTVALG	420
DPKWNVDNTA	FTEPWGRPON	DGPALRSIAI	LKIIDIYIKQS	GTDLGAKYPF	QSTADIFDDI	480
VRWDLRFIID	HWNSSGFDLW	EEVNGMHFFT	LLVQLSAVDR	SLSYFNASER	SSPFVEELRQ	540
TRRDISKFLV	DPANGFINGK	YNYIVETPMI	ADTLRSGLDI	STLLAANTVH	DAPSASHLPF	600
DIDDPAVLNT	LHHLMLHMRS	IYPINDSSKN	ATGIALGRYP	EDVYDGYGVG	EGNPWWLATC	660
AASTTLYQLI	YRHISEQHD	VVPMNNDCSN	AFWSELVFSN	LTTLGNDEGY	LILEFNTPAF	720
NQTIQKIFQL	ADSFVLKLKA	TWEQTGN				747

SEQ ID NO: 106 moltype = AA length = 621
 FEATURE Location/Qualifiers
 source 1..621
 mol_type = protein
 organism = *Aspergillus niger*

SEQUENCE: 106

NVISKRATWD	SWLSNEATVA	RTAILNNIGA	DGAUVSGADS	GIVVASPSTD	NPDIYFTWTR	60
DSGLVLKTLV	DLFRNGDTS	LSTIENYISA	QAIVQGISNP	SGDLSSGAGL	GEPKFNVDTE	120
AYTGSWGRPQ	RDGPALRATA	MIGFGQWLLD	NGYTSTATDI	VWPLVRNDLS	YVAQYWNQTG	180
YDLLEVNGSS	FFTIAVQHRA	LVBGSAPATA	VGSSCSWCDS	QAPEILCYLQ	SFWTGSFILA	240
NPDSSRSKAD	ANTLLGLSIH	TFDPEAACDD	STFQPCSPRA	LANHKEVDS	FRSIYTLNDG	300

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LSDSEAVAVG RYPEDTYNG NPWFCLTLAA AEQLYDALYQ WDKQGSLEVT DVSLDFFKAL 360
YSDATGTYSY SSSTYSSIVD AVKTFADGFV SIVETHAASN GSMSEQYDKS DGEQLSARDL 420
TWSYAALLTA NNRRNVVPSA SWGETSASSV PGTCAATSAT GYTSSVTVTS WPSIVATGGT 480
TTTATPTGSG SVTSTSKTTA TASKTSTSTS STSCTTPTAV AVTFDLTATT TYGENIYLVG 540
SISQLGDWET SDGIALSADK YTSSDPLWYV TVTLPAGESF EYKFIRIESD DSVEWESDPN 600
REYTVPQACG TSTATVTDTW R 621

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SEQ ID NO: 107      moltype = AA  length = 593
FEATURE            Location/Qualifiers
source             1..593
                   mol_type = protein
                   organism = Aspergillus oryzae

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SEQUENCE: 107
VQPVLRQATG LDTWLSTEAN FSRQAILNNI GADGQSAQGA SPGVVIASPS KSDPDYFYTW 60
TRDSGLVMKT LVDLFRGGDA DLLPIIEEFI SSQARIQGIS NPSGALSSGG LGPEKFNVDL 120
TAFTGAWGRP QRDGPALRAT AMISFGEWL V ENSHTSIATD LVWPVVRNDL SYVAQYWSQS 180
GFDLWEEVQG TSFPTAVASH RALVEGSSFA KTVGSSCPYC DSQAPQVRCY LQSFWTGSYI 240
QANFGGGRSG KDINTVLGSI HTFDPQATCD DATFQPCASR ALANHKVSTD SFRSIYAINS 300
GRAENQAVAV GRYPEDSYN GNPWFLLTLA AEQLYDALY QWDKIGSLAI TDVSLPFFKA 360
LYSSAATGTY ASSTTVYKDI VSAVKAYADG YVQIVQTYAA STGSMAEQYT KTDGQSQTAR 420
DLTWSYAALL TANNRRNAV V PAPWGETAAT SIPSACSTS ASGTYSVVVI TSWPTISGYP 480
GAPDSPCQVP TTVSVTFVAV ATTVYGESIK IVGSISQLGS WNPSSATALN ADSYTTDNPL 540
WTGTINLPAG QSFEYKFIRV QNGAVTWESD PNRKYTVPST CGVKSQVQSD VWR 593

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SEQ ID NO: 108      moltype = AA  length = 579
FEATURE            Location/Qualifiers
source             1..579
                   mol_type = protein
                   organism = Rhizopus oryzae

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SEQUENCE: 108
ASIPSSASVQ LDSYNYDGST FSGKIYVKNI AYSKKVTVIY ADGSDNWNMN GNTIAASYS 60
PISGSNYEYW TFSASINGIK EFYIKYEVSG KTYYDNMNSA NYQVSTSKPT TTTATATTTT 120
APSTSTTTPP SRSEPATFPT GNSTISSWIK KQEGISRPFAM LRNINPPGSA TGFIAASLST 180
AGPDYYYAWT RDAALTSNVI VVEYNTLTSG NKTILNLKLD YVTFSVKTQS TSTVCNCLGE 240
PKFNPDSAGY TGAWGRPQND GPAERATTFI LFADSYLTTQ KDASYVTGTL KPAIFKDLDY 300
VUNVWSNGCF DLWEEVNGVH FYTLMVMRK G LLLGADFAKR NGDSTRASY SSTASTIANK 360
ISSFWVSSNN WIQVSQSVTG GVSKKGLDVS TLLAANLGSV DDGFPTGSE KILATAVAVE 420
DSFASLYPIN KNLPSTYLGNS IGRYPEDTYN GNGNSQGNSS FLAVTGYAEL YYRAIKEWIG 480
NGGVTVSSIS LPFFKKFDSS ATSGKKYTVG TSDFNLLAQN IALAADRFLS TVQLHAHNNG 540
SLAEFDRRTT GLSTGARDLT WSHASLITAS YAKAGAPAA 579

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SEQ ID NO: 109      moltype = AA  length = 644
FEATURE            Location/Qualifiers
source             1..644
                   mol_type = protein
                   organism = Clostridium thermocellum

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SEQUENCE: 109
MANTYFNDAI IGNSGMLVCL TRNGELTRLF WPNIDYPQHF EKMATGIFYT GQKNSTSWFY 60
EDNWHHTQYY VEDTNILKTI CEDGGRGLRV EQTDFVLKDR DVMVRRYVIE NIGPNEVDLG 120
FVQYSSTVST TPELRSTLFD FNVDAIHYR HNYIYISISS SEVVQFQLGN NAFDCARYTE 180
LYGYDSIGMM KDGAMSFNIG KIEPBGKKT F NLFIASHTL KGVKELVRWC RKMNVDEEYE 240
KTRKYWLDLF KNARLIVTGD KNIDNLYKRS ILVFKLMSDE RTGGLLASAE IDEGFTRCGR 300
YAYCWGRDAA FITGALDTAG LTEAVDKFYQ WAVMTQDDDG SWQORYHMDG NLAPSWGLQI 360
DETGTLIWGM LKHVEVTKNI DFLKSMWESI KKGVEFLTRF IDSDTGLPAP SYDLWEERVG 420
EHTYSSAAVY AGIKAGAEAA RILGASEELI EKWEKAASDM KASIEKNFWR DEAGRFRISV 480
RTKLPNWGSE HSPYTTVIKV NEKGYFRDVT LEDWTIDVSL LGVSIIPGVF DVHDERVKKT 540
VEAIERALTS HPVGGIKRYE NDNYIGGNPW VLATLWVALY YIEIKEYEKA KDYLWATKS 600
CTALGLLPEQ VSKDNGEPCW VIPLTWSHAM YVLVLALKE AGVL 644

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SEQ ID NO: 110      moltype = AA  length = 644
FEATURE            Location/Qualifiers
source             1..644
                   mol_type = protein
                   organism = Clostridium thermocellum

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SEQUENCE: 110
MQKSYNNNAI TGNSSMLACF SERAELLRLF WPDIDYIQL DKMFLGLFEK NKTGSTVWLN 60
DIRCEHHQYV LPSDNIKNM VTNFFDGYKV VLYDFVHPM DVLVRRFEIE NLRGESREL 120
LMSFSAATSS DSEVACSLFD FMNEALVHYK PDSYIAVTS IPVYQFQIGN NANDAAVNTY 180
LYGKDDIGMM KDAISWDLG VFQPHAVKTT NVYLCAADTL KSCKALVRRV KTVGGLTAFR 240
ETGRYWKDYL EKTTLKSGN TLLDDLYKRS LLVFRMLYSK KSGGLMAAPE VDEYFTKCGK 300
YAYCWGRDAA FITGALDIG LCESVDHFKY WAVNVQDEDG SWQORYHMDG NLGPCWGLQV 360
DETGTIIWGM LNHNYNTKNT DFLKSVWDSV KAAADFLVRV IDSETGLPRP SFDLWEERYG 420
EHAYSSASVC AGLKSASEMA RILGKPSQY I QWETTADSI KKAIVKYFWK EDYRRFIRSI 480
RVKLNFGQGE PSSDTMLIKV NPKGYVRDVT KEDWIVDVSL VGLGIPPEIF ELNDPMLRDT 540
VSLIEQVLTA QGVGGIKRYE NDTYIGGNPW ILTTLWIALY HAKSGNYKKA KEYLIWAASG 600
KTELGLLPEQ INRDTGKPEW IIPLTWSHAM YVHVYSELIN AGVL 644

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SEQ ID NO: 111      moltype = AA  length = 608
FEATURE            Location/Qualifiers

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-continued

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source          1..608
                mol_type = protein
                organism = Arxula adeninivorans

SEQUENCE: 111
DSCHTFTLAN SPPDDKAVAL SSYSYCGGYL SASAFVKNLS YDKLVTLYWT NADNKSTPLN 60
AGSLDYVKAA SDDQSWELWS LNVTTVPDGV DALLNITYVA ASIGKTNSQQ LNVQVEATGD 120
PIPTPQIPTI YKPYASPSDF SDDITNWLKP SNDSQTGIAK SFLFNNINIP GAAPGTVIAA 180
QSYSEPDYAY TWVRDASLVM DVVNRLYSSA KSEEKRLYE KILFQYAKAG AQEQNDPTAI 240
SGMGEPKFYL NNTAFTGSWG RPQNDGPATR AITLIEFANA YLANGGSQDT VREQLYDSK 300
YPQVAPIKDD LQFVASNWSS PSPDLWEEEEE SAHFYTRLVQ RKALLLGADF ANDMGDHELS 360
DKLKTQASKL SDTLPEFWDS ARQLILYEYG PVLRGKYSYK DISVVLGVMH GYANDNVFSY 420
TNDQILATAY QVSTSFLDVY KVANTTSDES GKPLGIPVGR YPEDVYDVG GTSQGNPWYLT 480
TMAMAEFLYR SVQEFEDAGS IISDTSLEPF WKYFASSVDH KAGAKYNKND QSPKTSLSKL 540
TGWGDAPFMR AKYHTPSSGH MSEEFNRTTG EPRGAKDLTW SYASLLSAAF AREELRNQKN 600
YLTNVADL 608

SEQ ID NO: 112      moltype = AA length = 595
FEATURE            Location/Qualifiers
source              1..595
                    mol_type = protein
                    organism = Hormoconis resinae

SEQUENCE: 112
APTELKARDL SSFIASERAI ALQGALNNIG PDGSAPVGAG AGFVVASPSK ANPDYFYTWS 60
RDSALTLKMI IDEFILGNTT LQTIIIEQYIH AQAVLQTVSN PSGTFLPDGV GLGEPKFMVD 120
GTRFNGPWGR PQRDGPALRA IALMTYSNWL IKNGQFAEAK TKIWIPIAND LSYVGQYWNQ 180
SGFDLWEETY ASSFFTQIQN HRALVEGAQL AHDLGVTCTG CDQAPEVLCF LQSFWNGKYI 240
VSNINVNNGR TGLDGNILG AISTFDIDAY CDSPTLQPC H SGLANFKVL TDTFRNLYTI 300
NAGIPEGQGV AVGRYAEDVY MGGNPWYLIT TAAAEFLYDA VAQWKARHVL TVDETSLAFF 360
KDIYPEVTVR EYKSGNANSP FAQIMDAVTA YADSYVAIAE KYIPSNGLS EQFNRTGTGP 420
LSAIDLTSY AAFITMSQRR AGQYPSWGS RNALPPPTTC SASSTPGIYT PATAAGAPNV 480
TSQCQVSITF NINATTYYGE NLVIGNSSD LGAWNIDAY PLSASAYTD RPLWSAAIPL 540
NAGEVISYQY VRQEDCDQPY IYETVNRTLT VPACGGAATV TDDAWMGPGV SSGNC 595

SEQ ID NO: 113      moltype = AA length = 601
FEATURE            Location/Qualifiers
source              1..601
                    mol_type = protein
                    organism = Aureobasidium pullulans

SEQUENCE: 113
LSPESIQRER ATGSLSSWLS SENTVALQGV LNNIGASGSK ASGASAGVVV ASPSKSNPDY 60
FYTWTRDSAL VFKAIVDQLI AGNKSLEPLI QQYISAQAKL QTVNNPSGGL CSGGLAEPKF 120
EVDLTPFTGA WGRPQRDGA LRATAMIAYS RYLIANGNTT TVNNIIWPIV QNDLSYVTQY 180
WNQTTFDLWE EINSSSFFT AVQYRALVEG NNLATQLGKS CPNCVSQAPL VLCFLQSYWT 240
GSYALSNTGG GRSGKDANSI LTSIHIFDPA ASCDSTTFQP CSDKALANHK VVTDSPRSYI 300
SINQGIAGQS GVAVGYPED SYYNGNPWYL NTFAAAEQLY DAVYQWKIG SISITSISLP 360
FFKDVYSSAA VGTYSSTVT FTSIVNAVQT YADSYMSIAQ KYTPSNGALS EQYNRADGTP 420
LSAVDLTSY AAFLTAYNAR ANVLPASWGA SSAKLPNSCS SGSATGPCAA ATNTNWGNPG 480
SPSTGTPTTT TGGSCPTPTS IAVTFNEQKT TSYGENIYIV GSIPALGNWN TANAVALSAS 540
KYTSSNPLWT VTINFATGTS FNYKYIKKAQ DGSVTWESDP NRSYTVTGNC AGTATENDSW 600
R 601

SEQ ID NO: 114      moltype = AA length = 718
FEATURE            Location/Qualifiers
source              1..718
                    mol_type = protein
                    organism = Bacillus subtilis

SEQUENCE: 114
MVSIRRSFEA YVDDMNIITV LIPAEQKEIM TPPFRLETEI TDFPLAVREE YSLEAKYKYV 60
CVSDHPVTFG KIHCVRASSG HKTDLQIGAV IRTAAFDEF YYDGELGAVY TADHTVFKVW 120
APAATSAAVK LSHPNKSGRT FQMTRLEKGV YAVTVTGDH GYEYLFICN NSEWMTVDQ 180
YAKAVTVNGE KGVVLRPDQM KWTAPLKPFS HPVDAVIYET HLRDFSIHEN SGMINKGKYL 240
ALTETDTQTA NGSSSGLAYV KELGVTHVEL LPVNDFAGVD EEKPLDAYNW GYNPLHFFAP 300
EGSYASNPHD PQTRKTELKQ MINTLHQHGL RVILDVVFNH VYKRENSPFE KTVPGYFFRH 360
DECGMPSNGT GVGNDIASER RMARKFIADC VVYWLEEVN DGFREFDLGI LDIDTVLYMK 420
EKATKAKPGI LLFGEGWDLA TPLPHEQKAA LANAPRMPI GFFNDMFRDA VKGNTFHLKA 480
TGFALGNGES AQAVMHGIA GSSGKALAPI VPEPSQSINY VESHDNHTFW DKMSFALPQE 540
NDSRKRSRQR LAAAILLAQ GVPIHSGQE FRFTKQGVEN SYQSSDSINQ LDWDRRETFK 600
EDVHYIRRLI LRSKAHAPFR LRSAAIDIQH LECLTLKEHL IAYRLYDLDE VDEWKDIIVI 660
HHASPDSEVW RLPNDIPYRL LCDPSGFQED PTEIKKTAVV NGIGTVILYL ASDLKSFA 718

SEQ ID NO: 115      moltype = AA length = 710
FEATURE            Location/Qualifiers
source              1..710
                    mol_type = protein
                    organism = Bacillus licheniformis

SEQUENCE: 115
MPGISRPFEA YLDEMRTITV LVPKSRASSC SPPFLLEDQ GERIELSVKA QVELEEQFKY 60
VLESSCTVPF SVVHKVCCCE SVWTDLQIGS VTRSAAFKA FFYDGRGAF YSKGSTLFKV 120
WAPTASAAAI KLEDPDSLQT NTPQMRRKK GVFEVTVEGD LNGWSYLYEL YVNGKPLLTV 180

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DPYAKAVTAN	GEKGVVLDPE	EVKVEKHRAP	RLHSPCDAVI	YEVHIRDFSI	HEDSGMRHKG	240
KVVAFTEDGT	ETSGGFSTGI	AYLKELGVTH	IEVLPPHDFP	GVDELSPDQS	YNWGYNPLHF	300
NAPEGSYSLD	PQNPCKRITE	LKTMIQSLHK	HGFSVIMDAV	YNHVYKRETS	PFEKTPVPGYF	360
FRHNEYGFPS	DGTGVNDIA	SERLMVRKYI	LDSVRYWLEE	YDVGIRFDL	MGILDITVR	420
QISTLAENVK	PGVPLFGEGW	DLNTPLDGQ	KATLQNAKGV	PAVGFFNDRF	RNAVKGSTFE	480
LSDRGYALGD	TGKKAALQHG	IAGSPGFLQP	AQSINYVECH	DNHTFWDKMA	LCFEEDADTK	540
RLRQRLAVSI	VLLSQGVFPL	HAGQEFCTRK	NGDSNSYRS	DDINKLDWEK	RAELCEDVEY	600
VRQLIRLRRS	HPAFRLQKEE	EVKEHLSFMD	GTGEVTAYKL	KNIAAIDPWN	EIIIVHCPFA	660
KKETLKLDPQ	KQYLLHCDPF	TFPNGKVQAE	KRLRLNGIGT	YVLYEPKGIF		710

SEQ ID NO: 116 moltype = AA length = 918
 FEATURE Location/Qualifiers
 source 1..918
 mol_type = protein
 organism = Oryza sativa

SEQUENCE: 116

MAVGEECAAA	VASQGFVSDA	RAYWVTRSLI	AWNVDQDTS	LFLYASRDAT	MHVSDGAIHG	60
YDSKIELEPE	HASLPDNVAE	KFPFIRSIRT	FRVPSSVDVT	SLVKCQLAVA	SYDAHGRHQD	120
VTGLQLPGVL	DDMFAYTGPL	GAVFSDKDVD	LYLWAPTAQD	VRVCFYDGP	GPLLQTVQLK	180
ELNGVWSVTV	PRYRENQYYL	YEVKVYHPST	SQVEKCLADD	PYARGLSANG	TRTWLVDINS	240
ETLKPASWDE	LSDEKPNLES	FSDISIYELH	IRDFSAHDS	VDCNSRGGFR	AFTFQDSAGI	300
RHLRLLSAAG	LTHVHLPLSF	HFASVDDNKS	NWKFVDEAQL	AKLPPGSDEQ	QAAIVSIQOE	360
DPYNWGYDPV	LWGVKPGSYA	SNPDGSPRII	EYRQMVQALN	RIGLRVMDV	VYNHLDSSGP	420
FGVSSVLDKI	VPGYLLRRNV	NGQIENSAAM	NNTASEHFMV	DRLIVDDLNL	WAINYKVDGF	480
RPDLMGHIMK	STMTVMISIC	TISTIIKIKD	VFADTLIRAK	SAIRSLTRDV	HGVDGSKIYL	540
YGEGWDFGEV	AQNKRGINAS	QINMSGTGIG	SFNDRIRDSV	NGGNPFGNPL	QQGFSTGLFL	600
EPNGYYQONE	ADTRRELATY	ADHIQIGLAG	NLKDYLVRTH	TGEAKKGSDI	YTFDGSFVGY	660
TSSPVETINY	VSAHNDNTEF	DIVSIKTPIG	LSIDEKCRIN	HLASSMIALS	QGIPFFHAGD	720
EILRSKSLDR	DSYNSGDWFN	KLDFTYETNN	WGVGLPPRDK	NEENWHLIKP	RLENPSFRPL	780
KNHILSVFDN	FVDILKIRYS	SPLFRLSTAS	DIEQRVRFHN	TGPMVMVPGVI	VMSIKDAQNE	840
KCKMAQLDKN	FSYVVTIFNV	CPHEVSIIEH	DLASLGLLEH	PIQVNSSDAL	VRQSAYEASK	900
GRFTVPRRTT	AVFVQPRC					918

SEQ ID NO: 117 moltype = AA length = 963
 FEATURE Location/Qualifiers
 source 1..963
 mol_type = protein
 organism = Triticum aestivum

SEQUENCE: 117

MPMPMRMTLL	RHLSPAPALP	NPRRSSASSP	QRRPARARPP	PLHSARATAL	RARRTPMAAG	60
ETGASVSUSA	AEAEAEATQA	FMPDARAYWV	TSDLIAWNVS	EQEASVYLY	ASRTAAMGLS	120
PSNGGIQGYD	SKVELQPESA	GLPETVTQKF	PFISSYRAFR	VPSSVDVASL	VKQQLVIASF	180
GADGKHVDVT	GLQLPGVLDD	IFAYTGPLGA	VFREDSVSLH	LWAPTAQDVS	VCFDFGPGAP	240
VLETVQLKES	NGVWSVTGPR	EWENRYLYE	VDVYHPTKAQ	VLKCLAGDPY	ARGLSANGAR	300
TWLVDINNET	LKPASWDELA	DEKPKLDSFS	DITIYELHIR	DFSADHGTVD	SDSCGGFRAF	360
AYQASAGMOH	LRKLSDAQLT	HVHLLPSFHF	AGVDDIKSNW	KFVDECKLAT	FPPGSDMQQE	420
AVVAIQEEDP	YNWGYNPVLW	GVPKGSYASD	PDGSPRIIEY	RQMVQALNRI	GLRVVMDVVY	480
NHLDSSGPGC	ISSVLDKIVP	GYVVRDRTNG	QIENSAAMNN	TASEHFMVDR	LIVDDLNLWA	540
VNYKIDGFRF	DLMGHIMKHT	MMRAKAAALQ	LTRDAHGVDG	SKIYLYGEGW	DFAEVARNQR	600
GINGSQLNMS	GTGIGSFNDR	IRDAVNGGNP	FGNPLQGGFN	TGLFLEPNGF	YQGNEADTRR	660
SLATYADQIQ	IGLAGNLRDY	VLITHTGETK	KGSEIHTFDG	LPVGYTSSPI	EIINYVSADH	720
NETLFDVISV	KTPMNLSDNE	RCRINHLASS	MMALSQGIPI	FHAGDEILRS	KSIDRDSYNS	780
GDWFNKLDFT	YETNNWVGVL	PPSEKNEDNW	PLMKPRLENP	SFKPAKGHIL	AALDSFVDIL	840
KTRYSSPLFR	LSTASDIKQR	VHFHNTGPSS	VPGVIVMGIE	DARDEKPEMA	QLDANFSYVV	900
TVFNVCPHEV	SMDIPALASM	RLELHPVQVN	SSDALVGKSV	YEAATGRFTV	PRRTVSVFVE	960
PRC						963

SEQ ID NO: 118 moltype = AA length = 647
 FEATURE Location/Qualifiers
 source 1..647
 mol_type = protein
 organism = Clostridium phytofermentans

SEQUENCE: 118

MDEFWNSIDG	EKQYYIDGND	LGCTYTNRST	KLKVVAPTAS	MVVVNLYQNG	NAGKPYITEI	60
MKKEESGIWS	VCLLGDLLEG	YYTYLTVTDG	QTKEAVDPYA	RTTGLNGKRA	MILDLEKTNP	120
TGFLEDTKPK	FDSFLDAVIY	ELHIRDLSE	SDSGIKEGK	DLGLTELNTR	NSDGLTTGLS	180
HILDLGVTHI	HLRPCFVYAS	VEENSSIFN	WGYDPENYNV	VEGYSYTNPY	DGAVRVKEFK	240
TLVQSLHENG	LRVIMDVYIN	HTMKTEESNF	NKIVPDYYR	KVGDKFSDAS	ACGNETASER	300
LMVRKFIVDS	LIYWAKYHI	DGFRFDLMGI	HDIETMNEVR	KVLQIDPSI	ILYGEWVGVG	360
DSPLPAGQRA	MKANMSMLPG	IAAFSDDFRD	GLKGSVFLAE	EKGATGDSGD	KKESVKGFGV	420
ASTLHPQIDY	KKVNSYDSPW	ALEPAQCINY	VSAHDNYTLW	DKIACCKED	TYEIRVKKNK	480
LCAAIVFTSQ	GIPFLQAGEE	MLRNKPSSEI	AGEFVENSYN	SSDSVNCIKW	SNKANVIDVV	540
SYYEGILIRF	KEHKALRMQS	AKKISKRLTF	LPEEREDVIS	YLIQGDVLVK	TLCVIYNSSE	600
EKVITIRLPES	DWTVIIDGNN	SGVEPLYEVK	GTTVEVEPIS	CMVLVKD		647

SEQ ID NO: 119 moltype = AA length = 1764
 FEATURE Location/Qualifiers
 source 1..1764
 mol_type = protein

-continued

organism = <i>Streptomyces avermitilis</i>									
SEQUENCE = 119									
ATPPAPSPSDA	KLAAEAPARHD	ATREQYFVFM	PDRFANGDTS	NDKGGTLGSR	LSTGYDPTDK	60			
GFYQGGDLKG	LTRKLDYIKG	LGTTSIWLAP	IFKNQPVQGT	GKIDASAGYHG	YWITDFTKVD	120			
PHFGTNKDLK	TLISAKAHAK	MKVFPDVIAT	KHADVDVYEE	KSYGLYKGRG	FPYLTLKDRGP	180			
FDAGAYTDGP	RKFPADVGDG	FPRTPAVAAN	HTNPAKVPSWL	NDPMTYHNRG	DSTFAGESST	240			
HGDFSGLLDDL	WTEREPVEVRG	MEKIIYEKVV	DFGIDGFRID	TVKHHVTEFW	TQWATLADAY	300			
AKKRKGDDFF	MGEVYSADT	SVTSPYVTQG	RLDSTLDFFP	QDAARYSYQS	GGSAKKLASV	360			
FGDDYKYTTD	KANAYEQYTF	LGNHDMGRIG	YFLNQDNPKA	TDIAELLKRDR	LANELMFLSR	420			
GNPVVYYGDE	QGFTSGSGDK	DARQTMFAGS	VADYLDDEIE	GTDRGHASDA	YDTSAPLYKE	480			
IAALSKLRLK	NPALADGIQT	ERYAADGAGV	YAFSRDTDART	GTEYVVAVNN	ADKASATAFA	540			
TGSADATAFK	IHGDDTVLGS	DADKKITVTV	PAGAAVVLKA	AGRPSTPAAK	PSLTLKAPDA	600			
GATGTVELSA	DVDGGRLNRV	VFAAQVGAAN	WRTLGSADHA	PYRVQTQTIG	DVPAGTALRY	660			
KAVVIDAAGR	TASATAASTT	GTPPAAEPTP	ASSRDYAIHV	KYKPDGDDYD	WRLYANGDLA	720			
DGESTTWPAG	HDFVGRDAAG	AFAYVKLKPG	ASTVNFLVID	IDRGDDKVSAD	RTIDVTKAGE	780			
VWVEQKGETV	RTERPDYPAQ	DKTKAVIHXY	DARGDLTGWG	LHWTGTGAATP	TDWSKPLEPV	840			
TRDAYGAVFE	VPLTDGATSL	SYIIHKGDKE	DLSDADRSLD	TADGHEVWLL	NGQENHLLPP	900			
PAGSAALADD	TSKAVWIDR	NTVAMNGSDA	AASTQLSSLR	DGSIAVKDGS	LTSDDEBRWL	960			
LSKTSLTDQA	KAAFPHLKGS	TAMSWDPRDR	DRVREALAGS	VVASQRAANG	AVLAAATGVQL	1020			
AGVLDDLDLY	TKADLGPFTF	GGHPTLAWNA	PTAQSVSLEL	DGAHVMRKRN	NATGVMSVTG	1080			
PASWKGKPYR	YVVKWAPTVP	RKVVTNKVTD	PYSVALTTDS	ERSLVLDLDD	RS LAPSGWSS	1140			
LKKPKAVPLR	DAEIQELSHR	DFSVADARTVP	AKDRGYTLAF	TDKNSDGSRH	LRQLAESGYS	1200			
YVHLLPAFDI	ATIAEKKSGH	QATDCDLSAY	AADSEKQOEC	LTAVAAKAVR	NWGYDYPHFT	1260			
VPEGSYATDA	NGTRRTVEFR	RMVKSINQDG	LRVVMDEVYIN	HTAAAGQAGT	SVLDRIVPGY	1320			
YQRLLDAGSV	ATSTCCANTA	TENAMNGKLV	VDSVLTWAKE	YKVDGFRFPL	MGHQPKANIL	1380			
AVRKALDALV	VAKSDVGDKK	IILYGEKGNF	GVGLADARFV	QATQKNMAGT	GIATFSDRAR	1440			
DAVRGGGPPD	ADPGVQGGFG	GLYTDPNSSD	ANGTPAEQKA	RLLYHQDLIK	VGLSGNLAKY	1500			
RFTDSSGKEV	TQSGVDYNGT	GAGYADAPGD	ALAYADAHND	ESLFDALTYK	LPKGTPAGDR	1560			
ARMQVLAAMT	AALAGQSGSS	QAGSLDLLRS	SLDRNSYDSG	DNYNALHWNK	QYKNGGFRGL	1620			
PMAADNKS KW	PYATPLLTSTV	KVGC DQIEGT	SAGYQDLLRI	RTTEPDFSLS	TAGQVQSKLT	1680			
FYLSGKDEPT	GVI TMLKLDG	VVVFNATPDQ	QEQTVAALAG	KDYALHPVQA	AGADPIVKA	1740			
SPYTSKGMFA	VPGRTVAFPS	OVAR				1764			

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SEQ ID NO: 120      moltype = AA  length = 1079
FEATURE             Location/Qualifiers
source              1..1079
                   mol_type = protein
                   organism = Klebsiella pneumoniae
```

SEQUENCE: 120							
SSSPSGSGPG	SPGNPNPGBT	PGTPDPQDVV	VRLPDVAVPG	EAAQASANQA	VIHLVDIAGI		60
TSTPADYAT	KNLYLWNNET	CDALSA PVAD	WNVDS'TPTGT	SDKYPGYVIV	PLTKESGCIN		120
VIVRDKGTNKL	IDSDLRVSFG	DFTDRTVSNI	AGNSAVDYSR	ADAFRAAPGV	ALADAHVWDK		180
TLTLLWPGEN	KPIVLRYLAI	SKKVAADSNG	EFTDKYVKLT	PTTYQSQVSM	RPFHLSAYPA		240
FKLPPDVND	ELLQGETVSI	SAESDGILSS	AQVQTAGVL	DTDYAAAAEA	LSYGAGQLDS		300
HVPSRVWAPT	AQQVELLVHS	ADKKVVASHP	MTRDSASGAW	SQGGSGLDMKA	AFRYAMTYAV		360
GQTSFRKEQY	VEFDPYAHSL	STNSEYSQVV	LTRDASALKPE	GWGDGLTPHA	QKTKADLAKM		420
TIHESHIRDL	SAWDDQTPAE	LRGKYLA LTA	QESNMVQHKL	QLSASGVTHI	ELLPVFDDLAT		480
VNEFSKDVA	QDFPPSRCLC	INSAVKSSEF	AGCYDSGSTV	EEVLTLQOLQN	DSKDN PQVQA		540
LNTLVAQTDS	YNWGVDPPHY	TPEBGSATFD	PEGTARIKEF	RTMIQAIGLD	LGMNVMDVVU		600
YNHTAAGAAPT	RDTSLMDKIV	PWYYQRNLNT	TGSVESATCC	SGDAPEHQMF	AKLIADSLAV		660
WTTDYKIDGP	RPSVLMGYHPK	AQILSAWERI	KALNPDIYFF	SEDGWSNQSD	RFEIASQINL		720
KGTGITGFSD	RLRDARVGGG	PPDSGDALRQ	NQGVGSGAGV	LPNELTSMTD	DQARHLADLT		780
RKMAGNALND	FLVIDDKGAG	KKGSEIDYNG	APGGXAA DPT	EVFNVSXKHD	NQTLWDMDISY		840
CLAQEADLDT	FVRMQAVSLA	TVMLGQGI AF	DQCGSELLRS	KSFSTRPVDS	GDWFNRVDYS		900
LQDNMYNVGM	PRSSDDGSGNY	DIARVKDAV	ATPGETELKQ	MTAFYQELTA	LRKSSPLFTL		960
FAGATVMQVR	IIFRNTGADQQ	TGLLVMTIDD	GMQAGASLDS	VRDGVTVAIN	AAPSRTLQD		1020
GAGTSLQLSA	DOQAAGASDA	ASGVVOAMDG	SVTLPAWSVA	VLELPOGESO	GAGLPVSSK		1079

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SEQ ID NO: 121      moltype = AA  length = 546
FEATURE             Location/Qualifiers
source              1..546
                    mol_type = protein
                    organism = Rhizomucor pusillus
```

[illegible]

```
SEQ ID NO: 122      multype = AA  length = 481
FEATURE             Location/Qualifiers
source              1..481
                   mol type = protein
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organism = *Bacillus licheniformis*

SEQUENCE: 122

VNGTLMQYFE	WYTPNDGQHW	KRLQNDAEHL	SDIGITAVWI	PPAYKGTSSQA	DVGYGAYDLY	60
DLGEPHQKGT	VRTKYGTKE	LQSAIKSLHS	RDINVYGDVV	INHKGADAT	EDVTAVEVDP	120
ADNRNVISGE	HLIKAWTHFH	FPGRGSTYSD	FKWYWHFDG	TDWDESRKLN	RIYKFQKGTW	180
DWEVSNFEGN	YDYLMYADID	YDHPDVVAEI	KRWGTWYANE	LQLDGFRLDA	VKHIKFSFLR	240
DWVNVHREKT	GKEMFTVAEY	WSNDLGALEN	YLNKTNFNHS	VFDVPLHYQF	HAASTQGGGY	300
DMRKLLNGTV	VSKHPLKSVT	FVDNHDTPQG	QSLSTVQTW	FKPLAYAFIL	TRESGYPQVF	360
YGDMYGTKGD	SQREIPALKH	KIEPILKARK	QYAYGAQHDY	FDHHDIVGWT	REGDSSSVANS	420
GLAALITDGP	GGAKRMVYGR	QNAGETWHD	TGNRSEPVVI	NSEGWGEFHV	NGGSVSIYVQ	480
R						481

SEQ ID NO: 123 moltype = AA length = 484

FEATURE Location/Qualifiers

source 1..484

 mol_type = protein

 organism = *Aspergillus niger*

SEQUENCE: 123

LSAAEWRTQS	IYFLLTDRFG	RTDNSTTATC	DTGDQIYCGG	SWQGIINHLD	YIQGMGFTAI	60
WISPIEQLP	QDTADGEAYH	GYWQQKIYDV	NSNFGTADDL	KSLSDALHAR	GYMLMVDVVP	120
NHMGYAGNGN	DVDYSVDFPF	DSSSYFHPYC	LITDWDNLTM	VQDCWEGDTI	VSLPDLNTTE	180
TAVRTIWDW	VADLVSNYSV	DGLRIDSLE	VEPDFFPQGY	EAAGVYCVGE	VDNGNPALDC	240
PYQKVLGVL	NYPIYWQLLY	AFESSSGSIS	NLYNMIKSA	SDCSDPTLLG	NFIENHDNPR	300
FASYTSDYSQ	AKNVLSYIFL	SDGIPIVYAG	EEQHYSGGKV	PYNREATWLS	GYDTSAELYT	360
WIATTNAIRK	LAISADSAYI	TYANDAFYTD	SNTIAMRKGT	SGSQVITVLS	NKGSSGSSYT	420
LTLSGSGYTS	GTKLIEAYTC	TSVTVDSSGD	IPVPMASGLP	RVLLPASVVD	SSSLCGSGSR	480
LYVE						484

SEQ ID NO: 124 moltype = AA length = 586

FEATURE Location/Qualifiers

source 1..586

 mol_type = protein

 organism = *Aspergillus tamarii*

SEQUENCE: 124

ATPADWRSQS	IYFLLTDRFA	RTDGSTTAAC	NTEDRKYCGG	TWQGIIDKLD	YIQGMGFTAI	60
WITPVTGQLP	QHTAYGDAYH	GYWQQDIYSL	NENYGTADDL	KALSSALHER	GYMLMVDVVA	120
NHMGYDGAGA	SVDYSVFKEP	NSQEYFHSFC	LIQNYEDQTQ	VENCWLGDNT	VSLPDLDTTK	180
DEVKNEWYDW	VGLTVSNYSI	DGLRVDTVKH	VQKDFWPQYN	KAAGVYCIGE	VLDGDPAYTC	240
PYQDVMDGVL	NYPIYYPLLN	AFKSTSGSMN	DLNMMINTVK	SDCPDSTLLG	TFVENHDNPR	300
FASYTNDIAL	AKNVAAFIIL	NDGIPIIYAG	QEQUYAGGND	PANREATWLS	GYATDSELYK	360
LIASANAIRS	HAISKDTGFV	TYKNWPIYKD	DTTIAMRKGT	DGSQVVTILS	NKGASGDSYT	420
LSLGDGTGYK	GQQLTEVIGC	TTVTGSDGK	VPVPMAGGLP	RVLYPTEKLA	DSKICSSSGA	480
TSPGGSSGSV	EVTDFVYATT	VYQNIYITG	DVSELGNWTP	ANGVALSSAN	YPTWSATIAL	540
PADTTTIQYKY	VNIDGSTVIW	EDAISNREIT	TPASGTYTEK	DTWDES		586

SEQ ID NO: 125 moltype = AA length = 603

FEATURE Location/Qualifiers

source 1..603

 mol_type = protein

 organism = *Acidomyces richmondensis*

SEQUENCE: 125

LTPAEWRAQS	IYQVLTDRFA	LTNGSTTAPC	NLNEYCGGTW	QGIINKLDYI	QGMGFTAIWI	60
SPVVENIPAS	DNTADGESYH	GYWAQRIYEV	NPNFGSAADL	EALSEAIHAR	GYMLMVDIVT	120
NHMGYDGCST	CVDYSVDFPF	DNQSYFHPFC	LINYNATSI	QVCWEGDNIV	SLPDLRTEDS	180
DVLGMWETWI	QQLVANYSID	GLRVDMSQOV	DQAFWQPFMS	AAGDLYAVGE	VFNGDPTYTC	240
PYQQLPLGVL	NYPAYYWIWQ	AFESTSGSIG	NLVNGINEMK	NDCLDPTLLG	SFLENHDNPR	300
FPSYTSYSL	DKNAIAFAML	QDGIPIVYEG	QEQUYSGGSV	PNNREDIWS	GYSTTSELYT	360
FIKIIAIRT	QALTKDSSYL	TYKAYPVYSD	SQTIAMRKGE	TYPIISVFTN	SGASGSAYS	420
TLSSSDTGFS	ENQSITELLT	CTVSTTDSGG	NLVVNISGL	PRVYYPTSAI	SGSTVCAEST	480
STLTSSPIS	TSTSDCTTAS	SVAVTFDET	TTYGETIKL	SGSISQLGDW	NTQDAILLSA	540
ADYKSTDNVW	FVTINLPAGI	VFQYKYINVD	SDGDVTWEAD	PNHTYTVSAT	CATAATIHT	600
WQN						603

SEQ ID NO: 126 moltype = AA length = 588

FEATURE Location/Qualifiers

source 1..588

 mol_type = protein

 organism = *Aspergillus bombycis*

SEQUENCE: 126

ATPADWRSQS	IYFLLTDRFA	RTDGSTTATC	NTEDRKYCGG	TWQGIIDKLD	YIQGMGFTAI	60
WITPVTGQLP	QDTAYGEAYH	GYWQQDIYAL	NENHGTADDL	KALSSALHER	GYMLMVDVVA	120
NHMGYDGAGA	SVDYSVFNP	SSQDYFHSFC	LIENYDDQTQ	SENCWLGDNS	VSLPDLDTTK	180
DEVKNEWYEW	VGNLVSNYSI	DGLRVDTVKH	VQKDFWPQYN	EAAGVYCIGE	VLNGDPAYTC	240
PYQDVMDGVL	NYPIYYPLLN	AFKSTSGSMN	DLNMMINTVK	SDCPDSTLLG	TFVENHDNPR	300
FASYTNDIAL	AKNVAAFIIL	NDGIPIIYAG	QEQUYAGGND	PANREATWLS	GYSTDSEIYK	360
LIASANAIRN	HAVSTDGTFV	TYKNWPIYKD	DTTIAMRKGT	DGSQIVTILS	NKGASGDAYT	420
LSLGNTRYTA	GQQLTEVIGC	TTLTVGSDGN	VPVPMAGGLP	RVLYPTEKLG	DSKICSSSGR	480
GATSPGSSSG	SVETFDVYA	TTYVQMIYI	TGDVSELGNW	TPANGVALSS	ANYPTWSATI	540
ALPADTTIQY	KYVNIDGSTV	IWEDAISNRE	ITTPASGTYT	EKDTWDES		588

-continued

SEQ ID NO: 127 moltype = AA length = 570
 FEATURE Location/Qualifiers
 source 1..570
 mol_type = protein
 organism = *Alternaria* sp.

SEQUENCE: 127

ADTSAAKRSRS	IYFVLTDRIA	RSSSDTGGGS	CSNLGNYCGG	TFKGLSKLD	YIKNLGFDIAI	60
WITPVVANS	GGYHGYAQD	LYAVNSNYGT	AADLKSIVNT	AHSGKIYVMV	DVVANHMGQG	120
AISGNRPEPL	NQDSSYHSAC	DINYSSQTSI	EQCRIANLDP	LNTQSSQIRS	LLNTWISWLV	180
NEYSFDGVRI	DTVKHVEKDF	WPGFASAAAGV	YSIGEVWDGN	PTYLAEYARL	MPGLLNATY	240
YPMNNFYQQK	GSSQALVDM	NTVRDTFPDP	SALGTFLDNH	DNNRWLNQKN	DVTLLKNALA	300
FVILSRGIPI	VYVGTEQGYA	GGADPANRED	LWRSSFNTNA	DLYQAIIKLN	AARTSAGGLA	360
GNDHTHLYVS	SNAYAWSRAN	GNLVVLTTNA	SGSGNAQHCF	NTQKANGRWT	NVYNGGATVT	420
ADSGNICVN	VANGPEVLL	VSTATPTSAT	PTSNPSPTTL	LTSTACPTS	VSVSFTHRVT	480
TVFGDTIKIT	GNTAQLGNWN	PSNGVALSAA	SYTSSNPIWT	LTLPLPAGSA	IQYKFKVSS	540
GGTVTWESDP	NRSYSVPGCQ	ASASVSSQWQ				570

SEQ ID NO: 128 moltype = AA length = 551
 FEATURE Location/Qualifiers
 source 1..551
 mol_type = protein
 organism = *Rhizopus microsporus*

SEQUENCE: 128

SPITIRSQNS	NDWSSRVIYQ	LLTDRFAKT	DDQSPCSDLG	NYCGGSFQGI	INHLDYIAGM	60
GFDIAIWISPI	PQNAQGGYHG	YWATNFSAIN	SNFGSSNDLK	KLVAQAHAHN	MYVMLDVVAN	120
HVGTPTSTPMN	YNGYTFNQGS	YYHSYCDINY	NDQTSVEQCW	LSGLPDLNTE	NDYVVNTLYS	180
TVSNWISSEYG	FDGIRIDTVK	HVRKDFWDGY	VKAAGVFATG	EVLHGSVSYV	APYQSHVPSL	240
INYPYLYPIY	DVFTKAATMT	RLKSGYNDIQ	SGGFSNLNLL	LNFDNHDNP	RLLSKADQSL	300
VKNALTYSM	IQGIPIVYVG	TEQSYKGGND	PNNREPLWTS	GYSSSEMYQ	FIKQVIQIRK	360
GSNATVTMDI	DQADNVVYFQ	RGNYLAVVNN	YGQGSTNSVT	VKSGSFADGT	VLKDVFSGAT	420
ATVKNKSITF	QLQNGNPVAV	SPQGATSPGG	SGSGSEVTFD	VYATTVYQGN	IYITGVDVSEL	480
GNWTPANGVA	LSSANYPTWS	ATIALPADTT	IQYKYVNIDG	STVIWEDAIS	NREITTPASG	540
TYTEKDTWDE	S					551

SEQ ID NO: 129 moltype = AA length = 565
 FEATURE Location/Qualifiers
 source 1..565
 mol_type = protein
 organism = *Syncephalastrum racemosum*

SEQUENCE: 129

VPLSIIDKRA	GVTTLSKRAA	ADDWKSRSIY	QIVTDRFARS	DGSTSGCGDL	SNYCGGDYKG	60
IQNQLDYIAG	MGFDIAIWIS	IPENTDGGYH	GYWAKNFEAL	NTNFGSADDL	KALVTAAGHK	120
GMVVMLDVVA	NHAGPTSGGD	YSGFTFDSAS	NYHAQCIDIY	ENQTSIEQCW	VADNLPDINT	180
EDDTIVSKLH	SIVSDWVTY	DFDGIKIDTV	KHIRKDFWSG	YEEAAGVFAT	GEVFDGDAAY	240
VGPYQDQLSS	LINYPYLYAI	RDVFTAGSGF	SRISDMLSSI	NSNFKDPSVL	TTFVDNQDNA	300
RFLSVKSDTS	LYKNALAFIT	LTEGIPVYVY	GTEQGGFKGN	DPNNREVLWT	SNYDTSSDLY	360
KFIKIVNDQV	RQKSNKTVKL	NVDVGTNTYA	PTHGKNLIVV	NNYSGGSTAS	VTVKVGDSDIA	420
DGTLKLVDAVS	NITATVSGGS	ITFSLNNGLP	ALFVPSSGAT	SPGGSSGSVE	VTFDVYATTV	480
YGQNIYITGD	VSELGNWTPA	NGVALSSANY	PTWSATIALP	ADTTIYQKYV	NIDGSTVIWE	540
DAISNREITT	PASGTYTEKD	TWDES				565

SEQ ID NO: 130 moltype = AA length = 583
 FEATURE Location/Qualifiers
 source 1..583
 mol_type = protein
 organism = *Rhizomucor pusillus*

SEQUENCE: 130

ATSDDWKGKA	IYQLLTDRFG	RADDSTSNCS	NLSNYCGGTY	EGITKHLDIY	SGMGFDIAWI	60
SPIPKNSDGG	YHGYWATDFY	QLNSNFGDES	QLKALIIQAAH	ERDMYVMLDV	VANHAGPTSN	120
GYSGYTFDDA	SLYHPKCTID	YNNQTSIEQC	WVADELDPID	TENSDNVAIL	NDIVSGWVGN	180
YSFDGIRIDT	VKHIRKDFWT	GYAEAAGVFA	TGEVFNGDPA	YVGPYQKYL	SLINYPMYYA	240
LNDVFPVSKK	GFSRISEMLG	SNRNAFEDTS	VLTTFPVDNHD	NPRPLNSQSD	KALFKNALTY	300
VLLGEGIPIV	YVGSEQGFSG	GADPANREVL	WTTNYDTSSD	LYQFIKTVNS	VRMKSINKAVY	360
MDIYVGDNAY	AFKHGDALVV	LNNYSGGSTN	QVSFVSQKGF	DSGASLMDIV	SNITTTVSSD	420
GTVTFNLDKG	LPAIFTSATG	GTTTATPTG	SGSVTSTSKT	TATASKTSTS	TSSTSCCTPT	480
AVAVTFDLTA	TTTYGENIYL	VGTSIQGLDW	ETSDGIALSA	DKYTSDDPLW	YVTVTLPAGE	540
SFEYKFIRIE	SDDSEVESD	PNREYTPQA	CGTSTATVTD	TWR		583

SEQ ID NO: 131 moltype = AA length = 553
 FEATURE Location/Qualifiers
 source 1..553
 mol_type = protein
 organism = *Dichotomocladium hesseltinei*

SEQUENCE: 131

QPVNITKRAS	AADWRSRAIY	QVLTDRTFART	DGSTSGCSNL	SNYCGGTFCG	ITNKLDIYAN	60
LGFDIAIWIS	IPNTNSPGGYH	GYWATDFYGI	NSNFGSSNDL	KELVNAHAHN	GMVVMLDVVA	120
NHAGPTSN	GYGTFGSSG	LYHNRCSINY	NDQRSIEQCW	VADDLPDINT	ENNDNVNKP	180
NIVSTVWKTY	GFDIAIRIDTV	KHVRKDFWPG	YTSAGVFAT	GEVFDGNPSY	VADYQNYMES	240

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LINYPYLYAL	NDVFASGYSF	SRLSNQRVAN	YHAFKDVSVL	PIFIDNHDNP	RFLNKKNDIA	300
QFKNALTYVL	LGEGIPVYVY	GSEQAYAGGA	DPANREALWS	SGFSTNSDMY	QFIAKLNRVR	360
QKSNKSVYMD	LDVQNNVYAF	MHGKSLVVLN	NFGNGASRQV	TVNVGAQVAS	NTRLTDVVS	420
TSVTVSGSSV	TFTINNGLPA	VFTVSGATSP	GGSSGSEVET	PDVYATTVYG	QNIYITGDVS	480
ELGNWTPANG	VALSSANYPT	WSATIALPAD	TTIQYKYVNI	DGSTVIWEDA	ISNREITTPA	540
SGTYTEKDTW	DES					553

SEQ ID NO: 132 moltype = AA length = 552
 FEATURE Location/Qualifiers
 source 1..552
 mol_type = protein
 organism = Lichtheimia ramosa

SEQUENCE: 132

RPFVKRATAD	DWRDRAIYQI	LTRDFARSDG	STDNCSDLN	YCGGNYQGII	QQLDYIEGMG	60
FDAIWISPIIP	ANADGGYHGY	WATDFESLND	HFGSQDDLKA	LVDAAAHERGM	YVMLDVVANH	120
AGPTNNGDYS	GYTPGSSDLI	HPQCSIDYSN	QNSIEQCWVA	DNLPDIDTEN	DSNVNKLHDI	180
VSTWVSTYGF	DGIRIDTVKH	VRKDFWPGYA	SAAGVFATGE	VYDGDQTYVG	AYQDYLDSSL	240
NYPLYALND	VFASGQGFQR	LSNQRNAIMS	AFKDVSVLTS	FVDNHDVARF	LSTKNDQALF	300
KNALAFVLLG	ETIPVVYVGS	EQGFAGGADP	ANREALWSSG	YDTSSDLYQF	IATINNVRQ	360
KSGKKVYMDL	DVQDNVYAFM	HGDALVVLNN	YSGGASNQVS	VNVGAQVAES	TSFTDAISGT	420
SITVSSGSVT	FTLDNGNPAI	FVPAGATSPG	GSSGSEVETF	DVYATTVYGO	NIYITGDVSE	480
LGNWTPANGV	ALSSANYPTW	SATIALPADT	TIQYKYVNID	GSTVIWEDAI	SNREITTPAS	540
GTYTEKDTWD	ES					552

SEQ ID NO: 133 moltype = AA length = 587
 FEATURE Location/Qualifiers
 source 1..587
 mol_type = protein
 organism = Penicillium aethiopicum

SEQUENCE: 133

ARTADWKPRS	IYQTMTRDFA	RTDGSTTSPC	NTKAGLYCGG	TWRGTIDHLD	YIQGMGFDAV	60
MISPIIENIE	GRVDYGEAYH	GYWPLNLDNL	NSHFGTHQDL	LDLSDALHSR	GMYFMMDTVI	120
NNMAYITNGS	DPATDIDYSV	FTPFNNADYF	HPYCTMNWSV	PAIAQRCTNG	DDTVALPDLF	180
TEHEDVQQLL	IKWANKAIKT	YSIDGLRIDA	AKHVNPDFLR	KFSDGVDIFM	TGEVLEGSVS	240
IMEDYQSNYI	NSLPNPIYF	EILSAFTNGN	TSQLAIAVEN	MRVAIPDVNA	MASFSENHDK	300
PRIASYNDDM	SIKNNVLVFT	MLPDGIPMIY	QQEQHLKGD	GVPHNREAIW	LSKYDTEAEL	360
YKLIAKLNRI	RNHAGYLGSD	YFEDATHPIY	QGSSELAFTK	GVQGRQVMV	LSNQPSTSGR	420
YALDLAVSYN	AGTELMVDLN	CNNYTVDNQG	VLRVMDKGE	PRVFFPRKYM	EGSGLCGYSG	480
ATSPGGSSGS	VEVTFDVYAT	TVYQQNIYIT	GDVSELGNWT	PANGVALSSA	NYPTWSATIA	540
LPADTTIQYK	YVNIDGSTVI	WEDAISNREI	TPASGTYTE	KDTWDES		587

SEQ ID NO: 134 moltype = AA length = 597
 FEATURE Location/Qualifiers
 source 1..597
 mol_type = protein
 organism = Subulispora sp.

SEQUENCE: 134

LTPAEWGSQS	IYQVLTDRFA	LTDGTTTASC	DLNTYCGGTW	LGIQNHLDYI	QGMGFATAWI	60
SPIVTNIAGD	SVDGDSYHGY	WAQDITTVNS	AFGTEQDLIN	LSAALHERGM	YLMVDVVNNH	120
MGYLGCCTCV	DYSEYTPFNE	ESYHHPYCPT	DYSNLTSIQV	CWEGDNIVSL	PDLRTEDSDV	180
RSMWYDWTIP	LVAKYSIDGL	RMDSAEHVEK	SFWPGWVSAS	GVYNVGEVDE	GDPTIFPDWL	240
NYIDGTLNYP	AYYWITQAFV	STSGSISNLV	TGVNQLKASM	KTSTFGSFLE	NHDQPRFPSL	300
TSDTDLAKNA	IAPAMLADGV	PIVYGGQEQG	YSGGGVPNDR	EPLWTSGYST	TSAGYTFIKT	360
INAVRHIAVT	QDTAYVAYQA	YPIYSDSRVI	AMKSSSVLAV	FSNIGSSGSG	YSITLPAGAF	420
AASQALTDV	SCQTYTADAS	GGLTFTFGQA	PSVYATASL	AGSGLCGTTG	TGGSTGTTTA	480
SETGGSSPTS	TACASVPVTF	NEKVTTVVGE	TIKISGSVAA	LGDWATGSVA	ALSAASYTSS	540
NPQWDVTISF	APGTVIEYKY	INVASSGAVT	WEADPNHTYT	VPASCATAAV	VSDTWQT	597

SEQ ID NO: 135 moltype = AA length = 601
 FEATURE Location/Qualifiers
 source 1..601
 mol_type = protein
 organism = Trichoderma paraviridescens

SEQUENCE: 135

LTAQWRSQS	IYQVLTDRFS	QTNGATNSAC	NAGNQVYCGG	TWQGIKLNLD	YIKSMGFTAI	60
WISPVVENLA	GNSADGEAYH	GYWAQDIYQV	NTNFGSAADL	RALSEALHNA	GYMLMVDIVT	120
NHMGYLGCGT	CWQYNTFNPF	NSQSYHHPFC	LINFNSSNMT	QIQNCWEGDN	TVSLPDLATE	180
NANVLMSWQT	VITQLVANYT	IDGLRMDSCF	ELNYGYFEFF	QSSANVYIVG	EVNDGPAIV	240
CPYQKNYGLN	TLNTPAYYWI	TQAFQSTSGS	ISNLVNLNT	MKSECSDDTL	LGSFMENHDN	300
PRFPSLTSDI	SLAKNAIAFT	MLADGIPIIY	EGEQHLNGG	GVPNNREAIW	LSGYSTSAVL	360
YTHIKALNQi	RSQAIKQNSA	YVTTQAQAVT	SDSSTIVTRK	GSTGSQIVGV	FSNKGANGNS	420
YTLTLPSADT	GFTSNEQVVE	ILSCTAYTTD	SSGNLAVAMA	GGLPRVFYAR	SSLSGSGICP	480
NLGSGGGTPT	STPPTSCTAI	PVTTFDEKVT	TFGQTIKIAG	DISALGNWNT	ANAVTLTSAAN	540
YTSSNPLWAI	TLNLAPGVQV	EYKYINVAQN	GGVTWEADPN	HTYTVPSACT	AQPTVANTWQ	600
G						601

SEQ ID NO: 136 moltype = AA length = 598
 FEATURE Location/Qualifiers
 source 1..598

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mol_type = protein
organism = Byssosascus striatosporus

SEQUENCE: 136
LSADDWRAQS IYQLLTDRFA LTNGSTTAPC DTEEQIYCGG SWQGIIDKLD YIQGMGFTAI 60
WISPVVENLS GDSADGEAYH GYWAQNVYEV NPNFGATSDL VALSQVVHDK GMYLMLDVVT 120
NHMGYLGCGT CVDYSVFNPF NEESYYPHFC LIDYDNTTSI EVCWEGDNIV SLPDLRTEDS 180
DVLSTWESWV TELVSNYTV D GIRLDSTEEL DQAFLLPPFES AAGVYIVGEV DNGDPAVVCP 240
YQEYVSGVLN YPAYYWITQA FESTSGSIGN LVNGINTMKS DCSDTSLGGS FLENHDQPRF 300
ASLTSDISLA KNAIAFSMLQ DGIPIVYAGE EQHYSGGAVP NDREALWLSG YPTSSTLYTW 360
ITSLNQIRSH AIATNSSYLT YNAYPVYSDD STIVMRKGFA DNQIVSVYTN QGADATAYTL 420
DLPSTDTGFT ASQSLVEIGG CTTTATDDSG NLAVAMASGL PRIYYPAGL SGSGVCGQLG 480
SGGGTPTSTP PTSCTAIPVT FDEKVTTFG QTIKIAGDIS ALGNWNTANA VTLSAANYTS 540
SNPLWAITLN LAPGQVVEYK YINVAQNGGV TWEADPNHTY TVPSACTAQP TVANTWQG 598

SEQ ID NO: 137      moltype = AA length = 615
FEATURE            Location/Qualifiers
source              1..615
                    mol_type = protein
                    organism = Aspergillus brasiliensis

SEQUENCE: 137
LSAAEWRTQS IYFLLTDRFG RTDNSTTATC NTGDQIYCGG SWQGIINHL D YIQGMGFTAI 60
WISPIEQLP QDTS DGEAYH GYWQQKIYDV NSNFGTADDL KSLSDALHAR GMYLMLDVVP 120
NHMGYAGSGN DVDYSVDFPF DSSSYFHPYC LITDWDNLTM VQDCWEGDTI VSLPDLNTE 180
TVVRTIYDW VADLVSNYSV DGLRIDSLE VEPDFPGYQ EAAGVYCVGE VDNGNPALDC 240
PYQYLDGVL NYPIYQQLY AFESSSGSIS DLYNMIKSA SDCSDPTLLG NFIENHDNPR 300
FAYYTSYDYSQ AKNVLSYIFL SDGIPIVYAG EEQHYSGGDV PYNREATWLS GYDTSAELYT 360
WIATTNAIRK LAIAADSSYI TYANDPIYTD SNTIAMRKGT SGSQVITVLS NKGSSGSSYT 420
LTLSGSGYTS GTKLIEAYTC TSVTVDSNGD IPVPMASGLP RVLLPASVVD DSSLCGSGS 480
STSTTTSTAT ATTTSTKTST SSSSSSSSCT ASATAIPITF EELVTTTYGE EIYLSGSISQ 540
LGDWDTSDAV KLSADDYTSS NPEWSVTVT L PVGTTFEYKF IKVESGGSVT WESDPNREY 600
VPECSSGETV VDTWR 615

SEQ ID NO: 138      moltype = AA length = 604
FEATURE            Location/Qualifiers
source              1..604
                    mol_type = protein
                    organism = Penicillium subspinulosum

SEQUENCE: 138
ALSAEWRTQS IYFLLTDRFG RTDNSTTATC DTGDQIYCGG SWQGVINHL D YIQGMGFTAI 60
WISPIEQLP QDTS DGEAYH GYWQQKIYDV NSNFGTADDL VALSDALHAR DMYLMLDVVP 120
NHMGYDGDGD DVDYSVDFPF DSSSYFHPYC LITDYDDIEM VQDCWEGDTI VSLPDLNTE 180
TVVQDIYAW VADLVANYSV DGLRIDSLE VQPAFFPAYQ SAAGVYCVGE VDNGLPTLDC 240
PYQYLDGVL NYPIYQQLY AFESSSGSIS DLYDMINSVA SDCSDPTLLG NFIENHDNPR 300
FAYYTSYDYSQ AKNVASFIFL SDGIPIVYAG EEQHYSGGDV PYDREATWLS GYSTTAEELYT 360
WIATTNIRK LAISLDDYI TYVNDPFYTD ENTIAMRKGT SGLQVITVLS NLGADGSAYT 420
LTLSGSGYDS GTDLIEVYTC TSVTVDSNGD IAVPMESGLP RVFLPESSIK DSDLCSGTTT 480
TTTSTAATAT ATSTSTCTAA TEVSIIIFEL VTTTYGEEIY LSGSISELGS WDTSDALELS 540
AANYTSSNPE WYLEVTLFVG TSFEYKFIMI ESDGTVVWES DPNRSYTVPS ACSGAVETVV 600
DTWR 604

SEQ ID NO: 139      moltype = AA length = 606
FEATURE            Location/Qualifiers
source              1..606
                    mol_type = protein
                    organism = Penicillium antarcticum

SEQUENCE: 139
LTPAEWRSQS IYFMLTDRFG RSDNSTTAA C NVSDRTYCGG TWQGIINHL D YIQGMGFTAI 60
WITPVTEQLP QDTGDGEAYH GYWQQNIYEV DSNLGTAA DL LALSEALHAR GMYLMVDVVA 120
NHMGYAGAGS SVEYSVFHPF SSSSYFHSYC LISNYDDQSN VEDCWLGD TI VSLPDVDTTQ 180
TAVQTLWYDW IGDLVSNYSI DGLRIDTVKH VQKSFWP GYN DAAGVYCVGE IPDGDPAYTC 240
DYQNYMDGVL NYPIYQQLY AFQSSSGSIS DLYDMINSVK SDCADPTLLG NFIENHDNPR 300
FASYTSYDYSQ AKNVISFLFL SDGIPIIYSG EEQHYSGGAD PANREATWLS GYSTTAEELYK 360
YIATTNIRK AAVSADSSYI TTKNVFPYQD SHTLAMKKG S SASPVITVLS NYGSSGSSYT 420
LTLSGSGYSS GTNLMEYTC TSVTVDSNGN IAVPMASGLP RVLM LASSAS SICASSTTS 480
TATVATQTTT LTTTGTSTCTQ ATVLPVLFKE LVTTTYGQNV YISGSISQLG SWDTSSAIAL 540
SASSYNSSNP LWQVAITLPV GTSFQYKFLE KTTGSTTIQW ESDPNRSYTV PTGCVGTAT 600
AIATWR 606

SEQ ID NO: 140      moltype = AA length = 609
FEATURE            Location/Qualifiers
source              1..609
                    mol_type = protein
                    organism = Penicillium coprophilum

SEQUENCE: 140
LTPAEWRSQS IYFLLTDRFG RTDNSTTANC NVNDRVYCGG TWQGIINQLD YIQGMGFTAI 60
WITPVTKQLS QNTGDGTSYH GYWQQDIYV NPNHGTSDDL LALS KALHAR GMYLMVDVVA 120
NHMGYAGAGN NVDYSVFTPF NSASYFHSYC LISNYNDQSN VENCWLGD TT VSLPDLDTTQ 180
SSVQTLWNWV DGLRVDTVKH VQKSFWPAFN RAAGVYSVGE VPDGSPSYTC 240
DYQKYMDGVL NYPMYPLLR AFQSTSGSIS DLYNMIGTLS STCADSTLLG NFIENHDNPR 300

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FPSYTSDSYQ AKNVLSFLFL SDGIPIVYSG QEQHYSGGSD PANREALWLS KYSTTAEELYK 360
YIATTNKIRK AAVAADSSYI TSKNVAFYQD SHTLAMKKGS GSSPVITVLS NAGSSGSSYT 420
LYLSGSGYSS GTQLMELYTC TSVTVDSSENK IAVPMASGLP RVFVLASSVS NSGLCGSSTP 480
TTTATTATTA TQTTTATTTA GGCTQATALP VLFKELVTTT YGQDIYISGS ISQLGTWDTT 540
KAVALSADSY TSSNPLWQAT ITLPVGTTFQ YKFIKKANGA ITWESDPNRS YTVPTGCSGS 600
TATVTASWK 609

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SEQ ID NO: 141      moltype = AA length = 602
FEATURE            Location/Qualifiers
source             1..602
                  mol_type = protein
                  organism = Penicillium olsonii

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SEQUENCE: 141
LTPAEWRSQS IYFLLTDRFG RDDNSTTATC NTGDRTYCGG TWQGIINQLD YIQGMGFTAI 60
WITPVTEQLS ANTGYGTAYH GYWQQDIYEV NPNHGSSADL KALSAALHAR GMYLMVDVVA 120
NHMGVNGIGS SVDYSVFNPF SSSSYFHSYC LISNYNDQSN VENCWLGD TT VSLPDLDTTQ 180
TAVQTIWNEW ITDLVSNYSI DGLRIDTVKH VQKSFWPFGFN DAAGVYCVGE IFDGNPSYTC 240
DYQNYLDGVL NYPIYYQLLY AFQSTSGSIS DLYNMINTVA SDCADSTLLG NFIENHDNPR 300
FPSYTGDSYQ AKNVLSFLFL SDGIPIVYSG QEQHYSGGSD PANREALWLS KYSTTAEELYK 360
WIATTNKIRK LAVSADSSYI TSKNSPFYQD SHTLGMKKGS VITILSNNGA SGSSYTLTSL 420
SGSYSSGSKL MELYTCTTIT VDSSGNIPVP MVSGLPRLAI PASSIGSNGL CGSTTSPTTT 480
AATQTTTATT TGTCTQATAL PVLFKELVTT SYGQNVYISG SISQLGNWDA SSAIALSASS 540
YTSSNPLWQV TITLPVGTGF EYKFIEKSSG SATATWESDP NRSYTVPTGC AGTTATVTAT 600
WR 602

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SEQ ID NO: 142      moltype = AA length = 621
FEATURE            Location/Qualifiers
source             1..621
                  mol_type = protein
                  organism = Penicillium vasconiae

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SEQUENCE: 142
LTAAEWRTQS IYFLLTDRFG RTDNSTTATC SVSDRIYCGG SWQGIINHL D YIQGMGFTAI 60
WITPVTEQLS QDTGDGEAYH GYWQQDIYEV NTNYGTAADL LALSKALHRS GMYLMVDVVA 120
NHMGVDGAGN TVDYSVFNPF DSSSYFHSYC EISDYSNQTN VEDCWLGD TT VSLPDLDTTL 180
SSVQTIWYNW VTELVSNIYSI DGLRIDTVKH VQKSFWPFGYN SAAGVYCVGE VFDGDPAYTC 240
PYQSYLDGVL NYPIYYQLLY AFESTSGSIS SLYNMINSVA SDCSDPTLLG NFIENHDNPR 300
FASYTSDYSQ AKNVISFIFF SDGIPIVYAG QEQHYSGGSD PANREATWLS GYDTTATLYK 360
YITSTNKIRS LAISKDTAYI TSKNNAFYTD SNTIAMKKGS SGSQVITVLS NRGSSGSSYT 420
LTLGSGGYSS GTQLMEMYTC TAVTVDSNGN IAVPMASGLP RIYMLASSAC SICSSSCSTT 480
TTTSTTSTST TASTLKTITT TTSATSTTS TSCTQATALP VLFKEIVTTS YGQNIYISGS 540
ISQLGSWDTT NAVALSADQY TSSNNLWYVV VTIPVGTSTFE YKFIEETSGS STITWESDPN 600
RSYTVPTGCA GSTATVTATW R 621

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SEQ ID NO: 143      moltype = AA length = 615
FEATURE            Location/Qualifiers
source             1..615
                  mol_type = protein
                  organism = Penicillium sp.

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SEQUENCE: 143
LTAAEWRSQS IYFLLTDRFG RTDNSTTATC NVSDRIYCGG SWQGIINHL D YIQGMGFTAI 60
WITPVTEQLS QDTGDGEAYH GYWQQDIYEV NTNYGTAADL LALSKALHRS GMYLMVDVVA 120
NHMGVDGAGN TVDYSVFNPF DSSSYFHSYC EITDYSNQTN VEDCWLGD TT VSLPDLNTTL 180
SSVQTIWYDW VAALVSNYSI DGLRIDTVKH VQESFWPEYN SAAGVYCVGE VFDGDPAYTC 240
PYQNYLDGVL NYPIYYQLLY AFESTSGSIS DLYNMINSVA SDCSDPTLLG NFIENHDNPR 300
FASYTSDYSQ AKNVLSFIFF SDGIPIVYAG QEQHYSGGSD PANREATWLS GYDTSAEELYT 360
WITSTNKIRS LAVSKDTAYI TSKNDAFYTD SNTIAMKKGS GGSQVVTVLS NRGSSGSSYT 420
LTLGSGGYSS GTKLMEMYTC TAVTVDSNGN IAVPMASGLP RVYMLASSAC SICSSACSTT 480
TTSSTTSTAT TTSTTLKTTT TTTSTSTCTA TALPVLFEKI VTTSYGQNIY ISGSISELGD 540
WDTSTNAVALS ADQYTSSNNL WYVVVTIPVG TSFEYKFIEE TSGSSSITWE SDPNRSYTVP 600
TGCAGSTATV TATWR 615

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SEQ ID NO: 144      moltype = AA length = 606
FEATURE            Location/Qualifiers
source             1..606
                  mol_type = protein
                  organism = Heterocephalum aurantiacum

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SEQUENCE: 144
LTAAEWRSQS IYFLLTDRFA RTDGSTTAAC NLSQRAYCGG SWQGIINHL D YIQGMGFTAI 60
WITPVTKQIE ASTSDGTAYH GYWQQDIYNI NSHYGTADDL RALSALHRS GMYLMIDVVA 120
NHMGYPGAGT SVDYSIFTPF GSSSYFHSYC QITDYNQSN VENCWLGDNV VSLPDLNTQN 180
SNVRNLWYDW VEELVANYSV DGLRVDTVKH VEKDFWPSYN AAAGVYCVGE VFHGDPAITC 240
PYQNYMDGVL NYPIYYQLLY AFQSSSGSIT DLYNMINSVA SDCKDPTTLG NFIENHDNPR 300
FPSYTSMSQ AKSVIAFLFL SDGIPIIYAG QEQHYSGGAD PNNREAIWLS GYSTSTSLYQ 360
FISSTNSIRK LAISKDSYFL TSRNNPFYTD SNTIAMRKGS SGSQVITVLS NKGSSGNSYT 420
LTLTNHGYSS GAQLTELYTC SSIQVASSGG LAVPMASGLP RVLVPSSWIQ GSGLCGGGST 480
TTTTTATTTT TTTSTSSCA AATSLAVFVN ELVTTYGVEN IFIAGSISQL GSWDTGKSA 540
LSASQYTSNN PLWTATVSLP VGTSPQYKFI KKEPDGQVWV ESDPNRSYTV PAGCAGTTQT 600
VNTSWR 606

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SEQ ID NO: 145 moltype = AA length = 602
 FEATURE Location/Qualifiers
 source 1..602
 mol_type = protein
 organism = Neosartorya massa

SEQUENCE: 145

LTPAEWRSQS	IYFLLTDRFG	REDNSTTAAC	DVTQRLYCGG	SWQGIINHLD	YIQGMGFTAI	60
WITPVTEQFY	EDTGDGTSYH	GYWQQNIYEV	NYNYGTAQDL	KNLADALHAR	GYMLMVDVVA	120
NHMGYDGAGN	TVDYSVFTPF	DSSSYFHPYC	LISDYSNQTN	VEDCWLGDTT	VSLPDLDTTD	180
TTVRTIWDW	VKGLVANYSI	DGLRIDTVKH	VEKDFWPGYN	DAAGVYCVGE	VFSGDPTYTC	240
PYQNYLDGVL	NYPIYYQLLY	AFESTSGSIS	NLYDMINSVA	SDCADPTLLG	NFIENHDNPR	300
FASYTSDYSQ	AKNVISFIFP	SDGIPIVYAG	QEQHYSGGAD	PANREAVWLS	GYSTSATLYS	360
WIASTNRIRK	LAISKDAAYI	TSKNNPFYYD	SNTLAMRKGS	IAGAQQITVL	SNKGSSGSSY	420
TLSLSGTGYS	AGASLVEMYT	CTTLTVDSSG	NLPVPMASGL	PRVLVPSSWV	SGSGLCGSGS	480
TTTTTTTATA	TTTACTSATA	LPIVFEEVVT	TTYGENVYLT	GSISQLGNWD	TSSAIALSAS	540
KYTSSNPEWY	VTVTLPGVTS	FQYKFFKKES	DGSIVWESDP	NRSYTVPTGC	AGTTVTVSDT	600
WR						602

SEQ ID NO: 146 moltype = AA length = 588
 FEATURE Location/Qualifiers
 source 1..588
 mol_type = protein
 organism = Penicillium janthinellum

SEQUENCE: 146

ATPAQWRSQS	IYFMLTDRFA	RTDGSTTAPC	DTSQRAYCGG	TWQGIIDKLD	YIQGMGFTAI	60
WITPVTGQLD	GDTGDGTAYH	GYWQQDIYSL	NSNYGTASDL	KALASALHAR	GYMLMVDVVA	120
NHMGYNGAGN	TVDYSVDFAF	NSNQYFHSYC	EVTNYSNQTN	VEDCWLGDTT	VSLPDLNTEL	180
SSVQSIWYNW	VGSLVSNYSI	DGLRVDTVKH	VQKDFWPGYN	KAAGVYCVGE	VFDGDASYTC	240
PYQEVMDGVL	NYPMYYPLLR	AFQSTSGSMS	DLNMMINTVK	STCSDSTLLG	TFVENHDNPR	300
FASYTNDMSL	AKNVAAFTIM	ADGIPIIYAG	QEQHYSGGSD	PANREAVWLS	GYNTDSALYK	360
LIAKNVAIRS	VAISQSASYV	TYKNYPIYQD	ASTLAMRKGS	SGTQTITVLS	NRGASGSQYT	420
LSLGNTGYST	GTTLTEIITC	AKITVDSSGN	VPVPMASGEP	RILYPSSSIK	GSAICASSGR	480
GATSPGGSSG	SVEVTFDVYA	TTYVGQNIYI	TGDVSELGNW	TPANGVALSS	ANYPTWSATI	540
ALPADTTIQY	KYVNIDGSTV	IWEDAISNRE	ITTPASGTYT	EKDTWDES		588

SEQ ID NO: 147 moltype = AA length = 589
 FEATURE Location/Qualifiers
 source 1..589
 mol_type = protein
 organism = Aspergillus brasiliensis

SEQUENCE: 147

ATPAEWRSSQ	IYFLLTDRFA	RTDNSTTASC	DLSARQYCGG	SWQGIINQLD	YIQGMGFTAI	60
WITPVTAQIP	QDTGYGQAYH	GYWQQDAYAL	NSHYGTADDL	KALATALHSR	GYMLMVDVVA	120
NHMGHNGTGS	SVEYSVYNPF	NAKKYFHNLC	WISNYDNQTN	VEDCWLGDNT	VALPDLDTTR	180
TDVKNMWDW	VKSLVSNYSV	DGLRVDTVKN	VQKNFWPGYN	NASGVYCIGE	VFDGDASYTC	240
PYQDLDGVL	NYPMYYPLLR	AFKSTTGSIS	DLNMMINTVK	STCKDSTLLG	TFIENHDNPR	300
FANYTSDMSL	AKNVATFTIL	ADGIPIIYAG	QEQHYSGGND	PYNREATWLS	GYKTTSELYT	360
HIAASNKIRT	HAIKQDSGYL	TYKNYPIYQD	TSTLAMRKGY	NGTQTITVLS	NLGASGSSYT	420
LSLPGTGYYA	GQKITEIYTC	TNLTVNSNGS	VPVPMKSLPL	RILYPTDKLV	NGSSFCSSSG	480
RGATSPGGSS	GSVEVTFDVY	ATTVYGQNIY	ITGDVSELGN	WTPANGVALS	SANYPTWSAT	540
IALPADTTIQ	YKYVNIDGST	VIWEDAISNR	EITTPASGTY	TEKDTWDES		589

SEQ ID NO: 148 moltype = AA length = 588
 FEATURE Location/Qualifiers
 source 1..588
 mol_type = protein
 organism = Aspergillus westerdijkiae

SEQUENCE: 148

ATPAQWRSQS	IYFLLTDRFA	RDDGSTTATC	NTEDRKYCGG	TWQGIIDQLD	YIQGMGFTAI	60
WITPVTAQLT	EDTKYGDAYH	GYWQQDIYSL	NENYGTADDL	KALADALHER	DMYLMVDVVA	120
NHMGYAGAGD	SVDYSVFNPF	NSQDYFHSFC	LIQDYNDQTQ	SEDCWLGDNS	VSLPDLDTTK	180
SEVQDIWDW	VGGLVSNYSI	DGLRIDTVKH	VQKEFWPGYN	DAAGVYCIGE	ILDGDASYTC	240
PYQEVLDGVL	NYPIYYPLLN	AFKSTSGSIS	DLNMMINTVK	SDCPDSTLMG	TFIENHDNPR	300
FASYTDDIAL	AKNVAAFTIL	ADGIPIVYAG	QEQHYAGGED	PANREATWLG	KYNTDSSELYK	360
LIAASNAINR	HAISTDKEYV	NYKNYPIYKD	DSTIAMRKGF	DGAQIITVLS	NQGSSESSYT	420
LSLGDTFGSS	GDKLTEIYTC	TAVTVDSGDK	VPVPMDDGGAP	RALFPTEKLS	GSSLCSGSGR	480
GATSPGGSSG	SVEVTFDVYA	TTYVGQNIYI	TGDVSELGNW	TPANGVALSS	ANYPTWSATI	540
ALPADTTIQY	KYVNIDGSTV	IWEDAISNRE	ITTPASGTYT	EKDTWDES		588

SEQ ID NO: 149 moltype = AA length = 476
 FEATURE Location/Qualifiers
 source 1..476
 mol_type = protein
 organism = Hamigera avellanea

SEQUENCE: 149

ATPADWRSRS	IYFILTDRFA	RTDGSTTAEC	DTSARAYCGG	TWRGIINKLD	YIQNMGFTAI	60
WITPVTAQLP	GSTGHGSAYH	GYWQQDIYSL	EPNYGTADDL	RALASALHER	NMYLMVDVVA	120
NHMGWAGSGD	SVDYSVFNPF	DSADYFHPYC	LISNYEDQTE	VENCWLGDNT	VALVDLDTTR	180
SDVQNIWYEW	VDSLGVNYSI	DGLRIDTVRH	VQKDFWPGFN	DAAGVYSVGE	VFSGDTAYTC	240

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PYQEVLDGVL	NYPIIYPLLR	AFQSTSGSIN	DLNMMINTVK	SDCADSTLMG	TFLENHDNPR	300
FASYTSDVAL	AKNAIAFTIL	SDGIPIIYAG	QEQHYSGGND	PANREAVWLS	GYSTDSELYS	360
FVAVTNQIRN	YAIQDQGYV	TWKNVPIYQD	TSTLAMRKGT	DGSQVITVLS	NLGASGSSYT	420
LTLGSGSYSS	GQQLTEIFSC	ATVTVDSSGN	IPVPMGSGQP	KVFYPTAGLG	GSGICQ	476

SEQ ID NO: 150 moltype = AA length = 586
 FEATURE Location/Qualifiers
 source 1..586
 mol_type = protein
 organism = Hamigera avellanea

SEQUENCE: 150

ATPADWRSRS	IYFILTDRFA	RTDGSTTAEC	DTSARAYCGG	TWRGIINKLD	YIQNMGFTAI	60
WITPVTAQLP	GSTGHGSAYH	GYWQQDIYSL	EPNYGTADDL	RALASALHER	NMYLMVDVVA	120
NHMGWAGSGD	SVDYSVFNPF	DSADYFHPYC	LISNYEDQTE	VENCWLGD TN	VALVDLDTTR	180
SDVQNIWYEW	VDSLGVNGYSI	DGLRIDTVRH	VQKDFWPGFN	DAAGVYSVGE	VFSGDTAYTC	240
PYQEVLDGVL	NYPIIYPLLR	AFQSTSGSIN	DLNMMINTVK	SDCADSTLMG	TFLENHDNPR	300
FASYTSDVAL	AKNAIAFTIL	SDGIPIIYAG	QEQHYSGGND	PANREAVWLS	GYSTDSELYS	360
FVAVTNQIRN	YAIQDQGYV	TWKNVPIYQD	TSTLAMRKGT	DGSQVITVLS	NLGASGSSYT	420
LTLGSGSYSS	GQQLTEIFSC	ATVTVDSSGN	IPVPMGSGQP	KVFYPTAGLG	GSGICQSSGA	480
TSPGGSSGSV	EVTDFVYATT	VYQNIYITG	DVSELGNWTP	ANGVALSSAN	YPTWSATIAL	540
PADTTIQYKY	VNIDGSTVIW	EDAISNREIT	TPASGTYTEK	DTWDES		586

SEQ ID NO: 151 moltype = AA length = 576
 FEATURE Location/Qualifiers
 source 1..576
 mol_type = protein
 organism = Meripilus giganteus

SEQUENCE: 151

RPTVFDAGAD	AHSLHARAPS	GSKDVIIQMF	EWNWDSVAEE	CTNFIGPAGY	GFVQVSPPQE	60
TIQGAQWWT	YQPVSYLTG	KRGDRSQFAN	MITTCHAAAGV	GVIVDTIWNH	MAGVDSGTGT	120
AGSSPTHYNY	PGIYQNQDFH	HCGLEPGDDI	VNYDNAVEVQ	TCELVNLADL	ATDTEYVRGR	180
LAQYGNLLES	LGADGLRLDA	SKHIPVGDIA	NILSRLRSRV	YITQEVIFGA	GEPITPNQYT	240
GNQDVQEFY	TSALKDAFLS	SGISNLQDFE	NRGWVPGSGA	NVFVNVHDE	RNGASLNNNS	300
PSNTVVTATI	FSLAHPYGTP	TILSSVDGFT	NTDAGAPNNN	VGTCSTSGGA	NGWLCQHRWT	360
AIAGMVGFRN	NVGSAAALNNW	QAPQSQQIAF	GRGALGFVAI	NNADSAWSTT	FTTSLPDGSY	420
CDVISGKASG	SSCTGSSFTV	SGGKLTATVP	ARSAIAVHTG	QKGGSGATPT	SAPSTTPTSG	480
TVSMTFAEQA	TTTTGENIFL	VGSISQLGNW	NPASAIALLS	AAYPWTSVSV	NIPAGTTFQY	540
KFIRKETDGS	VWVESDPNRQ	ATAPASGTTT	LTSSWR			576

SEQ ID NO: 152 moltype = AA length = 575
 FEATURE Location/Qualifiers
 source 1..575
 mol_type = protein
 organism = Cerrena unicolor

SEQUENCE: 152

RPLNASNALD	ARAPSGAKSV	IIQMFETWTD	SVAAECTNFI	GPAGYGFVQV	SPPQETIQGD	60
QWWTDYQPV	YILTSKRGR	DQFAAMIDTC	HDAGVKVIVD	TIWNHMAGVE	SGTGvagssF	120
THYNYPGIYQ	NQDFHHCGL	SGDDIVNYDN	AQEVQTCELV	NLADLATETD	YVRGRLAEG	180
NDLLSLGADG	LRLDAAKHIA	VGDLANIIGR	LNSTPYITQE	VIFGSGEPIT	PNQYTGNQDV	240
QKFRYTSALK	DAFLNGDISS	LQDFENRGWV	AGSGANVFVT	NHDTERNGNS	LNNNSPNNAY	300
TLAMIFSLAH	PYGTPSILSS	YSGFTDIDAG	APNGGAGTCS	SGSGSNGWLC	QHRWTAVAGM	360
VGERNTVGS	ALNNWVSPQS	SQIAFGRGAL	GFVAINNNGD	TWSTFTTSL	PDGTYCDVIT	420
GTSSSGSCTG	SSFTVSGGTF	TANVAARDAV	AIHTGATGTG	SGSGNTSTGS	GGTTSDTVSV	480
SFAETATTTT	GENIFLVGSI	SQLGAWDPAS	AIALLSSASYP	TWTVTVTLP	GTTFEYKFIR	540
KETDGSVVWE	SDPNRQATT	SSSDAATTTT	STSWR			575

SEQ ID NO: 153 moltype = AA length = 562
 FEATURE Location/Qualifiers
 source 1..562
 mol_type = protein
 organism = Physalacria cryptomeriae

SEQUENCE: 153

APFLELQVRA	PSSTKQVIIQ	MFEWTWDSVA	AECTDFIGPA	GYGYVQVSPP	AEHITGDQWW	60
TDYQPVSYIL	TSKRGNRSQF	ANMVTTCHTA	GVLVIADALF	NHMAGIESGT	GTAGSSFTHY	120
DYPGIYQTQD	PHHCLTSGD	DISDYSSQAQ	VQTCELVNLA	DLATDTEYVR	AKLASANDL	180
ISLGVNGLRL	DAAKHIATDD	IKNILSRLSS	TVYITQEVIF	GSGEPTVPSM	YTQNGDVQEF	240
RYTSTLQSAF	SGGDISQLQN	LDSKGWIAGT	SANVFVANHD	TERGSSSLNY	KSSSNTYVTA	300
TIFSLAHPYG	TPTILSSYEF	SDTDAGSPNG	GAGTCSTTGG	ANGWLCQHRW	VAFSGMVGPH	360
NNVGTASLTN	WVSPQSNQIA	FERSGKGFVA	INNADSAWTA	TFTTSLAAGS	YCDVITGTSN	420
GSACSGTSYT	VSGGSFSAIV	AARSAVAIHT	GATGSGSGGG	GGSTGSAVIT	FQETATTTLG	480
ENIFLVGSI	QLRTWAPASA	IALSSASYPT	WSVTVSIPAG	TTFEYKFIRK	ESDGSVVWES	540
DPNRSATASS	SASTQTILTS	WR				562

SEQ ID NO: 154 moltype = AA length = 578
 FEATURE Location/Qualifiers
 source 1..578
 mol_type = protein
 organism = Lenzites betulinus

SEQUENCE: 154

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RPASTVPHGA	ETRS�DARAP	SGSKDVIIQM	FEWTWDSVAA	ECTNFIGPAG	YGFVQGSPPQ	60
EHIQGAQWWT	DYQPVSYTLT	SKRGDRTSFA	NMIQTCHTAG	VGVIQVDTLFN	HMAGVDSGTG	120
VAGSSFTHYN	YPGIYQNDQF	HHCGLEPGDD	IVNYDNAVEV	QTCELDNLAD	LATETEVVRG	180
RLAQYGNDDL	SLGADGMRLD	AAKHIAVGDI	ANILSRLNRT	VYITQEVIFG	AGEPITPNQY	240
TGNGDVQEFR	YTSALQDAFL	NSGIANLQVL	ENRGWVPGSG	ANVFVTNHD	ERNGASLNNN	300
SPSNTYVTAM	IFSLAHPFGT	PSILSSYSGF	TDTDAGAPNG	GVGTCSGSGG	TNGWLCQHRW	360
TAVAGMVGFR	NQVGSAALGN	WQSPQSQQIA	FGRGALGFVA	INNADSAWSA	TFTTSLPDGS	420
YCDVISGQTS	GSTCTGSSFT	VSGGSLSATV	PARSAIAVHT	GQKGTGSGSG	TGTGGGGSTG	480
SGNVAVNFAE	TATTTFGENI	FVVGSISQLG	TWNTANAIAL	SSPSYPTWTV	SISIPAGTTF	540
QYKFIRKETD	GSVVWESDPN	RQATAPASGS	TTLTSTWR			578

SEQ ID NO: 155 moltype = AA length = 571
 FEATURE Location/Qualifiers
 source 1..571
 mol_type = protein
 organism = *Trametes lujbarskyi*

SEQUENCE: 155

RPATFDAADA	RSVQPRAPSG	SKDVIIQMFE	WTWDSVAAEC	TNFIGPAGYG	FVQGNPPQEH	60
IQGDQWWTDY	QPVSYILTSK	RGDRTAFAAN	ISTCHAAGVG	VIVDTIFNHM	SGVDSGTGVA	120
GSSSFTHYNP	GIYQNDQFHH	CGLEPGDDIV	NYDNAVEVQT	CELENLADLA	TDTEYVRGRL	180
AQYANDLLSL	GADGLRLDAA	KHIPTGDIAN	ILSRLNRSVY	ITQEVIIYDG	EPITPNQYTG	240
NGDVQEFRYT	TALKNAFLGG	GISSLQSFEN	LGWVPGTGAN	VFVTNHDTER	NGNSLNNNSP	300
SNTYVTAMIF	SLAHPYGTPT	ILSSYSGFTN	TDAGAPNGGT	GTCGSGGGAN	GWLCQHRWTA	360
VAGMVGFRNN	VGSAALTNWQ	SPQSQQIAFG	RGALGFVAIN	NADSAWSTTF	TTSLLPDGSG	420
DVVSSTSSNG	GCTGSSFSVS	GSSLTATVPA	RSALIAHTGE	TGSGSNSGGG	SGSGGTVTIN	480
FAETATTTFG	ENIFVVGSI	QLGWNPNANA	IALSSASYPT	WTVSVSVVPA	TTFEYKFIRK	540
ETDGSVVWES	DPNRSDTAPA	SGTQTITTSW	R			571

SEQ ID NO: 156 moltype = AA length = 451
 FEATURE Location/Qualifiers
 source 1..451
 mol_type = protein
 organism = *Bacillus subtilis*

SEQUENCE: 156

GPAAASAETA	NKSNELTAPS	IKSGTILHAW	NWSFNTLKH	MKDIHDAGYT	AIQTSPINQV	60
KEGNQGDKSM	SNWYWLQPT	SYQIGNRYLG	TEQEFKEMCA	AAEEYGIKVI	VDAVINHTTF	120
DYAAISNEVK	SIPNWTGHNT	QIKNWSDRWD	VTQNSLLGLY	DWNTQNTQVQ	SYLKRFLERA	180
LNDGADGFRF	DAAKHIELPD	DGSYGSQFWP	NITNTSAEFQ	YGEILQDSAS	RDAAYANYMD	240
VTASNYGHSI	RSALKNRNLG	VSNISHYAYD	VSADKLVTWV	ESHDTYANDD	EESTWMSDDD	300
IRLGWAVIAS	RSGSTPLFFS	RPEGGNGGVR	FPGKSQIGDR	GSALFEDQSI	TAVNRFHNVN	360
AGQPEELSNP	NGNNQIFMNQ	RSGHGVVLAN	AGSSSVSINT	PTKLDPGRYD	NKAGAGSFQV	420
NDGKLTGTIN	ARSAVLYPD	DIEIRCNTFF	Q			451

SEQ ID NO: 157 moltype = AA length = 633
 FEATURE Location/Qualifiers
 source 1..633
 mol_type = protein
 organism = *Bacillus subtilis*

SEQUENCE: 157

GPAAASAETA	NKSNELTAPS	IKSGTILHAW	NWSFNTLKH	MKDIHDAGYT	AIQTSPINQV	60
KEGNQGDKSM	SNWYWLQPT	SYQIGNRYLG	TEQEFKEMCA	AAEEYGIKVI	VDAVINHTTS	120
DYAAISNEVK	SIPNWTGHNT	QIKNWSDRWD	VTQNSLLGLY	DWNTQNTQVQ	SYLKRFLDRA	180
LNDGADGFRF	DAAKHIELPD	DGSYGSQFWP	NITNTSAEFQ	YGEILQDSAS	RDAAYANYMD	240
VTASNYGHSI	RSALKNRNLG	VSNISHYASD	VSADKLVTWV	ESHDTYANDD	EESTWMSDDD	300
IRLGWAVIAS	RSGSTPLFFS	RPEGGNGGVR	FPGKSQIGDR	GSALFEDQAI	TAVNRFHNVN	360
AGQPEELSNP	NGNNQIFMNQ	RSGHGVVLAN	AGSSSVSINT	ATKLDPGRYD	NKAGAGSFQV	420
NDGKLTGTIN	ARSAVLYPD	DIAPKPHVFL	ENYKTGVTHS	FNDQLTITLR	ADANTTKAVY	480
QNNNGPETAF	KDGDQFTIGK	GDPFGKTYTI	MLKGTNSDGV	TRTEKYSFVK	RDPAKATIG	540
YQNPNHWSQV	NAYIYKHDGS	RVIELTGSWP	GKPMTKNADG	IYTLTLPADT	DTTNAKIVFN	600
NGSAQVPGQN	QPGFDYVLNG	LYNDSGLSGS	LPH			633

SEQ ID NO: 158 moltype = AA length = 487
 FEATURE Location/Qualifiers
 source 1..487
 mol_type = protein
 organism = *Schwanniomyces occidentalis*

SEQUENCE: 158

KPIFLSKRDA	GSSAAAARWS	ESIQQLVTDR	FARTDGSTSA	TCNTGDRVYC	GGTFQGIIDK	60
LDYIQGMGFT	AIWISPVVEQ	IPDDTGYGYA	YHGYWMKDIY	AINSNFGTAD	DLKNLSNELH	120
KRNMKLMVDI	VTNHYAWNGA	GSSVAYSNNY	PFNQQSYPHD	YCLITNYDDQ	TNVEDCWEGD	180
NTVSLPDLRT	EDSDVSIFN	LWVAELVSNY	SIDGLRIDS	KHVDSEFPYS	FQSAAGVYLL	240
GEVYDGPAY	TCPYQNYMSG	VTNYPLYPPM	LRFFQGTSSNS	VDELNAMISS	LESDCCKDITL	300
LGNFIENHDQ	PRLPSYTSDS	ALIKNAIAFN	LMSDGPIIY	YGQEQGYSGS	SDPNNREALW	360
LSGYSTSNGY	YKLISSENQI	RNQAIIYKDSK	YTTYWSDVLY	ASGHVIALQR	GADDQRIYSV	420
FNNLGSSSGQ	TVTFSTKYS	GEKVVDVLT	QTSYANSNST	LTVSISGGAP	RIYAPASLIA	480
NSGICNF						487

SEQ ID NO: 159 moltype = AA length = 583
 FEATURE Location/Qualifiers

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source                1..583
                      mol_type = protein
                      organism = Rhizomucor pusillus

SEQUENCE: 159
ATSDDWKGA  IYQLLTRDFG  RADDSTSNCS  NLSNYCGGTY  EGITKHLDIYI  SGMGFDAIWI  60
SPIPKNSDGG YHGYWATDFY  QLNNSNFGDES  QLKALIQAAH  ERDMYVMLDV  VANHAGPTSN  120
GYSGYTFGDA SLYHPKCTID  YNDQTSIEQC  WVADELPDID  TENSNDVAIL  NDIVSGWVGN  180
YSFDGIRIDT VKHIRKDFWT  GYAEAAGVFA  TGEVFNGDPA  YVGPIYQKYL  SLINYPMYA  240
LNDVFSKSK  GFSRISEMLG  SNRNAPEDTS  VLTTFVDNHD  NPRFLNSQSD  KALFKNALTY  300
VLLGEGIPIV YYGSEQGFSG  GADPANREVL  WTTNYDTSSD  LYQFIKTVNS  VRMKSNAKAV  360
MDIYVGDNAY AFKHGDALVV  LNNYSGSGSTN  QVSFVSXGKF  DSGASLMDIV  SNITTTVSSD  420
GTVTFNLKDG LPAIFTSATG  GTTTTATPTG  SGSVTSTSKT  TATASKTSTS  TSSTSTCTPT  480
AVAVTFDLTA TTTYGENIYL  VGSISQLGDW  ETSDGIALSA  DKYTSSDPLW  YVTVTLPAGE  540
SFEYKFIRIE SDDSVIEWESD  PNREYTVPQA  CGTSTATVTD  TWR  583

SEQ ID NO: 160      moltype = AA  length = 616
FEATURE             Location/Qualifiers
source              1..616
                    mol_type = protein
                    organism = Aspergillus niger

SEQUENCE: 160
AEWRTQSIYF  LLTDRFGRTD  NSTTATCDTG  DQIYCGGSWQ  GIINHLDYIQ  GMGFTAIWIS  60
PITEQLPQDT  ADGEAYHGYW  QQKIYDVNSN  FGTADDLKS  L  SDALHARGMY  LMVDVVPNHM  120
GYAGNGNDVD YSVFDPDPDS  SYFPHYCLIT  DWDNLTMVQD  CWEGDTIVSL  PDLNNTTETAV  180
RTIWDVWVAD LVSNSYVDGL  RIDSVLEVEP  DFFPGYQEEA  GVCYCVGEVDN  GNPALDCPYQ  240
KVLGDVLNYP IYWQLLYAFE  SSSGSISNLY  NMIKSVASDC  SDPTLLGNFI  ENHDNPRFAS  300
YTSYYSQAKN VLSYIFLSDG  IPIVYAGEEQ  HYSGGKVPYN  REATWLSGYD  TSAELYTWIA  360
TTNAIRKLAI SADSAITYYA  NDAFYTDSNT  IAMRKGTSGS  QVITVLSNKG  SSGSSYTLTL  420
SGSGYTSGTK LIEAYTCTSV  TVDSSGDIPV  PMASGLPRVL  LPASVVDSSS  LCGSGRRTTT  480
TTTAATSTSK ATSSSSSSSA  AATSSSCTA  TSTTLPIITE  ELVTTTYGEE  VYLSGSSISQL  540
GEWDTSDAVK LSADDTSSN  PEWSVTVSLP  VGTTFEYKFI  KVDGEGSVTV  ESDPNREYTV  600
PECGNGSGET VVDTWR  616

SEQ ID NO: 161      moltype = AA  length = 512
FEATURE             Location/Qualifiers
source              1..512
                    mol_type = protein
                    organism = Bacillus stearothermophilus

SEQUENCE: 161
APFNGTMMQY  FEWYLPDDGT  LWTKVANEAN  NLSSLGITAL  WLPPAYKGT  RSDVGYGVYD  60
LYDLGEFNQK  GTVRTKYGTK  AQYLQAIQAA  HAAGMQVYAD  VVFDHKGAD  GTEWVDAVEV  120
NPSDRNQEIS  GTYQIQAATK  FDPFGRGNTY  SSFKWRWYHF  DGVWDDES  LSRIYKFRGK  180
AWDWEDVTEF  ISGDYLTAEW  TKFDFPGRGN  TYSDFKRWY  HFDGVDWDQS  RQFQNRIFYK  240
RGAWDWEVD  SENGNYDYL  YADVMDHPE  VVNELRRWGE  WYTNTLNLDG  FRIDAVKHIK  300
AFDMRTLMTN  TLMKDQPTLA  VTFVDNHDE  PGQALQSWVD  PWFKPLAYAF  ILTRQEGYPC  360
VFYGDYYGIP  QYNIPSLKSK  IDPLLIARRD  YAYGTQHDYL  DHSDIIGWTR  EGGTEKPGSG  420
LAALITDGP  GSKWMYVGK  HAGKVFDLT  GNRSDVTIN  SDGWGEFKN  GGSVSVVWPR  480
KTTVTIAR  ITTRPWTGEF  VRWTEPRLVA  WP  512

SEQ ID NO: 162      moltype = AA  length = 483
FEATURE             Location/Qualifiers
source              1..483
                    mol_type = protein
                    organism = Bacillus halmapalus

SEQUENCE: 162
HHNGTNGTMM  QYFEWHLND  GNHWNRLRDD  ASNLRNRGIT  AIWIPPAAWK  TSQNDVG YGA  60
YDLYDLGEFN  QKGTVRTKY  TRSQLESIAH  ALKNNGVQVY  GDVVMNHKGG  ADATENVLAV  120
EVNPNNRNQE  ISGDYLTAEW  TKFDFPGRGN  TYSDFKRWY  HFDGVDWDQS  RQFQNRIFYK  180
RGKAWDWEVD  SENGNYDYL  YADVMDHPE  VVNELRRWGE  WYTNTLNLDG  FRIDAVKHIK  240
YSFTRDWLTH  VRNATGKEMF  AVAEFWKNDL  GALENYLNKT  NWNHVSVDVP  LHYNLYNASN  300
SGGNVDMAKL  LNGTVVQKHP  MHAVTTFVDNH  DSQPGESLES  FVQEWFKPLA  YALILTREQG  360
YPSVFYGDY  GIPTHSVPA  KAKIDPILEA  RQNFAYGTQH  DYFDHNNIIG  WTREGNTTHP  420
NSGLATIMSD  GPGGEKMYV  GQNKAGQVWH  DITGNKPGTV  TINADGWANF  SVNGGSVSIW  480
VKR  483

SEQ ID NO: 163      moltype = AA  length = 478
FEATURE             Location/Qualifiers
source              1..478
                    mol_type = protein
                    organism = Aspergillus oryzae

SEQUENCE: 163
ATPADWRSQS  IYFLLTDRFA  RTDGSTTATC  NTADQKYCGG  TWQGIIDKLD  YIQGMGFTAI  60
WITPVTALPL  QTTAYGDAYH  GYQQDIYSL  NENYGTADDL  KALSALHER  GMYLMVDVVA  120
NHMGVDGAGS  SVDYSVFKPF  SQQDYFHPFC  FIQNYEDQQT  VEDCWLGENT  VSLPDLDTTK  180
DVVKNEWYDW  VGLSVSNYSI  DGLRIDTVKH  VQKDFWPGYN  KAAGVYCIGE  VLDGDPAYTC  240
PYQNVMDGVL  NYPIIYPLL  AFKSTSGSMD  DLYNMINTVK  SDCPDSTLLG  TFVENHDNPR  300
FASYTNDIAL  AKNVAAFIIL  NDGIPIIYAG  QEQHYAGGND  PANREATWLS  GYPTDSELYK  360
LIASANAIRN  YAIKSDTGFV  TYKNWPYIKD  DTTIAMRKGT  DGSQIVTILS  NKGASGDSYT  420
LSLSGAGYTA  GQQLTEVIGC  TTVTVGSDGN  VVPVPMAGGLP  RVLVPTKELA  GSKICSSS  478

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SEQ ID NO: 164 moltype = AA length = 483
 FEATURE Location/Qualifiers
 source 1..483
 mol_type = protein
 organism = *Bacillus amyloliquefaciens*

SEQUENCE: 164

VNGTLMQYFE	WYTPNDGQHW	KRLQNDAEHL	SDIGITAVWI	PPAYKGLSQS	DNGYGPYDLY	60
DLGEFQQKGT	VRTKYGKSE	LQDAIGSLHS	RNVQVYGDV	LNHKAGADAT	EDVTAVEVNP	120
ANRNQETSEE	YQIKAWTDFR	FPGRGNTYSD	FKWHWYHFDG	ADWDESRKIS	RIFKFRGEGK	180
AWDWEVSSSE	GNIDYLMYAD	VDYDHPDVVA	ETKKWGIWYA	NELSLDGFPRI	DAAKHIKFSF	240
LRDWWQAVRQ	ATGKEMPTVA	EYWQNNAGKL	ENYLNKTSFN	QSVFDPVLFH	NLQAASSQGG	300
GYDMRRLLDG	TVVSRHPEKA	VTFVENHDTQ	PGQSLSTVQ	TWFKPLAYAF	ILTRESGYPO	360
VFYGDMYGTK	GTSPEKIPSL	KDNIEPILKA	RKEYAYGPQH	DYIDHPDVIG	WTREGDSSAA	420
KSGLAALITD	GPGGSKRMYA	GLKNAGETWY	DITGNRSDTV	KIGSDGWGEF	HVNDGSVSIY	480
VQK						483

SEQ ID NO: 165 moltype = AA length = 583
 FEATURE Location/Qualifiers
 source 1..583
 mol_type = protein
 organism = *Rhizomucor pusillus*

SEQUENCE: 165

ATSDDWKGA	IYQLLTRDFG	RADDSTSNCS	NLSNYCGGTY	EGITKHLDI	SGMGFDIWI	60
SPIPKNSDGG	YHGWWATDFY	QLNSNFGDES	QLKALIAAH	ERDMYVMLDV	VANHAGPTSN	120
GYSGYTFDDA	SLYHPKCTID	YNNQTSIEQC	WVADELPDID	TENSDNVAIL	NDIVSGWVGN	180
YSFDGIRIDT	VKHIRKDFWT	GYAEAAGVFA	TGEVFNPDPA	YVGPYQKYL	SLINYPMYA	240
LNDVFSKSK	GFSRISEMLG	SNRNAFEDTS	VLTFVDNHD	NPRPLNSQSD	KALFKNALT	300
VLLGEGPIV	YVGSEQGFSG	GADPANREVL	WTTNYDTSSD	LYQFIKTNS	VRMKSNAKAV	360
MDIYVGDNAY	AFKHGDLV	LNNYSGSGTN	QVSFVSQGF	DSGASLMDIV	SNITTTVSSD	420
GTVTNFKLDG	LPAIFTSATG	GTTTTATPTG	SGSVTSTSKT	TATASKTSTS	TSSTSTCTPT	480
AVAVTFDLTA	TTTTYGENIYL	VGSISQLGDW	ETSDGIALSA	DKYTSSDPLW	YVTVTLPAGE	540
SFEYKFIRIE	SDDSVIEWS	PNREYTVQA	CGTSTATVTD	TWR		583

SEQ ID NO: 166 moltype = AA length = 623
 FEATURE Location/Qualifiers
 source 1..623
 mol_type = protein
 organism = *Kionochaeta ivoriensis*

SEQUENCE: 166

LSPAGWRQSS	IYQVMTDRFA	RTDGSTIASC	DTSQQAYCGG	TWQGLINKLD	YIQGMGFTAV	60
WISPMVHOMA	GATSDGESYH	GYWAQDINTV	NSAFGTAADL	KALSAALHAR	GMVLMVDVVP	120
NHFAYDGGGT	CVDYSIFNPF	NSESYFHPFC	LIDYSNTTSI	QVCWEGDNTV	SLPDLRTEDS	180
DVRSIWNWSI	SSVISTYGV	GLRVDQAQV	ETSFVAGFEA	AAGVYMGVEV	FNGDPTYVTP	240
PQDYMDGVNL	YPAYWITQA	FESTSGSISN	LANGMNTMKS	LAKNTSLGS	FLENHNDNPR	300
PSLTSDMSLA	QNAIAFTMLM	DGIPIIYQGG	EQHFSGGGVP	SDREAVWLSG	YPNDTTLAY	360
ITKLNAVRSW	AIAKDSSYLA	YMAYPVYTD	HTIAMRKGD	GYQVISVYTN	VGASGSSYSV	420
TLTSADTGFT	ASQSVVDLVG	CKTYTADSTG	SLSLSLTGGI	PIILYPAASL	TGNTICTSTG	480
GTTTTATPTG	SGSVTSTSKT	TATASKTSTS	TSSTSTCTPT	AVAVTFDLTA	TTTTYGENIYL	540
VGSISQLGDW	ETSDGIALSA	DKYTSSDPLW	YVTVTLPAGE	SFEYKFIRIE	SDDSVIEWS	600
PNREYTVQA	CGTSTATVTD	TWR				623

SEQ ID NO: 167 moltype = AA length = 588
 FEATURE Location/Qualifiers
 source 1..588
 mol_type = protein
 organism = *Aspergillus niger*

SEQUENCE: 167

LSAAEWRTQS	IYFLLTDRFG	RTDNSTTATC	DTGDQIYCGG	SWQGIINHLD	YIQGMGFTAI	60
WISPIEQPL	QDTADGEAYH	GYWQQKIYDV	NSNFGTADDL	KSLSDALHAR	GMVLMVDVVP	120
NHMGYAGNGN	DVDYSVDFPF	DSSSYFHPYC	LITDWDNLTM	VQDCWEGDTI	VSLPDLNTE	180
TAVRTIWDW	VADLVSNYSV	DGLRIDSLE	VEPDFFPGYQ	EAAGVYCVGE	VDNGNPALDC	240
PYQKVLGVL	NYPIYQQLLY	AFESSSGSIS	NLYNMIKSA	SDCSDPTLLG	NFIENHNDNPR	300
FASYTSDYSQ	AKNLIAYTFL	SDGIPVYAG	EEQHYSGGKV	PYNREATWLS	GVDTSALYLT	360
WIATTNAIRK	LAIADSAYI	TYANDAFYTD	SNTIAMRKGT	SGSQVITVLS	NKGSSGSSYT	420
LTLGSGGYTS	GTKLIEAYTC	TSVTVDSSGD	IPVPMASGLP	RVLLPASVVD	SSSLCGGSGR	480
GATSPGGSSG	SVEVTFDVYA	TTVYQNIYI	TGDVSELGNW	TPANGVALSS	ANYPTWSATI	540
ALPADTTIQY	KYVINDGSTV	IWEDAISNRE	ITTPASGTYT	EKDTWDES		588

SEQ ID NO: 168 moltype = AA length = 588
 FEATURE Location/Qualifiers
 source 1..588
 mol_type = protein
 organism = *Aspergillus oryzae*

SEQUENCE: 168

ATPADWRSQS	IYFLLTDRFA	RTDGSTTATC	NTADQKYCGG	TWQGIIDKLD	YIQGMGFTAI	60
WITPVTALPL	QTTAYGDAYH	GYWQQDIYSL	NENYGTADDL	KALSSALHER	GMVLMVDVVA	120
NHMGYDGPSS	SVDYSVFPFP	NSASYFHPFC	FIQWWDNQTO	VEDCWLGDN	VSLPDLDTTK	180
DVVKNEWYDW	VGLSVSNYSI	DGLRIDTVKH	VQKDFWPQYN	KAAGVYCIGE	VLDGDPAYTC	240

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PYQEVLDGVL	NYPIYYPLLN	AFKSTSGSMD	DLYNMINTVK	SDCPDSTLLG	TFVENHDNPR	300
FASYTNDIAL	AKNVAAPFIL	NDGIPIIYAG	QEQHYAGGND	PANREATWLS	GYPTDSELYK	360
LIASANAIRN	YAIKDTGFV	TYKNWPIYKD	DTTIAMRKGT	DGSQIVTILS	NKGASGDSYT	420
LSLSGAGYTA	GQQLTEVIGC	TTVTVDSSGD	VPVPMAGGLP	RVLYPTEKLA	GSKICSSSGR	480
GATSPGGSSG	SVEVTFDVYA	TTYGQNIYI	TGDVSELGNW	TPANGVALSS	ANYPTWSATI	540
ALPADTTIQY	KYVNIDGSTV	IWEDAISNRE	ITTPASGTYT	EKDTWDES		588

SEQ ID NO: 169 moltype = AA length = 583
 FEATURE Location/Qualifiers
 source 1..583
 mol_type = protein
 organism = *Penicillium canescens*

SEQUENCE: 169

LTPAEWRSQS	IYFMLTDRFG	RSDNSTTAAC	NVSDRTYCGG	TWQGIINHLD	YIQGMGFTAI	60
WITPVTEQLP	QDTGDGEAYH	GYWQQNIYEI	DSKLGTAADL	LALSALHAR	GMVLMVDVVA	120
NHMGVAGSGN	SVDYSVFNPF	SSSSYFHSYC	LISNYDDQSN	VENCWLGDIT	VSLPDLDTTQ	180
TAVQTIWYDW	IADLVSNYSI	DGLRIDTVKH	VQKSFWPQYN	DAAGVYCVGE	IFDGDPAITC	240
DYQNYMDGVL	NYPIYYQLLY	AFQSSSGSIS	DLYNMINSVK	SDCADSTLLG	NFIENHDNPR	300
FASYTSDYSQ	AKNVISFLFL	SDGIPIIYSG	QEQHYSGGAD	PANREATWLS	GYSTTAELYK	360
YIATTNKIRK	LAVSDSSYLT	TKNVPFYQDS	HTLAMKKGSS	ASPVITVLSN	YGSSGSSSYTL	420
SLSGSGYSSG	TKLMEMYTCT	SVTVDSGSGI	AVPMASGLPR	VLMLASSANS	LCGSSGATSP	480
GGSSSGSEVT	PDVYATTVYG	QNIYITGDVS	ELGNWTPANG	VALSSANYPT	WSATIALPAD	540
TTIQYKYVNI	DGSTVIWEDA	ISNREITTPA	SGTYTEKDTW	DES		583

SEQ ID NO: 170 moltype = AA length = 583
 FEATURE Location/Qualifiers
 source 1..583
 mol_type = protein
 organism = *Acidomyces acidothermus*

SEQUENCE: 170

LTPAQWRGQS	IYQVLTDRFG	RTDDSTTAAC	DVNDYCGGSW	QGIINHLDYI	QDMGFSIAWI	60
SPVVENLVGD	TQDGSAYHGY	WAQNIYALNP	NFGTVSDLVA	LSAALHQRGM	YLMVDVVTNH	120
MGYDGCDCGV	DYSVFTPFNS	QSYFHPFCLI	DYNNSTSIKV	CWEGDNIVSL	PDMRTEDSDV	180
ATEWNTWISE	LVSNSYIDGL	RIDSAQQVDN	AFPPFPQAAA	GGIHVLGEVF	NGDPNVVCPY	240
QDFMSGVLNY	PAYYYITQAF	QSTSGSISNL	VNGINQMKT	CTDTLLGSGF	LENHDNPRFP	300
SYTSDLSLDK	NAITFTILQD	GIPIIYEGQE	QHYSGGTVPN	NREAIWLSGY	DKSAPLYTWI	360
ASVNQIRNQA	IFKDSNYLTY	MAWPIYSDAS	TIAMRKGFDP	LQIISVYSNK	GASAAASYTIS	420
LESSTTGFTA	NEALVEVMSC	TTYTTDGSNG	LAVTISGGLP	AVFYPKAQLA	GSGICGATSP	480
GGSSSGSEVT	PDVYATTVYG	QNIYITGDVS	ELGNWTPANG	VALSSANYPT	WSATIALPAD	540
TTIQYKYVNI	DGSTVIWEDA	ISNREITTPA	SGTYTEKDTW	DES		583

SEQ ID NO: 171 moltype = AA length = 586
 FEATURE Location/Qualifiers
 source 1..586
 mol_type = protein
 organism = *Kinochaeta ivoriensis*

SEQUENCE: 171

LSPAGWRQQS	IYQVMTDRFA	RTDGSTIASC	DTSQQAYCGG	TWQGLINKLD	YIQGMGFTAV	60
WISPMVHQMA	GATSDGESYH	GYWAQDINTV	NSAFGTAADL	KALSAAHAR	GMVLMVDVVT	120
NHFAYDGCCT	CVDYSIFNPF	NSESYPHPPC	LIDYSNTTSI	QVCWEGDNTV	SLPDLRTEDS	180
DVRSIWNWSI	SSVISTYGV	GLRVDSAQV	ETSPWAGFEA	AAGVYVMGEV	FNGDPTVYTP	240
FQDYMVGVLN	PAYYITQAF	FESTSGSISN	LANGMNTMKS	LAKNTSLGSG	FLENHDNPRF	300
PSLTSDMSLA	QNAIAFTMLM	DGIPIIYQGG	EQHFSGGGVP	SDREAVVLSG	YPNDTTLTAY	360
ITKLNAVRSW	AIAKDSSYLA	YMAYPVYTD	HTIAMRKGD	GYQVISVYTN	VGASGSSYSV	420
TLTSDTGTFT	ASQSVVDLVG	CKTYTADSTG	SLSLSLTGGI	PIILYPAASL	TGNTICTSGA	480
TSPGGSSGSV	EVTDFVYATT	VYQNIYITG	DVSELGNWTP	ANGVALSSAN	YPTWSATIAL	540
PADTTIQYKY	VNIDGSTVIW	EDAISNREIT	TPASGTYTEK	DTWDES		586

SEQ ID NO: 172 moltype = AA length = 585
 FEATURE Location/Qualifiers
 source 1..585
 mol_type = protein
 organism = *Aspergillus terreus*

SEQUENCE: 172

LTPAEWRSQS	IYFLLTDRFG	RTDNSTTAAC	DTSDRVYCGG	SWQGIINQLD	YIQGMGFTAI	60
WITPVGTQFY	ENTGDGTSYH	GYWQDDIYDL	NYNYGTAQDL	KNLANALHER	GMVLMVDVVA	120
NHMGYDGAGN	TVDYSVFNPF	SSSSYFHPYC	LISNYDNQTN	VEDCWLGDIT	VSLPDLDTTS	180
TAVRNIWYDW	VADLVANYSI	DGLRVDTVKH	VEKDFWPQYN	SAAGVYCVGE	VYSGDPAYTC	240
PYQNYMDGVL	NYPIYYQLLY	AFESSSGSIS	DLYNMISSVA	SSCKDPTLLG	NFIENHDNPR	300
FASYTSDYSQ	AKNVITFIFL	SDGIPIVYAG	QEQHYSGGSD	PANREATWLS	GYSTSATLYT	360
WIATTNQIRS	LAISKDAGYV	QKANNPFYSD	SNTIAMRKGT	TAGAQTIVTL	SNKGASGSSY	420
TLSLSGTGYS	AGVTLVETYT	CTTVTVDSG	NLPVPMTSGL	PRVFPVSSWV	NGSALCNGAT	480
SPGGSSGSVE	VTFDVYATT	YQNIYITGD	VSELGNWTPA	NGVALSSANY	PTWSATIALP	540
ADTTIQYKYV	NIDGSTVIWE	DAISNREITT	PASGTYTEKD	TWDES		585

SEQ ID NO: 173 moltype = AA length = 565
 FEATURE Location/Qualifiers
 source 1..565
 mol_type = protein

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organism = Thamnidium elegans

SEQUENCE: 173
VPLSVLDKRN  GVTTLSKRAA  AADWKSRSIY  QLVTDRFGRS  DGSTSACGDL  SNYCGGDYKG  60
IQNQLDYIAG  MCFDAIWISP  IPENTDGGYH  GYWAKDFEKL  NTNFGSADDL  KALVTAAGHK  120
GMYVMLDVVA  NHAGPASGGD  YSGFTFSSAS  NYHPQCTIDY  DNQTSVEQCW  VADDLPDINT  180
EDDTIVSKLH  SIVSDWVTY  DFDGIRIDTV  KHIRKDFWSG  YEEAAGVFAT  GEVFDGDAAY  240
VGFPYQQLSS  LINYPLYAY  RDVFSAGSGF  SRISDMLSTI  KSNFKDPSVL  TTFVDNQDNA  300
RFLSVKSDMS  LYKNALAFIT  LTEGIPVVY  GTEQGFGKGD  DPKNREVLWT  SNYDTSSDLY  360
KFIKIVNNDV  RQKSDKTVT  DVDVGTNTYA  FTHGKNLIV  NNYGSGSTES  VTVKVGDSVA  420
DGTKLVDAVS  NITATVSGGS  ITFSLKDGLP  ALFVPSSGAT  SPGGSSGSVE  VTFDVYATTV  480
YGQNIYITGD  VSELGNWTPA  NGVALSSANY  PTWSATIALP  ADTTIQYKYV  NIDGSTVIWE  540
DAISNREITT  PASGTYTEKD  TWDES  565

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SEQ ID NO: 174      moltype = AA  length = 574
FEATURE            Location/Qualifiers
source             1..574
                   mol_type = protein
                   organism = Meripilus giganteus

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SEQUENCE: 174
RPTVFDAGAD  AHSLLHARAPS  GSKDVIIQMF  EWNWDSVAEE  CTNFIGPAGY  GFVQVSPPQE  60
TIQGAQWWT  YQPVSYTLTG  KRGRSQFAN  MITTCHAAAGV  GVIVDTIWNH  MAGVDSGTGT  120
AGSSPTHYNY  PGIYQNDQFH  HCGLEPGDDI  VNYDNAVEVQ  TCELVNLADL  ATDTEVVRGR  180
LAQYGNLDS  LGADGLRLDA  SKHIPVGDIA  NILSRLRSRV  YITQEVIFGA  GEPITPNQYT  240
GNGDVQEPFY  TSALKDAFLS  SGISNLQDFE  NRGWVPGSGA  NVFVNNHDE  RNGASLNNNS  300
PSNTVVTATI  FSLAHPYGT  TILSSVDGFT  NTDAGAPNNN  VGTCTSGGA  NGWLCQHRWT  360
AIAGVMGFRN  NVGSAALNNW  QAPQSQQIAP  GRGALGFVAI  NNADSASTT  FTSLPDGSY  420
CDVISGKASG  SSCTGSSFTV  SGGKLTATVP  ARSAIAVHTG  QKGGGGGATS  PGGSSGSSEV  480
TPDVYATTYV  GQNIYITGDV  SELGNWTPAN  GVALSSANYP  TWSATIALPA  DTTIQYKYVN  540
IDGSTVIWED  AISNREITTP  ASGTYTEKDT  WDES  574

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SEQ ID NO: 175      moltype = AA  length = 671
FEATURE            Location/Qualifiers
source             1..671
                   mol_type = protein
                   organism = Chaetomium megalocarpum

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SEQUENCE: 175
LYINGSVTAP  CDSPIYQGE  ILKAIELARP  FSDSKTFVDM  PTIKPVDEVI  AAFSRLSQPL  60
SNNSELTAFL  AENFAPAGGE  LEEVPISELE  TDPSFLNKL  DVDIKEFVGK  VIDIWPDLTR  120
RYAGPNCSCQ  CANSFIPVNR  TFFVAGGRFR  EPYVWDSYWI  LEGLLRTGGA  FTQISKNMIE  180
NFLDFVETIG  FVPNGARIY  LNRSQPPLLT  LMVKTYVDYT  NDTSILERAL  PLLVKEVEFW  240
TNNRTVSITA  TDGKEYTLNR  YSVNNNQPRP  ESYREDYVTA  NNASYAQS  IIPVKTPLN  300
ETEKAELYSN  LASGAESGWD  YTAARWLKTP  DAARDVYFPL  RSLNVRISV  VDLNSILYQN  360
EVIISEYLEQ  AGNSSEAKRF  ADAAEQRSEA  MYALMWNATH  WSYFDYNLTD  NSQRFVFPAD  420
ADTAPSDQTA  APPGQQLVFD  IAQLYPPFTG  AAPASLKAMP  LAVQNAVYARV  ARMLDTKAGA  480
IAATNLRTQ  QWDQPNVWPP  LQHILMKGLL  NTPATFGTED  PAYAHTQDLA  LRLAQRYLDS  540
TFCTWYATGG  STSATPQLQ  AAPGATGTMF  EKYADDATNV  AGGGGEYEVV  EGFGTNGVL  600
IWAADVFGDR  LVRPDCGNIT  AAHTHSSEKR  TVAGASRPDA  GASGVRERR  AVELDPWDAA  660
WTKMFGRSKL  R  671

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SEQ ID NO: 176      moltype = AA  length = 662
FEATURE            Location/Qualifiers
source             1..662
                   mol_type = protein
                   organism = Lecanicillium psalliotae

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SEQUENCE: 176
LYQNGSVIAP  CDSPIYCHGD  ILKEVELARP  FTDKSTFVDM  PAKKPLAEIQ  AAFDKLEKPL  60
SNTALNEFL  STYPEDAGGE  LKAVSKDKLK  TDAKFVDKLN  DTVIKEFVNK  VIDIWPDLTR  120
EYAGSATNCT  NCPNSFIPVN  RTFVVAGGRF  REPYYWDSY  IVEGLLRTGG  AFVDITKNTI  180
ENFLDLIEQ  GFIPNGARLY  YLNRSQPPLL  SQMVKNYISY  TNDTDILKRA  LPILVKEHEF  240
FMNNSRVEIT  VENKTYTLNR  YAVSNTQPRP  ESFREDYNTV  NNNSYYAASG  IIPVKTPLN  300
ESEQATLYAN  LASGAESGLD  YTTAKWSTNP  RDSMEDYFYP  LRSLNIMNIV  PVDLNSILYG  360
NEKAISEFYN  ITGNSSEADS  WSKKAAERQE  AIQAVFNET  LYSYFDYNRT  SSSQHIYIPS  420
DDDTQSFENA  TAPAGMQEVF  TVTQFYPPFM  GAAPDYIRNN  PHAVKTAYSR  IAKYLELKPG  480
GIPSSNLRSG  QQWDQPSVWP  PLMHVLMKGL  VNTPATFGKE  DPAYKDVHKL  ALTLGQRYLD  540
STFCTWYATG  GSTSETPQLS  GLSDSDVGIM  FEKYDDTSIN  HAGGGGEYEV  VEGFGWSNGV  600
LMWVADTFGN  ELKRPDCGNI  TAAHVHFGKR  SVSAVELSSR  DAQRVKKFGR  RAEGRMKVPG  660
VL  662

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SEQ ID NO: 177      moltype = AA  length = 637
FEATURE            Location/Qualifiers
source             1..637
                   mol_type = protein
                   organism = Doratomyces sp.

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SEQUENCE: 177
LYTNGSIIAP  CDSPIYCHGD  ILHQVQLAHP  FPDKSTFVDM  PAIKSVDEIQ  AAFDKLDKPL  60
KNDTALQNFL  AENFAEAGHE  LAEVDPSELS  TDPFRFLDKV  DTVIHEFTQK  VIDIWPDLTR  120
SYSPSGAGSD  CPDCPNFLP  VNRTFVAGG  RFRPYYWDS  YWIVEGLLRT  GGDFLGVSRL  180
IENFLDFVD  DFGFVNGAR  RYVLNRSQPP  LLSLMVKSYV  EQTNDTEILD  RAVPLLIKEY  240
EFWTKNRTVE  VPFNETITL  NQYNVDNTQP  RPEYREDYI  TATNASYSSA  SGEVYEEVEK  300

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-continued

LNETQRATVY	RNLATGAESG	WDYTSRWMAR	PRDAVEDVVF	PLRSLNIIIEI	VPVDLNSILY	360
ANELAIAEFV	RLAGADDCCE	EAAAWREALAE	KRSADMHRML	WNDTLHSYFD	YNLTSSSQNV	420
YVPADNDTAP	FERPGGTPEG	SQVFFSAAQF	YPFWTGAAPS	SLRDDPAAVQ	LAYSRSVASYL	480
DLRAGGIPAT	NLRTGQQWDQ	PSVWPPLMHI	LTSGLLNTSP	ASASDDDDPS	YAPTRDLALS	540
LAQRYLDSTF	CTWYATGGST	SETPQLDGFT	DEDKGAMFEK	YDDSATNVAG	GGGEYEVVEG	600
FGWTNGVLIW	MVDTFGDELS	RPDCGDIEAA	NAHPARK			637

SEQ ID NO: 178 moltype = AA length = 601
 FEATURE Location/Qualifiers
 source 1..601
 mol_type = protein
 organism = *Mucor moelleri*

SEQUENCE: 178

IQNHTSFSCD	SPIYCEGDIL	HTVQLAKIFS	DSKTFVDMPT	SKSESQVIEA	FKAIGGRNAT	60
IAQVQQFLNE	NFLTAGTEVK	RLTNITIPEL	NWIDNITDPD	YRGWISKLNQ	AWSNLTFTFD	120
TSVLCQDCAT	STLPVSRPFV	VPGGRPFREFY	YWDSFFVIKG	LLSDQVELA	KNMILNFFDF	180
IDTYGFIPNG	ARIYYLNRSQ	PPFLTQMVFA	FWEKTSKDEF	MTNALPFLDK	EYNFWMTNNS	240
ISVPDPKPNP	KKYKMNHYVT	LNTSPRPESY	VEDYNTVNNG	TDYSQAVKLQ	LYADIAAGAE	300
TGWDYSSRW	RQKHPAPNQT	EGYEMLRSIN	TANIVPIDLN	SLLNWNTTML	AEWHDRFGEK	360
SKSKKKSAYY	QTQAKKRLDA	MEKLMWNPDT	YTYDYDNLTS	SSQNLEFTPA	NLPPIWLGLAL	420
PNKVMKNKTE	LARVDEETEN	ALRKYPGILT	TSYNTTMCQW	DWPNGWPPLS	YVAMEGMNKV	480
EESLNGKKGS	KQDGKTYTTS	RGTSLARLSF	TLAERYAASA	YCGWYKTGGS	IPGILNKIDG	540
VADDGHMFEK	FDVNTIGISG	SQGEYVSQTG	FGWTNGIALW	IFDTYANLTA	PDCNQTTITLN	600
I						601

SEQ ID NO: 179 moltype = AA length = 650
 FEATURE Location/Qualifiers
 source 1..650
 mol_type = protein
 organism = *Phialophora cyclaminis*

SEQUENCE: 179

LYINGSVIAP	CDSPIYCHGE	LLKAIELARP	FSDSKTFVDM	PTRKPVDEVV	AAFEKLSKPV	60
VNDTKLGEFL	QANFLPAGGD	LSDFPPGSLE	TDPKFLDNIN	DTVIREPTEA	VIDIWPDLTR	120
RYVGASNCSTG	CANSFIPVNR	TFVVAGGRFR	EPYYWDSYWI	IEGLLRTGGN	FVQISKNIIE	180
NFLDFVETIG	FVPNGARIYY	LNRSQPPLLT	AMVKAYLEHT	NDTKILDRAV	PLLIKEYNFW	240
ITNRSVTLTG	PDGNEYTLQR	YSVNNNQPRP	ESYREDYNNTA	NNKSYYSTSG	IYIPENVSLN	300
DSEKKELYN	LATGAESGWD	YGTWLSRPA	DALRDVYFPL	RYVKTRDLVP	VDLNAILYQN	360
EASISSFLYI	QGNDTGAHAF	AKLAAARQKA	MTAVLWNATL	PSYFDYNMTS	SSHFSFIPVD	420
DDATSIETAT	APRGFQQIFH	VAQLYPFWTG	AAPASLRENP	LAVQRAFSPV	SQMLEEKAGS	480
IPATNFYTGQ	QWDEPNVWPP	LQHIIMEGLR	NTPATFGEDD	PIYQGVQDLA	LRVAQRYLDS	540
TFCTWYATGG	STSLTPKLAG	LTDNAKGIMF	EKYGDNSTNV	AGGGGEYEVV	EGFGWTNGVL	600
IWAVDTFGNK	LKRPDGDIQ	PAHVEARGLO	RRAVELDKWD	AQWVKRFGAK		650

SEQ ID NO: 180 moltype = AA length = 654
 FEATURE Location/Qualifiers
 source 1..654
 mol_type = protein
 organism = *Thielavia arenaria*

SEQUENCE: 180

LYINGSVIAP	CDSPLYCHGD	ILKAIELARP	FQDSKTFVDM	PTIRPVEEVI	AAFNRLSQPL	60
SNNSELNAPL	QANFAPAGGE	LEEVPSELS	TNPVFLDKLN	DTVIKEFVAK	VIDIWPDLTR	120
RYAGPGNCSE	CADSFIPVNR	TFVIAGGRFR	EPYYWDSFWI	LEGLLRTGGA	FTEISKNMIE	180
NFLDFVETIG	HIPNGARIYY	LNRSQPPLLA	GMVKNYVDYT	NDTSILERAL	PLLIKEYEFW	240
TNNRTVQVTA	SDGKTYTLHR	YVNNNQPRP	ESYREDWITA	NNASYAASG	IYIPVKTPLN	300
ESEKAALYSN	LASGAESGWF	YGTFRFLMRPE	DAARDIYFPL	RHLNVRDMVT	VDLNAILYQN	360
EVIISEYLEQ	AGNNTAERF	ASAARQRSEA	MYALMWNETL	WSYFDYNLTS	NSQYTFIPAD	420
VNATAAEKAN	APEGQKVIHF	IAQLYPFWTG	AAPDQLKNNP	LAVQKAYSPV	AEMLDIKPGA	480
IPATNFITGQ	QWDQPNVWPP	LMHVLMAGLL	NTPPTFGEDD	PAYQAVQALA	LRLGQRYLDS	540
TFCTWYATGG	ETSQTPRLQG	VSPDATGTMF	EKYADNAINV	AGGGGEYEVV	EGFGWTNGVL	600
IWAADVFFANG	LKRPDGDNIT	AAHTHNGAKR	AVELHPRDAA	WTKKFGKRAL	KKRA	654

SEQ ID NO: 181 moltype = AA length = 661
 FEATURE Location/Qualifiers
 source 1..661
 mol_type = protein
 organism = *Thielavia antarctica*

SEQUENCE: 181

LYVNGSVTAP	CDSPLYCQGE	ILKAIELARP	FADSKTFVDM	PTLRPLDDVI	AAFRKLSQPL	60
SNSSELNAPL	AANFAPAGGE	LEEVPKSEL	TKPAFLDKVE	DVVIKEFVGK	VIDIWPDLTR	120
RYAGPGNCTE	CANSFIPVNR	TFVVAGGRFR	EPYYWDSYWI	LEGLLRTGGD	FTKISRNIIE	180
NFLDFVDTIG	FVPNGARIYY	LNRSQPPVLT	LMVKTYVDYT	NDTSILERAL	PLLIKEHEFW	240
TNNRSVSVEV	DGKHTHTLNR	FVNNNQPRPE	SYREDWITAN	NASYAASGI	IYIPVKTPLNG	300
TQQAELYANL	ASGAESGWDY	TSRWLKTSPD	AARDVYFPLR	SLNVINTIPV	DLNSILYQNE	360
VIISEYLEQA	GNKSGAEKFS	DAARQRSEAM	YALMWNATNW	SYFDYNLTSS	GQNTHTFPADA	420
DATAAETTSS	PAGTQLLFSV	SQLYPFWTGA	APQQLKSNPL	AVTQAYARVA	AMLDKAGAI	480
PATNLLTGQ	WDEPNVWPP	QHVLMQGLLN	TPATFGADDP	AYQATQALAL	RLAQRYLDS	540
FCTWYATGGS	TSATPRLAGV	SAGAEGSMFE	KYGDSTNVA	GGGGEYEVVE	GFGWTNGVLI	600
WAADVFFAGKL	QRPECGDIEA	AQTHGGAARR	GLGMERRAIE	LDPWDARWTK	AFGKRALRRR	660
A						661

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SEQ ID NO: 182 moltype = AA length = 667
 FEATURE Location/Qualifiers
 source 1..667
 mol_type = protein
 organism = Chaetomium sp.

SEQUENCE: 182

LYINGSVTAP	CDSPLYCQGE	ILKAIELARP	FSDSKTFVDM	PTIKPVDDVI	AAFSRLSQPL	60
SNNSELNAFL	AENFAPAGGE	LEEVPESELE	TDPAPFLDKLE	DTTIKEFVTK	VIDIWPDLTR	120
RYAGPGNCSK	CANSFIPVNR	TFVVAGGRFR	EPYYWDSYWI	LEGLLRTGGA	FTEISKNIIE	180
NFLDFVDTIG	FIPNGARIYY	LNRSQPPLLT	LMVKTYVDYT	NDTSILERAL	PLLIKEHEFW	240
TNNRSVAITA	ADGKQYLQOR	YVVMNNQPRP	ESFREDYITA	NNASYAASG	IIPVKTPLN	300
ETEKAELYSN	LASGAEGWD	YTARWLKTPN	DAARDVYFPL	RSLNVIGMVP	VDLNSILYQN	360
EVIIAEYLQO	AGNSSEARRF	ATAAEKRSEA	MYALMWNSTH	WSYFDYNLTS	NSQRIFVPTD	420
ADADPVDQTN	APPGHQVLFD	IAQLYPFWTG	AAPASLKNNP	LAVQLAYARV	AHMLDTKAGA	480
IPGTNFRITG	QWDQPNVWPP	LQHVLMMGGLL	NTPTTFGEAD	PAYQEVQRLA	LRLAQRYLDS	540
TFCTWYATGG	STSDMPQLQG	VNPGATGTMF	EKYADNATNV	AGGGGEYEVV	EGFGWTNGVL	600
IWAADVFGEG	LTRPDCGNIT	AAHTHSGAKR	GLDGGEGGAG	GLWGRRRAVEL	DPWDARWTKM	660
FGRRKRE						667

SEQ ID NO: 183 moltype = AA length = 671
 FEATURE Location/Qualifiers
 source 1..671
 mol_type = protein
 organism = Chaetomium nigricolor

SEQUENCE: 183

LYINGSVTAP	CDSPLYCQGE	ILRAIELARP	FSDSKTFVDM	PTIKPLEEVI	AAFNQLTQPL	60
SNNSELNTFL	AENFAPAGGE	LEEVPKDELN	TDPGFLDKLN	DTTIREFVAK	VIDIWPDLTR	120
RYAGGGNCSE	CANSFIPVNR	TFVVAGGRFR	EPYYWDSYWI	LEGLLRTGGA	FIEISKNIIE	180
NFLDFVETIG	FIPNGARIYY	LNRSQPPLLT	LMVKTYVDYT	NDTSILERAL	PLLIKEHEFW	240
VNNRSVEITA	ANGQTYTLNR	YHVMNNQPRP	ESYREDYITA	NNGSYAASG	IIPVVRTPLN	300
ETEKAELYAN	LASGAESGWD	YTARWLKTPN	DAANDVYFPL	RSLNVRLGVP	VDLNSILYQN	360
EVIIAEYLQO	AGNLSLAQR	AEAAEQRSEA	MYALMWNATY	WSYFDYNLTS	NSQRIFVPLD	420
ADSRITETVG	APPGHQVLFD	IAQLYPFWTG	AAPANLKNNP	LAVQQAYS RV	ASMLDAKAGA	480
IPATNFRITG	QWDQPNVWPP	LQHILMMGGLL	NTPTTFGDS	PAYQHVRDLA	LRLAQRYLDS	540
TFCTWYATGG	STSDMPQLQG	VSPDATGTMF	EKYADNATNV	AGGGGEYEVV	EGFGWTNGVL	600
IWAADVFGDA	LKRDPDCGIE	AAHTHQAKKR	DVEGLERRAV	ELDPWDAAWT	KMFGRSKLRK	660
RGAGGQKRWV	S					671

SEQ ID NO: 184 moltype = AA length = 666
 FEATURE Location/Qualifiers
 source 1..666
 mol_type = protein
 organism = Chaetomium jodhpurense

SEQUENCE: 184

LYFNGSVIAP	CDSPLYCQGE	ILKAIELARP	FSDSKTFVDM	PTIKPVDEVI	AAFNRLSQPL	60
TNNSELNAFL	AENFAPAGGE	LEEVPKDELN	TDPKFLDKLE	DATIKFVAK	VIDIWPDLTR	120
RYAGASNCSE	CANSFIPVNR	TFVIAGGRFR	EPYYWDSYWI	LEGLLRTGGA	YTQISRNMLE	180
NFLDFVETIG	FIPNGARIYY	LNRSQPPLLA	MMIKNYVDYT	NDTSILDRLA	PLLIKEHEFW	240
INNRSVSITA	ADGKQYTLHR	YVVMNNQPRP	ESYREDYITA	NNASYAASG	IIPVKTPLN	300
ESEKAELYAN	LATGAESGWD	YTARWLKTPN	DAADVYFPL	RSLNVGMVS	VDLNSILYQN	360
EVIIAEYLER	AGNISAEERF	AAMAQQRSEA	MYALMWNSTH	WSYFDYNLTS	NSQRIFVPLD	420
DDSSSTAEQAN	SPPGHQVLFD	IAQLYPFWTG	AAPESLKSNP	LAVQLAYS RV	ARMLDTKAGA	480
IPATNFRITG	QWDQPNVWPP	LQHVLMMGGLL	NTPTTFGESD	PAYQNVRLA	LRLAQRYLDS	540
TFCTWYATGG	STSDMPQLQG	VSPDATGTMF	EKYADNATNV	AGGGGEYEVV	EGFGWTNGVL	600
IWAADVFDAG	LKKPDCGNIT	AAHTHSAKRG	LERRAVELDP	WDAAWTKMFG	RSKLRKREEG	660
RKRWLS						666

SEQ ID NO: 185 moltype = AA length = 670
 FEATURE Location/Qualifiers
 source 1..670
 mol_type = protein
 organism = Chaetomium piluliferum

SEQUENCE: 185

LYINGSVIAP	CDSPLYCHGE	ILKAIELARP	FSDSKTFVDM	PTIKPLDEVI	AAFSQLSQPL	60
SNNSELNAFL	AENFAPAGGE	LEEVSKDELQ	TDPTFLDKLD	DTTIKEFVSK	VIDIWPDLTR	120
RYVSSDCSG	CANSFIPINR	TFVVAGGRFR	EPYYWDSYWI	LEGLLRTGGA	FVDISRNIIE	180
NFLDFVETIG	FVPNGARIYY	LNRSQPPLLT	LMVKTYVDYT	NDTSILERAL	PLLVKEHEFW	240
TNNRSVSITA	NGKQYTLNRY	SVNNNQPRPE	SYREDYITAS	NESYAAESGI	IIPVVRTPLNE	300
TEKAELYSNL	ASGAESGWDY	TSRWLKTND	AANDVYFPLR	SLNVLGLVPV	DLNSILYQNE	360
VILAAYFEQA	GNSSEAEERFA	AAAEQRSEAM	YDLMWNATHW	SYFDYNLTSS	SQRIFVPLDD	420
GASTQEOSTS	PPGYQVLFDV	AQLYPFWTGA	APAALKSNPL	AVQHAYS RVA	DMLNTKAGAI	480
PATNFRITGQ	WDQPNVWVPL	QHIIMQGLLN	TPPTFGESDP	AYENTQSLAL	RLAQRYLDST	540
FTCTWYATGGS	TSDMPPQLGV	SAGATGTMFE	KYADDATNVA	GGGGEYEVVE	GFGWTNGVLI	600
WAADVFGDAL	KRPDCGNITA	ASTHEGATKR	DLRGLGRRRA	ELDPWDAAWT	KMFGRALKRK	660
REQGRETWN						670

SEQ ID NO: 186 moltype = AA length = 674
 FEATURE Location/Qualifiers

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source          1..674
                mol_type = protein
                organism = Myceliophthora hinnulea

SEQUENCE: 186
LYINGSVTAP CDSPIYCHGE LLKGVELAHP FVDSKTFVDM PTLKPVDEV L AAFSKLRQPL 60
SNNSELNNFL AEYFAPAGHE LEEVPDSELQ TDPKFLDKLE DRTIKEFVSK VIDIWPDLTR 120
RYAGPGDCSD CANSFIPVNR TFVVAGGRFR EPYYWDSYWI LEGLLRTGGA FTQISKNIIE 180
NFLDFIDTIG FIPNGARIYY LNRSQPPLLT RMVKSVDYDT NDTSILERAL PLLIKEHDFD 240
TNNRSVSVTA SNGKTYTLHR YHVENNQPRP ESYREDYITA NNGSYAASG IYYPVKTPLN 300
ETEKAVLYSN LASGAESGWD YTAARWLRVPD DAARDVYFPL RSLNVREIVP VDLNSILYEN 360
EVIIAGYLEK AGNSSEAKRF ASAAKQRSEA MYNLMWNATH WSYFDYNLTS NAQNIFIPAD 420
EDTAPFDRTA APPGKQVLFH IAQLYPFWTG AAPAHLKSNP LAVQKAYARV SRMLDSKKGA 480
IAATNYRTGQ QWDQPNVWPP LQHVLMOGLL NTPATFGESD PAYQGVQKLA LRLAQRYLDS 540
TFCTWYATGG STSDFPQLQG VSPDATGIMF EKYADSAINV AGSGGEYEVV EGFQWTNGVL 600
IWAADVFGNK LKRPDGNGIT AAHTHSEAKR SLGDGGLARR AVELDPWDAA WTKMFGRSKL 660
RRREAEDVRK RWSS 674

SEQ ID NO: 187      moltype = AA length = 659
FEATURE            Location/Qualifiers
source             1..659
                  mol_type = protein
                  organism = Chloridium virescens

SEQUENCE: 187
LYINGSVIAP CDSPLYCQGD ILKAIQLAQP FSDSKTFVDM PTTQPVQVI AAFNQLPQPV 60
SNNSELQNF LNTYFAPAGGE LTAEPKDLQ L TNPMLNKL N DTVIREFVTA VIDIWPDLTR 120
TYTGASNCTG CDSFIPVNR TFVVAGGRFR EPYYWDSFWI LQGLLRTGGA FTQISKNIIE 180
NFLDLVDQFG FVPNGARVYY LNRSQPPVLS EMVRTYVAYT NDTSILARAI PTLIKEHNF 240
MQNRTVNI TG ADGNTYTLNQ YHVNTQPRP ESYTEDYITA NNHSYYATSG IYYPETKPLN 300
DSEIANLYAN LASGAESGWD YGSRYLARNP DAAQDVYFPL RSLNVLNIVP LDLNSLLYQS 360
EQNIALFLQA TGNSSSEAEQW TSLAAQRQTA IHALMWNELT WSYFDYNLTS NGQNIYVPAD 420
KDATPADTAS APPGYQVLEF VAQFYPFWTG AATDELRRNP LAVRQAFTRV DAYLTAKAGG 480
IPATNLMTGQ QWDQPNVWPP LMHVLMQSL L DTPATFGAAD PSYAALQGLA LRLAQRYLDM 540
TFCTWYATGG STSQTPKLQG LGPDVAGTMF EKYADNATNI AGSGGEYTVV EGFQWTNGVL 600
IWAADTFGAQ LTRPNCNGIT AAHVTPGKRS VGLGRRAVEL HESDARWVKM FGSRAWRNA 659

SEQ ID NO: 188      moltype = AA length = 662
FEATURE            Location/Qualifiers
source             1..662
                  mol_type = protein
                  organism = Gelasinospora cratophora

SEQUENCE: 188
LYVNGSVTAP CDSPIYCYGE LLHQVELARP FSDSKTFVDM PTIKPVDEV L EAFSKLTLP L 60
SNNSELHFL NTYFGPAGGE LEAVPTDQLH VSPTFLDNVS DDVVKQFVNS VINIWPDLTR 120
KYVGAGEICT GCANSFIPVN RTFVVAGGRF REPYYWDSFW ILEGLLRTGG AFTEISKNTI 180
ENFLDLVEQI GFVPNGARLY YLNRSQPPLL TQMVRIYVEH TNDTSILERA VPILKKEWEW 240
WITNRTVEVT ADGKTYSLQR YHVDNNQPRP ESYSEDYITA NNNSYYATSG IYYPETTPLN 300
DTQKAQLYAN LASGAESGWD YSTRWLKNPN DAARDVYFPL RSLNVLEIVP VDLNSILYQN 360
EVTIGKFLAQ QGNKDEAEW AKKAEQRSEA MYKLMWNSTL WSYFDYNLTS SSQNIYVPAD 420
PQVFPFEKPS GTPGYQVLF SINQMFFFWT GAAPDQLKAN PLAVKLTFRD VKNYLDNKAG 480
GIPATNFVTG QQWDEPNVWP PLMHVLM DGL LNTPATFGED DPAYQETQNL ALRLAQRYVD 540
STFCTWWATG GSTSETPKLQ GLGSDAKGIM FEKYADNSTN VAGGGGEYEV VEGFQWTNGV 600
LIWAADKFGD KLKRPNCGDL TPANVGKRAV ELDAFDAKFT KKFARKGKLE KLKAKFKRRA 660
AI 662

SEQ ID NO: 189      moltype = AA length = 543
FEATURE            Location/Qualifiers
source             1..543
                  mol_type = protein
                  organism = Acidobacteriaceae bacterium

SEQUENCE: 189
QTTTTSAGLH DTLAYIKRTW HTLERSNKTL LKSADDDVKG QAGTLTLVVS QDVKPQAVNA 60
SLRRELPPAD KKRIVVRQLP EHPEAVEPAG LLYLPYPYV V PGGRFNEMYG WDSYFILLGL 120
VHDDDELALAK NMTDNFIYEI EHYGMILNAN RTYYLTRSQP PFLTQMILEV YRRTGDKWL 180
ASTLPAIEKY YAYWMREPHL TPETGLSRVW GGADTPAPEV VHGEKDAAGH NQYDRVREY 240
RTHNVATYDV SQYYDKATDR LKPLFYIADR AMRESGFDP SRYGPFSAI IHYDPVCLNS 300
LLYRMESDTA TILQLNRTS EARVWEKRAT QRAELVNRLM WNEEKGLYFD YDFITRRQSN 360
YHFVTTFYPL WAGIASRQQA DRVRKNLSIF ERAGGLQTS D YISGSQWDAP FGWAPLQIMT 420
VEGLRRYGFN EDADRISRKF INMVRDFEE HGTIKEKYDV VIGKSDLAAG LKFGYTSNEA 480
GFGWTNAAVV LFIEELAGER PLAASLDRES MPMLRQRHLS PQPSVWPFFS PQAPQYRRRD 540
PYR 543

SEQ ID NO: 190      moltype = AA length = 532
FEATURE            Location/Qualifiers
source             1..532
                  mol_type = protein
                  organism = Acidobacterium capsulatum

SEQUENCE: 190
GARSSLSPHA VAGQPIDAY IHTAWSTLTR SMSDCKSVAD PKLKSTPVLV LPRDLAVPAN 60
VAAMQKQCHV RVLRLPIVIT HFDQIRESQI ATPGLLYLPH PYVVPGRFRN EMYGWDSFFI 120

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LKGLLDDHHI	ALARGIVENF	FFETIAHYGGI	LNANRTYYFT	RSQPPFLSSM	IRAIYAAEVA	180
EGHTQAAHAW	LVEAYPYAVR	DHALWMSPIH	QAGNTGLARY	FDTGQGPVPE	MADDSTYYQD	240
VIRWLLAHPG	LHTGYLMHGS	PHLDAAARER	LAQLSCDPTL	SKVCARAHVH	GYWLTRSFYK	300
GDRAMRESGF	DTTFRFGPFS	GSTQHFAPVG	LNALLYKYER	DLAWMAAQLG	KPGEAAKWNS	360
EAETRRAEEM	HYLWNAQKKM	YFDYNFETHR	QSSYAFITTF	YPLWAGAADK	AQQQGVIASL	420
PLFEHPGGGLA	ISNHDSGVQW	DLPYGWAPTE	WMAVQGLLRA	DDQHDARRIA	SEFNRTVRTT	480
YQHDHAIYEK	YDVVNRSNDF	RVTAGYTQNV	VGFGWTNAVY	LEFQSLLAHP	GQ	532

SEQ ID NO: 191 moltype = AA length = 568
 FEATURE Location/Qualifiers
 source 1..568
 mol_type = protein
 organism = Acidovorax wautersii

SEQUENCE: 191

AAAVSSGLRI	SERHPAYSTN	PARVQVPGAP	GAPPSDHCTP	ADRYQELFVA	VQSQQIFEDS	60
KTFVDCGPTG	EPEDILAAYR	AEHAQPDFDL	ARFVAQHFTA	PQVAANDYVG	APGMALAEHI	120
DALWPVLTRK	PEDHPVRGSA	LPLAHPYVVP	GGRFAELYYW	DSYFTMLGLA	ATGRSELVQC	180
MTDNFARLID	AFGFVPNGTR	TYYLSRSQPP	LFAAMAELGA	LVGGPPVSHY	LPQLLQEHAW	240
WMDGLHVLHP	GEARRRVVSL	PGGEILNRYW	DDRDTPREES	WREDIETASA	VDRPSADVVR	300
DLRAAESGW	DFSTRWLRAP	DAANPASLHL	SQICTTDLPL	VDLNAFLYRM	EVSIKASQS	360
AGDRATATHF	HDLAHRREA	VNRLMWNEAE	GAYFDYDWR	GELRGCLTAA	TVVPLYAGMA	420
TEAQAAAVAR	AVRTHLLAAG	GLATTVCSSD	QQWDRPNGWA	PLQWMAVRGL	ERYGHKDLAL	480
EVRQRWIETV	RSVYQREGKL	VEKYAVGNGD	GAPLCGGGGG	EYPLQDGFGW	TNGVVQCWLD	540
PRYDTYAAAQ	TVYVGPTDDG	TADSLPEA				568

SEQ ID NO: 192 moltype = AA length = 529
 FEATURE Location/Qualifiers
 source 1..529
 mol_type = protein
 organism = Xanthomonas arboricola

SEQUENCE: 192

APMDTPVVNA	PAATPPTPDL	AYPELFQAVQ	SGELFDDQKH	FVDFLPLRDP	ALINADYLAQ	60
HDHPGFDLRK	FVDANFEESP	PVQTDAIRQD	TALREHIDEL	WPKLVRSQTH	VPPYSSLLAL	120
PHPYVVPVGR	FREYVYWDY	FTMLGLVKSG	ETTSLRQMLD	NFAYLIDTYG	HIPNGNRSYY	180
LSRSQPPFFS	YMVELQAGVE	GEAVYQRYLP	QLRKEYAYWM	QGSDELQPGQ	AARHVVRLLD	240
GSLLNRYWDE	RDTPRPEAWL	HDTRTAAEAG	DRPAAEVYRD	LRAGAESGWD	YTSRWLADGQ	300
NLRTTIRTTAI	VPIDLNSLLY	HLERTLAQAC	AQPGAECADQ	YAALALRRKQ	AIDAHLWNAA	360
GYADYDWQT	RKLSAQVTAA	ALYPLFTGLA	TDAHAKRTAS	TVRARLLRPG	GLATTAVKTG	420
QQWDEPNQWA	PLQWVAVDGL	RRYGEDALAR	TIGERFLTQV	QALFAREHKL	VEKYGLEADA	480
AGGGGGEYAL	QDGFGWNTGV	TLMLLNLYPD	TATKPAPAKR	ARKPEAAAR		529

SEQ ID NO: 193 moltype = AA length = 540
 FEATURE Location/Qualifiers
 source 1..540
 mol_type = protein
 organism = Kosakonia sacchari

SEQUENCE: 193

AEAQNTPPQP	DILLGPLFSD	VQTAKLFPPDQ	KTFADAVPKG	DPLMILADYR	MQRMQTSFDL	60
RHFVDVNFPL	PKEGEKYVPP	EGQNLREHID	GLWPVLTRTT	DSAGKWSLL	PLPKPYVVP	120
GRFREYVYWD	SYFTMLGLAE	SGHWDKIEDM	VTFNFAHEIDT	WGHIPNGNRS	YYLSRSQPPF	180
FSLMVELLAT	HDGDEALKTW	LPQMEKEYQY	WMEGADTLQP	GQANKRVVKL	SDGSVLNRYW	240
DDRDTPRPES	WLDDVTAKN	NPNRPATEIY	RDLRSAAASG	WDFSSRWMD	PNQLGTIRTT	300
SIVPVDLNL	MFMEKMLAR	GYQAAGDSAK	ASQYDALANA	RQKGIEANLW	NEKEGWYADY	360
DLKTKKVRNQ	LTAALYPLF	VNAAKDRAD	KVASAAKERL	LKPGGIATTT	VNSGQQWDAP	420
NGWAPLQWV	TAGLQNYDQ	KLAMEVSWRF	LTNVQHTYDR	EKKLVEKYDV	STTGTGGGGG	480
EYPLQDGFGW	TNGVTLKMLD	QICPKEKPCD	SVPQTQPAQQ	PAKVEPTSQ	PSKQQQAVAQ	540

SEQ ID NO: 194 moltype = AA length = 532
 FEATURE Location/Qualifiers
 source 1..532
 mol_type = protein
 organism = Enterobacter sp.

SEQUENCE: 194

DEQPAFQKNS	PDILLGPLFN	DVQSAKLPD	QKTFADAVPK	SDPLMILADY	RMQHTQSSFD	60
LRHFVEMNFT	LPAGEKEYVP	PAGQSLREHI	DDLWPVLTRT	TDKASNKWS	LLPLPKPYV	120
PGRPREVYV	WDSYFTMLGL	AESGHWDKIS	DMVDNFAWEI	DTFGHIPNGN	RSYLSRSQPP	180
PFFSMVVELL	ATHSDALKK	YRPQMEKEYA	YWMEGVDSLQ	PGQANQVRVK	LDDGSVLNRY	240
WDDRDTPRPE	SWLDDVNTAK	NNPNRPATEI	YRDLRSAAAS	GWDFSSRWMD	DPQKLGTIRT	300
TGIVPVDLNA	LMFMKEKLLA	RASQEAAGDS	AASKYEALAT	ARQKAIENHL	WNDKEGWYAD	360
YDLKSKKVRN	QLTAAALFPL	YVKAASQERA	DKVAAATSAR	LLKPGGITTT	TINSQQQWDA	420
PNGWAPLQWV	ATEGLQNYG	NKVAMDVTWR	FLKNVQHTYD	REKKLVEKYD	VSTGTGGGGG	480
GEYPLQDGF	WSNGVTLKML	DLVCPKEKPC	DSVPENQPAA	NDEAAPVKAS	AQ	532

SEQ ID NO: 195 moltype = AA length = 720
 FEATURE Location/Qualifiers
 source 1..720
 mol_type = protein
 organism = Saitozyma flava

SEQUENCE: 195

-continued

QNPASSSSSFS	PTPVTTMVPS	PTAALNATVP	GQGVYPPLQA	WCNNGGNDTF	CPGVLLQDVQ	60
LSGIFPDSTK	FVDKPTRGTL	NETLRTFASL	GNNLTVGQIE	GFVNNSPKGE	GLELSQVALE	120
GFNPNPAPLD	TISDPIYQGW	MSVVNSYWT	LIRETNRSAL	CNGDCESSLI	PLNNTIVVPG	180
GRYREIYYWD	SYWILIGLLE	SELTAYATDL	ISNFMDFIQT	YGFIPNGGRK	YYLNRSQPPV	240
FTQMLNTYVQ	RTGNTSILAR	GLPLAQHEL	WWENNRVISV	TSPYSNITRR	VAHFVAVNNTA	300
PRPEGVYVEDY	ETAFGASPAL	NETARGQLYS	DLAAGAESGW	DYSSRWCKQP	VINLTDNFPA	360
LRTINTAKII	PVDLNSLLAG	DHTLLANLYE	LYGNSSNANT	SVTSNSSQLV	PYHRQLAKNY	420
SDAVLDLHWD	AQKAWFYDFN	LTANARADII	HPGGTFALWQ	NITPPAVASN	ETYALEVVSG	480
ARFLLGKYTS	IPSVSTLIET	GLNWDFFNSW	PPHVYSSIKA	FETLGRAFPN	ASVLSNISIP	540
FSDVQKNQLG	LDESAIPAQ	ASTIGNSTSQ	IAEAQKQWP	QALSIEYANR	YMQAAPCSWY	600
STGGSIDGLL	TQLPLSELNM	TGTFQAGTTG	VMFEKFNVD	LDAAGGGGEY	KVQIGFGWTN	660
GVALWLGNF	GQYLPQPTCP	LIIPIEVQNG	MNSSVYQNGT	MTNSTKTQSY	IFQGHRIPRK	720

SEQ ID NO: 196 moltype = AA length = 723
 FEATURE Location/Qualifiers
 source 1..723
 mol_type = protein
 organism = Phaeotremella skinneri

SEQUENCE: 196

QTPTSSSSSS	FSATPVSTSV	PSPTVPLNSS	VIGQGLYPPA	QAWCNGGMND	TFCPGVLLQD	60
VELSGIFADS	KTFPDKPTVG	TLNSTLQAF	ALPANVTVGE	IETFVNQYFK	GEGLLEQVA	120
IEGFVANPAI	LNNITDPVFK	GWSTVNGYV	QLLIRQTNES	SLCNETSCAS	SLIPLNHTIV	180
VPGGRYREIY	YWDSFWILQ	LLKSELYTYS	WDLQNFMDL	IETYGFLPNG	GRKYLYNRSQ	240
PPVFIQMLDA	YIKVTGSSSI	LARALPIATT	ELAWSSNRT	IPVTSPTYGI	THLVAHYAVT	300
NSAPRPEGVY	EDFTTAMGAS	PALNDSARAE	LYSELASGAE	TGWDYSSRWC	RQPLLNLTND	360
NPALRTLNVK	AIIPVDLNSL	LAGDHALLAN	LYDLYSNSSS	SNSTSNSSSI	SNSSSQAYH	420
RQAAQNLTA	ILDHLWSPSK	SFFYDFNTSS	NSQSDIYTPA	SLFPLWQNT	PPALVGNETA	480
ALQLVSGVRY	LLGKYAGIPS	VATLLATGLN	WDFPNSWPPH	VYTSIKAFQT	LYRVNPNATV	540
LSNLTLSFAQ	VQAGQLGLSE	TGFQIQPAST	VGNTSLETAE	AKGKPWPVAL	GIEYANRYMQ	600
SAFCSWYSTG	GEISGVLTLQ	PLSDLNSTGT	FTAGQTGMVF	EKFNVTDPDA	AGGGGEYTVQ	660
TGFGWTNGVV	LWVAGEFGSL	LPAPTCLIP	IIEISGSSNS	SIYANGVTN	STALYRGYKI	720
PRK						723

SEQ ID NO: 197 moltype = AA length = 663
 FEATURE Location/Qualifiers
 source 1..663
 mol_type = protein
 organism = Trichoderma asperellum

SEQUENCE: 197

LYTNGSVIAP	CDSPIYCHGD	ILQDIQLAHP	FSDSKTFVDM	PAKRPLREIQ	SAFNNLSKPL	60
TNDSSLQVFL	ETYFADAGGE	LVEVPRSSLT	TNPAPLNKIN	DTVIREFVTK	VIDIWPDLTR	120
RYSGAVGNCS	TCPNFPIPVN	RTFVVAGGRF	REPYWDSYW	IEGLLRGTGG	AFVGIKNTI	180
ENFLDFIDRF	GFIPNGARLY	YLNRSQPPLL	SQMVRITYEH	TNDTSILKRA	LPLLVEKEDF	240
WIRNRTISVN	IANKTYTLQQ	YAVQNTQPRP	ESFLEDYVTA	NNRSYYATSG	IIYPENKHLN	300
STEMADLYAN	LATGAESGND	YTSRWLANPS	DAINDVYFPL	RSLNKEIIVP	VDLNSILYGN	360
ELTISQFYNR	TGDATAARAW	AERAANRSAA	IQAVFWNETL	FSYFDYNLTS	SSQYVVPAD	420
KDTISLDRQT	APSGKQVFFH	VGOFYFPFWT	AAPDYVKNNP	YAVTRAYDRV	TGYLDAQPGG	480
IPASNVQTGQ	QWDQPNVWPP	HHHILMQGLN	NVPATFTAQD	PSYQDIQNLS	LRLGQRYLDF	540
TFCTWLATGG	STSTPKLHGG	LSDQDVGIMF	EKYDDNSTNA	AGGGGEYEVV	EGFGWTNGVL	600
LWTADTFGNK	LKRQPCGNIT	AGHPAPASSS	SKRSVAVQLDA	WDARRVKKFG	KRTEGRVKRP	660
FLF						663

SEQ ID NO: 198 moltype = AA length = 674
 FEATURE Location/Qualifiers
 source 1..674
 mol_type = protein
 organism = Corynascus sepedonium

SEQUENCE: 198

LYINGSVTAP	CDSPIYQCGE	LLKAVELARP	FVDSKTFVDM	PTIKPVDEV	AAFSKLSLPL	60
SNNSELNAPL	YENFAQAGHE	LEEVPDSELE	TDKFLDKLE	DRTIKEFVKG	VIDIWPDLTR	120
RYAGPSNCTE	CANSFIPVNR	TFVVAGGRFR	EGYWDYSWI	VEGLLRTGGA	FTHISKNIIE	180
NFLDFVDTIG	FIPNGARIYY	LNRSQPPLLT	LMVKSIVDYT	NDTSILDRL	PLLIKEHEFF	240
MNNRTVSTIG	SNGKEYTLNR	YHVNNQPRP	ESFREDYITA	NNGSYYASSG	IIYPVKTPLN	300
ETEKAAALYSN	LATGAESGWD	YTSRWLGVP	DAARDVYFPL	RSLNVRDIVP	VDLNSILYQN	360
EVIIAEYLEK	AGNSSAAKRF	ATAAEQRSEA	MYSLMWNATH	WSYFDYNLTD	NTQHIFVPAD	420
EDTAPQDRIE	APPQQVFFH	IAQLYFPFWT	AAPASLKAMP	LAVQAYARV	ARMLDIKKG	480
IPATNVRTGQ	QWDQPNVWPP	LQHILMKGLL	NTPATFGKSD	PAYQSVQNLA	LRLAQRYLDS	540
TFCTWYATGG	STSDFPQLEG	VTPGATGMVF	EKYADNATNV	AGTGGEYEVV	EGFGWTNGVL	600
IWAADVFGNK	LKRPDCCGNIT	AAHTSSAKR	GLEENKLPRR	AVELDPWDAA	WTKMFGRSKL	660
RRREAEDVRK	RWMS					674

SEQ ID NO: 199 moltype = AA length = 674
 FEATURE Location/Qualifiers
 source 1..674
 mol_type = protein
 organism = Myceliophthora thermophila

SEQUENCE: 199

LYINGSVTAP	CDSPIYCHGE	LLKGVELAHP	FVDSKTFVDM	PTLKPVDEV	AAFSKLRQPL	60
SNNSELNPNL	AEYFAPAGHE	LEEVPKGELQ	IDPKFLNKLE	DRTIKEFVSK	VIDIWPDLTR	120

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RYAGPGDCSG	CANSFIPVNR	TFVVAGGRFR	EPYYWDSYWI	LEGLLRTGGA	FTQISKNIIE	180
NFLDFIDTIG	FIPNGARIYY	LNRSQPPLLT	RMVKSVDYDT	NDTSILERAL	PLLIKEHDFP	240
TNNRSVSVTA	SNGKTYTLHR	YHVENNQPRP	ESYREDYITA	NNGSYYAASG	IIYPVKTPLN	300
ETEKAVLYSN	LASGAESCWD	YTARWLVRPD	DAARDVYFPL	RSLNVREMPV	VDLNSILYEN	360
EVIIAEYLEK	AGNSSEAKRF	ASAAKQRSEA	MYNLMWNATH	WSYFDYNLTS	NAQNIFFVPAD	420
EDTASFDRYA	APPGQQVLFH	VAQLYFPWFG	AAPAHLKSNP	LAVQKAYARV	SRRLDTFKKGA	480
IAATNVRTGQ	QWDQPNVWPP	LQHVLMOGLL	NTPATFGESD	PAYQGVQKLA	LRLAQRYLDS	540
TFCTWYATGG	STSDFPQLQG	VSPDATGIMF	EKYADSATNV	AGGGGEYEVV	EGFGWTNGVL	600
IWAADVFGNK	LKRPDGCGIT	AAHTHSEAKR	SLGDGGLARR	AVELDPWDAA	WTKMFGRSKL	660
RRREAEDVRK	RWSS					674

SEQ ID NO: 200 moltype = AA length = 661
 FEATURE Location/Qualifiers
 source 1..661
 mol_type = protein
 organism = reesei GH37

SEQUENCE: 200

LYINGSVIAP	CDSPIYCHGD	ILREIELAHP	FSDSKTFVDM	PAKRPLSEIQ	TAFANLPKPL	60
RNDSSLQTFI	ASYFADAGGE	LIQVPRANLT	TNPTPLSKIN	DTVIEQFVTQ	VIDIWPDLTR	120
RYAGDAAVKN	CSSCPNSFIP	VNRTFVVAGG	RFREPYWDS	YWIVEGLLRT	GGAFVGIARN	180
TIDNPLDFIE	RFGFVPPNGAR	LYYLNRSQPP	LLSRMVKYVI	DHTNDTAILR	RALPLLKVEH	240
EFWTRNRTVD	VRVNNKTYVL	NQYAVQNTQP	RPESPREDFQ	TANNRSYYAA	SGIIYPATKP	300
LNESQIEELY	ANLASGAESG	NDYTARWLAD	PSDAMRDVYF	PLRSLNNKDI	VPVDLNSILY	360
GNELAIQAQY	QGTGNTTAAR	EWSSLAANRS	ASIQAVFWNE	TLFSYFDYNL	TSSSQNIYVP	420
LKDKAVALDR	QTAPPGKQVL	FHVGFYFPFW	TGAAPPEYLRN	NPFAVTRIFD	RVKSYLDTRP	480
GGIPASNVTN	GQQWDQPNVW	PPHMHILMES	LNSVPATFSE	ADPAYQDVVN	LSLRLGQRYL	540
DFTFCTWRAT	GGSTSETPKL	QGLTDQDVGI	MFEKYNDNST	NAAGGGGEYQ	VVEGFGWTNG	600
VLLWTADTFG	SQLKRPQCGN	IMAGHPAPSK	RSAVQLDMWD	ASRVKKFGRR	AEGRMGTLHA	660
W						661

SEQ ID NO: 201 moltype = AA length = 670
 FEATURE Location/Qualifiers
 source 1..670
 mol_type = protein
 organism = Chaetomium virescens

SEQUENCE: 201

LYINGSVTAP	CDSPLYCQGE	ILKAIELARP	FSDSKTFVDM	PTIKPLEEVI	AAFGRLLKQPL	60
SNNSELTAFL	AENFAPAGGE	LEEVPKSELH	TDPVFLNKLD	DAVVKEFVGK	VIDIWPDLTR	120
RYAGPGNCNS	CENSFIPVNR	TFVVAGGRFR	EPYYWDSYWI	VEGLLRTGGA	FVGITKNILE	180
NFLDFIETIG	FVPNGARIYY	LNRSQPPLLT	KMIKIYVDHT	KDTSILQRAL	PLLIKEHEWW	240
TNNRSVTVTG	PNGKTYTLNR	YHVNNNQPRP	ESFREDYITA	NNGSYYATSG	IIYPVKSPLN	300
ETEKDETYAN	LATGAESCWD	YTARWLRTPN	DAKDVFYFPL	RSLNVRNMIP	VDLNSILYQN	360
EVIIIEYLEQ	AGNKSEAQRW	FQAANQRSEA	MYALMWNATH	WSYFDYNLTS	NSQYIFIAND	420
EDATTAEQAN	SPPGQQVLFH	ISQLYPFWTG	AAPDQLKKNP	LAVQQAYYRI	ERMLNEKAGA	480
IPSTNFRGTQ	QWDEPNVWPP	LQHILMOGLL	NTPATFGTAD	PAYAAVQNLA	LRLAQRYLDS	540
TFCTWYATGG	STSTPQLQG	VSPGATGIMF	EKYADNATNV	AGSGGEYEVV	EGFGWTNGVL	600
IWAATFTGNK	LKRPDGCDIQ	AAHTHTDKKK	RWSVDGEVRA	REMAVELDP	WDAKWTMFG	660
QAKGRVGRRS						670

SEQ ID NO: 202 moltype = AA length = 508
 FEATURE Location/Qualifiers
 source 1..508
 mol_type = protein
 organism = Rhodothermus marinus

SEQUENCE: 202

QDRVACQVPL	PSVERIEAVR	AYIRQSWDVL	TRSHRDLLAA	VQDPKIEHEP	GTPWPLYIAA	60
TEDSVAVVHR	LQQLLPDSVL	QQIVLRVLPE	DPVAHLDEIH	PHGLLYLPEP	YVVPGGRFNE	120
MYGWDYSFIV	VGLLRDGRVD	LAKAMTDNHL	YQVRHYGKVL	NANRTYYLTR	SQPPFLSAMV	180
LAVYAHTQDR	DWLAAAVPLI	ERYYAYWTFP	PHLAGETGLS	RYYDLGEGPA	PEVVAGERDA	240
QGRTHYDRVR	EYYRMHEVTA	YDESLYYVAE	ADSLTPLFYK	GDRSMRESGF	DPSNRFGPFS	300
VDIIHYAPVG	LNALLYRMET	DLARIHEILG	DTAAAAAWRA	RAEARREERVD	RYLWDSERGL	360
YFDYNFRGTG	RSYVVFATTF	YPLWVGMAFP	EQAARVAANL	YLLEAPGGLL	TSTHISGSQW	420
DAPYGWAPLY	LIAVEGLRRY	GYDEAADRLT	AKFVSMIVED	FERTGVILEK	YDVVQRRSDV	480
ALRYGYTSNE	IGFGWTNAVF	AELLAQMD				508

SEQ ID NO: 203 moltype = AA length = 674
 FEATURE Location/Qualifiers
 source 1..674
 mol_type = protein
 organism = Myceliophthora sepedonium

SEQUENCE: 203

LYINGSVTAP	CDSPIYCQGE	LLKAVELARP	FVDSKTFVDM	PTIKPVDEVL	AAPSKLSLPL	60
SNNSELNAPL	YENFAQAGHE	LEEVPDSELE	TDKFLDKLE	DRTIKEFVGK	VIDIWPDLTR	120
RYAGPSNCTE	CANSFIPVNR	TFVVAGGRFR	EPYYWDSYWI	VEGLLRTGGA	FTHISKNIIE	180
NFLDFVDTIG	FIPNGARIYY	LNRSQPPLLT	LMVKSVDYDT	NDTSILDRLAL	PLLIKEHEFF	240
MNNRTVSTIG	SNGKEYTLNR	YHVENNQPRP	ESFREDYITA	NNGSYYASSG	IIYPVKTPLN	300
ETEKAAALYSN	LATGAESCWD	YTSRWLGVP	DAARDVYFPL	RSLNVRDIVP	VDLNSILYQN	360
EVIIAEYLEK	AGNSSAAKRF	ATAAEQRSEA	MYSLMWNATH	WSYFDYNLTD	NTQHIFVPAD	420
EDTAPQDRIE	APPGQQVFFH	IAQLYFPWFG	AAPASLKNAP	LAVQQAYARV	ARMLDIKKGA	480

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IPATNYRTGQ	QWDQPNVWPP	LQHILMKGLL	NTPATFGKSD	PAYQSVQNLA	LRLAQRYLDS	540
TFCTWYATGG	STSDFPQLEG	VTGATGVMF	EKYADNATNV	AGGGGEYEVV	EGFGWTNGVL	600
IWAADVFGNK	LKRPDGNGIT	AAHTHSSAKR	GLEENKLPRR	AVELDPWDAA	WTKMFGRSKL	660
RRREAEDVRK	RWMS					674

SEQ ID NO: 204 moltype = AA length = 657
FEATURE Location/Qualifiers
source 1..657
 mol_type = protein
 organism = Moelleriella libera

SEQUENCE: 204

LHTNGSLIAP	CDSPIYCYGD	ILKQVELARP	FADSKTFVDM	PGVKPLAEIQ	AAFDKLEKPL	60
RNNTALQDFL	KTYFADAGQE	LQEVPKSELK	TDPQLKTLN	DTVIREFVTK	VIDIWPDLTR	120
SYKGSNTNCT	DCPNSFIPIN	RTFVIAGGRF	REPYYWDSYW	ILEGLLRTRG	SFTQIAKNTL	180
ENFLDFVEQF	GFVPNGARIY	FLNRSQPPML	SQMVKLYIDH	TNDTAILQRA	LPLLVEKEHAW	240
WMNNRTVDVQ	VGGKTYKLNK	YAVSNTPEPR	ESYREDFETA	SNSSYYAQSG	IIPETKTKLN	300
DSQRAVLIAN	MATGGENGWD	FSSRWIANPS	DSVRDVYFPL	RTNNAQNVVP	VCLNSILYGN	360
EMTIGGFNPS	TGNTTAGQEW	AAKAKARSEA	MHATMWNETH	FSYFDYNLTS	SAQDVYTLAD	420
DDTSIYDNGT	LTGAPPGYQV	AFNGAQFYFP	WQGAAPTLYK	ENPQAVKTAY	ARVAQYLKVR	480
KGIPATNLK	AREQWDQPNV	WPPLMHILMQ	GLLNTPTPTFG	SSDPSYKSVR	SMALTLAQRY	540
LDSTFCTWYA	TGGSTSETPK	RPGLPEKDKG	IMFEKYADNS	IDIAGSGGEY	EVVEGFGWTN	600
GVLIWAADTF	ANELKRPDG	NGSSSSSTSS	AAKRGLSAVE	LHPADASRIK	RFSSSKG	657

SEQ ID NO: 205 moltype = AA length = 660
FEATURE Location/Qualifiers
source 1..660
 mol_type = protein
 organism = Acremonium dichromosporum

SEQUENCE: 205

IYVDGNITAP	CDSPVYCYGE	MLHQIQLAKP	FDDSKTFVDM	PALKPLSQIQ	AAFDRLDKPL	60
SNNSALNNFL	DEFFADAGGE	LSEVDKADLE	TDPVFLDKID	DEVVKEFTNK	VIDIWPDLTR	120
RYSGAAASNC	TNCPSSFIPL	NRTFVVAGGR	FREPYWDTY	WIEGLLRTRG	GSFTDISRDI	180
IFENFLDFVDQ	FGFVPNGARI	YYLNRSQPPV	LSRMVQAYIE	HTNDTDILDR	ALPLLMEKEH	240
FFSENRTIDI	EGPGTTYTLT	NRVDVRNNQP	RPESYSEDYE	TATNTSYSH	DSGIIYPETE	300
PLNDRTERANL	YSALASGAES	GWDYSSRWIA	RPRDAAEDVY	PPLRSLNTNN	IVPVDLNSIM	360
YANEMAIAGF	LNQTNASAA	AEWEELAYNR	SEAIHALMWN	ETYMSYFDYN	LTSAAQHIVV	420
PADDDVSTLE	SSTAPEGHMV	LFSVSQFYFP	WTGAAPSYIK	NNPFAIAQIY	KRVETLLDTR	480
KGIPATNFR	TGQWDQPSV	WPPLMHILMA	GLQNTPATFG	EDDPAYQVHV	EIALRIGQRY	540
LDSTFCTWRA	TGGATSETPQ	LEGFTDRDVG	IMFEKYADNS	TNIAGSGGEY	EVVEGFGWTN	600
GVLIWTVDEF	GNELKRPDG	DLEAADTTER	RKRSALQLAP	RDAQRTKKFG	KRAVERQPF	660

SEQ ID NO: 206 moltype = AA length = 667
FEATURE Location/Qualifiers
source 1..667
 mol_type = protein
 organism = Fusarium sambucinum

SEQUENCE: 206

LYVNGTVVAP	CDSPIYCHGD	ILEQVELARP	FSDSKTFVDM	PAIRPLSEIQ	KAFDELDKPL	60
RNNSALADFL	SENFADAGNE	LEEVPEDDELK	TDPKFLDNIN	DTVIREFTEK	VIDIWPDLTR	120
RYDQDSKNCS	DCPNSFIPVN	RSFVVAGGRF	REPYYWDSYW	IIEGLLRTRG	SFVNIKNTI	180
ENFLDFIEFY	GFVPNGARIY	YLNRSQPPLL	SQMVKIYIDH	TNDTDILERA	LPLLVEKEHEF	240
FMKNSRVPIY	YNTETMLNT	YVNSNTQPRP	ESYREDYITA	TNKSYYSTSG	EVVSGGEELS	300
FKQKETLYGN	LASGAESGLD	YTSKWIAPE	NAIRDNYFPL	RYLNTRNIIP	VDLNSILYGN	360
ETAIADFYEQ	TGNSASSEQW	REVAANRSYA	MHAFMWNELT	WSYFDYNLTS	KAQIYFPAD	420
NNTVSVDTED	APKGQVFFS	PTQFYPPFWLG	AAPDYLKNNP	YAVLSAYKRV	ATYLDKREGG	480
IPASNIETGQ	QWDQPNVWPP	MMHILMAGLE	KVPATFGIMD	PSFIEIRKLA	LRLGQRYLDS	540
TFCTWYATGG	STSETPKLES	VSDKEDGIMF	EKYADNATNT	AGGGGEYEVV	EGFGWTNGVL	600
IWAEEFNGNR	LTRPKCKNLE	TAHSSDKRDT	SAVMLHARDA	KHVKKFGRRK	RAEKAACKR	660
SSRLPHF						667

SEQ ID NO: 207 moltype = AA length = 660
FEATURE Location/Qualifiers
source 1..660
 mol_type = protein
 organism = Phoma sp.

SEQUENCE: 207

LYQNGSIIAP	CDSPIYCYGD	LLREIELAQP	FSDSKTFVDL	PTIRPLDEVL	RAFNNLTKPI	60
QNTTELNNFL	TTYFGEAGSE	LDALPKDQLE	TQPDFLDNVN	SSVIVNFTSQ	VIDIWPDLTR	120
RYVGAGNCTG	CVSSFI PVNR	TFVVAGGRFR	EPYWDSEFWI	IEGLLRTRGS	FTQIAENIIE	180
NFMDLVEELG	FVPNGARRY	ENRSQPPLLT	QMVRVYVEYT	QNYTLLEAL	PVLEQYEFW	240
VNNRSVTLEK	GGKNTYLLHY	NVSNTPRPE	SYREDYITAN	NLTYYNENGE	QFNASHPLND	300
TQKALLYAEL	ASGAETGWDY	SSRWLANPSD	AVNDDFFPLR	SLNVINTIPV	DLNSILYNE	360
ITIAEFHRR	GNYCAARQW	ELARNRSEAM	TALLWNAEY	SYFDYNLTSS	AQNIYTLADN	420
TSTPLSLAGA	PAGYQVQGL	SLQYFPWTGA	APDSIKGDPT	AIRRAFARVE	EALDTEAGAV	480
SATNLFTGQ	WDEPNVWPP	QYIAIQGLLN	TPLEVEDDD	EQTAEDYVWT	QNLALRLAQR	540
YTDSLFTCTWR	STGGATEEP	QLPGATGNGT	IFEKYSDEAI	NARGGGGEYT	VVEGFGWSNG	600
VLIWAVDNFG	QKLTPDCGN	ITAAAPPSTA	SKRKRSAYEI	HQRDAAWIKG	TKENKMFQKK	660

SEQ ID NO: 208 moltype = AA length = 737

-continued

FEATURE
source

Location/Qualifiers
1..737
mol_type = protein
organism = *Lentinus similis*

SEQUENCE: 208

LPQAVTPSST	SVSSQTVSTA	VPSPTASLTS	TLPSQIPLPP	KQDWCPSEIF	CAGELLQTVN	60
VAQLYPDPKT	FVDKPTARKS	QQVVSNFQNI	GGNSSNVTVG	AIEDFVNSNF	KEGGLELEPI	120
AFANFNPTPA	FLNNVSDPLV	KAWSQIVHGY	WTQLTRSTND	SALCPEGTES	GSCSSLIPL	180
NHTFVVPGR	FREQYYWDSY	WIVQGLLVSE	LYDIVNDTLQ	NFMDLEHIG	FIPNGGRIYY	240
LNRSQPPLFI	HMLTSYVQAS	GDTSILKRAL	PLAEKELAWW	SANRSVQVKS	PYTNATHNVY	300
RYHVTNTAPR	PESYYTDYIT	ANDPTLQTP	TEQQRADLYA	ELATGAESGW	DYSSRWLKEP	360
LAGGSNNTSP	ALRSLNIRSL	VPVDLNSILY	KAHLNLAALY	NNESHSAAS	NHTQAAIALR	420
EAILDLHWD	NKRAFYDFNI	TSNARNDI	FS AATFYPLWSG	IIPDEILSDG	SGATAFGAF	480
AVNMVLNRYN	GTFPVTFLET	GLQWDAPNAW	PPHQFIIIEA	LRNLPSNITN	TPLPSAPSGN	540
SYALIPAGQ	VGIAEGDLPG	QILSPGQAT	KTGPAADINT	LNGTVVNGN	ATSGEGWA	600
LQRELANRYI	ASALCSWHAT	GGEVFNLLPR	LSDSQLQITQ	SQNNTGNMFE	KFSINDIDSA	660
GRGGEYTVQA	GFGWTNGVVL	YLTHVFGDKL	VAPSCPVLVA	LSSNTATTSG	AVAQMTLPSS	720
LAVTVGAVVL	GFVGLVL					737

SEQ ID NO: 209

FEATURE
source

moltype = AA length = 683
Location/Qualifiers
1..683
mol_type = protein
organism = *Diaporthe nobilis*

SEQUENCE: 209

QTLGDIYYDG	DNIAPCSSAL	YCYGDILDSI	QRAKPFADSK	TFVDMPTRPV	LEEVRAYDQ	60
LTKPLQNNTE	LLDFLSNNFG	PAGQEVVPVD	PGSLGVNASF	LGGIANAVNR	EFTAEVIDLW	120
PNLTRWVNES	AVCAECDNSL	LSIKRPFIVA	GGRFPREPYW	DSYWLHGLL	RSGGNPTRIA	180
RNQIENFLDF	VEDYGFVPNG	ARVYYLNRSQ	PPLLAQMVRI	YIEQTGDATI	LDRAIPLLR	240
EHDFFMSNRT	VHVSFVGRNY	TLNRYNVANT	EPRPESYYED	YTQVNNASYY	ANDGRVPFPT	300
NTTQAEKDLQ	YKNLASGAES	GWDFSTRFMR	DPTIAANDTY	FPLASYNIIN	IIPVELNSIL	360
YWNEVTIAAF	LRQQQQQVFN	ATAEADADAW	DARAASRSEA	MYSVMWNETL	GGYDFDNLTS	420
GSQDVFWARD	ADSLPTEQAG	TEPGQQVVFN	IGQLSPFWTG	AAPRSLAGDP	AAVRRAFSRV	480
DEYLGSRKGG	IAPTNNFVSGQ	QWDQPSVWPP	HMHILMEALL	RTAEAGGESE	GGSEDWAWAQ	540
DLALRLGQRY	FDSAYCTWRA	TGGGTPSSPP	LANPPQDLGG	QMFEKYSQDS	LNEAGSGGEY	600
VVVVGFGWSN	GVLWVADTF	RDRLQTPACG	DLTTRGEAGK	MREKRMSENG	RHGDGSNAIS	660
AVKLDYFDAA	WTSENVGGLH	GVR				683

SEQ ID NO: 210

FEATURE
source

moltype = AA length = 763
Location/Qualifiers
1..763
mol_type = protein
organism = *Solicoecozyma terricola*

SEQUENCE: 210

QGNSSMSASN	ATSSSTGTGT	SSSAAAVMTA	VPMPAPLSS	PVEMLPLPPV	QPCWNGGENA	60
TYCPGSLMQL	VQLSGIYNDS	KTFPDKPTQY	NASVTYQAFD	ALPVNATVGD	VETFEVKYFK	120
GEGQELQTVQ	IQNFTQNPTF	LNVIDDELYK	GFVSTVNGYW	SLLVRQTNES	ALCTNGACES	180
SLIPLNRSFI	VPGGRYREIY	YWDSFWILEG	LLKSELYLYA	YNLENFMDL	IEKFGFLPNG	240
GRSYLNRSG	PPVVFQMLNA	YIQVTGNVSI	LTRALPIAET	ELQWWRNRT	ISVTSPTYGT	300
NYSVARYFVT	NSAPRPEGYV	EDITTAFFGN	PALNESARSA	LYAELASGAE	TGWDYSSRWC	360
KQPLNLNTDN	DPALRTLNV	QNPVVDLNSL	LCGDHVLNAN	LYEFYMNSTI	GGSGISSGG	420
GNSTIGMGAW	ANSTSGSNMT	SNSTSSSNMT	GLVQSHRMIA	KEFEAAINDL	MWDKQKLWY	480
DFNMTANARA	DVYHPGGLFP	LWQNTIPSDI	VGNDAFAAGV	FAGVRYMVGM	YPGPPAPASL	540
IQTGLNWDAP	NVWPPHYVTG	IKALETVLRI	NPNSSVVPNI	TLTDFTRIPT	GQLGLNQSGL	600
PPQASALGN	GTADLDQAIA	QMKGKFWPT	ALAIEFANRY	LQSAFCSWYS	TGGSIPGLLQ	660
QLSPQELNLT	GSLTTGSQAM	GNIFEKFNLT	NVDAAGGGGE	YTVQIGFGWS	NGVILHTAGE	720
YQYLVQPSG	PLIAIHETAN	ATSTNATKSN	NTMVFSGYRL	PHD		763

SEQ ID NO: 211

FEATURE
source

moltype = AA length = 734
Location/Qualifiers
1..734
mol_type = protein
organism = *Dioszegia cryoxerica*

SEQUENCE: 211

QSSSSSSFSF	TPVTTAVPSA	TAALNQTVSG	QGVYPPLQPW	CNAGENATYC	PGVILQDVQL	60
SGLFPDSKTF	VDKPTNGTQN	ATQQAQQQLG	NNITLGLAQ	FVNVSFRGEG	LELSQVPING	120
FVANPAAVNK	VSNPLYRQGV	STVNSYWSLL	IRETNQSAVC	TTQCESSLIP	LNVTIVVPGG	180
RYREIYYWDT	FWILEGLLKS	ELYTYAWDVL	QNFMDFVDVY	GFIPNGGRKY	YLNRSQPPVF	240
IQMLDAYVKA	TGNVTILERA	LPLASEEMRW	WINNRTQVT	SPFTGITRRV	YVFNVTNSAP	300
RPEGYVEDYE	AAPGAQPLT	EAQRGALYAE	LATGAESGWD	YSSRWCKQPV	INVTDLNLPAL	360
RTLNGRSIVP	VDLNSLQAGN	HALLARLYEV	YINASTTSNT	TAAQTIRANA	TQEIHALHKT	420
ANDYSQAVLD	LHWDPARAWF	YDFNLTSNSR	ESLYTPAGTF	ALWQNTIPPG	LEGNDTAALR	480
IASGARYLLG	RYGGIQGVSS	LLVTGLNWDF	PNSWPPHTYT	SIKAFQTLGR	LVGNASIVGN	540
ATIPFSQVAT	NQLGLNETQL	PPQDPALQGN	ASLTVPSARN	VSWPLALEIE	YASRYMQGAF	600
CSWYSTGGSI	SGLLTQLPVS	QLNATGTYYA	GQTGQMFEKF	NATDIDAAGG	GGEYTVQIGF	660
GWTVGVVLWL	ADNFGQYLPQ	PTCPLVVLSL	TNLNSTTAGN	STNGTSPAAP	GNATSLVAVS	720
ELQGIWEGQR	VS	RD				734

SEQ ID NO: 212

moltype = AA length = 1020

-continued

FEATURE	Location/Qualifiers	
source	1..1020	
	mol_type = protein	
	organism = Talaromyces funiculosus	
SEQUENCE: 212		
LPFNERVDQV	LRSEYVTSKL	DSRSTKPSKH GHTYQTQFLG VTWDQRNWR
ESRGSIANGY	IGLNVAGAGP	LFELDSPVDG DVINGWPLFS RRQTFAGLAG FYDLQPTNG
TNFPWLQYQ	DDSAISGVPH	WGGMVLDLGD GEYLDATVDN STISDYTTY DYKAGVLSWD
YKWTPKNANG	SFGISYKIFA	NKLDVNQAVV QLSITPSTNG SASVNVVIDG YAAVRTDFVS
SGNESDVVYT	AVKPNGVTNV	TAWIYTALDG DDAFDISSAA LVNDKPYVHQ NDSSIAQSVN
VTFTAGTTIT	INKFVGAAST	DAPFDPQSTA REAALSARRR GFDDLFRSHI SEWAQVMPDD
SVDDFTLANG	TLPNDFPIIE	SAVMAVVPY YLLQNTVGP
TSDSYAGQIF	WDADVMMQPG	LVAAPFESAK RITNYRAAKY SQALENAKTA YTSSQNQTWF
SPDAAIYSWT	SGRVGNCTAT	GPCWDYEHYH NGDIGISLVN EWVVSVDNET FKNKHFPIYN
SIATLYGDL	KKNGSYTTLT	NMTDPDEYAN NVDAGGYTMT LISQTLNAN AFRKQFGMNE
NTTWTMADN	ILLIRENDVT	LEYTTMNSV AVKQADVILS TFPLDYTKNY TTSALNDLD
YYALKQSPDG	PGMTYAIFSI	VANDVSPSGC SAYTYAQYSY DPYIRGPFFQ FSEQLLDDYT
INGGTHPAPF	TLTGHGGANQ	VVLYGYLGLR LLPDDMLHID PNLPPQIPSI KYRTFYWRGW
PIQAASNYTH	PLTIQATSYA	NKSIHVSVDH NTVNSTTYSL SANGSALVVP
NRQIGSINTV	AGNVVQCKSV	LSTDAYQKQY YPISAVDGAA STKWQPEFAA NISSLTVDLT
GSNVSSVSGF	YFDWAQAPPT	NITVLLHNS SAALASSGDK PGSSAVTLNI TISNPYNAST
YNANIILPS	SNSTNYTFPA	PVSKPRYATL FVQGNQALDE TDTKSGNGTG ATVAEWAILS
		1020
SEQ ID NO: 213	moltype = AA length = 1063	
FEATURE	Location/Qualifiers	
source	1..1063	
	mol_type = protein	
	organism = Hamigera avellanea	
SEQUENCE: 213		
ASPKSRINQC	LKKHAGQGS	DEETSSNVYQ TRFPGVTWDE DNWSLTTSVL DQGHYQSRGS
VANGYLGINV	ASVGFFFELD	IPLNGDVING WPLYSRRTQF ATISGFSDSQ PETNGTNFGW
LNQYGGESVI	SGVPHWSGLI	LDLGDGTYLD STVDNATLSG FTSSYDFKAG VLSWSYQWTP
EKGKGSYDIT	YRLPTNKLYV	NQAVVDMEIV PTVAGKASV NVIEGSSAVR TDFVESGEDD
GAIFSAVRWF	GISNVYATFY	ANLTVSDNVD LSSRTLVSNK PYVSTNESSI AQSVVDQFIP
GKSVRITKVF	GAASTDAFAN	PQETAKRAAS TAQTNGYLSK LNSHIAEWAS VMPDDSVEDF
SLPETGKLP	DEHIESAII	SVTNTYLLQ NTVGKNAIKA SSDAALNMDS ISVGGLTSDS
YAGLIFWDAD	IWMQFGLVAS	HPAAAEVFTN YRVAKYPQAV KNIEATAPASS KNQTNFSPSA
AAYPWTSGRY	GNCTGTGPCF	DYQYHLNGDI GLSMINQWV SGGTQTFRK HPPIYDASA
FFSNLVERNG	STWTLTNTMT	PDEYANHIDA GGYTMPLIAE TLLYANSFRK QFGVEPNETW
NBIAENVLVL	RTNGVTLEFT	SMNGSAPVKQ ADVLVVYPL DYNNNYSPED SLNDLDYYAN
KQSEDPAMT	WAIFSVVANE	ASPSGCSAYT YAQYAYPYA RAPFFQLSEQ MIDDASINGG
THPAYPFLTG	HGGANQVNL	GYLGLRLLPD NVIHVDPNLP PQIPHLKYRT FYWRGWPMAS
ASNYTHTTIQ	RAVNVPALST	ADQKFANVSI PVHVGPEHNA TVYRLPVNGT LTIPNRQIAS
KNTVAGNLIQ	CRPVESQND	QPGQFPISV DGASSTRWQP KHADNVSAVT VTFADDEEVS
LVSGFYFDWA	QAPPVDAVI	FHNSSLENPA SAFSPASNS SSESVTITL KNVEQSDPYD
PESDKLDIIA	IPTGNTTKVT	LPSAVPAARY ATLFITGNQA LGPEDIAAKN GTGATVAEWA
IVGQTSSATS	KRSIQTRKLQ	VRSGAALSGL GFAQRRRQSA EMY
		1063
SEQ ID NO: 214	moltype = AA length = 1020	
FEATURE	Location/Qualifiers	
source	1..1020	
	mol_type = protein	
	organism = Talaromyces ruber	
SEQUENCE: 214		
LPFNERVDHV	LRSHDLTSRL	HSRSAPKSNH GGTYQTQFTG VTWDQRNWR
ESRGSIANGY	IGLNVAGAGP	FFELDTAVDG DVINGWPLFS RRQTFAGLAG FYDLQPTTNG
SNFPWLQYQ	DDSVISGVPH	WGLVLVDLGN GEYLDATVDN STISDYTTY DYKAGVLSWH
YKWTPKNANG	SFEIKYKIFA	NKLDVNQAVV QLSITPSANG SASVANVIDG YSAVRTTEFVA
SGNESDAIFT	AVKPVGVSNV	TAWIYAALDG DDAFDSSAT LVNDKPYVHQ NDSSIAQSVN
VTFTAGTTIT	INKFVGAAST	DAPFDPQSTA REAALKARRR GFDDLFRSHI SEWAQVMPDD
SVDDFTLANG	TLPNDFPIIE	SAVMAVVPY YLLQNTVGP
TSDSYAGQVF	WDADVMMQPG	LVAAPFESAK RITNYRTAIY SQALENAKTA YTSSQNQTSTF
SSDAAIYSWT	SGRYGNCTAT	GPCWDYEHYH NGDIGISLVN QWVVSVDNET FKNTHFPIYN
SIATLYGDL	KKNGSYTTLT	NMTDPDEYAN NVDAGGYTMT LISQTLNAN AFRKQFGMDE
NTTWTMADN	VLLIRENDVT	LEYTTMNSV AVKQADVILS TYPLDYTKNY TTSALNDLD
YYALKQSPDG	PGMTYAIFSI	VANDVSPSGC SAYTYAQYSY DPYIRGPFFQ FSEQLLDDYT
ANGGTHPAPF	TLTGHGGANQ	VVLYGYLGLR LLPDNVLHID PNLPPQIPSV KYRTFYWRGW
PIQAASNYTH	PLTIQATSYA	NRSISVQVQ NTVNSTTYSL PANGSAIVVP
NRQIGSINTV	AGNIAQCKSV	LSTDAYQPGQ YPISAVDGAA STKWQPEFAA NISSLTVDLT
SSNASTVSGF	YFDWAQAPPT	NITVLLHNS NAPLTSSNSN GGNSTVSLNV TISNPYDASA
YNANVITLSS	SNSTNYTFPA	PVSKPRYATL FVQGNQALDE TDLKAGNGTG ATVAEWAVALS
		1020
SEQ ID NO: 215	moltype = AA length = 1055	
FEATURE	Location/Qualifiers	
source	1..1055	
	mol_type = protein	
	organism = Trichoderma lixii	
SEQUENCE: 215		
ATSNNRVSEC	LGRNGSGSTG	VHFSKNVYKT DPAVGTWDED NWLLSTTELK QGAPESRGS
		60

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ANGYLGINVA	SVGPFFELDT	EENGDISGWN	PLFSRRQSF	TIAGFWDSQP	VMNGTNFPWI	120
SQYGSDTAIS	GIPHWSGLIL	DLGGNTYLD	TVDNRTISNF	RSTYDYKAGV	LSWSYKWTPT	180
GNKGSFDISY	RIFANKLYVN	QAVVDLQVTA	SKNVEASIVN	VIDGFAAVRT	DFVESGEDGN	240
AIFSAVRPNG	VANVTAYVYA	DITGSGGVLD	SSRKIVHNKP	VVHANASSIA	QAVPVKFSAG	300
RAVRVTKFGV	GASSDAFKNP	KQIAKSAAAT	ALKNGYSKSL	NSHVTEWAAV	MPRESSVDSFA	360
DPKTGKLPND	NYIIDSALIA	VVNTYYLLQN	TVGKNGSKAA	NGAPVNVDSI	SVGGLTSDSY	420
AGQVFWADAL	WMQPGLLAH	PEAERIINY	RLARYGQAKE	NVKTSYAGSQ	NETFFSASAA	480
VFPWTSGRYG	NCTATGPCWD	YEYHLNGDIG	LALVNQWVNV	GDTKDFEKNL	FPVYDSIAQL	540
YGNLLKPNGT	AWTLTNMTDP	DEYANHVDAG	GYTMPPIAET	LQNANTFRKQ	FGIEQNKTWN	600
DMASNALVLR	ENGVTLEFTT	MNGSAVVKQA	DVIMVTFPLS	YTTNYTTEDA	LNDLDYYANK	660
QSPDGPAMTY	AFFSIVANEI	SPSGCSAYTY	AQYAFKPYVR	APFYQLSEQI	IDDSSINGGT	720
HPAYPPLTGH	GGANQVVLFG	YLGRLVLPDD	FIHIEPNLPP	QIPYLRRTF	YWRGWPIASW	780
SNYTHTTISR	ASGVAALDGA	DQRFAGKTIT	IHSGPEESPK	AYHLPVKGSV	VVPNKQIGSQ	840
QTYAGNLVQC	HAASSPDSYV	PGQFPAAVD	GATSTKWQPA	SADKLSSITV	SLDQEDVGSQ	900
VSGFHFDDWAQ	APPVNAIIIF	HNEAINDPAT	VLKSQKHNSN	YKVVSSLTNI	KQSNPYIKTT	960
DLDIATPIG	NTTNVTLSPQ	VAASRYASLV	IVGNQGLDHA	DVVGKNGTGA	TVAEWAIIGH	1020
SKGHTGAPGH	HGKRKLNLRA	AAAMSDPDSF	ARRRQ			1055

SEQ ID NO: 216 moltype = AA length = 1056
 FEATURE Location/Qualifiers
 source 1..1056
 mol_type = protein
 organism = Aspergillus cervinus

SEQUENCE: 216						
STFSHKNDRI	LKGLKRHGDH	YSRKSNTNST	DVYQTKFDGV	TWDDDNWLLT	TTALDQGDPR	60
SRGSIANGYL	GINVASVGPF	FELDTAENG	VISGWPLFSR	RQTFATIAGF	FDSQPTTNGS	120
NFPWLYQYGG	DSVTSIGVPHW	SGVLVDLGN	TYLDSITVDN	TIENFTSMYD	YKSGVLSWSY	180
TWVPAGNKG	FDIVYRLFAH	KLNVNQAVVD	MEITPSLSFN	ATVNVILDGY	SAVRTDFVES	240
GNDNGAIFTA	VRPWGISNVT	AYVYANLTGT	PNVDLSSRTI	VANKPYVHTN	ASSIAQAVNV	300
SCSPNETVRI	TKFVGAASSD	AFENPQETAR	QAVSAAIGAG	YLRSLGTHIA	EWASIFPDDS	360
VDRFTDPATG	KLPANKHIIN	SALIAVTNTY	YLLQNTVGKN	AIQAVSGAPV	NIDSISVGG	420
TSDSYAGQVF	WDADVWMQPG	LAASHPEAAQ	RITNFRAAKY	PQALANIETA	FTSSKNQTSF	480
SPSAAAFPTW	SGRFGNCTAT	GPCWDYEHY	NGDIGLSL	QWIASGDTQT	FRETHFPIYD	540
SIATMYSNIV	ERNGSYWTLL	NMTDPPDEYAN	QIDAGGFTMP	LISQTLN	AFRQQFGLDV	600
NQWSEIANN	VLVLNDNGVT	LEYTTMNGST	VVKQADVVDL	TYPLVYSNNY	TSQNSLDDLD	660
YYANQQSPDG	PAMTWAIFSI	VANDVSPSGC	SAFTYHQYSY	DPYARAPFFQ	LSEQLIDDA	720
TNGGTHPAFP	PLTGHGGANQ	VVIFGYLGLR	LLPDEAIHID	PNLPPQIPHV	AYRTFYWRGW	780
PISAQSNSTH	TTISRAMNAS	PLDTADSRFA	NVSIPIYVGT	ESNATVFPQLP	PTGPLTILNR	840
QNGFNNTIPG	NVAQCRPVYS	PDDYAPGQFP	IAAVDGATST	RWQPSANTS	SVTVTLPTDT	900
INSPVSGFVF	NWQQLPVYNA	TVIFHDDLLE	NPAATISSSG	NSSSYRVVMT	LTNIQQSSPY	960
NAQIAALDEI	TIPTGNTTIV	QLTNHAQTSR	YATLLISGNQ	GLGDTQDGVG	ATVAEWVILG	1020
QGGSSSSNS	NGKRKLGA	AAALSNGWTE	RRRRLI			1056

SEQ ID NO: 217 moltype = AA length = 1053
 FEATURE Location/Qualifiers
 source 1..1053
 mol_type = protein
 organism = Rasamsonia brevistipitata

SEQUENCE: 217						
LRSEARVAQV	VRAYSSSTGV	EAGHGNTATQY	ETRFPQVTD	QRHWRLKSTV	LDQGHFQSRG	60
STANGYVGIN	VASAGPFPEL	DTPVDGDIVN	GWPLFSRRQT	FATIAGFYDE	QPRTNNGSNF	120
WLYQDGESEV	ISGVPHWGL	ILDLGDGYL	DATVDDSTIS	DYSTVYDYKA	GILSWSYKWT	180
PKSSKGSFKI	SYRLFAHKL	INQAVVRMEI	TPSADTDATV	VNVLDGYSVA	RTDFVSGSKD	240
GDAVYSAVSP	WEVQNVYAYV	YAVLDGSDGV	DLSTLSLVNG	KPYVHTNESS	IAQSVNVRFR	300
AGKTVTVTKF	VGAASTDAFP	DPQQTAREAA	LAAKEGYDA	LLRSHVAEWA	AVMPESVD	360
FTYHNGTLP	DDFIVESAIM	AVVNPYYLLQ	NTVGENALRE	VSHAPVNEWS	ISVGGLTSDS	420
YAGLIFWDA	LWMHPLGLA	FPQAATRI	YRVAKYEQAR	QNAKTSPTGS	KNQTFWSDSA	480
AVYPWTSGRY	GNCCTGTGCPW	DYEYHLNGDI	GLSLINEWVA	SGDTKTFQES	YFPIYDSIAT	540
LYADLLQQNG	SHWTLTNMTD	PDEYANAVDA	GGYTMPLIAQ	TLLYANSFRQ	QFGAQPNSTW	600
TEMASNILFL	RENDITLEYT	TMNNSVQVKQ	ADVVLVITYPL	EYTTNYNAGN	ALTDLDYYAL	660
KQSPDGPAMT	YAFISIVANE	VSPSGCSVYT	YAQYSYDPYV	RPPFFQLSEQ	LVDDYTLNGG	720
THPAYPPLTG	HGGANQVVF	GYLGLRLLPD	NVIHIDPNLP	PQIPQVKYRT	FYWRGWPIQA	780
YSNYTHTTIG	RAADVLA	ADQRFANTTI	PVQVSGSNA	TVYQLPIDGV	LTVANRQVAS	840
TNTVAGNLAQ	CQPDVSSDSY	VPGQYPLAAV	DGAASTKWQP	SFAANVSSVT	VSLPESESGT	900
LVSIFYFDWA	QAPPVNAVTV	FHNNTVENPT	RYMSSPTFVT	HIDNITLSSP	YNAEANPAAT	960
ILLPSSNTTN	VTLAHPMPVP	RYATLFTITGN	QALSESEVQA	QNGTGATVAE	WAILASNPET	1020
GSSKRNLQLR	GVGRSALAGL	PRRANRKERR	TDV			1053

SEQ ID NO: 218 moltype = AA length = 1034
 FEATURE Location/Qualifiers
 source 1..1034
 mol_type = protein
 organism = Acremonium curvulum

SEQUENCE: 218						
LEERVSTLN	RHGVRSYNNR	RADNGDQNR	SNTSHPIYQ	TSFEGVTWDS	RNWRLLQGTVL	60
DQGHYQSRGS	IANGYFGIN	ASAGPFPEL	TPVDGDIVNG	WPLFSRRQT	AGLAGFWATQ	120
PTTNGTNFPW	LYQYGEDSPI	SGIPHWGGLV	LDLGDVYLD	ATVDNKTIKN	YRTTFDYKAG	180
VLTWDYTWSP	KRSGSGSFDI	YTSFLFANKL	INQAAVRLSI	RPSRDTKAKV	VNVLEGYAAV	240
RTDFVDSGKD	GDAISAVRP	GVVNVYAYV	YAVLDADSAV	DLSSAKVIDD	APVLYTNKST	300

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IAQSVNVEFK	ANQNVITITK	VGIASDADF	KPRDVAQAA	FAGKRRGYDD	ALRSHVSEWA	360
QVMPDSDVD	FSSDNGTLPD	DGFIIESSIM	AVVNPFFYLLQ	NTVGANALRR	VNNAAVNDYS	420
ISVGGLTSDS	YAGLVFWDAD	IWMQPGGLAAA	FPEAAQRITN	YRVALYPQAK	RNIKTAFQSS	480
KNKTRFSDDA	ALYPWTSGRW	GNCTASGPCF	DYEHHLNGDI	GIAFVNEWIT	SGDEKAFEEK	540
YFPIYDSIAT	AFANLLQKNG	TQWTLTNMTD	PDEYANHVD	GGFTMPLISQ	TLTYANLFRK	600
KFGKEENDTW	ADMAENVLIL	RENDVTLEYT	AMNNSVEVKQ	ADVVLNTFPL	DYTRDYAPSA	660
ALNDLDYYAL	KQSPDGPAMT	YAFISIVANE	VSPSGCSAYT	YAQYSYSPYL	RGPFHQLSEQ	720
LTDDFTTNGG	THPAYPFLTG	HGGANQVVL	GYLGLRIVPD	DKIHVDPNLP	PQIPQVKYRT	780
FYWHGWPIAA	KSNTYHTTIS	RATTIKELDT	ANKKYANASI	QVVVSGSKSA	KTYKLPANGS	840
SIVVANRKIG	TVNTLEGNMI	QCQPAQSFD	FVPGQFPISI	NDGAASTKWQ	PEFANNISAV	900
TVTVPASKSK	KISGFYFDWA	QAPPTNATVV	LHDEKMDNPT	MVLLPVSCKDD	DKKGSVARVN	960
VTISEPWSAK	DKSNFVIGLQ	GGTNTNFTFS	EPVAAKRYAT	LFIQGNQALD	KVDIKYKNGT	1020
GATVAEWGIL	SDDA					1034

SEQ ID NO: 219 moltype = AA length = 1015
 FEATURE Location/Qualifiers
 source 1..1015
 mol_type = protein
 organism = Talaromyces piceae

SEQUENCE: 219

ARVDQVLQAR	DTHHPSLHSS	SSYQTRFDGV	TWDQRNWRLQ	SRVLDQGHYQ	SRGVSANGYL	60
GINVASAGFF	FELDTFVDGD	VINGWPLFSR	RQTFAGLAGF	YDRQPTTNST	NYGWLNQYGD	120
ESVISGIPHW	SGLVLDLGDG	HYLDATVDSS	TISDYTTTTYD	FKAGVLSWDY	NWAPRQHGGG	180
NSSPHISYQL	FAHKLIDINQA	VVKLSITPSA	SGNASVNVVI	DGYSAVRTEF	VKSGTDGDAV	240
YSAVSPVGV	NMTAVVTVL	DGDEAFNLSS	AQLVTGKPYV	YQNDSSIAQS	VNVEFRAGET	300
VTITKPVGAA	STDAFADPRQ	TARDAALLAK	KKGFDDLLRS	HVSEWQVMP	DGSVDDFTRE	360
DGTLDPDEFI	IESAVTAVVN	PFYLLQNTVG	KNALQVRVNA	PVNDWSIPVG	GLSSDSYAGL	420
IFWDADVWMQ	PGLVAAPFES	AQRITNVRAA	MYRQARANIQ	SAFASSQNK	VFPDGAIIYP	480
WTSGRYGNCT	GTGPCFDY	HLNGDIAISL	VNQWVVS	ETFKNEHPPI	YDSIATMYAD	540
VLKKNGSFYT	LTNMTDPDEY	ANNKDAAGGFT	MPLIAKTLN	ANDFRKQFNM	EEENSTWNEKA	600
ASVQLLREND	VTLEYTTMNN	SVAVKQADVV	LMTFPLDYTA	NYSSSSALND	LDYYALKQSP	660
DGPGMTYAI	SIVANQVSPS	GCSAYTYAQY	SYYPYARAPF	FQLSEQLLDD	YTANGGTHPA	720
YPFLTGHGGA	NQVVLGYGL	LRLVPDETLY	IDPNLPPQIP	QVKYRTFYWR	GWPIQAASNY	780
THHTTIRRA	TAPLDTADEK	YTDEAITLHV	GQQSDGNKYQ	LPADGTPVT	ANRRVSSVNT	840
VKGNMIQCQP	VQSTASIQPG	QFPAAVDGA	ASTKWQPEFA	ANASSLTVEI	PSSGKKRKTV	900
SGFFPDWAEA	PPTNATVVLH	NNPVSEPTLS	SFEAGKDG	VRAPDIEISS	PYNASTYADE	960
IIIPQGNSTN	YTFSHVPFAP	RFATLFVQGN	QALDKVDVEN	KNGTGPMAE	WVILE	1015

SEQ ID NO: 220 moltype = AA length = 1022
 FEATURE Location/Qualifiers
 source 1..1022
 mol_type = protein
 organism = Penicillium sp.

SEQUENCE: 220

LPLEERVDRV	LRYSVSGSGL	EARHSKYTYQ	TQFDGVTWDQ	QNWRLSTVL	DQGHYSSRGS	60
IANGYIGLMV	AGAGPFFELD	APVDGDVING	WPLFSRRQTF	AGLGGFYDVQ	PTTNGSNYPW	120
LDQYGYDSVI	SGIPHWGGLV	LDLNGDYL	ATVDNSTISD	YTTTFDYKAG	VLSWDYNWTP	180
KNTSPGISYK	IFSKLDINQ	AVVQLSITPS	ANGTASVANV	IDGYAAVRTE	FVTSNGNDSA	240
LFTAVKPTGI	NNVTAWIYAV	LDGDDAFDFS	SATLVNKNKY	INQNDSSIAQ	AVDVEFTADS	300
TVTITKPVGA	ASTDAFADPQ	KTAKDAALAA	RIKGFEDLLR	SHVSEWQVMP	PDDSVDDFSL	360
ADGTLDPDDI	IESVMAVAV	NPYLLQNTV	GENALRRVND	APVNIWSIPV	GGLTSDSYAG	420
QIFWDADLWM	QPGVAAAPF	SAKRISNYRV	AKYPEALANT	NTSFAGSQNH	TTFSSDAIY	480
SWTSGRYGN	TATGPCWDYE	YHLNGDIGIS	LVNQWVTS	TETQNDLFP	IYNSVATLYA	540
LDLLKNGSY	TLTNMTDPDE	YANNVDAGGY	TMTLISKTL	NANAFRKHFG	LDQNSTWT	600
AENVLVIREN	DVTLEYTTM	NSVAVKQADV	VLSTFPLDY	MNYTTSDAVN	DLDDYALKQA	660
SDGPGMTYAI	FSIVADKVS	SGCSAYTYAQ	YSFDPYIRSP	FFQFSEQLDD	DYTTNGGTHP	720
APFPLTGHG	ANQVVLGYL	GLRLLPDNV	HIDPNLPPQI	PSVKYRTFYW	HGWPIQANSN	780
YTHHTTIRRA	TAPLDTADQ	TYSNTSITVQ	VGTSNGTYS	LRVDGTPLTI	KNRRIGSVQT	840
ISGNIAQCQP	VQSPDSYEPG	QYPLSVVDGA	SSTKWQPESA	ANISSVTVT	GSSYSPGNSS	900
NATASSILGF	YFNWAQAPPD	NITVVLHNAS	IPYLNASLSS	LSNTTTTTS	NITVSNPYDS	960
SADEKIIIVL	SSNTNTNYTF	SPVPVPRFAT	LFIQGNQGLD	QTDLEYGN	GATVAEWAIL	1020
SA						1022

SEQ ID NO: 221 moltype = AA length = 1020
 FEATURE Location/Qualifiers
 source 1..1020
 mol_type = protein
 organism = Talaromyces aurantiacus

SEQUENCE: 221

LFPQDRVDQV	LRYSYAKTLD	IRSAKSSSKH	GNTYETQPPG	VTWDQRNWRL	QSTVLDQGHY	60
ESRGSIANGY	IGLVNAGAGP	FFELDTFVDG	DVINGWPLFS	RRQTFAGLAG	FYDLQPTTNG	120
SNFPWLQYQ	DDSAISGVPH	WGGLVLDLGD	GQYLDATVDN	STVSDYKTTY	DFKAGVLSWD	180
YKWTPKSSNV	SFGISYKVF	NKLDVNQAVV	QLSITPSANG	SASVNVVIDG	YSAVRTDFVS	240
SGNESDAIYT	AVKPLGVS	TAWIYATLDG	DDAFDFSSAT	IVNKNKYVHQ	NDSSIAQSVN	300
VTFTAGTTVT	INKFVGAAS	DAFPDPRSTA	KEAALAGR	GYDDSFRAHI	SEWQVMPDD	360
SVDDFTLANG	TLPNDTFIE	SAMAVVNPY	YLLQNTVGPN	ALRRVNNAPV	NDWSISVGGL	420
TSDSYAGQVF	WDADVWMQPG	LVAAPFESAK	RITNYRAAIY	SQALANAKTA	YTSSQNQTSF	480
SSDAIISWT	SGRYGNCTAT	GPCWDYEH	NGDIGISLVN	QWVSGDNET	FKNTHFPIYN	540
SIATLYGDL	KKNGSYTTL	NMTDPDEYAN	NVDAGGYTMT	LISQTLNAN	AFRKQFGMD	600

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NTTWTEMADN	VLLIRENDIT	LEYTTMNSV	AVKQADVLS	TYPLDYTKNY	TTSAALNDLD	660
YYALKQSPDG	PGMTYAIFSI	VANDVSPSGC	SAYTYAQYSY	DPYIRGPFFQ	FSEQLDDYT	720
ANGGTHPAFP	FLTGHGGANQ	VVLYGYLGLR	LVPDDKLHID	PNLPPQIPSV	KYRTFYWRGW	780
PIQAASNYTH	TTIQRAITVA	PLSTADMTYA	NKSISVQVGQ	NTANSTTYSL	PVNGSALVIS	840
NRQIGSINTV	QGNIAQCKSV	QSMNGYQPGQ	YPISAVDGAA	STKWQPEFAA	NVSSLTVDLT	900
SSNASSVSGF	YFDWAQAPPV	NVTVVLHNST	SASLTSSAAQ	NGSSTVSLNI	TISNPYNASS	960
YDANVIELSS	SNTTNYRPPA	PVPKPRYATL	FVQGSQALDE	TDMKAGNGTG	ATVAEWAILS	1020

SEQ ID NO: 222 moltype = AA length = 1016
 FEATURE Location/Qualifiers
 source 1..1016
 mol_type = protein
 organism = Talaromyces pinophilus

SEQUENCE: 222

LPFNERVDQV	LRSSYSPKNLE	SRSTKHGNSY	QTQFSGVTWD	QRNWRLQSTV	LDQGHYESRG	60
SIANGYIGLN	VAGAGPFPEL	DTAVDGDVIN	GWPLFSRRQT	FAGLAGFYDL	QPTTNGSNFP	120
WLSQYGDDSA	ISGVPHWGGL	ILDLDGGEYL	DATVDNSTIS	DYTTTYDYKA	GVLSWDYKWT	180
PKNSKASFGI	NYKIFANKLD	VNQAVVQLSI	TPSANGSGSV	VNVIDGYSAV	RTDFVSSGNE	240
SDVIYTAVKP	VGNNVNTAWI	YAALDGDDEAF	DFSSAELVND	KPYVHQNDSS	IAQSVNVVAF	300
AGTTITINKF	VGAASTDAFP	DPQSTAREAA	MTARRRGFDD	LFRSHVSEWA	QVMPDSDVDD	360
FTLANGTLPN	DTFIIESAVM	AVVNPYYLLQ	NTVGANALRR	VNNAPVNDWS	IPVGGLTSDS	420
YAGQIFWDAD	VWMQPLVAA	FPESAKRITN	YRTAKYSQAL	ENAKTAYTSS	QNQTSFSSDA	480
AIYSWTSGRY	GNCTATGPCW	DYEHHLNGDI	GISLVNQWV	SGDNETFKNT	HPPIYNSIAT	540
LYGDLKKNG	SYTTLNMTD	PDEYANNVDA	GGYTMTLISQ	TLNANAFRKL	QFGMDENTTW	600
TEMADNILLI	RENDVTLEYT	TMNNSVAVKQ	ADVILSTFPL	DYTKNYTSTA	ALNDLDYYAL	660
KQSPDGPGMT	YAIFSIIVAND	VSPSGCSAYT	YAQYSYDPYI	RGPFFQFSEQ	LLDDYTANGG	720
THPAPFPLTG	HGGANQVVLV	GYLGLRLLPD	DMLHIDPNLP	PQIPSVKYRT	FYWRGWPIQA	780
ASNTHHTTIQ	RENDVTLEYT	ADPAYANTSI	SVSVGQNTAN	STTYSLPVNG	SAIVVPNRQI	840
GSINTVAGNI	AQCQSVLSTD	AYQPGQYPIS	AVDGAASTKW	QPEFAANVSS	LTVDLTSSNA	900
SSVSGFYFDW	AQAPPTNITV	LLHNSSSAAL	TSSSTHGGSS	SVSLNITISN	PYDASSYDAN	960
VIALSSSNTT	NYTFSAPVAK	PRYATLFLVQ	NQALDEDTDK	AGNGTGATVA	EWAILS	1016

SEQ ID NO: 223 moltype = AA length = 1069
 FEATURE Location/Qualifiers
 source 1..1069
 mol_type = protein
 organism = Talaromyces leycettanus

SEQUENCE: 223

TSANARINRC	VKKHAGGKTP	SGPSNNYQYT	RFPQVTWDQD	NWCLSTTTL	DQGHYESRGSV	60
ANGYLGIVNA	VGPFPEFDEL	PVDGDVINGW	PLFDRRMSFA	TISGFWDQOP	TTNGSNFPWL	120
YQYGESVIS	GVPHWGLIL	DLGDNTYLD	TVDSRTISGF	STTYDFKSGV	LSWSYQWTPA	180
GMNGSYNTTY	RLFAHKLIVN	QAVVDMVVS	STEAKATVNV	VIDGASAVRT	DFVESQDDG	240
AIYTAVRPWG	IANVTAYIYA	NITGSDNVDM	RSRALVTNKP	YVNGNASSIT	QAVNVHFTPG	300
KSVRITKPVG	GASSDAFSNP	QQIAKQACST	AQANGYVKSL	RSHVAEWASV	MPDSDVDDFT	360
PPSNGTLPAD	EYIIIESQIIS	VANTYYLLQN	TVGKNAINAS	SSTELNKDSI	AVGGLTSESY	420
AGMIFWDADV	WMQPLVASH	PEAAQRITNY	RVAKYPQAKA	NVATAYQSSK	NQTNFSPDAA	480
VYSWTSARYG	NCTATGPCWD	YEYHLNGDIG	LSIINQYVAS	GDTQTFKEKL	FPVFDVATL	540
YSNIVQKNGS	SWTLTNMTDP	DEYANQVDAG	GYTMPLIAQT	LLYANSFRQQ	FGLTENDTNW	600
ETAAQDVLVIR	ENGVTLEFTT	MNGSAVVKQA	DVVLDTYPLG	YTHNYGPTDA	LNDLDYYANR	660
QSPDGPAMTW	AIFSIVANQI	SPSGCSAYTY	AQYAFSPYAR	APFYQLSEQL	IDDASLNGGT	720
HPAYPPLTGH	GGALQVNLFG	YLGFRYLPDN	VIHIDPNLPP	QIPHIYRTFT	YWRGWPIATA	780
STYTHHTLRS	AWNVSLLDSA	DPKAFANASIP	VHVGLSNVNT	VYRLPVNGTL	TVPNRMVSGK	840
NTLAGNMVQC	RPVQSMQGYQ	PGQFPISVVD	GASSTKWQPL	YSANVSSVTV	TLSSSAVGKS	900
VNGFYFDWAQ	NPPVNAAVVF	HNSSPAQNPA	TTFSPDNPSA	SGNLYSVSVS	LKDIQLSDPY	960
DPATDLDLVI	AIPKGNITNV	TLSSPVPAAR	YATLFIQGNQ	ANSPAEEAAK	NGTGATVAEW	1020
AILGQEVQNN	GYGDQIEARR	LDVRGAAALS	GMGSFTQRRK	RKMILPRFD		1069

SEQ ID NO: 224 moltype = AA length = 1016
 FEATURE Location/Qualifiers
 source 1..1016
 mol_type = protein
 organism = Talaromyces variabilis

SEQUENCE: 224

ALSPGARVNO	VLAHNSLPP	SLTDNSTGSS	YKTRFDGVTW	DQRNWRLQSR	VLDQGHYEAR	60
GSVANGYIGI	NVASAGPFPE	LDTVPDGDVI	NGWPLFSRRQ	TFAGLAGFYD	LQPTTNSNTY	120
GWLNQYGDSE	VISGIPHWAG	LVLDLGNGDY	LDATVDNSTI	SDYTTTYDYK	AGILSWDYKW	180
TPRQNSSSFR	ISYQLFANK	DINQAVVKLS	ITPSKSGNAS	VNVVIDGYSA	VRTDFVKS	240
DGNAIYTAVS	PVGSVNTAW	VYAVLDGDKA	FDLSSPSRVT	GKPYIHQNES	SIAQAVNVEF	300
SAGKTVTIAK	FVGAASDAF	SNPQKKAKTA	ALDGKKKGFE	DLLRSHVSEW	AQVMPDSDV	360
DFTLANGTLP	DDSFIEQAV	TAVVNPYYLL	QNTVGKNALQ	RVNNAAPVNDW	SIPVGGLSSD	420
SYAGMIFWDA	DVWMQPLVLA	APFESAKRIT	NYRAAKYQQA	IANVKTAYSS	SQNETVFSNP	480
AAIFPWTSGR	YGNCTGTGPC	WDYEHHLNGD	IAISLVNQWV	VSGDTETFPQ	EHFPIYNSIA	540
TVYGDLLKKN	GSYYTLNMT	DPDEYANNKD	AGGYTMPLIA	NTLIKANEFR	RQFGMAENAT	600
WNEMAENLVI	IRENDVTLEY	TTMNNSVAVK	QADVVLRTFP	LDYTTNYSSS	AALNDLDYYA	660
LKQSPDGPGM	TYAIFSIVAN	EVSPSGCSAY	TYAQYSYDPY	ARAPFFQLSE	QLIDDYTTNG	720
GTHPAYPFLT	GHGGANQVVL	YGYLGLRLVA	DEILHIDPNL	PPQIPQVYTR	TFYWRGWPIQ	780
AASNYTHTTI	HRATTVAPLD	SAEKKYAKSA	ISVQVQQQAN	STTYKLPADG	TPLTVANRKY	840
GSTNTIKGNI	AQCQSVQSV	SVQPGQYALA	AVDGAASTKW	QPEFAANVSS	LTVSIPSGKT	900
SVSGFYFDWA	QAPPSNATVV	FHNNSVSNT	FSSFGSRKGA	RITHIDVKLS	NPYNASSNAD	960

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AIVIPSGNTT NTFNSNPVPA PRFATLFPVQG NQGLDKVDVQ NGNGTGATVA EWAILE 1016

SEQ ID NO: 225 moltype = AA length = 1050
 FEATURE Location/Qualifiers
 source 1..1050
 mol_type = protein
 organism = *Aspergillus niger*

SEQUENCE: 225
 LPGKNARISA SLKRHAGRDV PQTALNSTNV YQTKFSGVTW DEDHWLLTTT TPDQGHYQSR 60
 GSVANGYLG I NVANIGPFFE LDEPVNGDVI NGWPLYSRQ SFATISGFWD RQHTNGSNF 120
 PWLSQYGDD S VISGVPWHS G LILDLGDDTY LDATVDNRTI SNFKSTYDFK SGVLSWSYTW 180
 TPQGNKGSYA ITYRLFAHKL YVNRAVVDME ITPLTNGNAT VVNVLDGYAA VRTDFVASGQ 240
 EEGAIFSAVR PWGVNNVTAY VYATLDGSDS VDLSSRRIVT DKPVVSTNSS SVAQAVDVMF 300
 TANETVRITK FVGGAATDYF LATQETAKAA CLAGLADGYV KSLQSHVGEW ATIMHDHSDV 360
 RFTDPATGKL PEDSHIVDSA IIAVTNTYYL LQNTAGTNAI VAAGGIPNVV DSCAPGGLTS 420
 DSYGGQIFWD ADLWMQPLV ASHPESAQR F TNRYIALHYQ AQANIETAFT GSKNQTSFSS 480
 SAAIYPWTSG RFGNCTATGP CWDYQYHLNG DIGLAMINQW VASGDTAWFK NYLFPIYDAA 540
 ATLYSELVER NGSSWTLTNN TDPDEYANSI NAGGYTMPLI AETLQANKL RKQFGLPNE 600
 TWDEIAEDVL ILRENGVTLE YTSMNGSAVV KQADIVLNTF PLTYESDNYT ATNSLTDLDY 660
 YANKQSDAGP AMTYAIFAIV ASDVSPSGCS AFTYHQYSYA PYARGPWYQL SEQMIDDASI 720
 NGGTHPAPFF LTGHGGANQV ALYGYLGLRL HPDDTIYIDP NLPPQIPHIT YRTFYWHGWP 780
 ISAWSNYTH T TIQRDSSLAP LASADLLFSN VSIKVQVQGS TASADEATY YLPLSGALT V 840
 PNRMIGSVNT TPGNQVQCHP VYSPDAYEPG QPPISAVDGA TSTKWQPS TS DLTSLTVTLS 900
 TTAAGAEV SGGFYFDWSQA PPNLTVIFH DSPIGNPSTV FAAAGSNSTG YRVITSMSNI 960
 VQSKPYNAIS AEELNVVSI P TANTTITLD APVQKARYAT LLIAGNQANE TAGATVAEWV 1020
 ILGQNSTSS SAQAKRKMSA RSKATLAQLS 1050

SEQ ID NO: 226 moltype = AA length = 1056
 FEATURE Location/Qualifiers
 source 1..1056
 mol_type = protein
 organism = *Trichoderma reesei*

SEQUENCE: 226
 TTLVDRVTKC LSRHDGSDAE SHFSKNVYKT DFIGVTWDED NWLLSTTQLK QGAPEARGSV 60
 ANGYLGINVA SVGPFFEVD T EEDGDVISGW PLFSRRQSFA TVAGFWDAQP QMNGTNFPWL 120
 SQYGSDTAIS GIPHWSGLVL DLGGGYLDA TVSNKTISHF RSTYDYKAGV LSWSYKWTPK 180
 GNKGSFDISY RLFANKLHV N QAVVDMQVTA SKNVQASIVN VLDGFAAVRT DFVESGEDGS 240
 AIFAARVPRNG VANVTAYVYA DITGSGGVNL SSRKIVHNKP YVHANASSIA QAVPVKFAAG 300
 RTVRVTKFVG AASSDAFKNP KQVAKKAAAA GLSNGYTKSL KAHVEEWATV MPSSVDSFA 360
 DPKTGKLPA D SHIVDSAI I A VTNITYLLQN TVGKNGIKAV DGAPVNVDSI SVGGLTSDSY 420
 AGQIFWDADL WMQPGLVAH PEAARITNY RLARYGQAKE NVKTAYAGSQ NETFFSASAA 480
 VFPWTSGRYG NCTATGPCWD YEYHLNGDIG ISLVNQWV N GDTKDFEKNL FVPYDVAQL 540
 YGNLLRPNKT SHIVDSAI I A VTNITYLLQN TVGKNGIKAV DGAPVNVDSI SVGGLTSDSY 600
 DMASNVLVLR ENGVTLEFTA MNGTAVVKQA DVIMLTYP LS YGTNYSAQDA LNDLDYYANK 660
 QSPDGPAMTY AFFSIVANEI SPGGSAYTY AQNAFKPYVR APFYQISEQL IDASVNGGT 720
 HPAYPFLTGH GGAHQVLF G YLGLRLVPDD VIHIEPNLPP QIPYLRVTF YWRGWPI SAW 780
 SNYTHHTLSR AAGVALEGA DQRFARKPIT IHAGPEQDPT AYRLPVKGSV VIPNKQIGSQ 840
 QTYAGNLVQC HAASSPNYV PGQFPAAVD GATSTKWQPA SADKVSSITV SLDKEDVGS L 900
 VSGFHDWAQ APPVNATVIF HDEALADPAT ALASAHKHNS KYTTVTSLTN IELSDPYVST 960
 KDLNAIAIPI GNTTNTVLSH PVAASRYASL LIVGNQGLDP VDVKAKNGTG ATVAEWAIFG 1020
 HGKEHSGKPS SHSKRRLNVR TAATLSNPRS FMRRRL 1056

SEQ ID NO: 227 moltype = AA length = 19
 FEATURE Location/Qualifiers
 source 1..19
 mol_type = protein
 organism = *Saccharomyces cerevisiae*

SEQUENCE: 227
 MLSLKTLLCT LLTVSSVLA 19

SEQ ID NO: 228 moltype = DNA length = 185
 FEATURE Location/Qualifiers
 source 1..185
 mol_type = genomic DNA
 organism = *Saccharomyces cerevisiae*

SEQUENCE: 228
 agtgctttta actaagaatt attagcttt tctgcttatt tttcatcat agtttagaac 60
 accttatatt actgaatagt ttatgaatct atttaggttt aaaaattgat acagttttat 120
 aagttacttt tcaaaagact cgtgctgtct attgcataa gcactggaag gggaaaaaaa 180
 aggtg 185

SEQ ID NO: 229 moltype = AA length = 555
 FEATURE Location/Qualifiers
 source 1..555
 mol_type = protein
 organism = *Pycnoporus sanguineus* glucoamylase

SEQUENCE: 229
 QSSAVDAYVA SESPIAQGV LNNIGPNGSK AHGAKAGIV ASPSTENPDY LYTWTRDSSL 60
 VFKLLIDQFT SGDDTSLRGL IDDFTSAEAI LQQVSNPSGT VSTGGLGEPK FNIDETAFTG 120

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SEQUENCE: 234
MQLLRCSIF SVIASVLA                                     18

SEQ ID NO: 235      moltype = DNA length = 47
FEATURE            Location/Qualifiers
misc_feature        1..47
                    note = ARTIFICIAL DNA PRIMER
source              1..47
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 235
atgatgaaaa aataagcaga aaagactaat aattcttagt taaaagc      47

SEQ ID NO: 236      moltype = DNA length = 26
FEATURE            Location/Qualifiers
misc_feature        1..26
                    note = ARTIFICIAL DNA PRIMER
source              1..26
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 236
atgctttcgc ttaaaacggt actgtg                             26

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The invention claimed is:

1. A method of producing a fermentation product from a starch-containing or cellulosic-containing material comprising:

- (a) saccharifying the starch-containing or cellulosic-containing material; and
- (b) fermenting the saccharified material of step (a) with a recombinant yeast cell;

wherein the recombinant yeast cell comprises a heterologous polynucleotide encoding a trehalase having a mature polypeptide sequence with at least 95% sequence identity to the amino acid sequence of SEQ ID NO: 223.

2. The method of claim 1, wherein the trehalase has a mature polypeptide sequence comprising or consisting of the amino acid sequence of SEQ ID NO: 223.

3. The method of claim 1, wherein saccharification of step (a) occurs on a starch-containing material.

4. The method of claim 3, comprising liquefying the starch-containing material by contacting the material with an alpha-amylase prior to saccharification.

5. The method of claim 4, wherein liquefying the starch-containing material and/or saccharifying the starch-containing material is conducted in presence of exogenously added protease.

6. The method of claim 1, wherein fermentation and saccharification are performed simultaneously in a simultaneous saccharification and fermentation (SSF).

7. The method of claim 1, comprising recovering the fermentation product from the fermentation.

8. The method of claim 1, wherein the fermentation product is ethanol.

9. The method of claim 1, wherein the recombinant yeast cell comprises a heterologous polynucleotide encoding a glucoamylase or a protease.

10. The method of claim 1, wherein the recombinant yeast cell is a *Saccharomyces cerevisiae* cell.

11. The method of claim 1, wherein the trehalase has a mature polypeptide sequence with at least 97% sequence identity to the amino acid sequence of SEQ ID NO: 223.

12. The method of claim 1, wherein the trehalase has a mature polypeptide sequence with at least 98% sequence identity to the amino acid sequence of SEQ ID NO: 223.

13. The method of claim 1, wherein the trehalase has a mature polypeptide sequence with at least 99% sequence identity to the amino acid sequence of SEQ ID NO: 223.

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